# Technical Appendix:
# Auditing Controller Replacement after Environment-Model Change

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## Reader guide to the main argument

The main paper audits candidate exclusion in replacement planning. Its Fig. 1 distinguishes source strict and source-kind questions, encoded strict winning, public action and fixed raw solver calls. The main argument proceeds from the basic strict theorem and corollary proofs to Patrol's positive/negative argument, then the retained-duty proposition with its full proof and application to the actual export. Source-kind, case evidence and the conditioned cohort follow, before the exclusion contract and three saved-evidence checks. Use this supplement to inspect individual obligations; historical section labels remain stable.

| I want to check... | Start here |
|---|---|
| The preservation theorem | L.6 gives the complete relations, game-based definition of $W_C$, theorem/corollary proofs and corrected-guard proof. L.1 gives the retained-duty extension, its safety fixed point and example with two source states, also summarized after the main paper's complete proposition proof. |
| The Patrol continuation | S and U.1 give the exact source/rules, positive witness and export binding; AB.6 gives the route. U.6 gives the reduced calculation. AD.4–AD.6 give the same-entry all-history completion; U.5/U.8 retain full-helper UNKNOWNs. |
| The implementation comparisons | Vehicle: E/X and AD.3 for source proof, regions, initial quantifiers and target-specific Kind witnesses. Obstacle: AC.1–AC.4 for strict regions, AC.5–AC.7 for positive witnesses/public controls, AE for path scope, AG for the completed raw pair and AF for its distinct history. |
| The reader tasks | Vehicle guide: saved loss membership and target predecessors. Patrol guide: positive policy, all-response countercertificate and typed export premises. Obstacle source and completed-pair guides: source certificate, exact query, sole typed $0 \to 1$ SYS release and false/true records (AC.7, AG). |

These guides refer to the accompanying release [18]; its paper reading map identifies the matching main paper, supplement and document source ZIP. The tasks check saved evidence within each guide's stated scope; document compilation does not rerun scientific computations.

**Definitions retained in the technical sections.** The main paper now defines $W_C$ directly through one source strict-winning strategy that maintains $C$ at every induced source-legal finite prefix, including the initial state, with its evolving memory. Section L.6 retains the equivalent $C \cap \mathrm{Win}_{\mathrm{str}}(M_C)$ formulation: $M_C$ adds $C$ at both ends of each legal transition, while the outer $C$ covers the length-zero prefix and excludes outside-$C$ vacuous dead-end winners. Replacing $C$ by $F$ gives the corresponding $W_F$ definition. The initial-query quantifiers are in L.6; the all-history Kind quantifiers and sufficient source-witness constructions are in AC.5 and AD.1–AD.3. AD.2 also retains the general fair-ENV-selector lemma and full proof moved from the main paper. The

main summarizes Patrol's all-history completion and Vehicle's per-target reachability; their full constructions remain in AD.4–AD.6 and X/AD.3. AC.7 gives the phase/rank certificate obligations now summarized for Obstacle B.

For statewise strict queries over $D_{\text{query}} \subseteq D$, preserving each queried source winner requires $W \cap D_{\text{query}} \subseteq W_R(\cdot, 0)$. The right side is $W_C$ or $W_F$ under the applicable hypotheses; a strategy may leave $D_{\text{query}}$ and must continue to maintain its response option. Initialized strict synthesis instead requires equality of the Boolean decisions $I_M(W)$ and $I_M(W_R(\cdot, 0))$, with unchanged initializers and initial $b = 0$. Section L.6 defines $I_M$ by $\forall$ENV$\exists$SYS: for an initially realizable source, each allowed ENV initial value needs at least one allowed surviving SYS response. Preservation of every winning initial response, or of the whole source-winning region, is sufficient but not generally necessary. A sound negative oracle also needs the stated query/objective correspondence. This is a reasoning contract, not an implemented online gate.

**Unfavorable results and history.** J and AB.4–AB.5 retain the stopped 59-model selection, outcomes and shared dependencies; Y and AA retain separate controlled comparisons. K and the archive directory below locate saved records and their checking scope. All historical UNKNOWNs remain. A and L.2 preserve the earlier normalized cooperative derivation; those results are not prerequisites silently added to the main strict-preservation theorem.

**Release and archive scope.** This supplement supplies proofs, exact source bindings, saved proof objects and bounded software studies. The accompanying document and evidence release is `controller-replacement-audit-20261026-v8`. Its paper reading map identifies the matching PDFs and document source ZIP, which rebuilds the documents without rerunning scientific computations. AF and AG identify separate experimental archives and their checking scope; those archives are not part of the document source ZIP.

The repository's `paper/`, `results/`, `tool/`, `reproduce/` and `docs/` directories organize the paper, evidence, code, scoped checks and version guidance. Bundles are in `results/archives/`, with guides in `results/guides/`. The five added packages below preserve earlier scientific results and UNKNOWNs; the fifth adds the positive source certificates and matched ordinary-mode controls in AC.7. The existing continuation release contains selected Patrol, manager, reduced-query and fixed-lasso records; it does not distribute the complete local Section U archive. The full contract-contrast, equivalent-old-source and U.4 runtime records retain their local scope. Section K retains the historical archive map.

Fixed-version paths below are relative to `https://github.com/research-artifact-archive/seams2027` at their stated immutable tags. Earlier evidence tags and layouts remain unchanged. Archive names are literal; fresh output directories must be new or empty as each entrypoint requires.

**Archive starting points.** Patrol's continuation release contains saved semantic replay for the Patrol, reduced-query, manager and fixed-lasso records (U–V), without new native execution. The separate Vehicle188 package supplies saved full-region membership and initialization checks (E), without a native solver rerun. Entry 1 supplies saved source and arena checks for Guard and certificates (L, N, P–Q); certificate revalidation is separate from new synthesis. Entry 2 supplies saved Lift graph replay and a separate kit that regenerates one matched pair (H, K). Entry 3 rechecks 76 saved files and exact external sources for earlier Patrol and extended Lift (S, R.7), without a controller or native solver rerun.

### Added packages: cohorts, Obstacle and historical bindings

In the accompanying release, the five archive paths below are relative to `results/archives/`. Their package identifiers are distinct from supplement section labels. Each guide gives entry points and the scope of checking; a static integrity check is not a scientific replay.

| Package | Archive | Guide in `results/guides/` |
|---|---|---|
| AA: 44-arm cohort | `v71-compiled-cohort-evidence-20261026.zip` | `V71_COMPILED_COHORT.md` |
| AC: Obstacle | `v71-obstacle-evidence-20261026.zip` | `V71_OBSTACLE_CERTIFICATES.md` |
| Z: historical Patrol | `v71-historical-patrol-source-20261026.zip` | `V71_HISTORICAL_PATROL.md` |
| AD: public Kind | `v71-public-kind-gui-evidence-20261026.zip` | `V71_PUBLIC_KIND_GUI.md` |
| R6: positive / control | `v71-r6-positive-certificates-and-controls-20261026.zip` | `V71_R6_POSITIVE_CERTIFICATES_AND_CONTROLS.md` |

**Compiled variants (Section AA).** Extract `aa_compiled44/`; start with `README.txt` and `COHORT_INDEX.json`. The conditioned eleven-source roster has 44 arms: 43 saved native candidates, 40 accepted results and four UNKNOWNs. `SOURCE_RESTORATION.json` binds the required primitives and shared checkers to exact members of the unchanged Core and FixedSevenNative archives. The `replay_saved.py` adapter for the 40 accepted arms is unexecuted after relocation and has no external wall/RSS supervisor; it excludes the four UNKNOWNs. Packaging adds no fresh-host replay.

**Obstacle certificates and reached states (Section AC).** Extract `ac_obstacle8cert98/`; start with `README.txt` and `published_update/direct_certificate_check/amendment_01/FINAL_CERTIFICATE_SUMMARY.json`, `FINAL_CERTIFICATE_TABLE.csv`, and `ALL98_FIXED8_MEMBERSHIP.json` in that directory. Eight accepted theorem-based certificates preserve the fixed order of all 98 boundary states. The earlier eight independent exported-game fixed-point attempts remain UNKNOWN: two source and six helper arms. The separate initial certificate protocol retains one serialization UNKNOWN and seven unattempted cases. Later certificates do not relabel those attempts. `SOURCE_RESTORATION.json` and `restore_sources.py` provide a hash-bound local recipe for omitted distributor specifications and the ordinary projection. Neither restoration nor relocated certificate replay was performed for publication.

**Two public Kind observations (Section AC.6).** Extract `ad_gui_kind_ab/`; start with `SCIENTIFIC_DISPOSITION.json`, `gui/A/RESULT.json`, `gui/B/RESULT.json`, and the associated `COMPLETION_AUDIT_PUBLIC.json` records. A is the unchanged old authored-initial query; B fixes the updated point $r2404$. Each public action returned false once. The source-kind bridge is retained in `theory/SOURCE_KIND_LEMMA.txt` and its adoption record. The preparation exporter's supervisor remains UNKNOWN with an unrecorded OS exit code; the later saved-predicate gate is separate. Raw screenshots and AX captures are omitted; authored observations, exact logger XML and source-snapshot hashes are retained. A manual console transcription is not raw process output.

**Positive witnesses and ordinary-mode controls (Section AC.7).** Extract `r6_positive_certificates_and_controls/`; start with `SCIENTIFIC_DISPOSITION.json` and the fifth guide above. `certificates/A.json`, `certificates/B.json`, `checkers/check_certificate.py` and

theory/INDEPENDENT_CHECKER_RATIONALE.md supply the direct-source positive witnesses and their checker. Acceptance and separate ENV-totality receipts are under c1/stages/; qualification records are in c1/stages/checker_qualification/qualification/. These positive witnesses are distinct from AC.4's saved-region/theorem certificates; the earlier eight full-fixed-point UNKNOWNs remain.

c2/A/RESULT.json and c2/B/RESULT.json record ordinary-GR(1) true answers, one action each; prior_kind/A/RESULT.json and prior_kind/B/RESULT.json retain the previous selected Kind false answers. Console transcriptions and logger XML accompany the records. SOURCE_RESTORATION.json binds the five exact input files through distributor/member hashes and documented projection/initializer changes. Upstream source bodies, archived source bodies, raw screenshots and AX captures are omitted from public redistribution. An authored visual console transcription is not raw stdout. The guide specifies the package's validation scope; no fresh-host scientific replay is inferred from these authoring records.

For the joint static check of the four earlier packages only, extract them into one fresh parent directory, retaining aa_compiled44/, ac_obstacle8cert98/, z_historical_patrol/, and ad_gui_kind_ab/. Then run:

```
cd /path/to/fresh-parent/ad_gui_kind_ab
python3 -I -S -B verify_candidate.py
```

This checks the adjacent AC scientific files and all three sibling assembly manifests. The unexecuted saved-predicate adapter requires AC scientific inputs only; that smaller dependency set is insufficient for the complete static verifier. No new GUI action or synthesis occurs.

**Historical Patrol (Section Z).** Extract z_historical_patrol/; read README.txt, historical/ORIGINAL_ARTIFACT_PATROL_EVIDENCE.json and historical/AUDIT.json. The archived report records 1670358790.spectra,TRUE,FALSE for Filename,4FP,R+4FP. Here R+4FP means reduction plus four fixed points, not this paper's preserving $R^+$ construction. The package binds that historical report and its source; it adds no artifact execution or current-default/GUI equivalence. Third-party source text and runtime binaries are omitted; SOURCE_RESTORATION.json gives exact distributor/member pins.

### Existing continuation release: selected evidence

Use continuation-evidence-20261024-v1 [1]. The guide identifies archives/continuation-evidence-20261024-v1.zip; inside it start with REVIEWER_START_HERE.txt and claim_to_evidence.json. The 41 original records retain their exact bytes and source-member ledger. quick_verify.py checks transport identity and recorded properties only; its PASS is not semantic replay.

For the saved Patrol, sequential-manager, and reduced-query checks, use:

```
python3 -B "$BUNDLE_ROOT/replay/replay.py" \
  --bundle "$BUNDLE_ROOT/replay" --out "$FRESH_REPLAY"
```

Here BUNDLE_ROOT is the absolute extracted archive root and FRESH_REPLAY is a new or empty external output directory. Python 3.9+ and the standard library suffice; optimized Python is rejected. Expected result: PASS_PORTABLE_SAVED_EVIDENCE, with stages patrol, manager, and reduced. The reduced stage binds the saved winning output and all 4,800 layout memberships; it does not recompute the native winning fixed point. No native synthesis, extraction, manager-worker execution, or full-helper retry occurs; full-helper UNKNOWN is unchanged. Historical source/runtime and worker-provenance checks omitted from replay are listed in replay/OMITTED_HISTORICAL_CHECKS.json.

Qualification is an author-host fresh-directory sandbox check with workspace and network access denied, not an external reproduction.

For the separately qualified fixed lasso in Section V, use:

```
python3 -I -S -B \
  "$BUNDLE_ROOT/lasso/replay/replay_lasso.py" \
  --bundle "$BUNDLE_ROOT/lasso/replay" \
  --out "$FRESH_LASSO_OUTPUT" --timeout-seconds 60
```

This Python 3.10+/POSIX standard-library check expects `PASS_SAVED_LASSO_REPLAY`. It checks 23 public semantic payloads, nine metadata projection values and 19 DAGs, not the original 25-pin local verification. Original Spectra and `final_region` are hash-only omitted; source-value equality is assembly-time evidence and original paths/runtime provenance are not revalidated. It evaluates one fixed initialized transformed-arena lasso, not a returned controller.

Older Guard and matched-Lift kits remain separate; use `REPLAY_COMMANDS.txt` and Entries 1–2 for their restoration steps and qualification boundaries. The full local R5 archive is not distributed.

### Separate evidence: Vehicle188 native strict pair

The separately packaged Vehicle188 records are identified by `vehicle188-native-evidence-20261024-v1` [2]. Inside the bundle, start with `README.txt`, `MANIFEST.json`, `TRANSFORMATIONS.json`, and `OMITTED_HISTORICAL_CHECKS.json`. For a saved-evidence check using Python 3.10+ and its standard library:

```
python3 -I -S -B "$BUNDLE_ROOT/replay.py" \
  --bundle "$BUNDLE_ROOT" --out "$FRESH_RESULT_JSON"
```

The result file must be new and outside the immutable bundle. Expected status:

```
PASS_SAVED_IDENTITY_MEMBERSHIP_INITIALIZATION
```

The check verifies transport integrity, exact saved model/function identity, the full fixed source state and helper-false target, saved winning-region membership/counts, and quantified initialization. It does not recompute a winning fixed point, run a native solver, or replay the separate source-policy certificate. Historical complete-memory, process and loaded-class checks remain recorded evidence. The native experiment itself is described in Section E; its two calls are distinct from this saved-data check. The bundle preserves encoded stdout, with no raw-stream hash claim. Local qualification uses an isolated directory with workspace and network access denied, including positive and corrupted-input checks; it does not establish external reproduction.

### 1. Guard, source certificates, and finite studies

Use tag `certificate-finite-evidence-20261023-v2` [3] and its `CERTIFICATE_FINITE_EVIDENCE.md`. The two downloads are `archives/three-study-evidence-20261023-v2.zip` and `archives/intruder-certificate-evidence-20261023-v1.zip`. From that repository version, Python 3.10+ and its standard library suffice for the following *integrity-only* check:

```
python3 -I -S -B tools/verify_current_distribution.py
```

It checks the version's 302-file distribution and 22 downloads, not any scientific result. Do not substitute the older version-bound certificate verifier for this v2 metadata check.

**Guard and the finite-study tree.** After extracting the three-study archive, set `BUNDLE` to its `three-study-evidence-20261023-v2/saved-study-inputs/` directory. Within it, inspect `evidence/experiment/v58_submission_20261023/guard_audit/RESULT_01.json`: the complete audit checks 16 source and 64 helper current/next assignments, with safety iteration $8 \to 4 \to 4$. The same tree contains `request_protocol/results_01/RESULT.json` and `guard_census/results_01/RESULT.json` under `evidence/experiment/v58_submission_20261023/`: six request cells, seven modeled episodes, and eight Humanoid policies. Humanoid cooperative admission is negative.

For a new saved-study replay, use an absolute Python path, a fresh external replay directory, and a POSIX host providing process groups, `ps`, and resource limits. No JDK is needed. The current inner manifest hash is `6c020cb00df8b6bc4d3be8ba0e2a0edeb61abddc83e4dc2a416440d03ef4b806`. The binding entry point is:

```
"$PYTHON" -I -S -B "$BUNDLE/prepare_three_study_replay.py" \
  --bundle-root "$BUNDLE" \
  --expected-manifest-sha256 6c020cb00df8b6bc4d3be8ba0e2a0edeb61abddc83e4dc2a416440d03ef4b806 \
  --replay-root "$FRESH_ROOT" --python "$PYTHON" \
  --confirmed-root-controlled-binding --ordinary-usage-gate-confirmed
```

Inspect the generated binding and run authorization, then follow the three `science_commands` and comparison command in the generated `REPLAY_PLAN.json`, as specified by the bundled README. The retained v1 `REPLAY_SUMMARY.json` records three executions, 418 comparisons, and 66 request events. The v2 package is a metadata privacy derivative with 133 inner files; it retains `ASSEMBLED_NOT_REPLAY_QUALIFIED` and was not newly replayed. `PRIVACY_AMENDMENT.json` explains the changed manifest identity.

**Intruder proof objects.** Within `seams2027-intruder-certificate-evidence-v1/`, the nested `seams2027-intruder-certificate-replay-v1/README.md` and `REPLAY_PLAN.json` give six exact commands: source-semantics checks, full original/repaired certificate verifiers, and two rejection suites. Expected certificate sizes are 276,829 labelled states for one allowed original initializer $(0, 7)$ and 407,172 states covering all eight repaired initializers. The source certificates concern joint strict/cooperative winning, not the complete strict-winning region. Python 3.10+ and a suitable supervised POSIX host suffice; no Java, network, or new synthesis is used. Section N states the countercertificate and repair limits.

## 2. Matched Lift: saved replay or fresh native generation

**Saved pair.** Use tag `lift-causal-evidence-20261021-v1` and `LIFT_CAUSAL_EXTENSION.md`. The archive is `archives/lift-causal-extension-public-20261021-v1.zip`, with root `seams2027-lift-causal-extension/`. Starting in that repository version:

```
python3 -I -S -B tools/verify_lift_causal_extension.py \
  --archive archives/lift-causal-extension-public-20261021-v1.zip \
  --extract-to ../lift-causal-evidence
python3 -I -S -B ../lift-causal-evidence/seams2027-lift-causal-extension/source_restoration/
    fetch_lift_causal_sources.py \
  --output-root ../lift-causal-sources --cache ../lift-source-cache
python3 -I -S -B ../lift-causal-evidence/seams2027-lift-causal-extension/consumer/replay.py \
  --bundle ../lift-causal-evidence/seams2027-lift-causal-extension \
  --sources ../lift-causal-sources --out ../lift-causal-replay
```

Python 3.10+ and its standard library are sufficient. Source restoration retrieves fourteen exact inputs; no JDK is needed. Key members are `evidence/CAUSAL_COMPARISON_CLOSURE.json`, `evidence/SUCCESS_CONTROLS_RESULT.json`, and `evidence/adaptation_episode_04/EXECU`

TION_RECEIPT.json. Expected results are 2,048 typed states, 28 saved memory sets (57,344 memberships), and 32 graph judgments on sixteen arms. The matched baseline has 154 states/1,170 edges and is strict but non-cooperative; the variant has 154/1,176 and satisfies both. The status is PASS_SAVED_SEMANTICS_REPLAY. This recomputes saved finite-set and graph/event checks; it does not repeat native production, extraction, or episode execution.

**Fresh matched pair.** Use tag native-regeneration-evidence-20261021-v1, NATIVE_REGENERATION.md, and archive archives/lift-native-regeneration-public-20261021-v1.zip. Extract with

```
python3 -I -S -B tools/verify_native_regeneration.py \
  --archive archives/lift-native-regeneration-public-20261021-v1.zip \
  --extract-to ../native-regeneration-inspection
```

The kit root is seams2027-lift-public-regeneration-kit/. Change into the extracted kit root before the following commands. Its README.md defines the configuration values and run-specific review/authorization; configure_run.py is the entry point:

```
"$PINNED_PYTHON" -I -S -B configure_run.py \
  --jdk-home "$JDK_HOME_FOR_THIS_RUN" --python "$PINNED_PYTHON" \
  --workspace-deny-root "$EXCLUDED_WORKSPACE" \
  --deny-probe "$EXISTING_EXCLUDED_FILE" --output "$FRESH_RUN" \
  --deadline-utc "$UTC_DEADLINE"
"$PINNED_PYTHON" -I -S -B "$FRESH_RUN/launch_public.py" execute
```

Execution requires the reviewed generated authorization, an installed pinned JDK 17.0.19 (565-entry inventory), pinned Python, and macOS sandbox-exec. Runtime binaries are not bundled. The qualified configuration fetched 165 public inputs (113,412,497 bytes), restored fourteen sources, compiled six custom Java sources to nineteen matching classes using 164 JARs, and made one native solver invocation with two extractors sharing the same game and memory. Four graph checks recover the same 154/1,170 and 154/1,176 pair; 36 model/memory BDD equalities and 176 supervised processes are recorded. NATIVE_REGENERATION_VALIDATION.json and evidence/validation/native-regeneration-20261021/qualification_01/run_01/PAIRED_EVIDENCE.json are repository-level records. This qualifies one same-host configuration, not another adaptation episode or the six dormant controls. Three metadata members were subsequently sanitized; the full final 32-file serialization was not executed again.

## 3. New Patrol and extended Lift saved-evidence kit

Use tag pending-diagnostic-evidence-20261023-v1 [4] and download archives/seams2027-20261023-pending-diagnostic-evidence-v1.zip. Its 525,798 bytes have SHA256 151a8d04517d853aff771c2eb338fb72aba11ab21cfc2f43eb9c1e09ed7a398d. The outer root pending-diagnostic-evidence-v1-delivery/ contains eight files: the delivery manifest, README, validation attestation, inner ZIP, and four validation records. Extract its pending-diagnostic-evidence-v1.zip; commands below run inside the resulting pending-diagnostic-evidence-v1/ directory.

**Exact members and prerequisites.** Patrol inputs are primary277/sources/new.spectra, primary277/sources/old.spectra, and their primary277/sources/imports/DwyerPatterns.spectra. The principal certificates are primary277/graphs/C_CERTIFICATE.json and primary277/graphs/WC_COUNTERCERTIFICATE.json; actual checks are primary277/actual/SOURCE_AST_AUDIT.json, primary277/actual/GRAPH_AUDIT.json, and primary277/actual/HANDOFF_AUDIT.json. Extended Lift uses lift/actual/RESULT.json, lift/actual/EVENTS.jsonl,

and lift/actual/EVENT_AUDIT.json. The README and REPLAY_SCOPE.json separate original execution from consumer verification. Python 3.10+ and its standard library, two exact external Lift source files, and fresh output outside the kit are required. No JDK, compiler, native solver, or BDD package is loaded.

Obtain the fourth archive in Entry 2 and the exact upstream HTML at https://arxiv.org/html/2609.12231v1 separately. The HTML must have 1,624,775 bytes and SHA256 78a3adba587d3179008afefa9ef7cecc9bd9eacc030d64c64a30f9171955b8ee. The offline restoration helper checks both dependencies and reconstructs lift_new.spectra and lift_old_no_b1.spectra:

```
python3 -B lift/source_restoration/restore_pending_sources.py \
  --fourth-archive /path/to/lift-causal-extension-public-20261021-v1.zip \
  --paper-html /path/to/2609.12231v1.html --output ../restored-lift
python3 -B tools/verify_manifest.py --bundle . \
  --manifest-sha256 a49211ac9a882a0270c9744f7e7338db917fbde94a5030ebf1507808cdd8caaf
python3 -I -B tools/compare_saved_semantics.py --bundle . \
  --manifest-sha256 a49211ac9a882a0270c9744f7e7338db917fbde94a5030ebf1507808cdd8caaf \
  --restored-lift ../restored-lift --output ../saved-recheck-01
```

**Expected result and qualification boundary.** The final status is PASS_RELOCATED_SAVED_EVIDENCE_RECHECK, with PRIMARY277_RECHECK.json, LIFT_RECHECK.json, PUBLIC_QUALIFICATION.json, and REPLAY_RECEIPT.json. Patrol checks 77 translated clauses against 61,435 typed support assignments, old/new graphs of 140/277 and 849/12,240 states/edges, and 278 saved events with one old and 67 new commits. Lift checks nine episodes and 180 saved events, eighteen old and 96 new commits, three retained-native dead ends, and six fair self-loops. Forty-three negative controls are rejected and one positive fairness control passes. The 76-file inner kit retains its assembly-time statuses; the outer validation receipt records one later successful author-host check on macOS 15.7.9, arm64, Python 3.10.9, using exact already available source copies. Fresh retrieval and restoration were not rerun for that check, and the original workspace remained accessible. Neither controller was executed anew, no native AST was regenerated, and saved BDD predicates were not reevaluated. Historical process/object identities remain historical evidence. This is not independent external replication.

The nine-episode extension is separate from the six-episode native handoff package at tag pending-native-handoff-evidence-20261023-v2 [5]. Its archive pending-native-handoff-evidence-20261023-v1.zip and README retain the ten-stage bounded execution recipe: six episodes, 98 events, twelve old and thirty new commits, 44 semantic comparisons, and sixteen mutation fixtures (Section R before R.7).

**Research workflow.** The added native Patrol and Lift contract-sensitivity studies are indexed by v64_native_continuation_20261023/RESULT_SUMMARY.json and v64_lift_contract_contrast_20261023/CONTRAST_SUMMARY.json in the local experiment snapshot. Section U.3 adds v64_old_spelling_control_20261023/; Section U.4 adds v64_admission_runtime_20261023/RESULT_SUMMARY.json and its REPORT.md. These paths are relative to the local experiment/ directory. Section U identifies the native exports, certificates, internal checks, negative results, resource-limited attempts, and actual paused-admission runs. The U.4 directory preserves both predeclaration amendments, all three trials including the first Lift setup failure, and the original process receipts whose started_utc field was sampled at receipt completion. Its monotonic elapsed durations remain valid. These complete experiment directories remain local; selected saved Patrol records and the later sequential manager evidence are now distributed in the continuation release above. That public subset does not publish all U.1–U.4 inputs, runtime records, source specifications,

or omitted provenance checks.

**Historical appendix directory.** Section A gives the round-to-alternating derivation; B–C cover source comparisons and archived-label scope. Section D covers saved released arenas and safety-subgame checks; J reports strict-preservation loss calculations, including resource UNKNOWN. For continuation witnesses, E gives Vehicle's source proofs, saved-model evidence and completed native strict queries, whereas F is source-only and residual. Published-replacement evidence is in G–H; H includes the Lift T diagnosis and original maintenance episode, followed by a separate ten-boundary noninitial handoff census; it also separates Lift P's correspondence failure from its direct graph audit. Section I gives the constructed Choice/Toggle pair. Section K maps the claims to existing saved-evidence archives. Section L gives the full retained-constraint proof and counterexample, cooperation characterization, implementation details, mechanism controls, and detailed decision table. Section L.6 gives the complete literal-preservation proofs, corrected-guard proof, and propagation, initialization, and Guard examples. The main paper gives full proofs of the basic strict theorem and corollary, then closes Patrol's positive/negative argument before the retained-duty proposition, its full proof, the existing example with two source states, and explicit safe-entry condition $F$. Application to the actual export precedes source-kind, case evidence and the conditioned cohort; the exclusion contract and three saved-evidence checks follow. The main paper's Fig. 1 maps the five objects used in these queries and procedures. L.6 retains technical statements and examples. Section L.1 gives the relational formulation, safety fixed point and retained-constraint proof; L.2 also retains the full normalization lemma and proof moved from the body. A and L.2 give the normalized cooperative derivation and completed-round proof. The remaining sections provide variants, row-level evidence, and reproduction details. Section M reports the complete eleven-entry Elevator residual census; the fixed preference was inactive throughout. Section N separates the unchanged Intruder source obstruction from its controlled repair and preserves the earlier bounded UNKNOWN records. Section O gives the matched Port extraction and Polling fairness control. Section P reports the constructed request-enable update, all six routes and seven episodes. Section Q gives the complete eight-policy Humanoid census. Section R gives a separate native Lift handoff study with genuinely pending work, actual stored nonzero ranks, and a newly enabled input. Section S adds the pending patrol update, exact conversion-loss certificate, and actual source execution. Section R.7 adds the separate extended Lift continuations. The opening evidence index introduces the current continuation release and its selected saved-replay scope; Section K preserves the earlier immutable public archives and qualification boundaries. Section T collects the detailed Lift configuration and secondary evaluation material omitted from the main paper. Section U adds the actual native Patrol continuation checks, preserving the old-arena negative and direct-solver UNKNOWN results, the Lift contract-sensitivity control with its unchanged-extraction outcome, the isolated equivalent-old-source control, and complete-graph admission executed at paused software boundaries with measured stage costs. U.5 preserves the full-helper diagnostic UNKNOWN; U.6 gives the separately completed theorem-reduced native query, and U.7 collects the Patrol decisions. U.8 retains an unrestricted full-helper query and its UNKNOWN outcome; U.9 executes the sequential Lift candidate-selection loop. Section V records the primary-theorem premise, supported selected route, and one fixed saved-arena lasso that refutes the raw language-inclusion premise. Sections W and X give further source/extraction details and the complete Vehicle saved-region, cooperation and initialized-reachability audit. Section Y reports the fixed-seven native route controls. Section Z records the historical public Patrol report and its distinct driver and limits. Section AA records all 44 arms of the conditioned compiled-cohort comparison, with separate native candidates, adopted audits and terminal UNKNOWNs. Section AB collects the supporting output diagnosis, comparison

table and historical/reuse details moved out of the main narrative. Section AC gives the published Obstacle pair: strict-region certificates, switch roster, source-kind lemma and two fixed public Kind checks (AC.5–AC.6). Section AC.7 adds positive rank/response witnesses and matched ordinary public controls. The opening claim-to-evidence map also locates AD–AG: AF retains the interrupted diagnostic; AG gives the distinct completed exact-B pair. All earlier UNKNOWN records remain.

**Cooperation notation in this supplement.** In model-specific statements, $K \subseteq W$ denotes states admitting a strict-winning source strategy that, after every source-legal prefix, permits an ENV-safe, ENV-fair continuation with that same strategy and its reached memory. This legal-prefix property is distinct from Kind's all-history requirement stated in Section AC.5. The main paper uses the property directly in its case proofs without the set symbol $K$.

## A. Round-to-alternating correspondence

**Source fixed point and one-edge predecessors.** We restate the vector operator of [6, Eq. (15)], using local bound variables and explicit grouping. On the bipartite arena $H = (Q_0, Q_1, \delta_0, \delta_1)$, put $Q = Q_0 \cup Q_1$. For $T \subseteq Q$, its one-edge predecessors are

$$\begin{aligned} \mathrm{Pre}_\exists(T) &= \bigcup_{k=0}^{1} \{q \in Q_k : \delta_k(q) \cap T \neq \varnothing\}, \\ \mathrm{Pre}_1(T) &= \{q \in Q_0 : \delta_0(q) \subseteq T\} \cup \{q \in Q_1 : \delta_1(q) \cap T \neq \varnothing\}, \\ \mathrm{CondPre}_{\mathrm{edge}}(U, R) &= \mathrm{Pre}_\exists(U) \cap \mathrm{Pre}_1(U \cup R), \end{aligned}$$

as in [6, Eqs. (1), (4), and (5)]. For assumption sets $a_1, \ldots, a_m \subseteq Q$ and guarantee sets $g_1, \ldots, g_n \subseteq Q$, with $m, n \geq 1$, define

$$\begin{aligned} \Omega_{ji}(\mathbf{Z}, Y, X, U) &= (g_j \cap \mathrm{Pre}_1(Z_{j^+})) \cup \mathrm{Pre}_1(Y) \\ &\qquad \cup \big((Q \setminus a_i) \cap \mathrm{CondPre}_{\mathrm{edge}}(U, X \setminus a_i)\big), \\ \mathcal{T}_j(\mathbf{Z}) &= \mu Y.\, \bigcup_{i=1}^{m} \big(\nu X.\, \mu U.\, \Omega_{ji}(\mathbf{Z}, Y, X, U)\big), \\ \mathbf{Z}^* &= \nu \mathbf{Z}.\, \big(\mathcal{T}_j(\mathbf{Z})\big)_{j=1}^{n}, \qquad j^+ = (j \bmod n) + 1. \end{aligned}$$

Here the outer greatest fixed point is simultaneous; $Y$ is local to component $j$, and $X, U$ are local to branch $(j, i)$. The source defines the final region as $\bigcup_{j=1}^{n} Z_j^*$ [6, Appendix B.1]. These operators take one alternating edge. The following lemma instead substitutes the completed-round operators $C$ and $P$ for $\mathrm{Pre}_1$ and $\mathrm{Pre}_\exists$, respectively, so its conditional predecessor is $P(U) \cap C(U \cup R)$. The semantic soundness theorem uses its language-inclusion premise and mode/rank strategy, while completeness quantifies over all system strategies [6, Theorem 3 and Appendix B.2]; the following lemma transfers final regions only, not that strategy construction.

**Lemma 1** (Round and alternating fixed points)**.** *Let $H = (Q_0, Q_1, \delta_0, \delta_1)$ be a finite total bipartite arena, where $Q_0 = V \neq \varnothing$, environment edges go from $Q_0$ to $Q_1$, and system edges go from $Q_1$ to $Q_0$. Use nonempty assumption and guarantee justice lists, and place every justice label only on $Q_0$. Replace the one-edge predecessors in the vector formula of [6, Eq. (15)] by the completed-round operators $C(X) = \forall\delta_0\, \exists\delta_1\, X$ and $P(X) = \exists\delta_0\, \exists\delta_1\, X$, with $\mathrm{CondPre}(U, R) = P(U) \cap C(U \cup R)$. Then each final round vector component equals the $Q_0$ restriction of the corresponding alternating component. Thus their unions have the same completed-state region.*

*Proof.* We read the third term of the inner body in Eq. (15) as $(Q \setminus a) \cap \mathrm{CondPre}(U, X \setminus a)$, where $Q = Q_0 \cup Q_1$ and $a \subseteq Q_0$ is the selected assumption. For $X \subseteq Q_0$ and $S \subseteq Q_1$, put

$$\begin{aligned} R(X) &= \{t \in Q_1 : \delta_1(t) \cap X \neq \varnothing\}, \\ A(S) &= \{s \in Q_0 : \delta_0(s) \subseteq S\}, \\ E(S) &= \{s \in Q_0 : \delta_0(s) \cap S \neq \varnothing\}. \end{aligned}$$

Then $C = AR$, $P = ER$, and the lift $L(X) = (X, R(X))$ is an order embedding that preserves unions. On pairs, the one-edge operators are $\mathrm{Pre}_1(X_0, X_1) = (A(X_1), R(X_0))$ and $\mathrm{Pre}_\exists(X_0, X_1) = (E(X_1), R(X_0))$. Totality gives $L(V) = Q_0 \cup Q_1$.

Fix lifted outer parameters $Z = L(z)$ and $Y = L(y)$, a guarantee $g$, and an assumption $a$. Write $N = V \setminus a$, $T = g \cap C(z)$, and $B = T \cup C(y)$. Temporarily assume $y \subseteq B$. For fixed alternating $X = (x_0, x_1)$, the inner least fixed point has equations

$$U_0 = B \cup \big(N \cap E(U_1) \cap A(U_1 \cup x_1)\big), \qquad U_1 = R(y \cup U_0).$$

The second equation follows because justice is absent on $Q_1$ and $R(U_0) \cap R(U_0 \cup S) = R(U_0)$ for every reserve $S$. Every fixed point has $U_0 \supseteq B \supseteq y$, so $U_1 = R(U_0)$. Conversely, a fixed point of $u \mapsto B \cup (N \cap P(u) \cap A(R(u) \cup x_1))$ lifts to a fixed point of this paired system. This correspondence preserves and reflects order, hence transports the least fixed point. The resulting outer $X$ map has image in the lifts. Its fixed points therefore correspond, again in both order directions, to those of

$$J(x) = \mu u.\big[B \cup \big(N \cap P(u) \cap C(u \cup x)\big)\big].$$

Consequently the alternating branch is $L(\nu J)$.

The round branch uses instead

$$I(x) = \mu u.\big[B \cup \big(N \cap P(u) \cap C(u \cup (x \cap N))\big)\big].$$

Pointwise $I \leq J$, so $\nu I \subseteq \nu J$. For $x^* = \nu J$, every output of the inner body lies in $B \cup N$, whence $x^* \setminus N \subseteq B$. Each inner fixed point contains $B$; on every $u \supseteq B$ we have $u \cup x^* = u \cup (x^* \cap N)$. The two inner maps at $x^*$ thus have the same fixed points and the same least fixed point. Hence $I(x^*) = J(x^*) = x^*$, giving the reverse inclusion and $\nu I = \nu J$.

It remains to justify $y \subseteq B$ along the local $\mu Y$ iteration. It holds at $y = \varnothing$. Let $x_a$ be the final branch for assumption $a$, and put $t = \bigcup_a x_a$. Every $x_a$ contains $B$, so $y \subseteq B \subseteq t$. At its nested fixed point,

$$x_a = B \cup \big((V \setminus a) \cap P(x_a) \cap C(x_a)\big).$$

Monotonicity therefore gives $t \subseteq B \cup C(t) \subseteq T \cup C(t)$. Thus the required invariant holds for the next $Y$ value. Since $R$ preserves unions, the alternating update is $\bigcup_a L(x_a) = L(t)$. Induction and finite stabilization establish correspondence of the local least fixed points. Finally, the simultaneous outer vector iteration starts at the lifted top vector; the target in component $j$ is $g_j \cap C(z_{(j \bmod n)+1})$ on $Q_0$ and empty on $Q_1$, where $n$ is the number of guarantee components. Applying the local argument at every update proves componentwise correspondence at the final outer greatest fixed point. □

**Scope.** This lemma identifies final fixed-point values; it does not identify all inner approximants or extracted ranks. Appendix B.1 of [6] defines the vector formula's region as the union of its final components, so componentwise correspondence also identifies the completed-state restriction of that union. Before applying the lemma, empty justice lists are padded by $V$ in rounds and by $Q_0$ in the alternating arena. This gives nonempty index sets on both sides. The padding also

preserves the objective because $Q_0$ is visited infinitely often on every play. If $V = \varnothing$, the region is empty directly; no initialized game theorem is invoked. For the arena $H_V$, totality follows from $V = \nu X.(\mathrm{Alive}_E \cap \mathrm{CPre}(X))$: $\delta_0$ retains every original-legal input, and $\delta_1$ retains all original-safe responses into $V$, at least one for each such input. The lemma is algebraic and does not require justice normalization; normalization supplies the premise for the separate semantic soundness theorem.

## B. Source comparisons

**Comparison with the published preservation proof.** The Guard counterexample uses Boolean input $x$, response $y$, trivial initial and justice predicates, and $\rho_e = \rho_s = x'$. Section 4.1 of [7, p. 4] writes the converted clause forms as $\rho_n^e \vee \mathrm{next}(\mathit{envHelp})$ for each safety assumption and $\rho_n^s \vee \mathit{envHelp}$ for each safety guarantee. We denote *envHelp* by $b$ and its next value by $b'$, conjoin the respective clauses, and include helper persistence. Theorem 1 of [7, p. 5] takes $M'$ to be the specification obtained by reducing $M$, and concludes: "Then $M$ is realizable if and only if $M'$ is realizable." The literal reduction has environment safety $\rho_e^R = (\rho_e \vee b') \wedge (b \Rightarrow b')$ and system safety $\rho_s^R = \rho_s \vee b$.

The gap is specifically on the edge into the first helper-true state. Appendix A of [7, pp. 12–13], in the second part of the proof of Theorem 1, reasons that system safety holds after the helper is set to true and writes the relaxed environmental safety using an unprimed helper. Section 4.1 instead guards environmental safety with the *next* helper. On an edge $b = 0, b' = 1$, the input is therefore already released while the system response is not. The later failure of environmental fairness cannot discharge that edge's strict safety obligation. The Sec. 4.1 clause forms as written therefore do not satisfy this stated equivalence. Appendix A's different environmental guard and the released semantic correction are outside this counterexample's scope. It does not by itself refute the separate well-separation statement in Theorem 2.

**Existing semantic correction.** In the released `PitermanSemanticReduction.java`, version `1.0.0.202402072123`, `doReduction` lines 28–29 add `envHelpIsTrue.id()` and `envHelpIsFalse.and(envHelpPrimeIsTrue)` to system safety through two successive `orWith` calls. These are exactly the guards $b$ and $\neg b \wedge b'$ analyzed in the main paper. The same calls appear at lines 56–57 of the saved source snapshot [8]. The method returns without adding a helper when environmental safety is universally true. This source comparison does not identify the configured syntactic arena with the complete semantic variant.

Table 1 compares these forms with the saved syntactic arenas and their applicable strict-preservation results.

**Raw recurrence for the Justice control.** For the Justice control in Section L.2, $D$ is the typed carrier, $D_0 = \{\neg p\}$, $\rho_e = (e' = p)$, $\rho_s = \neg p$, $A = \mathbf{GF}e$, and $G = \mathbf{GF}\neg p$. The all-zero play satisfies $G$ but violates $A$, witnessing the missing premise already in the original game. The raw single-justice four-fixed-point recurrence returns $D_0$: its predecessors satisfy

$$\mathrm{CPre}(T) = \mathrm{EPre}(T) = \neg p \wedge \big(T(0,0) \vee T(0,1)\big),$$

and the outer iteration is $D \to D_0 \to D_0$. With either helper guard, its raw region is $\neg p \vee b$, and its initial answer is true. This is a diagnostic of a missing applicability condition, not a counterexample to the conditional theorem of [6].

Table 1: Helper variants and applicable strict-preservation results. The first-flip column describes original-variable safety on $b = 0, b' = 1$; $\top$ means released. Complete semantic variants guard all safety constraints. A syntactic route requires complete relation checks after all translations, including monitor updates. These results alone do not establish raw cooperative-core verdict transfer or returned-graph cooperation.

| Variant | First-flip ENV / SYS | Applicable result | Evidence and scope |
|---|---|---|---|
| Kind [7, Sec. 4.1] literal relations | $\top$ / $\rho_s$ | Main exact-continuation theorem: exact surviving strict region; full proof in Section L.6. | The 59-model study measures literal strict loss. The matching Vehicle arena loses a cooperative continuation while preserving initialization. |
| Kind [7, App. A] proof notation | $\rho_e$ / $\rho_s$ | Neither safety duty is released on the first flip: the environment guard uses current $b$. Outside Guard's scope. | Source comparison in Supplement B; not identified with the configured syntactic route. |
| Complete corrected semantic variant | $\top$ / $\top$ | Proposition L.4 in Section L.6: strict-region and initial preservation. | Existing released semantic guard, analyzed as complete semantic relations. |
| Released syntactic route | Arena-specific | Helper arenas: Supplement L, Proposition L.1, after relation checks. Checked no-helper arenas: original-model identity. | All eight completed helper and five identity cases preserve strict winning. Separately selected Humanoid 044 loses strict initialization; it already lacks cooperative initialization (Sections D, Q, and T). |

**PCar464: initial impossibility with auxiliary memory.** The source clauses used below are explicit here. Write $\tau$ for `throttle`, $o$ for `obstacle`, $z$ for `sideSense`, and $q_b, q_f$ for `spec_b2f`, `spec_f2b`. The declared domains are $o \in \{\mathsf{CLEAR}, \mathsf{BLOCKED}\}$, $z \in \{\mathsf{CLEAR}, \mathsf{P_O}, \mathsf{BLOCKED}\}$, $\tau \in \{\mathsf{BWD}, \mathsf{FWD}, \mathsf{STOP}\}$, and $\mathit{steer} \in \{\mathsf{LEFT}, \mathsf{RIGHT}, \mathsf{CENTER}\}$. Both direction monitors start in $S0$ and range over $\{S0, S1, S2\}$. Their exact source updates on these enum values are:

| | next $q_b$ | | | next $q_f$ | | |
|---|---|---|---|---|---|---|
| current monitor | BWD | FWD | STOP | BWD | FWD | STOP |
| $S0$ | $S1$ | $S0$ | $S0$ | $S0$ | $S1$ | $S0$ |
| $S1$ | $S1$ | $S2$ | $S0$ | $S2$ | $S1$ | $S0$ |
| $S2$ | $S2$ | $S2$ | $S2$ | $S2$ | $S2$ | $S2$ |

Each subtable uses its own current monitor as the row and the current throttle as the column. No claim about illegal bit encodings is needed. For the SYS-owned Boolean response auxiliaries $r_B, r_Z$, initialization sets both to true, and system safety includes

$$r'_B \leftrightarrow (o = \mathsf{CLEAR}) \vee (r_B \wedge \tau \neq \mathsf{BWD}),$$
$$r'_Z \leftrightarrow (z = \mathsf{CLEAR}) \vee \big(r_Z \wedge \neg(\tau = \mathsf{FWD} \wedge \mathit{steer} = \mathsf{CENTER} \wedge z = \mathsf{BLOCKED})\big).$$

The four environment justices are

$$\mathbf{GF}(q_b = S0 \vee q_b = S1), \quad \mathbf{GF}(q_f = S0 \vee q_f = S1), \quad \mathbf{GF}r_B, \quad \mathbf{GF}r_Z.$$

The physical system clauses require

$$\mathbf{G}(o = \mathsf{BLOCKED} \Rightarrow \tau \neq \mathsf{FWD}),$$
$$\mathbf{G}((o = \mathsf{CLEAR} \wedge z = \mathsf{CLEAR}) \Rightarrow \mathit{steer} = \mathsf{CENTER}),$$
$$\mathbf{GF}(\tau = \mathsf{FWD} \wedge \mathit{steer} = \mathsf{CENTER}).$$

Original environment safety is unrestricted. The response-pattern updates belong to system safety even though their justices belong to the environment. These clauses transcribe the unchanged source `PcarLTL_464_PCar_fixed.spectra`, lines 3–15 and 22–70; Supplement K's C14 entry links it and the saved initializer argument.

PCar also admits a source-level impossibility argument from its declared initializer, independently of the extracted graph and our region computation. Suppose a strict-winning, legal-prefix cooperative strategy exists, with arbitrary private memory. Start with an allowed environment valuation with $\mathit{sideSense} = \mathsf{CLEAR}$, and choose

$$\mathit{obstacle}_{t+1} = \mathsf{CLEAR} \iff \mathit{throttle}_t = \mathsf{BWD},$$
$$\mathit{sideSense}_t = \mathsf{CLEAR}.$$

This policy respects Mealy order and the unrestricted environment safety relation. Cooperation prevents either direction-change monitor from reaching its absorbing state $S2$, because its environment justice would then have no satisfying continuation. For $t \geq 1$, a blocked obstacle prohibits forward motion. A clear obstacle means the preceding throttle was backward, which puts the backward-to-forward monitor in $S1$; forward motion would then force $S2$. Thus forward motion occurs at most initially, violating the system justice requiring forward motion with centered steering infinitely often.

Nevertheless, every environment justice holds. Both monitors remain outside $S2$, and the side-sensor response holds continuously. For the other response auxiliary, a backward throttle at $t$ produces a clear obstacle at $t+1$ and a true auxiliary at $t+2$. Infinitely many backward choices therefore satisfy its justice. If $t$ is the final backward choice, the auxiliary is true at $t+2$ and remains true thereafter; with no backward choice, its true initializer is preserved. This contradicts strict winning and establishes cooperative initial infeasibility.

**Meaning of the published labels.** The archived SYN15 coverage CSV records TRUE/TRUE for Elevator386 and PCar464 at physical lines 20 and 33. Its columns are Filename, 4FP, and R+4FP; the README identifies them as realizability results underlying the paper's evaluation. The C-backed coverage driver consumes a Boolean, without a returned-controller certificate or per-run source hash. Section C gives the complete source association and aggregate checks; exact historical translated-model equality remains unestablished.

**Released route and later source continuity.** We execute release `1.0.0.202402072123`. Its packaged UI bytecode exposes the syntactic Kind reduction as a selectable option and initializes ordinary GR(1) as the default. The separate packaged semantic reduction is not invoked by this selected route; we do not report a Lift T outcome for that alternative. A separate audit at Spectra commit `c1ced71d` [8] found that its Java four-fixed-point core and both helper-reduction classes match the packaged release sources after removing only the leading licence comment and normalizing line endings. At this snapshot, the selectable Kind realizability route invokes the syntactic reduction before default translation and BDD generation, then selects a separated solver for ordinary contracts; ordinary GR(1) remains the initialized default. These facts establish source continuity at this snapshot, without establishing a current-build outcome. The CUDD wrapper and native interface have changed, and the surrounding solver hierarchy also differs. A matching generated-language/parser and dependency build was not qualified for this commit. Frontend provider and translation sources are present; their complete build and execution remain unverified.

## C. Published positive sources: attribution and semantic scope

**Population and attribution** We join the previously fixed cohort of all 59 authoritative independent v2 original-model certificates to the four archived `4fp_cover` tables [9]. The cohort was selected by availability of complete certificates, without filtering on region sizes or outcomes. For each CSV filename, we construct its exact archive path in the corresponding SYN suite and require that path to belong to the source unit's recorded alias set. We then verify that the archived bytes agree with the roster, local original source, import closure, and saved model input. This is an exact co-packaged-source association, not a per-run historical input receipt.

The CSVs contain 320 raw records: 48, 74, 12, and 186 in SYN15, SYN17, SYN20, and SYN23', respectively. SYN17 contains one `error` record and SYN20 one `an error occurred` record. The remaining 318 completed Boolean records have R+4FP-positive totals 29/48, 61/73, 10/11, and 93/186, matching Table 1 of [7]. Error records are not negative decisions.

Of our 59 source units, 23 match 25 records and 36 are unmatched. Of the matched units, 18 have positive labels (19 records) and five have negative labels (six records). Source unit 055 has two byte-identical aliases with positive records; unit 056 has two aliases with negative records. No matched record is an error. Unmatched units have no classification in these particular coverage tables; they are not published negatives. The seven units in Table 3 each correspond to one positive record. This bounded certified cohort is not a random sample, and the counts support no estimate of error prevalence in the published corpus. Table 2 summarizes the separate original-initial cooperation classification of these 23 matched source units.

**Initial predicates and witness counts** For a typed region $R$ we use

$$I_M(R) = \forall e : \theta_e(e) \Rightarrow \exists y : \theta_s(e, y) \wedge R(e, y),$$

Table 2: Archived labels versus original initial cooperation on the same 23 saved models. Entries count source units (CSV records). The 11 positive-labelled units with $I_M(K) = \mathsf{F}$ comprise seven independent obstructions and four using the proved correspondence. Two of the eleven (072 and 159) depend on the audited initializer convention. Historical C-model identity is unestablished; this is not an error-rate estimate.

| Archived R+4FP | $I_M(K) = \mathsf{T}$ | $I_M(K) = \mathsf{F}$ |
|---|---|---|
| Positive | 7 (7) | 11 (12) |
| Negative | 0 (0) | 5 (6) |

Table 3: Seven source units with archived R+4FP-positive labels and initial cooperative obstructions. Counts concern the saved original Java models: $n_E$ counts allowed initial ENV choices; $m_R$ counts choices missing a SYS/AUX response in $R$. All seven have $I_M(W) = \mathsf{T}$. PCar464's obstruction has a separate source proof; RoboticArm232's literal source has no legal successor. The initializer obstructions for 072 and 159 depend on the audited state-invariant translation convention (Section C.4).

| Source unit | $n_E$ | $m_V$ | $m_{W_V}$ |
|---|---|---|---|
| 032 Elevator386 | 4 | 1 | 1 |
| 058 PCar464 | 6 | 0 | 0 |
| 059 PCar552 | 6 | 6 | 6 |
| 060 PCar553 | 6 | 6 | 6 |
| 072 Airport145 | 10 | 8 | 9 |
| 159 RoboticArm227 | 125 | 124 | 125 |
| 164 RoboticArm232 | 15625 | 15625 | 15625 |

where $y$ includes every SYS-owned logical auxiliary. The number $n_E$ counts typed ENV valuations satisfying $\theta_e$. The missing count $m_R$ counts those valuations for which no such response exists.

All 59 certificates give $I_M(W) = \mathsf{true}$; 12 give $I_M(W_V) = \mathsf{false}$. Six of those 12 have published positive labels: 032, 059, 060, 072, 159, and 164. They also have $I_M(V) = \mathsf{false}$. The separate PCar464 (058) source argument establishes its initial cooperative obstruction despite $I_M(V) = I_M(W_V) = \mathsf{true}$. The separate $K$ cross-check classifies the same 23 saved models using the proved round-to-alternating correspondence and justice normalization. Of the 18 positive-labelled units, eleven fail $I_M(K)$ and seven pass. Besides the seven obstructions above, PCar variants 054, 055, 061, and 062 pass $I_M(W)$, $I_M(V)$, and $I_M(W_V)$ but fail $I_M(K)$. Their $K$ sizes are 16, 0, 42, and 16, respectively; none of their 2, 2, 6, and 2 allowed initial input choices has an initialized response in $K$. These four conclusions use the correspondence theorem; the previous seven do not require it. All five negatively labelled units also fail $I_M(K)$. A positive $I_M(K)$ certifies legal-prefix strategy existence in the saved model, not cooperation of a particular graph or the stronger all-history recovery property. The exact-loss calculation for corpus159 remains resource UNKNOWN; an initial obstruction does not fill that missing result. Among the five newly linked positive units (excluding the already linked 032), 059, 060, 072, and 159 belong to the existing ten cases failing $W \subseteq C$; 164 does not.

## C.1 RoboticArm232: contradictory next-state assumptions

The literal source `roboticarm_Robot_232.spectra` declares six environment variables $s_0, \ldots, s_5$ with the enumeration $\{\mathsf{RED}, \mathsf{GREEN}, \mathsf{BLUE}, \mathsf{EMPTY}, \mathsf{NOT}\}$ (lines 6–17), and an environment variable *task* with domain $\{\mathsf{SCAN}, \mathsf{SORT}\}$ (lines 13, 20–24). There are no system or auxiliary declarations.

The only declared initial assumption fixes $task = \mathsf{SCAN}$ (lines 30–31), so there are $5^6 = 15625$ initial ENV choices and the unique empty SYS response. Setting every $s_i = \mathsf{RED}$ is permitted.

Lines 69–83 define

$$strongAssumptions = \bigwedge_{i=0}^{2}(s_i \neq \mathsf{EMPTY} \wedge s_i \neq \mathsf{NOT}) \ \wedge \ \bigwedge_{i=3}^{5} s_i = \mathsf{NOT}.$$

Line 89 requires $\mathbf{GX}\mathit{strongAssumptions}$. Since $\mathsf{EMPTY} \neq \mathsf{NOT}$, every next $s_i$ must differ from $\mathsf{EMPTY}$. In contrast, line 107 requires $\mathbf{GX}\mathit{colorExists}(\mathsf{EMPTY})$, whose definition at lines 111–117 is $\bigvee_{i=0}^{5} s_i = \mathsf{EMPTY}$. Thus the same next state must contain an EMPTY value and contain none. No current state has an environment-safety-legal successor. In particular, the nonempty declared initial set has no legal infinite continuation, so no strategy can meet legal-prefix cooperation. The additional $\mathbf{G}(task' = \mathsf{SORT})$ assumption does not remove this contradiction. This is a direct propositional argument over the source formulas, independent of a region calculation or controller extraction.

Read-only checks of the saved original-model export provide a consistency check: its parsed and expanded ENV ASTs retain both opposing next-state constraints, its saved $\rho_e$ is false, and its independent certificate has $|V| = |W_V| = 0$. These stored results are not premises needed for the source proof. The exact co-packaged source has a positive record at physical line 46 of the SYN17 coverage CSV (header counted as line 1). This establishes the relation between the reported source label and the literal source contract; it does not reconstruct the historical parser, translation, reduction, C solver state, or returned controller.

## C.2 RoboticArm232: a direct raw-core calculation

For the preceding source, $\rho_e = \bot$, $\rho_s = \top$, and both justice lists are empty. Its system-response carrier contains the single empty assignment. The original arena is deadlocked: language inclusion holds vacuously, while nonblocking and cooperation fail.

Add the fresh environment helper $b$ and retain the original initializer with $\neg b$. Both the complete literal guard and its first-edge correction simplify to

$$\rho_e^R = (\bot \vee b') \wedge (\neg b \vee b') = b', \qquad \rho_s^R = \top.$$

The transformed typed carrier $D_R$ has $5^6 \cdot 2 \cdot 2 = 62500$ states, counting the colors, task, and helper. The initializer fixes task and helper, leaving the $5^6$ choices above. Every state has legal environment moves and a safe system response; every infinite play has $b = 1$ after its first step. The helper assumption $A = \mathrm{GF}\neg b$ is therefore impossible on this total arena. The unchanged empty system-justice list represents $G = \top$; padding it by one tautology preserves that meaning, as in the released `GR1Game` constructor. Padding is distinct from copying environment justices into the system objective.

Let $a = \{b = 0\}$ and write CPre and EPre for the round predecessors. The singleton raw fixed point of [6, Eqs. (9), (15)] is

$$\nu Z.\, \mu Y.\, \nu X.\, \mu U.\ \mathrm{CPre}(Z) \cup \mathrm{CPre}(Y) \cup \big((D_R \setminus a) \cap \mathrm{EPre}(U) \cap \mathrm{CPre}(U \cup (X \setminus a))\big).$$

At $Z = D_R$, totality gives $\mathrm{CPre}(D_R) = D_R$. Thus the body is constantly $D_R$, all inner fixed points are $D_R$, and the outer update fixes $D_R$. Every declared initial input has its empty system response in that region, so the initialized raw decision is positive. Directly in the alternating formula, a tautological guarantee makes the body the whole arena. With justice labels restricted to completed states, the inner least fixed point reaches those states and then all intermediate states, giving the same result.

Here $L(G)$ is nonempty and $L(A)$ is empty. The cited conditional cooperation guarantee therefore does not apply; this calculation does not refute it. Normalization changes the effective guarantee to $a$. Since every successor has $b' = 1$, both predecessors of $a$ and of $\varnothing$ are empty. The normalized outer sequence is $D_R, a, \varnothing$. Keeping the redundant tautology as a separate component also yields the empty final region. Original-arena viability instead rejects immediately because $V = \varnothing$; transformed viability is $D_R$ and cannot enforce helper fairness.

This is a hand derivation for the stated semantic pipeline on a public source, consistent with its archived positive label. Exact historical C-model identity and the historical decision's cause remain unknown. No native execution or returned-controller observation is claimed.

### C.3 PCar552 and PCar553: a one-edge obstruction

Both sources initialize *parkResult* = NA and *spec_policy* = SEEKING, with *spec_park* = false. Choose the allowed initial inputs *obstacle* = BLOCKED and *sideSense* = BLOCKED. The environment may keep those sensor values and choose *parkResult*′ = PARALLEL. The sensor-freeze assumption is met regardless of the initial throttle, and the only other direct ENV safety assumption is the current-state implication

$$spec_policy = \mathsf{SEEKING} \;\Rightarrow\; parkResult = \mathsf{NA},$$

which holds at the chosen initial state. The response patterns add SYS-owned monitor updates and ENV justices, not a restriction on this next ENV input.

System safety requires $spec_policy' = \mathsf{SEEKING}$ when the current obstacle is BLOCKED and the current policy is neither PARKING nor PARKED. Every safe response therefore reaches a completed state with policy SEEKING and parkResult PARALLEL. There the current-state ENV implication is false for every next input, so no legal continuation exists. A response that avoids the forced policy violates strict safety on the ENV-legal edge. Initial SYS choices that violate another current-state guarantee likewise cannot rescue a strict strategy. Thus no strict cooperative strategy exists for all declared initial inputs, including strategies with arbitrary private memory. This one-input argument does not derive the stronger saved missing count of six.

### C.4 AirportShuttle145 and RoboticArm227: initializer scope

For AirportShuttle145, the declared ENV initializer requires a closed door and no technician acknowledgement, but leaves station and stopButton unconstrained. Choosing station ONE and stopButton PRESSED falsifies the current-state assumption $atStation \Rightarrow stopButton = \mathsf{RELEASED}$, where *atStation* means station is not NONE. Every SYS/AUX initial response therefore has no ENV-safety successor. This explains an initializer obstruction, not all eight missing $V$ choices or the ninth missing $W_V$ choice.

For RoboticArm227, the declared initializer permits task SCAN and any of $5^3$ color triples. Its current-state assumption requires SCAN to imply $(s_0, s_1, s_2) = (\mathsf{EMPTY}, \mathsf{RED}, \mathsf{NOT})$. Each of the other 124 triples gives no first ENV-safety successor, independently of all SYS/AUX choices. This explains its saved missing $V$ count 124, but does not establish the additional $W_V$ obstruction for the remaining tuple.

These last two arguments depend on the declared initial predicate remaining separate from current-only ENV safety. A convention that incorporates those invariants into the initial predicate can exclude the witness inputs. This matters for the historical coverage route, whose saved preprocessing reclassifies ENV SAFETY as STATE_INV. The archived StateInvTranslator disassembly has guards for next/prime and for ENV formulas referring to SYS/AUX variables before initial strengthening.

The relevant Airport and RoboticArm227 current-only formulas have no SYS/AUX reference, so their witnesses may be removed. Both direct PCar552/553 ENV safety formulas refer to SYS variables; the sensor-freeze formula also contains next. These local observations do not prove equality of the full translated models.

**Limits of the historical inference** The raw CSVs report Boolean feasibility labels, not cryptographic per-run source identities or returned-controller certificates. The literal-source contradiction for RoboticArm232 remains a source-contract fact even if historical preprocessing changed the model's meaning. We neither infer the cause of its reported positive label nor claim that the historical C-backed model is identical to our saved Java model. The other saved-model obstructions and initializer-sensitive arguments retain their stated scope. No new native, BDD, fixed-point, or controller execution was used for this attribution and written-proof analysis.

**Elevator032: complete graph details** The checked counts are $|W| = |C| = 12$ and $|V| = |W_V| = |K| = 11$; exactly one of four initial inputs lacks a response in $K$. The released 14-node graph yields an original view with nine nodes, 32 edges, and all four initial nodes. Initial node 3, (`NONE`, `STOP`), is a dead end there; its native successors all enter helper-true states. The saved legal prefix $0 \to 9 \to 3$, with valuations (`LOWER`, `STOP`), (`LOWER`, `FWD`), and (`NONE`, `STOP`), also reaches it from another initial node. Restricting initialization alone would therefore not make this existing graph cooperative.

**PCar058: complete graph details** Its strict and nonblocking regions have $|W| = |C| = |V| = |W_V| = 1404$, but $K = \varnothing$, on a 1944-state carrier. All six initial input choices have a response in $W_V$. No helper is added: the syntactic pass returns before response-pattern expansion because there is no direct parsed ENV safety constraint. The native graph and original view both contain 96 nodes and 576 edges.

## D. Per-input safety-subgame cardinalities

For the eight helper-bearing outputs in the fixed 14-input comparison, $D$ is the full typed original state space, including every logical auxiliary; $U$ is the safety-winning region after the helper is set; and $F$ is the condition for a safe entry into $U$. Section L proves the decomposition summarized in the main paper; the full relation comparisons follow below. The five other completed outputs have no helper and preserve the original model identically. These rows do not include the separately selected Humanoid 044 or AutonomousVehicle 188.

**Complete fourteen-input roster.** The subset contains all available independently certified carriers of at most 4,096 states. Selection was fixed before an interrupted second extraction attempt, without selecting by region emptiness or controller outcomes; that later attempt did not change these fourteen inputs. Table 4 lists the fixed inputs used on all four paths. The identifiers and source paths come from the frozen `small_native_recovery/SELECTION.json`; the same fourteen identifiers appear once per path in the 56 saved rows of `PUBLIC_14X4_OUTCOMES.tsv` (Section K). Helper status is taken from the released-route records, with the incomplete SmartHome1 run remaining UNKNOWN. AutonomousVehicle 184 is distinct from the separately selected continuation witness AutonomousVehicle 188.

Table 4: The fourteen public inputs. Source suffixes are relative to `artifact/experiments/specs/`. "Helper" records the released route; UNKNOWN does not mean that no helper was added.

| ID | Source suffix | Helper |
|---|---|---|
| 002 | `example/HumanoidLTL_377_Humanoid.spectra` | No |
| 032 | `syn15/ElevatorLTL_386_Elevator.spectra` | Yes |
| 034 | `syn15/GyroLTLVar3_702_GyroAspect.spectra` | Yes |
| 035 | `syn15/GyroLTL_390_GyroAspect.spectra` | Yes |
| 036 | `syn15/GyroLTL_486_GyroAspect.spectra` | Yes |
| 037 | `syn15/GyroLTL_588_GyroAspect.spectra` | Yes |
| 038 | `syn15/GyroLTL_590_GyroAspect.spectra` | Yes |
| 043 | `syn15/HumanoidLTL_409_Humanoid.spectra` | No |
| 057 | `syn15/PcarLTL_394_PCar.spectra` | Yes |
| 058 | `syn15/PcarLTL_464_PCar_fixed.spectra` | No |
| 093 | `syn17/AirportShuttle_AirportShuttle_49.spectra` | Yes |
| 149 | `syn17/SimpleVehicle_VehicleSpec_6.spectra` | No |
| 184 | `syn20/AutonomousVehicle_sprint01_ver1.spectra` | No |
| 223 | `syn20/SmartHome1_sprint01_ver1.spectra` | UNKNOWN |

Table 5: All eight helper inputs in the fixed 14-input cohort. Full typed original states include every logical auxiliary. Exact function comparison gives $W_R(\cdot, 0) = W_F = W$ and $W_R(\cdot, 1) = U$ in every row. The five no-helper cases also preserve $W$.

| ID | $\lvert D\rvert = \lvert U\rvert$ | $\lvert F\rvert = \lvert C\rvert$ | $\lvert W\rvert = \lvert W_F\rvert$ |
|---|---|---|---|
| 032 | 12 | 12 | 12 |
| 034 | 640 | 208 | 208 |
| 035 | 40 | 14 | 14 |
| 036 | 240 | 60 | 58 |
| 037 | 960 | 120 | 116 |
| 038 | 1920 | 992 | 992 |
| 057 | 162 | 132 | 132 |
| 093 | 360 | 232 | 232 |

**Full relation checks** All 13 completed released arenas from the fixed 14-input comparison preserve the original strict-winning region. The remaining input keeps its resource UNKNOWN. We establish this by auditing the full saved relations and applying the safety-subgame decomposition proved in Section L, retaining every original typed variable and logical auxiliary.

Eight outputs introduce the helper. For each, the independently calculated safety region is $U = D$ and the safe-entry condition is $F = C$. Seven have $F \subsetneq D$, but every original winning state belongs to $F$, so $W_F = W$. Direct strict GR(1) solution of each full saved transformed arena agrees with $W_F$ at $b = 0$ and $U$ at $b = 1$. Five no-helper outputs instead satisfy complete original-model identity. All 13 original strict initial decisions remain true, and original regions match their prior independent certificates under complete typed name/value mappings across encodings. Table 5 gives per-input cardinalities.

We compare ENV and SYS safety separately over all typed current/next valuations in $(b, b') = 00, 01, 10, 11$ and on legal joint edges. Elevator matches both complete guards because its original system safety is true. The other seven helper cases match literal ENV safety everywhere and original SYS safety in 00 and 01, but retain nontrivial SYS constraints in 11: neither complete guard matches, even on legal joint edges. SYS differences in environment-impossible 10 are recorded separately.

Original initialization and ordered justices, including helper justice, match throughout the typed current domain. Thus the complete-literal theorem alone cannot establish preservation here.

Retained constraints therefore need not cause loss: here they permit indefinite safe responses from every helper-true state. These checks concern saved strict arenas, independently of the native raw four-fixed-point decision or cooperation of an extracted graph. The 14-input cohort is disjoint from the ten condition-violating models; its correspondence and preservation results do not transfer to them.

Direct and decomposed calculations share the GR(1)/BDD engine. Finite-set qualification on 902 synthetic games covers non-Boolean types, multiple justices and environmental dead ends; it supplies a second representation, not an independently compiled public model.

**A separate condition-violating input** We also checked Humanoid 044, selected before its released-model run as the smallest of the three previously observed initial flips. This outcome-informed case is separate from the fixed 14-input cohort. The full typed mapping retains both past-time auxiliaries and verifies the original relations against the earlier model. The released SYS relation again retains constraints at $b = 1$, yet $U = D$ and $F = C$. Direct solution and the generalized decomposition agree: $W_F = W_R(\cdot, 0) = \varnothing$ and $W_R(\cdot, 1) = D$. Thus all 2,387 original winning states are lost, including 1,211 through the invariant restriction beyond the immediate loss. The original initial decision changes from true to false; all eight allowed input choices lack a winning initialized response. Its original nonblocking region $V$ was already empty, so this is not rejection of a cooperatively feasible contract. This model-only experiment makes no claim about a native four-fixed-point verdict or an extracted controller.

## E. Public source inventory and a cooperative continuation witness

**Inventory and certified cohort.** The archive contains 449 source paths and 425 byte-distinct units: 423 non-library sources and two pattern libraries. Extraction used representative archive-path order, without prioritizing known outcome labels, with a fresh JVM per input and limits of 60 seconds, 512 MiB heap, and 128 MiB direct memory. After excessive debug output limited the first run, a declared rerun used reversible output compression with the same input order and computation limits. A BDD-file cap followed by a process-cleanup permission failure caused an integrity stop. Its saved outcomes are 70 completed original-model runs, 107 resource UNKNOWNs, 54 technical UNKNOWNs, one integrity UNKNOWN, and 193 unattempted inputs. These are not a completed evaluation of all 425. Of the 70 completed runs, the independent checker certified 59; two exceeded its 64-variable limit and nine failed public-field roster integrity checks involving array metadata. All 59 normally terminated PASS certificates enter the preservation cohort, without filtering by region sizes or outcomes. The checker shares the compiler-exported predicates and typed encoding; its independent ROBDD computation does not validate the compiler. The C59 reader-index entry (Supplement K) links the complete roster, the extraction receipt, the independent checker's protocol and frozen inputs, and its certification and stop records. These are saved records of the stopped run, not evidence of a completed whole-corpus rerun.

**Selection and scope.** Unit 188 is the unchanged public source `AutonomousVehicle_sprint02_ver1.spectra` from the Kind artifact. It is outside the fixed 59-model certified cohort. An initial exploratory lexical screen examined the 366 units outside the certified cohort, including both libraries and the eleven uncertified completed models. Eleven units were read in detail, and the retained witnesses were selected by their outcomes. These observations support no prevalence

estimate. The source has ten Boolean environment variables, system lane $\ell \in \{1, 2, 3\}$ and speed $v \in \{0, \ldots, 10\}$, no explicit initializer, and no pattern imports or logical auxiliaries. Its typed carrier has $2^{10} \cdot 3 \cdot 11 = 33792$ states.

**Source contract.** Write $o_i, p_i, g_i$ for obstacle, pedestrian, and green-light inputs in lane $i$, and let $f_i = g_i \wedge \neg p_i \wedge \neg o_i$ mean that lane $i$ is free. The environment safety relation is

$$(o_1 \Rightarrow \neg o_2') \wedge (o_2 \Rightarrow (\neg o_1' \vee \neg o_3')) \wedge (o_3 \Rightarrow \neg o_2'),$$

and its justice assumptions are $\mathsf{GF} f_i$ for all three lanes. The fourteen system safety clauses require stopping for a red light or pedestrian in the occupied lane, positive speed in a free occupied lane, speed below five at a crosswalk, no direct jump between the outer lanes, and the following responses to current obstacles. A blocked outer lane forces the next lane to be the middle lane. From a blocked middle lane, a next obstacle on side 1 forces side 3, a next obstacle on side 3 forces side 1, and two clear sides require an outer lane. The system justice is $\mathsf{GF} \bigvee_i (\ell = i \wedge f_i)$.

Let $s$ have $\ell = 2$, $v = 0$, $o_2 = 1$, $o_1 = o_3 = 0$, all lights green, no pedestrians, and no crosswalk. This state satisfies the current-state speed duties. The typed next input with $o_1' = o_3' = 1$ is excluded by environment safety but requires both $\ell' = 1$ and $\ell' = 3$ under system safety. Thus no typed system response exists and $s \notin C$. This obstruction uses only the next-lane clauses and is independent of the timing convention for unprimed invariants. The policy and initializer checks below use the saved translation, in which the seven speed duties are current-state obligations.

**A sufficient invariant and policy.** Put $f = \bigvee_i (\ell = i \wedge f_i)$ and define

$$R = (f \wedge v = 1) \vee (\neg f \wedge v = 0), \qquad P = R \wedge (\ell \neq 2 \vee \neg o_2).$$

The policy selects the next lane by these ordered rules:

1. From a blocked middle lane, choose side 1 if $o_1' = 0$ and side 3 otherwise. Environment safety makes the chosen side clear.
2. From a blocked outer lane, choose the middle lane. Environment safety makes the next middle lane clear.
3. Otherwise, if $o_2' = 0$, choose the middle lane.
4. Otherwise remain in the current outer lane; from the middle lane choose side 1.

Set the next speed to one exactly when the selected next lane is free, and to zero otherwise. These rules define one response to every typed input. From $R$, every original-legal input has a safe response that enters $P$. The current speed duties hold because the current state is in $R$, and the next speed rule establishes them at the successor. The lane rules meet all forced moves and exclude outer-to-outer jumps.

On $P$, an input with $f_2'$ always selects lane 2. The first rule is inactive there; the second and third rules choose lane 2 whenever it is free next. Consequently $\mathsf{GF} f_2$ implies the system justice. The all-clear input, with all lights green and no pedestrians, is environment-legal from every state and may repeat forever. It satisfies all environment justices. The same policy therefore has an assumption-fair continuation after every legal prefix, including prefixes from $s \in R$. This proves legal-prefix cooperation. Here the environment assumptions are entirely system-independent: cooperation follows automatically from strict winning. The example does not demonstrate a separate strategic difficulty in cooperation.

For every state in $P$, arbitrary typed inputs also admit some safe response: a blocked outer lane moves to the middle, a clear outer lane may stay, and an unblocked middle lane may move to side 1. The speed rule meets the state duties. Hence $P \subseteq C$. The legal policy preserves $P$, so it wins in the original game restricted by $C$. For every initial environment value, choose lane 1 with the speed rule. This lies in $P$ and satisfies the original initializer. Thus $I_M(K)$ and $I_M(W_C)$ are both true, although $s \in K \setminus W_C$. Membership in $K$ follows from the exhibited strict cooperative strategies and the legal-prefix characterization in Section L.2, using the round-to-alternating correspondence in Section A and the total-arena soundness and completeness theorem of [6]. No complete $K$, $W$, or $W_C$ region is needed for this argument.

**Reachability under one cooperative winning strategy.** The invariant policy itself avoids $s$. A second strategy uses one private bit, initially armed, and the same initial selector. If its first current state is all-clear, lane 1, speed 1, and its first next input equals the environment component of $s$, it chooses lane 2, speed zero. On every other first input it uses the base policy. It then disarms the bit and uses the base policy forever. The exceptional edge satisfies original environment and system safety. After that edge, the next legal base-policy response enters $P$. Thus every legal play enters $P$ within two responses, and changing this finite prefix does not change the justice argument. The all-clear continuation also establishes cooperation while the bit is armed. One allowed initial input and next input therefore reach $s$ under an initialized cooperative winning strategy. This is existential reachability under a specified strategy, not forced reachability, a synthesized native controller, or a recorded execution. The bit is strategy memory and is not added to the source model.

**Meaning for continuation.** The loss concerns re-synthesis that retains the full current state, including lane and speed. Fixing only the environment state allows a different initial system response and need not cause refusal. No source initializer was changed to manufacture an initial counterexample. There is no observed repair transaction or deployment benefit. The earlier full-region attempt remains recorded as resource UNKNOWN; the targeted certificate below did not complete it. A subsequent, separately bounded native pair now computes both full strict regions, as reported after the source-correspondence checks.

**Targeted saved-model evidence.** Two fresh model-only executions export the original and released transformed arenas before solver construction. The original has 33792 typed states and 32 Boolean encoding variables; the transformed arena has 67584 states and 34 encoding variables, adding only the environment helper. Full typed comparisons of the environment and system relations in all four helper quadrants agree exactly with the literal complete-relation equations. Original initialization and the ordered justice lists also agree, with the unique added $\mathsf{GF}\neg b$ assumption in position four. In particular, the system relation is unrestricted when the current helper value is $b = 1$. This is a separate case from the fixed 14-input comparison and its retained constraints.

On the fresh original predicates, a targeted checker verifies thirty obligations implementing the preceding proof. It establishes $|R| = 3072$, $|P| = 2560$, $|C| = 6656$, and singleton displayed states. The policy covers all $33792 \cdot 1024 = 34603008$ typed current/input pairs with exactly one response each. Safety and invariant entry are checked on every original-legal policy edge from $R$; universal totality alone is not treated as safety. Ordered justice identities, the all-clear fair continuation, the quantified initializer, the response obstruction at $s$, and the exceptional initialized edge all pass. A separate synthetic qualification has one positive case and nine deliberate predicate mutations, each producing exactly its expected failed obligations. It does not count as public-model evidence.

The target checker performs no winning-region fixed point or controller extraction. Combining its

invariant certificate with the independently audited full helper correspondence gives $s \in K \setminus W_R(\cdot, 0)$ and $I_M(K) = I_M(W_R(\cdot, 0)) = \mathit{true}$. The latter equality concerns strict initial feasibility in the transformed arena, not cooperation there or a native four-fixed-point verdict. The checks retain all original variables and their typed encodings.

**Independent source correspondence.** A separate audit directly encodes the unchanged source's fourteen system-safety clauses, three environment-safety clauses, and four ordered justice predicates. Since the source has no explicit initializers, both initial predicates are true on the typed domain. Its seven speed constraints remain current-state obligations; no implicit next-speed or initial restriction is added. An independent saved-DAG branch reader agrees on all eleven exported predicates, including the three typed-domain factors. Raw-domain checks include unused lane and speed encodings. The initial and justice predicates agree on all 33792 typed states. Decoded variable support justifies exact quotient comparisons of each safety relation over all $33792^2 = 1141899264$ typed current/next pairs: system safety uses 4460544 representatives of multiplicity 256; environment safety uses 64 of multiplicity 17842176. Unreachable and environment-illegal pairs are included. All differences are zero. Qualification uses 97 hand-derived source cases and 51 artificial boundary checks. This strengthens the source-to-original-predicate correspondence for this input; it proves neither general compiler correctness nor a complete winning region.

**Completed native strict pair.** Two preregistered calls [2] use the unchanged source and the fixed twelve-field state $s$ above, first on the full original arena and then on the full released syntactic-helper arena, testing target $(s, 0)$. Both use the released `GR1Game` three-fixed-point strict solver with JTLV, without replacing initializers or conjoining any state, transition, or layout restriction. There is one fresh JVM per arm, no retry, no four-fixed-point call, and no controller extraction. Both processes exit normally and report complete winning memory. The results are:

| Arena | Typed states | Full strict $W$ | Target in $W$ | Initializer |
|---|---:|---:|---:|---:|
| Original | 33,792 | 8,064 | Yes | True |
| Released helper | 67,584 | 40,448 | No | True |

The helper count includes both helper values; it is not a helper-false slice count. The target in the second row is $(s, 0)$. Initial feasibility is tested separately as $\forall e : \theta_e(e) \Rightarrow \exists y : \theta_s(e, y) \wedge W(e, y)$; every one of the 1,024 allowed initial ENV values has a winning SYS choice in each arm.

Before solving, full field/ownership/encoding, initializer, transition and ordered-justice comparisons bind each fresh model to its previously certified arena. A separate saved-DAG audit verifies these identities, target membership and initial quantification, using its own parser and Boolean operations. Loaded class origins and hashes are checked; the native solver, compiler/exporter and field metadata remain shared trusted inputs. Source cooperation and existential initialized reachability use the preceding policy proofs, not either native call.

Each call has a 120-second wall limit, 1-GiB sampled process-group RSS limit, 768-MiB heap and 64-MiB direct-memory allowance. Encoded stdout and stderr are each capped at 16 MiB, nonstream artifacts at 64 MiB per arm, and combined arm outputs at 128 MiB. Elapsed times are 2.288 and 4.185 seconds, respectively; these observations support no speed comparison. The complete ordered stdout is stored with the previously qualified lossless encoding. Encoded streams and their end receipts are checked; no expanded raw stdout file or raw-stream SHA-256 is claimed.

This pair supplies native strict-backend corroboration for one existing outcome-selected public source/state. The transformed negative is correct for its own arena; transferring it to source infeasibility would be unsound. It is not a selected Kind four-fixed-point or frontend/default run, a

wrong initialized verdict, a native cooperative controller, or an executed replacement. Historical UNKNOWNs, including Patrol's full helper queries, are unchanged.

## F. A residual-only public source witness

Unit 120, `ExitModule_ParkingSpec_194.spectra`, is another outcome-selected source outside the certified cohort. Its witness is source-level only; it has no newly generated or certified model. The environment valuation is front sensor zero, side sensor one, parallel parking status, side status NONE, black color, alignment true, and done false. The system valuation is STOP, STRAIGHT, velocity zero, and PARKING. All ten instantiated response-pattern monitors are at their initial state $S_0$.

A typed next front-sensor value of one simultaneously activates an exit guarantee requiring next velocity one and a STOP guarantee requiring next velocity at most the current zero. Both constraints refer explicitly to next velocity, so the conflict does not depend on the timing convention for unprimed invariants. The input is environment-illegal: stopping freezes the front sensor at zero. Thus the full residual state lies outside $C$.

On every legal input, choose PARKED, STOP, STRAIGHT, and velocity zero, then retain those values. Stopping freezes the relevant sensor, alignment, color, and done values; parallel status remains fixed and the unrestricted side-status input has no active effect. All ordinary guarantees hold, and the system parking justice holds from the first successor. The eight environmental response-pattern triggers require forwarding or reversing, so they remain inactive. In the two system response patterns, respectively, the response is already true at velocity zero or the trigger is false. Each monitor remains at $S_0$ according to the imported `pRespondsToS` update and meets its $\mathsf{GF}S_0$ justice. The three ordinary environment justices contain the stopping disjunct. Every legal prefix therefore has a fair continuation under this residual strategy, with monitor state retained.

The declared initializer requires alignment false, whereas this witness has alignment true. We establish neither its reachability from the original initializer nor initial cooperative feasibility. The literal semantic theorem gives residual loss for the stated monitor expansion. The compiler's auxiliary ownership and any support generated by the unused past-time definition require separate checks before this reasoning transfers to a saved released model. This source witness is not included in any computational cohort or count.

## G. A published learned repair tested for cooperative admission

We additionally test the preferred learned Arbiter solution in Appendix C.4.1 of [10]. The same declared reconstruction from the eleven-candidate cohort restores the omitted response-pattern declaration: 886 bytes, no imports. After inspecting prior results, we selected the smallest candidate with empty original $V, K$ among those previously positive under residual initialization. This outcome-informed case is separate from the fixed 14-input public-corpus comparison. Cooperative admission adds to the degradation study's ordinary-realizability objective, which we do not dispute or historically reproduce.

**Source obstruction.** The environment fields are $a, r_1, r_2$, the public system fields are $g_1, g_2$, and two system-owned response monitors are retained. The original environment safety is

$$E(s) = a \vee (\neg g_1 \wedge \neg g_2 \wedge \neg r_1 \wedge \neg r_2), \qquad \rho_e(s, e') = E(s), \quad \theta_e = \mathit{true}.$$

Because this current-only invariant refers to system variables, the audited StateInvTranslator leaves it as current safety; it neither strengthens the environment initializer nor globally primes the formula.

Ordinary system invariants contribute current initialization and next-state safety. The response monitors update from current requests and grants.

If $E(s)$ is false, no legal environment input exists. Otherwise every next input is legal, including $(a', r'_1, r'_2) = (0, 1, 0)$. If no safe response exists, strict winning fails. Every safe response instead reaches $t$ with $E(t) = \mathit{false}$, regardless of system values, auxiliary states, or private memory. That legal prefix has no infinite legal continuation, hence no assumption-fair continuation. Thus no residual state admits a strict cooperative strategy. The nonempty original initializer then excludes initial cooperative realizability; this does not assert failure for artificial vacuous initializers.

**Complete output and independent check.** The configured released job returns a complete 112-state, 1,176-edge native graph with eight initial states covering all initial environment choices. The reachable helper-false view contains 56 states and 280 edges, preserving native IDs, rank memory, both monitors, and initial choices. Its original safety and input coverage pass; strict liveness holds, but 21 reachable states have no assumption-fair continuation. An independent implementation directly evaluates the saved decision DAGs against source-coded equations on all 128 original states and 16,384 state pairs. It checks the native printer, ordered justices, 1,792 native input candidates and 448 original-view input candidates. Bitset transitive closure independently establishes the graph properties. It imports no production BDD or fairness checker; its result was sealed before comparison with that checker, after which the state, edge, justice and SCC sets agreed. Both checks share compiler exports; this source-equation check is specific to the contract, not a general compiler proof.

**Two attempts and limits.** The first native attempt stopped at an 8-MiB stdout cap and remains UNKNOWN. A separately recorded followup kept the source, runtime, classes, synthesis settings, extractor, printer and checker unchanged. Native stream caps rose from 8 to 64 MiB, the wall bound fell from 60 to 10 seconds, and the output postcheck rose from 64 to 128 MiB; checker limits stayed fixed. It completed with all 11,927,997 stdout bytes retained, normal teardown and child reaping: one complete result from two attempts on one candidate. No transformed strict-winning region was calculated. Neither native acceptance nor the complete-literal $C = D$ result establishes preservation in this released arena. No repair learning, adaptation loop or deployment was executed; other cohort counts remain unchanged.

## H. The fixed eleven repair candidates and selected Lift behavior

The completed comparison covers all eleven replacements in Appendix C.3–C.4 of [10]: five preferred learned candidates and six trivial alternatives from five examples. Two Minepump alternatives use the unchanged listings. The other nine use the declared executable reconstructions appending the paper's omitted response-pattern declaration; Traffic Single P also lowers its direct eventuality to that pattern. Lift's one-floor assumptions and movement guarantees are unchanged from the published replacement listings. Its appended pattern includes auxiliary initialization, deterministic updates, and justice [10, Appendix C.2.1, C.3.3, and C.4.3]. These are correlated published candidates, not eleven independent applications. The constructed controls in Section I are excluded. The extraction intervention and modeled episode below are later followups on existing inputs; they do not alter this cohort's released-route outcomes or denominator.

**Traffic Single's direct-eventuality lowering.** The learned Traffic Single guarantee $\mathsf{G}(s \Rightarrow \mathsf{F}p)$, with $s = \mathit{car}$ and $p = \mathit{green}$, uses the paper's response pattern. Writing $q = 0$ for $S0$ and $q = 1$ for $S1$, its deterministic monitor is $q_0 = 0$, $q' = (q \vee s) \wedge \neg p$, with justice $\mathsf{GF}\neg q$. Every infinite

observable word has exactly one monitor run. If every trigger receives a current or later response, each pending obligation clears; either clearances recur infinitely or $q$ eventually stays zero. Conversely, an unanswered trigger makes $q$ permanently true from the next step, violating the justice. Hence the two forms have the same infinite observable trace language after projecting out $q$. This is the saved reconstruction argument; compiler correspondence, auxiliary ownership, and Mealy support remain separate obligations. The guarantee retains SYS ownership. The equivalence starts at the original initializer; continuation checks retain the reached monitor value.

**Initialization, correspondence, and output checks.** For contract-feasibility checks, every case uses its original initializer, including all logical auxiliaries. For the independently certified original cooperative region $K$, we check

$$I_M(K) = \forall e : \theta_e(e) \Rightarrow \exists y : \theta_s(e, y) \wedge K(e, y),$$

where $y$ includes the logical system monitors. All eleven have nonempty legal initial environment domains; eight pass this predicate. The saved region authority uses the original nonblocking arena and the normalized round/alternating correspondence of Section A. Neither a residual trace entry nor $K \neq \varnothing$ substitutes for initialization.

**Minepump P: strict initialization and the native refusal.** The saved original strict region contains 28 of the 32 typed states. Its three-node Boolean representation is

$$W = \neg \mathit{highwater} \vee \neg \mathit{methane} \vee \mathit{pump}.$$

The original environment initializer fixes highwater and methane to false; the system initializer fixes pump to false, the response monitor to $S0$, and the previous-pump auxiliary to false. Substitution into the saved $W$ gives $I_M(W) = \mathit{true}$, while the existing nonblocking and cooperative certificates give $V = K = \varnothing$ and $I_M(K) = \mathit{false}$. This is a static reading of the saved region and initial predicates, not a new solver run. The historical residual trace entry has highwater and methane true and pump false, so it lies outside $W$; it is not an original initial state. The configured route's native False cannot be transferred as a claim of original strict unrealizability. Cooperative rejection instead follows from the empty $V$; no complete current released strict-arena explanation of that raw refusal is asserted.

The released route and settings are specified in Sections L.3 and T; Section Y.1 distinguishes that selected route from the strict diagnostic queries. All eleven adopted candidate jobs retain the original initializers; earlier residual trace entries describe separate studies. The nine complete correspondence-gated views cover all original ENV initial choices. Lift P's direct graph audit and Minepump P's no-controller outcome are distinguished below. Complete native outputs are preserved in the study records. The Lift static addendum omits Lift P's historical 220,856,594-byte stdout, so that stream cannot be independently rehashed from the addendum. The correspondence-gated audit requires typed equality of initial, safety, auxiliary-update and ordered-justice predicates under identity or the helper-false mapping. The environment initializer is also checked over the full native typed domain. For the nine newly executed candidates, separate implementations check admitted graphs for original initial inputs, every legal next input, the selected safe response and private-memory successor, then strictness and cooperation. Saved comparisons require equal state/start/edge sets, ordered justice labels, SCC evidence and retained raw nodes, including ranks and auxiliaries. A failed correspondence or incomplete output remains UNKNOWN at this gate.

Independent hand-derived source equations are also compared with saved initialization, safety, and ordered justice predicates over the full typed domains, retaining logical auxiliaries. For Arbiter

P and Traffic S P, the prior audits use the helper-false view of the released model; a separate nine-candidate batch checks helper-free originals for every remaining candidate. These comparisons use the documented translation conventions. Source-to-original equality does not resolve Lift P's distinct released-model mismatch below.

Table 6: Correspondence-gated audit of the fixed eleven candidates. P denotes preferred/learned, T trivial, and S/U Traffic Single/Updated. Native False is the solver's negative verdict; dashes mean no controller. Question marks mean original correspondence failed. $^{b}$ denotes a reused prior baseline and $^{f}$ a separate Lift resource followup. This table retains the released-route baseline; the later extraction intervention and the post-hoc direct Lift P audit below are separate.

| Candidate | $I_M(K)$ | Native | View states/edges | Strict | Coop. |
|---|---|---|---|---|---|
| Arbiter P$^{b}$ | F | Graph | 56/280 | Pass | Fail |
| Arbiter T | T | Graph | 56/448 | Pass | Pass |
| Lift P$^{f}$ | F | Graph | ? | ? | ? |
| Lift T$^{f}$ | T | Graph | 154/1170 | Pass | Fail |
| Minepump P | F | False | — | — | — |
| Minepump T1 | T | Graph | 12/40 | Pass | Pass |
| Minepump T2 | T | Graph | 4/16 | Pass | Pass |
| Traffic S P$^{b}$ | T | Graph | 16/80 | Pass | Pass |
| Traffic S T | T | Graph | 12/80 | Pass | Pass |
| Traffic U P | T | Graph | 68/323 | Pass | Pass |
| Traffic U T | T | Graph | 56/416 | Pass | Pass |

**Attempt history and denominator.** Arbiter P reuses the complete followup in Section G; Traffic S P reuses its earlier 52-state/496-edge native baseline. Exact certificates and source-byte mappings are retained. Each remaining candidate received one declared native attempt. Six returned graphs passing both checks, Minepump P returned native false, and both Lift attempts stopped at the 64-MiB stdout cap, not the ten-second wall bound. Those two attempts remain resource UNKNOWN.

One separately authorized additional attempt per Lift candidate kept the sources, runtime, synthesis settings, extractor and audits fixed. Stream caps rose to 256 MiB and the per-case output threshold from 128 to 512 MiB; the ten-second native wall and 1-GiB RSS limits were unchanged. Both completed with full stdout: 220,856,594 bytes for Lift P and 219,540,020 for Lift T. The former still fails original correspondence; the latter has a certified strict-only view. At this gate, ten candidates return native graphs, nine have complete original-view certificates, seven pass both properties, two pass strictness only, and one returns native false. Followups do not add candidates or rewrite prior attempts.

**A separate original-initializer synthesis route.** After recording the historical outcomes above, we froze the same eleven source files, their order, and the settings for one additional route. It applies original-$V$ restriction and justice normalization $G \mapsto G \wedge A$, then the existing normalized four-fixed-point solver and unchanged generic graph constructor. It adds no helper and does not apply the final-choice preference used in the Lift intervention. The source initializers remain unchanged: every legal initial ENV choice must have a SYS/AUX response satisfying $\theta_s \wedge K$, and extracted controllers initialize their private rank to zero. Saved replay or residual entries are not substituted for these initial conditions. All eleven source ENV and joint initializers are nonempty.

Each source received one serial native process attempt, with concurrency one and zero retries. Eleven attempts led to eight solver/extractor calls; Arbiter P, Lift P, and Minepump P stopped before those calls on independently confirmed empty $V$. All eleven original arenas, including initializers, safety, typed auxiliaries, and ordered justices, match their source-bound saved comparators. The full $V$ and $K$ Boolean functions also agree, including all AUX valuations. The comparisons use exact exported predicates and shared parser-translated BDDs; they are not an independent compilation.

Table 7: Separate normalized-route results from the original initializer, in the fixed execution order. Init. counts graph initial nodes. Coop. is legal-prefix cooperation. Empty $V$ is an independently checked refusal before solver and extraction; its controller flags are null, shown as dashes. This table does not replace Table 6.

| Candidate | $I_M(K)$ | Outcome | States/edges | Init. | Strict | Coop. |
|---|---|---|---|---|---|---|
| Arbiter P | F | Empty $V$ | — | — | — | — |
| Arbiter T | T | Graph | 56/448 | 8 | Pass | Pass |
| Lift P | F | Empty $V$ | — | — | — | — |
| Lift T | T | Graph | 154/1176 | 2 | Pass | Pass |
| Minepump P | F | Empty $V$ | — | — | — | — |
| Minepump T1 | T | Graph | 12/40 | 1 | Pass | Pass |
| Minepump T2 | T | Graph | 4/16 | 1 | Pass | Pass |
| Traffic S P | T | Graph | 28/152 | 8 | Pass | Pass |
| Traffic S T | T | Graph | 20/120 | 8 | Pass | Pass |
| Traffic U P | T | Graph | 116/546 | 8 | Pass | Pass |
| Traffic U T | T | Graph | 84/640 | 8 | Pass | Pass |

All eight complete graphs pass the original-contract checks for initial coverage, legal-input coverage, safe responses, private-memory successors, strictness, and legal-prefix cooperation. The three refusals have false $I_M(K)$ and no controller verdict. This run has no UNKNOWN or unattempted row. Its graph certificates come from one original-contract checking implementation; the two-judge Lift experiment and the separate fourteen-input comparison retain their own scopes.

**Interpretation and limits.** The baseline and normalized jobs use the same original initializers. The new eight accepted graphs and historical seven passing outputs come from separate whole-route executions, not a matched extraction or success-rate comparison. Historical refusals, resource UNKNOWNs, and Lift P's released-model correspondence failure remain unchanged. For Lift T, the new certificate records zero copied source ENV justices and a 154-state, 1,176-edge graph with two initial nodes. Its equal size does not establish identity with the intervention graph; no graph-equivalence comparison was performed. The successful route supplies an existing alternative without the new preference, while the paired same-game, same-memory intervention still shows one sufficient local change. This whole-route comparison does not isolate the effects of $V$ restriction, normalization, helper removal, or extraction, and establishes neither preference necessity nor a general synthesis failure. It does not test Friendly's specialized strategy construction or add an eleven-case Slugs run. No new route output was installed or executed in the maintenance-to-service episode. The fixed correlated sources and single attempts establish no timing superiority or broader generality.

**Source clauses shared by Lift T and Lift P.** The Boolean environment inputs are $b_1, b_2, b_3, c$ and the system outputs are $f_1, f_2, f_3$. Write $m_i$ for the SYS-owned logical response monitor of button $i$, encoding $S0 = 0$ and $S1 = 1$. Let One($f$) mean exactly one true floor. The common environment

safety relation is

$$E_L = \mathrm{One}(f) \wedge \bigwedge_{i=1}^{3} \left[ (b_i \wedge f_i \Rightarrow \neg b_i') \wedge (b_i \wedge \neg f_i \wedge \neg c \Rightarrow b_i') \right].$$

One-floor consistency is thus a condition on the *current* outputs in environment safety. The movement guarantees are $\bigwedge_i (f_i \Rightarrow D_i)$, where

$$\begin{aligned} D_1 &= (f_1' \wedge \neg b_2 \wedge \neg b_3) \vee (f_2' \wedge b_2) \vee (f_2' \wedge b_3) \vee (f_1' \wedge c), \\ D_2 &= (f_1' \wedge b_1) \vee (f_2' \wedge \neg b_1 \wedge \neg b_3) \vee (f_3' \wedge b_3) \vee (f_2' \wedge c), \\ D_3 &= (f_3' \wedge \neg b_1 \wedge \neg b_2) \vee (f_2' \wedge b_1) \vee (f_2' \wedge b_2) \vee (f_3' \wedge c). \end{aligned}$$

Every button and cancellation guard here uses the current value. System safety conjoins these movement implications with

$$\bigwedge_{i=1}^{3} \left[ m_i' \leftrightarrow (m_i \vee b_i) \wedge \neg (f_i \vee c) \right].$$

In particular, it does not add $\mathrm{One}(f')$: a movement disjunct can permit extra true next floors. Initialization sets $b = 000$, $f = 100$, and $m = 000$, leaving $c$ unrestricted. The environment justice list is empty; the three system justices are $\mathbf{GF}\neg m_i$. Lift T has $\rho_e = E_L$. These are the clauses of `lift__trivial_1.spectra`, lines 3–81, shared with `lift__preferred.spectra` except for the latter's two added assumptions stated below. Supplement K's Lift T entry links the source, policy, positive graph, complete extension, and selected graph. The policy's private priority bit is separate from all three logical monitors.

**Lift T: feasibility and a failing selected graph.** The typed original carrier has 1,024 states, including three response monitors $m_i \in \{S0, S1\}$. The saved regions have $|V| = 384$ and $|K| = 316$. Both legal initial choices, distinguished by cancellation input $c$, have the unique initialized system/-monitor response in $K$. The native graph has 794 states and 13,874 edges; its original view has 154 states, 1,170 edges and two initial nodes. Both checkers establish strictness and reject cooperation.

Table 8 records the original-legal prefix $0 \to 10 \to 339 \to 438 \to 704 \to 680$. It first leaves both $K$ and $V$ on its last edge. All five prefix edges satisfy original safety and the exact current-state monitor updates. The final move is allowed by the next-$f_3$ disjunct of the current floor-2 move constraint. Its resulting two true floors violate the environment's current-state one-floor assumptions, making every next original environment input illegal. Nodes 680 and 708 are the only two original-view nodes outside both $K$ and $V$; each has zero original-view successors and 16 native successors, all helper-true. The original environment justice list is empty, but a finite dead end still fails the infinite-extension requirement.

For the same observed next environment tuple and the same required next monitor tuple $(S1, S0, S1)$, changing only next $f_2$ to false gives $(f_1', f_2', f_3') = (0, 0, 1)$. This satisfies source safety and lies in saved $K \cap V$. It is not a selected successor of node 704, and we assign it no replacement native rank or private memory. This one-step alternative is not a repaired controller or a composed strategy. It is also distinct from the later extraction experiment: at first creation of a bad node, the final native candidate retains floor 100, whereas the nearest-$Y$ progress filter in rank selection has already removed 001. On the displayed $704 \to 680$ edge, reuse of an existing node has already removed every continuable candidate. The diagnosis below distinguishes these stages.

Table 8: Recorded original-legal Lift T prefix, starting at native initial node 0. Bit strings use index order 1,2,3, with $S0 = 0, S1 = 1$; the helper is false throughout. The recorded native system-justice index $r$ names $\mathbf{GF}\neg m_{r+1}$, not a decreasing proof rank.

| Node | $b_1b_2b_3$ | $c$ | $f_1f_2f_3$ | $m_1m_2m_3$ | $r$ |
|---|---|---|---|---|---|
| 0 | 000 | 0 | 100 | 000 | 0 |
| 10 | 010 | 1 | 100 | 000 | 0 |
| 339 | 001 | 1 | 010 | 000 | 0 |
| 438 | 100 | 0 | 001 | 000 | 1 |
| 704 | 101 | 0 | 010 | 100 | 2 |
| 680 | 101 | 0 | 011 | 101 | 0 |

**A separate positive source policy.** A fixed, independently checked policy directly witnesses original initial feasibility. It always sets exactly one floor true and uses one private priority bit $d \in \{1, 3\}$, initially 3. From floor 1 or 3 it moves to floor 2 exactly when another floor is requested. At floor 2 it chooses a requested endpoint, stays if neither is requested, and uses $d$ if both are requested; a departure toward $j$ sets $d' = 4 - j$. Decisions use current buttons. Encoding $S1$ as true, every logical monitor updates as $m_i' = (m_i \vee b_i) \wedge \neg(f_i \vee c)$.

Every move satisfies an explicit source disjunct. The one-floor invariant keeps legal environment successors available. Take a state at time $t$ with $b_i$ true. If $f_i$ or $c$ already holds, the request is answered. Otherwise its button persists until arrival or cancellation. An endpoint request can be delayed by at most one trip to the opposite endpoint before priority reverses, so the requested floor is reached within four transitions from $t$ unless cancellation occurs first. Arrival or cancellation makes its monitor $S0$ in the next state, because its update reads the current $f_i$ and $c$. All three monitors therefore visit $S0$ infinitely often. Exhaustive construction from both original initial states (both values of $c$, with all three source monitors initially $S0$) produces 128 nodes and 1,020 edges, preserving every legal input, all ten original fields and the private bit. Direct source and independent finite-graph checks certify strictness and cooperation. This separate source-coded existence witness uses no native graph, rank, saved $K/V$ restriction or repair as a construction input; it makes no claim from arbitrary orphan monitor states or every state of $K$.

**Extension to the complete transformed model.** For an explicit cooperative suffix after any legal policy prefix, keep $c' = 1$ and choose $b_i' = b_i \wedge \neg f_i \wedge \neg c$ at every step. The two ENV button implications are satisfied, the policy preserves exactly one floor, and current cancellation clears all monitors within two transitions. Continuing these choices gives an infinite ENV-safe suffix; the source has no ENV justices. This is a semantic witness, not a claim that cancellation was used in the recorded workloads.

The same positive policy also extends to a strict strategy for the complete saved transformed model, with environment helper $h$. Keeping its exact source AUX updates and private priority, we checked all 49,152 prescribed transitions over every one-hot current floor, public input, AUX valuation, priority and current/next helper value. Every response satisfies both original system safety and complete native system safety. This includes all first-flip and post-flip inputs. On this one-hot carrier, which the policy preserves, the helper cannot revert on a native-legal edge. Initialization, domains and ordered justices were checked on all 2,048 native typed states. We do not approximate native system safety by merely disjoining the helper into the source relation, because AUX updates remain required.

Starting from either original initial input, a play that keeps the helper false is an execution of the

certified 128-state source controller. If the helper rises, it remains true, falsifying the environment justice $\mathbf{GF}\neg h$, while system safety continues. Thus the policy is native-strict and its helper-false view is cooperative. This establishes neither liveness from arbitrary orphan-AUX states nor cooperation after a helper flip. This source-policy argument alone does not attribute the selected controller's failure; the following paired experiment supplies separate evidence about extraction choices.

### Lift T: controlled extraction and modeled replacement

For this Lift T instance, changing one extraction preference while holding the solved game and fixed-point memory fixed produces a strict, cooperative original-contract controller. A subsequent constructed execution uses the checked controller after an environment-model weakening. Both are followups designed after inspecting the baseline failure, not held-out validation or a general synthesis correction.

**Baseline and fixed-point agreement.** A fresh run reproduces all 23 saved semantic artifacts of the released baseline, including its complete 794-state, 13,874-edge native graph and 154-state, 1,170-edge original-contract view. Both graph auditors again accept strictness and reject cooperation. A behavior-preserving observer uses the same model and memory objects; complete graph equality, semantic equality of the saved model and memory predicates, and unchanged variable order establish observation parity.

A separate finite-set implementation evaluates the inspected raw four-fixed-point recurrence on all 2,048 transformed typed states. It uses the saved transition predicates and their original move quantifiers, without an *Alive* restriction or justice normalization. Its region contains all 2,048 states, exactly matching the native region. The effective 12 $X$ slices, 12 $Y$ slices, three $Z$ components and final winning set also agree on all 57,344 state memberships. The recurrence uses Python integer sets; a shared saved-BDD decoder only reads predicate membership. These checks establish numerical agreement on this input, not compiler independence, the raw core's semantic soundness premise, or a general progress-memory theorem.

**The transformed strict game permits the dead end.** Node 680 belongs to the raw region and satisfies the original response condition $C$. A focused check of its saved residual strategy also witnesses membership in the actual transformed strict-winning region $W_R(\cdot, 0)$. All 16 legal next inputs flip the helper; none can keep it false. The residual graph has 321 states and 5,136 edges. At every reached node it covers exactly every legal input with a unique safe response, retaining all monitors and private memory. Every successor has the helper true and violates the added environment justice, so every infinite legal play satisfies the strict liveness implication. This uses the complete saved relations, rather than assuming that the syntactic arena matches the complete-literal theorem. The check reuses the saved-BDD decoder; it performs no new native generation or region solution. Thus the game itself permits this non-cooperative behavior. The paired intervention below shows that the same upstream data also permit an accepted output; it does not identify extraction as the unique cause.

**Where the continuable alternative is lost.** Here the original environment has no justice, and $\mathit{Alive}(s) = \exists e' : \rho_e(s, e')$ equals the current one-floor condition. The trace records two first creations of bad nodes and two later reuses. At first creation, rule a's raw successors have six floor patterns, including the live 001 and 100 alternatives. After its nearest-$Y$ choice, the final five-pattern candidate still contains 100 but not 001; `satOne` selects the dead 011. Later rule a prefers the existing dead node before the final choice. Thus the source-level 001 alternative above and the final-stage 100

alternative have different roles. The trace contains no dropped events or predicate references; it does not by itself certify a fair continuation after any counterfactual choice.

**Single-axis intervention and complete-controller checks.** Immediately before transition `satOne`, the experimental constructor intersects its final candidate with an actual helper-false input-availability preference. It uses this intersection when nonempty, otherwise retaining the native candidate. With helper $h$, the preference is

$$h(s) \ \vee \ \exists e' \, [D_e(e') \wedge \rho_e^{\text{native}}(s, e')[h = 0, h' = 0]].$$

Without a helper it uses the supplied native environment relation. The helper-false relation is not universally identical to the original environment relation: Section N gives a static translation counterexample. Lift's complete original-contract graph checks below establish its instance-specific result independently of that universal interpretation. This leaves helper-true candidates available. It changes no source clause, initializer, solver, stored $X/Y/Z$ set, variable order, rank rule, or earlier existing-state preference. There is no hardcoded floor constraint or node ID.

One solver call supplies the same model and memory objects to the released and experimental extractions. Their initial, transition, ordered-justice and effective memory predicates remain semantically equal; the baseline again matches the saved graph. Both complete outputs are checked by the production and separately implemented graph auditors, whose mathematical checks are unchanged and whose metadata gates explicitly identify the nonrelease variant. The production reader reconstructs saved predicates as BDDs; the other auditor evaluates the exported decision DAG with a separate reader and compares it to separately encoded source equations. Each run checks all 1,048,576 original current/next state pairs for correspondence and 2,464 typed input obligations on its 154-node original view. The second auditor imports neither the production checker nor its certificate and uses a separate reachability-based fairness test. Both still depend on the same compiler exports, typing information, and recorded translation convention; implementations were mutually visible. The shared decoder limitations for the fixed-point and residual-strategy checks above are distinct from this graph-audit pair. Table 9 reports the original-contract views. The experimental full native graph has 778 states and 13,624 edges; 726 of its 13,624 final candidates are refined, with no empty-preference fallback. This is one code-level change affecting many choices, not four patched witness edges.

Table 9: Paired extraction results on Lift T and six prespecified successful controls. Each cell gives original-view states/edges and strict/cooperative verdicts; P/F mean pass/fail. Refinements count candidate sets changed by the preference. These are overlapping followups, not additional members of the eleven-candidate cohort.

| Input | Released | Experimental | Refinements |
|---|---|---|---|
| Lift T | 154/1170, P/F | 154/1176, P/P | 726 |
| Arbiter T | 56/448, P/P | 56/448, P/P | 0 |
| Minepump T1 | 12/40, P/P | 12/40, P/P | 0 |
| Minepump T2 | 4/16, P/P | 4/16, P/P | 0 |
| Traffic S T | 12/80, P/P | 12/80, P/P | 0 |
| Traffic U P | 68/323, P/P | 68/323, P/P | 0 |
| Traffic U T | 56/416, P/P | 56/416, P/P | 0 |

The six controls were declared before their new executions and retain both verdicts with zero refinements and zero fallbacks. They establish preservation of those successes when the preference is

dormant, not generality when it is active. For Lift, unchanged game, raw region and memory suffice for this constructor to return a checked cooperative view after changing concrete extraction choices. One-step availability alone is no fairness proof; the positive conclusion comes from complete-controller checks. No sole-defect, missing-$W$ cause, or general synthesis-repair claim follows.

**A constructed environment change and exact handoff.** The old contract is Lift T with the additional environment restriction that all current and next buttons are false. A source-coded old controller stays at floor 100 and performs the actual response-monitor updates. Its two reachable cancellation states and four edges pass strictness and cooperation under this maintenance contract. Two modeled plants execute four old-controller steps, with next cancellation values $0, 1, 1, 0$. Removing only the maintenance restriction then exposes the original Lift T environment. Both plants pause at the existing state $b = 000, c = 0, f = 100, m = 000$.

After this change, a fresh paired native run generates both candidates, followed by four fresh graph judgments; no cached candidate is installed. The fixed order is released controller, then experimental controller. One admission policy accepts the first strict candidate; the other requires both strictness and cooperation. Each also requires exact agreement between the current full source valuation and an allowed initial native node. Installation preserves every source field and logical monitor and initializes the new controller's private memory to that matching node. The handoff therefore uses an already initial-compatible paused state; it is not arbitrary-state migration or a reset of pending requests.

**Executed outcomes.** Both arms execute the five inputs of the known failing prefix, in $(b, c)$ notation: $(010, 1), (001, 1), (100, 0), (101, 0), (101, 0)$. The strict-only policy installs the released controller and reaches floor 011 after these five post-change steps. Independent enumeration finds all 16 possible original next inputs illegal. The strict-and-cooperative policy rejects that candidate, installs the experimental one, and reaches floor 100 instead. It then executes the lexicographically smallest legal input at each step, with false ordered before true in $(b_1, b_2, b_3, c)$. Within the declared 32-step continuation bound, four more transitions give native nodes $680 \to 524 \to 544 \to 434 \to 434$. The final self-loop has $b = 000, c = 0, f = 001, m = 000$, meeting all three system justices; there are no original environment justices. Native IDs are local to each graph: experimental node 680 is not the released graph's dead node 680.

A separate no-replacement counterfactual applies $(010, 0)$ twice to the stationary old controller under the weakened contract. The first step is legal; the second proposed response violates the movement guarantee and is rejected. Thus keeping this old controller does not solve the changed model.

The independent episode audit imports neither the episode runner nor its transition implementation. It checks the 47-event log, source semantics, exact handoff, installed-controller hashes and actual native responses, and binds the four fresh full-graph judgments. It performs 24 environment and 24 system source checks, including the rejected counterfactual step, and 42 native-edge/response checks. All six supervised processes, comprising the outer execution, one native pair and four judges, complete with verified cleanup. The executed loop witnesses one infinite continuation; the separate complete-graph certificates justify the controller properties. This is a constructed in-process episode with a known-failure prefix, not hardware deployment, newly learned repair, arbitrary memory transfer, or a runtime-cost benchmark.

**Bounds, attempts, and provenance.** Table 10 gives the declared successful-run limits. Stream bounds apply separately to stdout and stderr. RSS caps apply to the owned process groups, not a

measured aggregate tree; output bounds are sampled termination thresholds with final checks, not exact quotas. The episode's outer final `wait4` resource value can include waited descendants on this host. Native and checker children retain a 16-MiB per-artifact limit; the outer episode's per-file limit is 1 GiB so it can capture the already bounded native streams. Each launch requires 1 GiB of free disk space.

Table 10: Bounds for the accepted diagnostic and episode runs. The episode includes serial child processes with their own limits.

| Process | Timeout (s) | RSS (GiB) | Stream (MiB) | Output (MiB) |
|---|---|---|---|---|
| Fresh baseline | 10 | 1 | 256 | 512 |
| Observer / paired extraction | 30 | 1 | 512 | 768 |
| Finite-set / memory comparison | 120 | 1 | 1 | 32 |
| Production graph judge | 60 | 1 | 8 | 64 |
| Independent graph judge | 180 | 1 | 8 | 64 |
| Outer modeled episode | 600 | 2 | 8 | 1024 |

**Observed durations of the completed episode.** Table 11 extracts recorded `elapsed_seconds` from the successful episode's process receipts. These are supervised wall durations including capture, finalization, and I/O; the timeout column is the declared cap, not an observation. The outer run contains the paired native generation and four serial graph judgments, together with modeled steps and orchestration. Its 28.36 s must not be added to those child durations. The separate saved-event replay occurred later and is outside that outer run. No solver-only or extraction-only time is inferred from the paired process.

Table 11: Recorded process durations for one completed constructed Lift T episode. Nested rows are components of the outer run; these observations are not a latency or scaling benchmark.

| Process | Observed wall (s) | Timeout (s) |
|---|---|---|
| Outer modeled episode | 28.36 | 600 |
| Paired native generation | 10.82 | 30 |
| Released: production judge | 0.12 | 60 |
| Released: separate judge | 8.60 | 180 |
| Experimental: production judge | 0.15 | 60 |
| Experimental: separate judge | 8.35 | 180 |
| Later saved-event replay | 0.11 | 120 |

These measurements describe one completed run of the three-floor example after the three retained technical UNKNOWN attempts below. They are not a statistical sample, online adaptation latency, hardware-normalized comparison, worst-case guarantee, or a scaling result beyond the bounded models. Process-group RSS caps remain limits; they are not measured aggregate-tree memory consumption.

The first observer attempt exceeded its 256-MiB stdout limit and remains UNKNOWN; a separately recorded followup raised only stream and output-tree limits to 512 and 768 MiB. The episode likewise retains three earlier attempts as technical UNKNOWNs. Attempt 01 stopped when the outer single-child supervisor detected a nested driver; the followup called the existing driver mains in the episode process while each native/checker child retained its own supervisor. Attempt 02's stream capture hit the inherited outer 16-MiB file limit; the next attempt raised that outer limit to

1 GiB. Attempt 03 produced the complete event log and five valid inner processes, but its final `wait4` RSS value of 557,793,280 bytes exceeded the outer 512-MiB cap, so it remains resource UNKNOWN. Attempt 04 raised the outer RSS cap to 2 GiB, retained all inner and output limits, and completed; its independent episode audit passes. Scientific inputs, candidate order, handoff, common input prefix and acceptance criteria stayed fixed. Each followup has a separate plan, output directory and receipt; no automatic retry rewrites an attempt.

The author-side records bind source, class, model, graph, checker, process and event-log identities. Their entry points are `CAUSAL_COMPARISON_CLOSURE.json`, `SUCCESS_CONTROLS_RESULT.json`, and `adaptation_episode_04/EXECUTION_RECEIPT.json`, followed by `adaptation_episode_04/independent/AUDIT_RESULT_attempt01.json`. These distinguish the original released result, experimental result and executed episode. The fourth public archive in Section K preserves them with explicit original/copy identities. Its consumer replays saved semantics and the event audit; it does not regenerate the native producer or execute a new adaptation episode.

### Lift T: executed noninitial handoffs from a frozen cohort

This separate followup executes existing native-controller prefixes to noninitial source and monitor states before replacing the controller. It preserves the preceding maintenance experiment and its distinct fresh-generation protocol. The new experiment changes admission at all ten declared boundaries; its twenty executed continuations all reach fair lassos. The rejected candidates' cooperation defects are established by complete rooted graphs, not by failures on these executed traces.

**Construction, selection, and frozen denominator.** The new contract is the same Lift T source used above. The constructed old contract appends only the environment invariant $\mathsf{G}\neg b_3$; its original initializers, system clauses, response monitors, and justice lists are unchanged. Operationally, the old environment relation requires both current and next $b_3$ to be false. The change removes exactly that restriction at an atomic pause.

This design was chosen after inspecting the known Lift failure, saved controllers, and all three single-button restrictions. Restricting $b_2$ still exposed a known non-one-hot native node. Restricting $b_1$ offered unanswered-monitor states but lacked matching rank-zero targets for the full proposed cohort and needed additional memory qualification. That qualification and a separate genuinely pending cohort are supplied in Section R; the present historical experiment and its outcomes are unchanged. The $b_3$ restriction supplied a small complete cohort and existing target mappings. The restriction, every prefix, all four candidates, their order, and both admission policies were then frozen before the new graph checks and executions. This is an outcome-informed mechanism study on one known source, not a held-out source or an independent benchmark.

From native node 0, the old mode reaches 26 private nodes. The cohort is *every* one of these nodes with a nonzero source response monitor: ten private nodes representing eight source valuations. Equal source values with different reached ranks remain separate rows. Both checking routes independently verify the cohort's completeness. All ten source states lie outside the original initializer and belong to the previously certified original region $K$; membership is checked in the saved region, without solving it again. These observations do not add contracts to the eleven-replacement, fourteen-public-input, or 59-model populations, or alter the separate eleven-state Elevator census and its $0/836$ refinements.

**Exact state and memory boundary.** The ten retained source fields are $b_1, b_2, b_3, c,\ f_1, f_2, f_3,$ and `SYS_CONSTRAINT.0.pRespondsToS.state` through `SYS_CONSTRAINT.2.pRespondsToS.state`. Write their values as $(b, c, f, m)$, with each triple ordered by floor and $S0 = 0, S1 = 1$. Every

row has one $S1$ monitor whose corresponding floor response already holds. The update

$$m_i' = (m_i \vee b_i) \wedge \neg(f_i \vee c)$$

therefore clears that boundary monitor on the next transition; other current requests can set other monitors. The experiment preserves nonzero logical history at installation, but no unanswered obligation already represented by an $S1$ monitor. Rows 03 and 07 do contain unanswered current triggers whose corresponding monitors are still $S0$.

Table 12 lists all source values and actual saved private roots. The retained arm keeps the reached controller object, node, and rank. A fresh arm creates a different private instance at a matching node whose *stored* rank is zero; it does not overwrite a rank, reapply the source initializer, or reset a monitor. The source object and all ten of its values remain unchanged during the switch. Rows 09–10 retain rank zero already, but retention and fresh initialization still denote different controller-instance operations. Node IDs are local to each graph, even when their numbers agree.

Table 12: Complete frozen noninitial roster. Old gives native node/rank; N, Z, and P give actual rank-zero roots in the released, normalized, and preference graphs. Every source valuation is noninitial.

| ID | $(b, c, f, m)$ | Old | N | Z | P |
|---|---|---|---|---|---|
| 01 | $(010, 0, 010, 010)$ | 728/1 | 626 | 94 | 626 |
| 02 | $(010, 1, 010, 010)$ | 729/1 | 627 | 95 | 627 |
| 03 | $(110, 0, 010, 010)$ | 730/1 | 646 | 102 | 646 |
| 04 | $(110, 1, 010, 010)$ | 731/1 | 647 | 103 | 647 |
| 05 | $(100, 0, 100, 100)$ | 656/2 | 492 | 68 | 492 |
| 06 | $(100, 1, 100, 100)$ | 657/2 | 493 | 69 | 493 |
| 07 | $(110, 0, 100, 100)$ | 716/2 | 580 | 84 | 580 |
| 08 | $(110, 1, 100, 100)$ | 717/2 | 581 | 85 | 581 |
| 09 | $(010, 0, 010, 010)$ | 626/0 | 626 | 94 | 626 |
| 10 | $(010, 1, 010, 010)$ | 627/0 | 627 | 95 | 627 |

**Old-mode qualification and forty complete suffix checks.** Before admission or runtime execution, both the source-equation route and the saved-model route certify the old controller from both original initial nodes, 0 and 1 (the two cancellation values). They agree on all 26 reached nodes and 120 edges and establish complete legal-input coverage, system safety, strict liveness, and cooperation. A failure, UNKNOWN, or disagreement at this gate would stop the direction without changing the restriction or prefixes.

Each of the ten boundaries then receives four immutable graph/root candidates in this fixed order: retain the old native instance; create a fresh native instance; create a fresh normalized instance; create a fresh preference instance. Each route checks all forty complete reachable suffixes under the unchanged new source relation. These are explicitly rooted judgments, not original-initializer judgments. At every reached node, the check covers each original-legal next input with exactly one saved safe response, preserving monitor updates and actual private successors. Missing or ambiguous responses are UNKNOWN rather than an invented edge. Strictness and cooperation use complete reachable-graph justice checks, not bounded traces.

Both routes agree on all forty complete judgments in Table 13; none is UNKNOWN. All candidates are strict. The retained and fresh-native suffixes each have 154 nodes and 1,170 edges and fail cooperation, reaching native dead-end nodes 680 and 708. Each normalized and preference suffix has

154 nodes and 1,176 edges and passes cooperation. For example, the native suffix from boundary 01 contains $728 \to 339 \to 438 \to 704 \to 680$; this graph witness was not the executed continuation. Reinitializing native private memory alone does not fix the rejected graph. The existing normalized alternative already passes, so this experiment does not establish necessity of the preference.

Table 13: All forty rooted candidate outcomes. Each cell gives strict/cooperative verdicts (P: pass, F: fail). Both checking routes agree, and all forty coverage and safety checks pass.

| ID | Retain old | Fresh native | Fresh normalized | Fresh preference |
|---|---|---|---|---|
| 01 | P/F | P/F | P/P | P/P |
| 02 | P/F | P/F | P/P | P/P |
| 03 | P/F | P/F | P/P | P/P |
| 04 | P/F | P/F | P/P | P/P |
| 05 | P/F | P/F | P/P | P/P |
| 06 | P/F | P/F | P/P | P/P |
| 07 | P/F | P/F | P/P | P/P |
| 08 | P/F | P/F | P/P | P/P |
| 09 | P/F | P/F | P/P | P/P |
| 10 | P/F | P/F | P/P | P/P |

**Actual step, pause, admission, and continuation.** For each boundary, each of the two policies creates an old native instance at node 0 and actually calls its step/commit operations along the fixed prefix in Table 14. It does not load the endpoint as a substitute for execution. The same modeled plant and source object then pause atomically, remove only $\mathsf{G}\neg b_3$, perform admission, and resume. Strict-only admission accepts the first complete strict candidate with coverage. Strict-and-cooperative admission also requires cooperation. Both require the independently agreed exact source/root/rank binding. If no candidate qualifies, the plant remains paused; no such abstention occurs in these twenty executions.

At all ten boundaries, strict-only admission retains the reached native instance. Requiring cooperation rejects both native candidates and installs the existing normalized graph in a new private instance at the specified rank-zero root. Thus all ten decisions differ and zero are equal. The preference candidate qualifies in the certificates but is never selected because normalization precedes it. There are forty candidate judgments per checker and twenty runtime episodes; the fresh native and preference candidates are not additional executed arms.

After admission, the fixed policy offers the lexicographically smallest source-legal next input in $(b_1, b_2, b_3, c)$ order, false before true. It enumerates the sixteen typed input valuations and checks legality before committing a unique saved response. Each suffix is bounded by 32 transitions and stops at the first repeated complete private state, dead end, rejected step, or UNKNOWN. A repeated state is a fair lasso only when its cycle meets every justice: Lift has no original environment justices and requires each response monitor to be $S0$ infinitely often.

All twenty executions complete: 56 old-prefix commits and 44 post-change commits, recorded in 302 events. Each suffix reaches a fair self-loop in two steps, except both policies at boundaries 03 and 07, which take three. There are no executed failures, incomplete outcomes, or UNKNOWNs. No post-change offered input sets $b_3 = 1$. Consequently these traces show successful bounded execution and different admission decisions, but no observed failure avoidance, improvement in service success, or activation of a newly allowed third-floor request. A retained graph can have the observed fair lasso while still failing cooperation elsewhere. The complete rooted certificates establish that difference.

Table 14: All twenty executed outcomes. Each row executes the old prefix twice, once per policy. The second and third sequences begin at the installed root and end with an actually traversed self-loop. Every suffix is a fair lasso; ID numbers refer to Table 12.

| ID | Old native prefix | Strict-only suffix | Cooperative suffix |
|---|---|---|---|
| 01 | $0 \to 8 \to 728$ | $728 \to 336 \to 336$ | $94 \to 16 \to 16$ |
| 02 | $0 \to 8 \to 729$ | $729 \to 336 \to 336$ | $95 \to 16 \to 16$ |
| 03 | $0 \to 8 \to 730$ | $730 \to 656 \to 0 \to 0$ | $102 \to 104 \to 0 \to 0$ |
| 04 | $0 \to 8 \to 731$ | $731 \to 336 \to 336$ | $103 \to 16 \to 16$ |
| 05 | $0 \to 8 \to 730 \to 656$ | $656 \to 0 \to 0$ | $68 \to 0 \to 0$ |
| 06 | $0 \to 8 \to 730 \to 657$ | $657 \to 0 \to 0$ | $69 \to 0 \to 0$ |
| 07 | $0 \to 10 \to 410 \to 716$ | $716 \to 626 \to 336 \to 336$ | $84 \to 94 \to 16 \to 16$ |
| 08 | $0 \to 10 \to 410 \to 717$ | $717 \to 336 \to 336$ | $85 \to 16 \to 16$ |
| 09 | $0 \to 10 \to 410 \to 716 \to 626$ | $626 \to 336 \to 336$ | $94 \to 16 \to 16$ |
| 10 | $0 \to 10 \to 410 \to 716 \to 627$ | $627 \to 336 \to 336$ | $95 \to 16 \to 16$ |

**Independent checks, qualification, and trust scope.** The source route derives legal inputs, system safety, and monitor updates from separately written source equations, using its own graph analysis. The saved-model route decodes the pinned BDD exports and independently checks typed source equality before its graph analysis. Both retain the exported identities, original-initial markers, and actual ranks; native printer parity is checked where a printer exists. The normalized export has no native printer. Both routes share the exported candidate graphs and provenance, and their code was mutually visible; this is not a blind implementation experiment or proof of compiler independence.

A separate source-based event auditor imports neither the runtime nor its transition implementation. It reconstructs every old prefix, step/commit, pause, admission, installation, source update, private successor, and fair cycle. All twenty episodes pass, including exact source-object continuity and preserved nonzero boundary monitors. Outer checks bind the event PID to the owned child, check event sequence and clock order, and reconcile the process receipts. All six scientific processes complete once and are reaped with their process groups absent: two old-mode checks, two forty-root checks, one twenty-episode execution, and one event audit. No new synthesis, native extraction, compiler run, or region solution occurs.

Before each relevant stage, synthetic qualification includes 30 source-checker cases, 55 saved-model-checker cases, 37 runtime cases, 21 event-auditor unit cases, and 12 full-envelope cases, plus full twenty-episode positive and abstention fixtures. The latter establish the audit's handling of both installation and remaining paused, not additional scientific episodes. Code revisions and failed synthetic attempts remain recorded. The scientific cohort is unchanged and has one attempt per stage; no failed scientific row is omitted or retried.

The frozen bounds are 180 seconds per checker and 120 seconds for the runtime, at most two concurrent children, and a sampled 2-GiB RSS limit per child. The memory threshold is not an operating-system address-space cap. Timeouts or exceeded limits produce UNKNOWN with cleanup of the owned process group. These bounds are not measured adaptation latency or scaling results; the preceding maintenance episode's 28.36-second measurement does not describe this saved-graph experiment.

**Exact input identity and saved evidence.** Table 15 records source and target graph identities of the frozen author-side inputs; the public manifest records original/copy identities when paths are rewritten. The plan and stage-specific execution authorizations bind model predicates, printer

exports, the existing $K$ result, code, and declared bounds. Retention and fresh-native initialization use the same graph bytes with distinct root/instance policies. The constructed old source is the unchanged new source followed only by the declared $\mathsf{G}\neg b_3$ assumption.

Table 15: Exact frozen input identities (full SHA-256).

| Input | Bytes | SHA-256 |
|---|---|---|
| New Lift source | 2218 | 6a8f8220385130b2eb219ecfa33186b179d610ffcf805a30f9cab47ab204dae7 |
| Constructed old source | 2277 | 8b32937fd7b465d1a1052ca5a2bb576cac686383a7d54015f44fa21442ff88f5 |
| Released graph | 860323 | bbee6ff9a32c55691f9b7b58c162e4e2c2af02d22792967537751ccb9d148a1f |
| Normalized graph | 97767 | 50ea0e539eea65b0fec615200fd0dffa973caabf90d515421aa358556caa3ca4 |
| Preference graph | 843906 | b478d31d20641af7c0e97a17e9d7225d95dadb3880f2562bf889a7405d8fbba7 |
| Frozen plan | 35107 | 43ff75e0562bb6c554eb018917c78cd1cc721f1b5e18db43582cbbb3a8b22539 |
| Frozen roster | 12579 | 815e9484e48ae29813deb5077a6272da9ae9f77215262c0b19a4168d65614b5e |

The eighth public static snapshot [11] is identified by `noninitial-handoff-evidence-20261021-v1`, with `NONINITIAL_HANDOFF.md` and `NONINITIAL_HANDOFF_MANIFEST.json`. Its scientific entry records are `evidence/EXPERIMENT_PLAN.json`, `evidence/COHORT.json`, `evidence/ROOT_OLD_GATE_ADOPTION.json`, `evidence/ROOT_CANDIDATE_CERTIFICATES_ADOPTION.json`, `execution/runtime_01/RESULT.json`, `execution/runtime_01/EVENTS.jsonl`, and `execution/INDEPENDENT_EVENT_AUDIT.json`. The derived `SUMMARY.json` records the scientific adoption's observations and limits with its original hash; the private coordination record itself is omitted. These distinguish actual author-side execution from inspection of saved evidence. The older public regeneration/replay entry point does not execute or replay these new handoffs; its original counts retain their original scope.

This followup narrows evidence about executed noninitial source and monitor continuity with explicit target-memory initialization. It does not establish arbitrary state or private-memory migration, carryover of unanswered obligations already represented by $S1$ monitors, unseen failure coverage, asynchronous or hardware deployment, or an additional active extraction rescue. Its admission difference uses rooted cooperation certificates; it supplies no new practical decision requiring the exact $W_C/W_F$ strict-preservation conditions and no new refusal transfer. The existing Vehicle and preservation evidence remain separate.

**Lift P: the whole-model timing mismatch.** Let $B = b_1 \wedge b_2 \wedge b_3$, $\phi = \neg c \vee \neg B$, and $\psi = f_1 \vee \neg B$. The added source assumptions give the documented original translation

$$\rho_e^P = E_L \wedge \phi(b', c') \wedge \psi(b, f), \qquad \theta_e^P = (\neg b_1 \wedge \neg b_2 \wedge \neg b_3) \wedge \phi(b, c).$$

The mixed invariant $\psi$ is evaluated at the current state; the environment-only $\phi$ is evaluated initially and at the next input. Both initial cancellation values remain allowed. The reduced helper-false relation instead checks $\phi$ at the current input. The remaining environment clauses agree with the independently encoded source equations. With both helper values fixed false, all 2,048 assignments

to the eleven original bits in the union of environment-relation supports are checked. The relations differ on 32: 24 false-to-true and eight true-to-false.

One witness starts at the original joint initial state $(0,0,0,0;1,0,0;S0,S0,S0)$ and moves to $(1,1,1,1;1,0,0;S0,S0,S0)$ with helper false at both ends. Original environment safety is false, reduced environment safety is true, and both system relations are true. A control with the opposite environment-relation disagreement is also checked. Initializers and ordered justices pass their separate checks. This diagnosis exhausts the environment relation's support, not the full system-relation domain; system safety is checked at the declared witnesses. This blocks transfer of the native verdict at the whole-model correspondence gate. The earlier diagnostic export-schema failure is retained as an integration failure, not a semantic witness. A separate selected-graph audit follows.

**Post-hoc direct original-contract graph certificate.** We separately audit the same saved native graph using the original predicates, without transferring its native verdict. The view begins at both original initial nodes and retains helper-false responses to every original-legal input, with raw node identities, private ranks, and AUX values unchanged. All 16 input valuations are checked at each of its 136 reachable states. The 958 legal-input responses are unique and satisfy original SYS/AUX safety; no missing response is repaired and no unsafe legal edge is discarded. The resulting 958-edge view passes strictness but fails cooperation. All 16 failing states are environmental dead ends with zero original-legal inputs, rather than missing controller responses.

Hand-derived source predicates with bitset transitive closure agree with a separate original-BDD, value-decoding, and Kosaraju/reverse-reachability route on all saved states, edges, inputs, ordered justices, and SCC evidence. Both routes share the frozen graph and prior source qualification. The BDD route reuses the earlier saved-BDD decoder, and implementation code was visible across routes; this is not independent compilation or a blind implementation comparison. This certificate leaves the whole-model mismatch and the historical correspondence-gated table unchanged. It adds no native synthesis run, new candidate, or initially cooperative false-refusal example.

**A source obstruction independent of the selected graph.** From either legal initial input, $b = 000$, $f = 100$, and all monitors are $S0$. The environment chooses $(b', c') = (010, 0)$. Source system safety forces $f_1' = 1$. Any additional true floor immediately violates the environment's current one-floor assumption; otherwise the next state has $b = 010$, $c = 0$, $f = 100$, and all monitors $S0$. The environment can then choose $(b', c') = (111, 0)$: button persistence, button clearing, and the next-input invariant all permit this move. System safety forces $f_2' = 1$. An additional true floor violates the one-floor assumption; with only $f_2'$ true, the next state instead violates the mixed current-state assumption $f_1 \vee \neg(b_1 \wedge b_2 \wedge b_3)$. Thus every safe response yields an environmental dead end within two rounds. A strict controller that avoids the first dead end must answer the second legal input safely; hence no strict cooperative strategy exists from either original initial input, regardless of private memory. This written source argument is independent of the selected graph and does not recompute a winning region; it does not claim that the intermediate state lacks an existential continuation. The argument uses the documented translation of mixed ENV/SYS invariants as current-state environment safety. Initially $b = 000$, and both chosen inputs have $c' = 0$, so $\phi$ is true at every current and next valuation used here; its current/next timing does not affect this obstruction.

# I. A declared pair separating region membership from selection

The pair was fixed before native execution and is separate from the eleven published candidates. Choice has one environment Boolean $e$, one system Boolean $p$, no logical auxiliary, and

$$\theta_e = \neg e, \quad \theta_s = \neg p, \qquad \rho_e = (e' \Rightarrow p),$$
$$\rho_s = \top, \quad A = \mathbf{GF}e, \quad G = \mathbf{GF}\neg p.$$

Toggle changes only system safety to $p' \leftrightarrow \neg p$.

**Original source proof.** From every one of the four original typed states, policy $p' = \neg p$ satisfies system safety and $G$ on every play in either contract. After every finite original-legal prefix, the environment can choose $e' = p$ forever; this is legal and makes $e$ alternate. Thus this same policy is strict and cooperative from every state, yielding $W = V = K = C = D$ and nonvacuous $I_M(K) = \mathit{true}$ for both contracts. In Choice, policy $p' = 0$ instead satisfies $G$ itself but forces $e' = 0$ after reaching $p = 0$, so no assumption-fair suffix remains. Every state of that bad policy still belongs to $K$.

**Audited complete transformed relations.** For helper $h$, both saved native models satisfy

$$\theta_e^R = \neg e \wedge \neg h, \qquad \theta_s^R = \neg p,$$
$$\rho_e^R = ((e' \Rightarrow p) \vee h') \wedge (\neg h \vee h'), \quad \rho_s^R = \rho_s \vee h,$$
$$A^R = \mathbf{GF}e \wedge \mathbf{GF}\neg h, \qquad G^R = G.$$

An independent saved-DAG check establishes these equations on all eight typed native states and all 64 current/next pairs in each case. Field ownership, domains and initialization are checked on the full typed native model; justice identities and order are checked at the original, preconstructor and solver phases. Hence the same $p$-toggle policy satisfies native system safety and $G$ unconditionally, and its helper-false restriction has the cooperative source behavior just proved. The transformation does not exclude this good original-view policy. This does not demand cooperation in the full transformed arena after an environment-chosen helper flip.

**Actual selected outputs.** Choice returns three native states and seven edges. Its original view has only node 0, with $(e, p) = (0, 0)$, rank 0, and a self-loop. It satisfies $G$ and fails cooperation while remaining inside $K = V = D$. Toggle returns seven native states and 18 edges. Its original view has nodes $0 = (0, 0)$, $1 = (0, 1)$ and $6 = (1, 0)$, all rank 0, with edges $0 \to 1$, $1 \to 0$, $1 \to 6$ and $6 \to 1$. It passes both checks. Each case received one native attempt; complete typed source correspondence and independent graph checks succeeded for both. Saved state, edge, justice, SCC and raw-node comparisons agree exactly.

The pair shows that $I_M(K)$ and membership of every reached state in $K$ do not certify the particular output's cooperation. It does not attribute that selection separately to the raw fixed-point core or the concrete constructor. Both unnormalized original arenas admit legal $G$-fair, $A$-unfair plays, so they do not refute the conditional language-inclusion theorem of [6]. The pair adds no public candidate, learned repair, adaptation episode or deployment result.

# J. Exact strict-preservation losses in the certified public cohort

We compute $C$ from each of the 59 certified original models by quantifying every typed next system and auxiliary value, followed by every typed next environment value. The quantifier ranges include

environment values disallowed by $\rho_e$. We recheck the saved independent $W$ function and preserve exact typed domains.

Forty-nine inputs satisfy $W \subseteq C$ and ten do not. The latter are one AMBA variant, three Humanoid variants, two PCar variants, one AirportShuttle variant, and three robotic-arm variants. By the main paper's exact preservation corollary, the complete semantic literal helper fails to preserve the entire original strict-winning region on these ten models.

Table 16: Exact loss for all ten condition-violating public inputs. $R$ is existential reachability through original legal joint edges from the original initializer. Every original initial decision is true. The last column reports the literal complete-semantic initial decision $I_M(W_C)$. Dashes retain the one resource-limited calculation; they are not zeros. IDs identify source variants in the fixed artifact roster.

| ID | Source family | $\lvert W\rvert$ | $\lvert W \setminus C\rvert$ | $\lvert W_C\rvert$ | $\lvert W \setminus W_C\rvert$ | $\lvert R \cap (W \setminus W_C)\rvert$ | $I_M(W_C)$ |
|---|---|---|---|---|---|---|---|
| 003 | AMBA | 34575 | 6144 | 28431 | 6144 | 2048 | True |
| 044 | Humanoid | 2387 | 1176 | 0 | 2387 | 1581 | False |
| 045 | Humanoid | 11232 | 5112 | 8 | 11224 | 7294 | False |
| 046 | Humanoid | 42312 | 37632 | 2388 | 39924 | 2667 | False |
| 059 | PCar | 5196 | 1600 | 3596 | 1600 | 200 | True |
| 060 | PCar | 5196 | 1600 | 3596 | 1600 | 200 | True |
| 072 | AirportShuttle | 4601344 | 4070400 | 530944 | 4070400 | 49352 | True |
| 159 | RoboticArm | 447213 | 355043 | — | — | — | Unknown |
| 178 | RoboticArm | 892496 | 361284 | 531212 | 361284 | 24648 | True |
| 179 | RoboticArm | 892496 | 361284 | 531212 | 361284 | 100140 | True |

We evaluate all ten condition-violating inputs, without selection by resulting loss or initial decision. Nine exact calculations complete and one reaches the fixed BDD resource limit. Table 16 reports $W_C$, exact loss, and the original quantified initializer. Three Humanoid variants change from $I_M(W) = \mathit{true}$ to $I_M(W_C) = \mathit{false}$. Their exact losses strictly exceed $W \setminus C$ by 1211, 6112, and 2292 states, respectively. Thus an immediate response-totality check alone misses losses propagated by the invariant requirement. For variant 045, $W_C$ even remains nonempty (eight states), while none of its 20 allowed initial input choices has an initial response in that region. Region nonemptiness therefore does not restore the lost initial decision.

The other six completed inputs retain strict initial feasibility. Every completed loss contains states existentially reachable by original-legal joint execution from the declared initializer. This establishes a possible continuation-state boundary, not reachability under a particular winning controller or an observed runtime execution.

All ten affected original contracts already have $I_M(W_V)$ false, and hence $I_M(K)$ false. Every strict cooperative strategy wins inside $V$, so this necessary condition excludes cooperative initial realizability independently of the normalized four-fixed-point calculation. We additionally intersect the independently certified $K$ with the exact loss $W \setminus W_C$ for every completed calculation. The intersection is empty in all nine: seven have $K = \varnothing$; the two PCar variants each have 1636 cooperative states, all preserved. Thus these public results demonstrate strict-region loss and three strict initial flips, but no loss of a certified cooperative state within this cohort. They do not demonstrate rejection of an initially cooperative-realizable public contract. Guard establishes that separate possibility. The tenth calculation remains UNKNOWN. Moreover, these counts concern the complete semantic literal equations. They do not by themselves transfer the theorem to the released syntactic route, whose helper-true constraints require separate analysis.

**Detailed literal-cohort summary** The 59-model cohort demonstrates strict loss under the complete literal equations, but no certified cooperative-state loss or rejection of an initially cooperative contract. Of these models, 49 satisfy $W \subseteq C$ and ten do not. Nine of the ten exact loss calculations complete; one reaches the fixed BDD limit. Three Humanoid variants lose strict initial feasibility, with loss propagating beyond $W \setminus C$. All ten already fail $I_M(W_V)$, excluding cooperative initialization. Every completed loss calculation has $K \cap (W \setminus W_C) = \varnothing$: seven have empty $K$, and two PCar variants retain their cooperative states.

**Detailed released-arena comparison** All fourteen comparison inputs satisfy $W \subseteq C$. Thirteen released-arena analyses complete and establish strict-region preservation; one is resource UNKNOWN. Of eight completed helper arenas, Elevator matches both complete guards because original system safety is true. The other seven match the literal environmental relation in every helper quadrant and original system safety at $(b, b') = (0, 0), (0, 1)$, while retaining nontrivial system constraints at $(1, 1)$. These seven reproduce the literal first-flip timing but match neither complete guard. Five no-helper arenas instead have complete original-model identity. Original initialization and ordered justices are retained.

## K. Public evidence and consumer rechecks

The opening evidence index gives the current continuation-release entrypoints and their saved-replay limits. This section retains the detailed historical archive map, including the pending-diagnostic release, and each earlier qualification boundary.

The evidence-reader indexes `EVIDENCE_READER_INDEX.txt` and `EVIDENCE_READER_INDEX.json` are public at documentation version `intruder-bounded-evidence-20261021-v1`. This version adds the bounded Intruder static snapshot while retaining eight earlier scientific archives, the constructed Guard source, and selected Minepump records [11]. Those frozen public indexes map eighteen historical claim groups to archive members: inputs, source listings where provided, saved results, checkers, and expected outcomes. The index adds navigation, not scientific validation or support for every historical rerun. The later certificate-and-finite-study release described below adds two public archives while retaining these historical versions. The earlier version `saved-evidence-20261020-v1` supplies three downloads, `SHA256SUMS`, `RELEASE_MANIFEST.json`, and a README with integrity and source-restoration commands. The public archive hashes differ from the historical ZIP hashes: 97 third-party text copies are omitted and retrieved from their original distributors by hash-checked scripts. All retained archive members preserve their bytes from the earlier anonymous evidence packages. These packages distinguish original execution identities from anonymized or compressed copies in their manifests and ledgers. Source restoration is required before checking those packages' complete file trees or using their closed-package recipes. Neither the download check nor restoration runs a scientific calculation; saved results and qualification limits remain unchanged.

**Core.** `saved-evidence-public-20261020-v1.zip` covers the 59-model preservation analysis, the same 14 inputs across four paths, and the 11 published replacements. Start with `PAPER_EVIDENCE_INDEX.tsv`, `PUBLIC_14X4_OUTCOMES.tsv`, and `READER_GUIDE.txt` inside `seams2027-saved-evidence/`. The index links the groups' saved inputs and results. `CONSUMER_RECIPE.txt` describes one representative Arbiter initialization recheck, not a rerun of every cohort or native synthesis. Historical correspondence-gated outcomes, including Lift P's UNKNOWN, remain unchanged. The Lift T entry separately links its 128-state positive source policy, 49,152-assignment complete extension, and failing selected graph, with their saved checks. It records prior author-side rechecks; that source-policy

extension has no new consumer recheck. The C14 entry also locates the unchanged PCar source and saved initializer argument.

The R11 entry also links repository files `evidence/minepump-original-strict-records.txt` and its JSON companion. They provide selected existing full-$W$ equality records, original-file hashes, and bindings to the public model and initial predicates used in Section H's Minepump inference. These excerpts support static inspection; the full historical auditors and their dependency closure are not added, and no new execution is claimed.

The historical Guard entry links the unchanged 115-byte source at `evidence/transition_guard.spectra` to its saved input digest, native refusal, helper-false predicate certificate, and later initialization-correspondence certificate in Core. The source strategy, observed refusal, and correspondence checks have distinct roles (Section L); neither certificate computes a cooperative winning region or identifies the refusal mechanism. The new three-study archive adds the full 16/64-assignment saved-arena check described in Section L, which identifies the first-flip obstruction and matches the saved raw region on this fixture. The historical archives and population counts remain fixed.

The C14 entry links the Slugs run receipt and its per-input certificates, which identify the saved original models and exported inputs. The export certificates record exact BDD round-trip equality for original initialization, safety, and justice predicates, preserving typed variable ownership, original initialization quantifiers, and Mealy move order. This checks the encoding of shared translated models, not the Spectra compiler independently; returned-graph checks and native refusals retain their separate scopes.

**Lift causal comparison and episode.** The fourth download, `lift-causal-extension-public-20261021-v1.zip`, has root `seams2027-lift-causal-extension/`. The repository's `LIFT_CAUSAL_EXTENSION.md` and `CLAIM_TO_FILE_INDEX.json` give the commands and exact members. First run the outer integrity/extraction tool, then restore the 14 exact upstream-derived sources using its hash-checked helper. Finally, `consumer/replay.py` recomputes the raw finite-set four-fixed-point region and all 28 observed memory sets, runs both mathematical graph checkers on 16 saved arms (the seven diagnostic pairs and the episode's fresh pair), and replays the episode's saved source/edge/event audit. A fresh extraction with Python 3.10.9 on macOS passes with `PATH` and `JAVA_HOME` empty. The consumer requires Python's standard library, not Java. This is saved-input replay, not new synthesis, extraction or episode execution. Same-object observations, process completion and class identities remain archived producer evidence, checked for consistency but not regenerated. Actual producer bytecode and large process streams are omitted with explicit hashes; this fourth archive alone does not provide a qualified native-regeneration entry point. The original/copy ledger records metadata sanitization separately from preserved graph and predicate content. Failed attempts and their result/process summaries remain. The prior three archives retain their own qualification limits.

**Fresh public-source producer qualification.** The fifth download, `lift-native-regeneration-public-20261021-v1.zip`, has root `seams2027-lift-public-regeneration-kit/`. Start with the repository's `NATIVE_REGENERATION.md`, its manifest and validation record, and the kit's `README.md`. They give integrity, configuration and execution commands, exact runtime identities, limits and expected outcomes. A bounded run of the exact distributed control files fetches the scientific source, checker, profile, and dependency inputs anew from public URLs. Its 165 requests retrieve 113,412,497 bytes. Selective extraction excludes historical models, controllers, and BDD results from the scientific inputs; restoration recovers 14 source files. Compilation of all six custom

Java sources produces 19 classes with the expected byte identities, using the fixed order of 164 JARs. Class-origin checks cover 47 probe classes and the 34 custom/released classes observed in the native execution. One solver invocation supplies the same game and memory to the two extractors. Both complete-graph judges again report baseline 154 states/1,170 edges, strict and non-cooperative, and variant 154 states/1,176 edges, strict and cooperative. All 176 supervised processes complete and are reaped; a separate read-only audit checks the sealed inputs, provenance, graph comparisons, and process records. Historical comparison occurs only after the scientific processes finish.

This qualifies the executed local placement on an installed, pinned JDK 17.0.19 and Python executable under macOS. The JDK is not restored from a public runtime archive; Python's standard library, operating system, platform libraries, and certificate store remain host supplied. It is not a fully hermetic or cross-platform reproduction. It does not regenerate the six inactive controls or execute another adaptation episode. An earlier local run and this qualification run have separate receipts. The latter uses a fresh generated configuration; distributed executable controls retain their executed bytes. Later receipt-path anonymization changes only presentation metadata, with separate original/copy identities and no scientific-content changes. The kit itself was copied locally, not fetched from public hosting during qualification. Anonymous qualification records alongside the kit ZIP identify the executed and distributed manifests and controls. Each later placement requires its own configuration, review and execution record.

**Focused saved-state certificate.** The kit also supplies `focused/check_node680.py` with a separate bounded command. Its executed copy verifies the public saved model and graph, the original/copy ledger and the permitted digest bridge before checking every node's typed valuation, rank and predicate bindings. It checks all 2,048 valuations of the sole environment justice, which is exactly $\neg h$. At saved node 680, all 16 original source inputs have three safe responses, but no original-legal next input remains; all 16 native-legal inputs flip $h$. The 321-state, 5,136-edge residual covers every legal input safely and keeps $h$ true thereafter. This proves actual transformed strict winning from this full helper-false valuation by falsifying environment justice forever. It neither equates the entire raw core with that winning region nor identifies extraction as the sole cause. The recheck uses the shared public BDD decoder and saved inputs; it performs no compilation, synthesis or new episode execution.

**Normalized route on the eleven original contracts.** The sixth download, `normalized-eleven-public-20261021-v1.zip`, has root `seams2027-normalized-eleven-evidence/`. Start with the repository's `NORMALIZED_ELEVEN.md`, manifest, validation record and results index. The archive's `README.md` gives exact commands. Its source-only helper reconstructs the eleven omitted paper-derived specifications from versioned publisher HTML and checks the listing and reconstructed bytes; restoration is separate from the consumer. The release qualification uses separately prepared exact sources and does not claim a new download of them.

With those sources in a separate directory, `consumer/check_saved.py` first checks the complete archive manifest, fixed source roster, ten original scientific module identities, and current model, result and controller bindings. It copies the inputs to a fresh work directory and checks all 33 model-to-source bindings. The unchanged graph checker then checks eleven arms: eight complete graphs satisfy strictness and legal-prefix cooperation; three retain empty-$V$ refusals with null controller flags. It also compares 22 original arenas and 22 full typed $V/K$ functions against their saved comparators, and checks eleven lossless stdout captures. Every new semantic-certificate field must agree with its saved counterpart, apart from six listed top-level provenance fields. The graph checks retain logical AUX, original initialization quantifiers, safety predicates and ordered justices. Each new arm and its

Table 17: Support availability and replay status in the manuscript and saved archives. One item may serve several evidence roles described in the main paper. An index entry does not establish an independent rerun.

| Evidence status | What it supports and what remains separate |
|---|---|
| Written derivations | Proofs and source arguments printed here; cited primary statements still require external comparison. |
| Shared-export recomputations | Separately implemented region operations on the same compiler-exported predicates; not independent compiler validation. |
| Author-side saved checks | Recorded region, relation, and complete-graph results; the index locates their inputs and checkers. |
| Consumer rechecks | One core initialization, three Vehicle scopes, Lift raw/memory, 32 graph judgments and episode-log checks, plus the separate normalized-route checks below; saved-input replay, not native synthesis or a new episode. |
| Qualified regeneration | Exact distributed controls fetch public scientific inputs, rebuild all custom sources, generate one Lift pair and run four graph checks; same pinned host runtime, not a new episode or control cohort. |
| Focused certificate | Saved node 680 belongs to transformed strict winning; typed predicate bindings and a complete safe residual are checked, not the entire winning region. |
| Static inspection only | Lift P's direct graph and source-obstruction addendum; no qualified relocated rerun and no rehash of the omitted historical stdout. |

two comparators bind the same exact source file.

The bounded wrapper uses Python's standard library and writes separate receipts without replacing historical evidence. Its portable source binder checks exact supplied source bytes and replaces only the original checker's profile-loading and frozen-source bindings. It does not repeat the private raw/diff/profile binding pipeline or independently compile the source semantics. All ten scientific modules retain their bytes; the wrapper adds input binding and orchestration. This is one graph-judge implementation on shared compiler-exported predicates, not two independent graph judges or producer regeneration. No new native synthesis, extraction, compilation or adaptation episode is performed. The original/copy ledger and exact JSON-pointer allowlist separate presentation-path changes and current derived-file identities from unchanged scientific content and historical execution identities.

**Residual-entry static snapshot.** The seventh download, `residual-census-evidence-20261021-v1.zip`, has root `residual-census-evidence/`. Start with the repository's `RESIDUAL_CENSUS.md` and the archive's `SUMMARY.json`. It contains the eleven exact entries, 22 complete graphs, 44 retained graph certificates, model and memory BDDs, paired diagnostics, and process-completion records for Section M. All pairs are identical and strict and cooperative; the preference makes no refinements among 836 examined candidates. The original initializer verdict remains false. The archive preserves original reference sources and identifies the omitted upstream-derived extractor through its existing source-restoration recipe. It supplies no qualified relocated replay or fresh-host regeneration. Its manifest and complete original/delivered hash ledger distinguish metadata-only location changes from historical execution bytes; an explicit omissions list identifies excluded host and process files.

**Noninitial-handoff static snapshot.** The eighth download, `noninitial-handoff-evidence-20261021-v1.zip`, has root `noninitial-handoff-evidence/`. The repository's `NONINITIAL_HANDOFF.md` and the archive's `SUMMARY.json` describe the constructed Lift handoff census: ten reached

private nodes, eight source/history valuations, forty rooted candidates checked by two routes, and twenty executed episodes independently audited from 302 events. All ten decisions change when cooperation is required; all twenty fixed continuations are fair lassos. No post-change input sets $b_3$ true. The archive includes saved graphs, models, BDDs, certificates, execution records, reference code and qualification evidence. The paper-derived source listing is omitted with its existing fourth-archive restoration reference; the added old-mode guard is specified separately. Original execution hashes and delivered-file identities remain distinct after location metadata is anonymized. This is static evidence: the earlier public replay entrypoint does not replay these handoffs, and no relocated replay or fresh-host execution is claimed.

**Bounded Intruder static snapshot.** The ninth download, `intruder-bounded-evidence-20261021.zip`, has root `seams2027-intruder-bounded-evidence/`. The repository's `INTRUDER_BOUNDED_EVIDENCE.md` and `INTRUDER_BOUNDED_MANIFEST.json` identify the frozen selection, unchanged source, two bounded native attempts, and independent checks historically reported in Section N. All three arms of those bounded attempts remain UNKNOWN; later source and repaired-controller certificates are supplied separately, not as contents of this earlier archive. The complete captured output set includes the incomplete 16-MiB relation; neither that file nor the producer's 14 recorded $V$ iterations is a certified result. The original-model correspondence PASS concerns the separately pinned historical original export. The archive preserves original and delivered identities, with metadata anonymization and explicit references for omitted upstream-derived constructor listings. It is a static evidence snapshot with no relocated replay claim.

**Vehicle verification.** `verification-addendum-public-20261020-v1.zip` separates P1 (policy, initialization, and specified continuation loss), P2 (original/released relations), and P3 (eleven original-source predicates). Entry points under `seams2027-verification-addendum/` are `P1_READER_RECIPE.txt`, `P2_READER_RECIPE.txt`, and `SOURCE_READER_RECIPE.txt`. The index links inputs, expected outputs, and adoption records: three bounded scopes for one Vehicle source; P3's eleven predicates are not the eleven replacement candidates. That historical full-region resource UNKNOWN remains recorded; Section E separately reports the later completed native strict pair.

**Lift static evidence.** `lift-direct-static-public-20261020-v1.zip` contains the separate direct original-contract graph certificate and source obstruction. Under `seams2027-lift-direct-static-evidence/`, start with `SCOPE.json`, `root/SELECTED_CONTROLLER_ROOT_ADOPTION.json.gz`, and `root/SOURCE_OBSTRUCTION_PROOF_V2.txt`. The index identifies both saved-graph routes and their comparison. This archive supports static inspection, not a qualified relocated scientific rerun; the omitted historical stdout cannot be independently rehashed from it. The direct certificate does not resolve whole-model verdict transfer.

**New public certificate and finite-study archives.** Two additional archives are available at `certificate-finite-evidence-20261023-v2`. The release preserves the earlier immutable versions cited above. Its `CERTIFICATE_FINITE_EVIDENCE.md` gives the two downloads and integrity command; both public ZIPs were retrieved and matched byte-for-byte to the verified local archives. `intruder-certificate-evidence-20261023-v1.zip`, rooted at `seams2027-intruder-certificate-evidence-v1/`, contains the full original and repair-G certificate payloads and verifier closure. Its outer README, manifest and replay summary identify the six relocated checks and the exact limits detailed in Section N. The original 276,829-state certificate and repaired 407,172-state certificate are now delivered in full; earlier local-record indexes alone are not their delivery mechanism.

`three-study-evidence-20261023-v2.zip`, rooted at `three-study-evidence-20261023-v2/`, contains the request protocol, Humanoid guard census, and Guard saved-arena audit. It supplies the frozen study inputs, scripts, complete saved results, and required runtime dependency closure. `PRIVACY_AMENDMENT.json` records a metadata-only privacy correction; no new replay was run. The historical `REPLAY_SUMMARY.json` records three successful relocated executions and 418 comparisons of the earlier bundle. These compare all scientific result fields, all six request certificates, every census arm, the complete decoded state/function tables, and all 66 events. Event process/object identifiers are symbolized only after checking continuity, actual installation changes, and monotonic clocks; numeric clock values and declared runtime provenance are excluded. The propagation control excludes only its relocated fixture path. Detailed anonymized binding, process, and comparison records are included. Inner pre-replay statuses remain historical. The scientific code, formulas, and frozen rosters are unchanged. These are author-run checks on the same host, not external replication or an operating-system sandbox. Scientific calculations use the Python standard library and included kernels; the supervisor also requires POSIX and `ps`. Native arenas are saved inputs, not newly regenerated outputs. Original and delivered hashes distinguish metadata anonymization from scientific content. Request policies and event records support Section P; all eight Humanoid arms, all 28 pair comparisons and the 18,816-state table support Section Q; the Guard inputs and complete equation checks support Section L. These are author-controlled studies with shared source representations and visible implementations. Their supplied evidence does not establish a generally better synthesizer or application prevalence.

# L. Formal details, route configuration, and decision evidence

Sections L.1–L.5 supply the full retained-constraint proof and counterexample, the detailed cooperation characterization, implementation identifiers, mechanism controls, and decision evidence. Section L.6 gives the complete literal-preservation theorem, full-region corollary, and corrected-guard proofs, with the propagation, initialization, and Guard examples. The main paper states and proves the central exact-continuation theorem and also states and proves the full-preservation corollary. L.6 retains both complete proofs with additional examples. The retained-constraint proof is in L.1; the complete normalization lemma and proof are in L.2; the cooperative construction and complete round correspondence are in A and L.2. These details support the summarized applicability conditions and the additional examples.

## L.1 Retained helper-true constraints

We use the main paper's finite, fully observed typed Mealy setting and strict semantics [12, 7], retaining all logical monitors in $D = E \times Y$. Let $W$ be the original strict-winning continuation region and $C(s) \equiv \forall e' \in E\, \exists y' \in Y : \rho_s(s, e', y')$. For $T \subseteq D$, let $M_T$ retain the original environmental relations and justice objectives while replacing system safety by $T(s) \wedge \rho_s(s, e', y') \wedge T(e', y')$. Write $W_T = T \cap \mathrm{Win}_{\mathrm{str}}(M_T)$; this outer intersection excludes vacuous winners outside $T$ at environmental dead ends. The original-initializer test is

$$I_M(T) \equiv \forall e \in E : \theta_e(e) \Rightarrow \exists y \in Y : \theta_s(e, y) \wedge T(e, y).$$

For relations $E_{ij}, S_{ij}$, the predecessor below uses $\forall e' \in E : E_{ij}(s, e') \Rightarrow \exists y' \in Y : S_{ij}(s, e', y') \wedge (e', y') \in Z$. The following gives the full retained-constraint hypotheses, statement, proof, and counterexample for the result summarized in the main paper.

**Preservation with retained constraints** When constraints remain at $b = 1$, preservation depends on both safe entry and continued safety. Let $E_{ij}(s, e')$ and $S_{ij}(s, e', y')$ be the transformed safety relations with current helper $i$ and next helper $j$. Assume $E_{00} = \rho_e$, $S_{00} = \rho_s$, and $E_{10} = \mathit{false}$. All original typed variables and logical monitors remain in $D$; the next helper is observed before the system responds. Let $A'$ and $G'$ be generalized Büchi objectives whose corresponding state predicates agree with $A$ and $G$ on $b = 0$ states, respectively. Require $A^R = A' \wedge \mathsf{GF}\neg b$ and $G^R = G'$ globally; the predicates may otherwise be arbitrary on $b = 1$ states. The relations in 01 and 11 may otherwise be arbitrary, with no next-system dependencies in their environmental component.

Let $U$ be the strict safety-winning region for the helper-true subgame, and let $F$ characterize safe entry to that region:

$$U = \nu Z.\ \mathrm{CPre}_{11}(Z), \tag{L.1}$$

$$F = \{s \in D \mid \forall e' \in E :\ E_{01}(s, e') \Rightarrow \exists y' \in Y :\ S_{01}(s, e', y') \wedge (e', y') \in U\}. \tag{L.2}$$

Here $\mathrm{CPre}_{11}$ uses $E_{11}$ and $S_{11}$ with the same typed quantifiers and dead-end convention as CPre. Define $W_F$ by replacing $C$ with $F$ in the definition of $W_C$, including the outer intersection with $F$.

**Proposition L.1** (Preservation with retained helper-true constraints)**.** *Under these hypotheses,*

$$W_R(\cdot, 1) = U, \qquad W_R(\cdot, 0) = W_F.$$

*Consequently, full original-region preservation holds exactly when $W \subseteq F$. If the original initializer is preserved and $b$ starts at zero, the transformed initial decision is $I_M(W_F)$.*

*Proof.* At $b = 1$, persistence keeps the helper true, so the added assumption fails and only strict safety remains. The fixed point $U$ supplies safe responses back into $U$; conversely, states with safety-winning residual strategies form a post-fixed point of $\mathrm{CPre}_{11}$. Thus $W_R(\cdot, 1) = U$.

From a winning helper-false state, each legal flip must enter $U$ safely. The same holds after every original-legal non-flip prefix, so all reached states satisfy $F$. Restriction to those histories wins $M_F$. Conversely, from $s \in W_F$, a winning $M_F$ strategy maintains $F$ until a flip, then enters $U$ and switches to a uniform safety-winning strategy, available in this finite, fully observed game. Without a flip, original duties hold; afterwards safety holds and the added assumption fails. Illegal inputs release strict duties. This proves $W_R(\cdot, 0) = W_F$.

Finally, $W_F \subseteq W \cap F$ gives necessity for preservation. If $W \subseteq F$, every state reached by an original winning strategy along original-legal histories lies in $W$, hence in $F$; that strategy wins $M_F$, proving sufficiency. The unchanged initializer gives $I_M(W_F)$. □

This is standard safety-game reasoning for a one-way helper, not a new solving algorithm. A nontrivial $S_{11}$ can still give $U = D$; even a proper $U$ can permit every winning active state to enter it. Conversely, the existence of an immediate $S_{01}$ response does not suffice if that response cannot reach $U$. For example, use one input value, a Boolean response $y$, true original initial and justice predicates, $\rho_e = \mathit{true}$, and $\rho_s = y'$. Then $W = C = D$. Let $E_{01} = E_{11} = \mathit{true}$, $S_{01} = \rho_s$, and $S_{11} = \neg y \wedge \neg y'$. Every state permits a flip response, but each response reaches $y = 1$, outside $U = \{y = 0\}$. Thus $F$ and the active winning region are empty, despite preserving the original first-flip relation. In the complete literal case $U = D$ and $F = C$; in the complete corrected case $U = F = D$.

## L.2 Cooperative existence and selection

**Cooperation of a fixed strategy** For a strategy $f$ on a total legal arena $H$, let $L_v(H, f)$ be its infinite plays from $v$. Legal-prefix cooperation requires

$$\operatorname{Pref} L_v(H, f) = \operatorname{Pref}\big(L_v(H, f) \cap L_v(H, A)\big). \tag{L.3}$$

Every legal prefix must have an assumption-fair suffix under the same strategy and its reached private memory. One fair play is insufficient. Input availability is a separate requirement: the infinite-extension obligation of [7, Definition 3] excludes finite environmental dead ends even with no justices.

**Negative conclusions under the all-history reading** Definition 3 of [7] quantifies over every environment strategy and every position of its play with the fixed winning system strategy. Restricting those positions to original-legal prefixes gives our requirement. A legal-prefix failure therefore refutes that stronger property for the same original-contract strategy and reached memory. For a complete helper-false view retaining every original-legal input, the witness also survives any completion on illegal inputs; helper-true escapes do not extend that fixed original-contract strategy. This negates one output, not the existence of another strategy. Contract-level negation additionally requires a separate argument excluding every strict cooperative strategy from the original initializer. Conversely, a positive legal-prefix certificate alone does not establish the all-history property.

**Applicability of the cooperative core** The Kind paper invokes the non-conflicting core's correctness [7, Theorem 3 and Appendix A]. The result of Majumdar et al. [6] has the sufficient premise

$$L_v(H, G) \subseteq L_v(H, A). \tag{L.4}$$

The established normalization $G \mapsto G \wedge A$ ensures this inclusion without changing $A \Rightarrow G$ [6]. Our normalized comparison applies it to the original nonblocking arena, before helper conversion. An invocation of the raw core must separately meet its premise.

**Original-contract comparison regions** Strict synthesis is restricted to the nonblocking safety region $V = \nu X.(\mathit{Alive} \wedge \mathrm{CPre}(X))$, where $\mathit{Alive}(s)$ means that some original-legal next environment input exists. This restriction maintains system safety and future input availability against all original-legal inputs; it imposes no environment justice. Normalized non-conflicting synthesis places the environment justices among the system justices before invoking its four-fixed-point core on $V$. Denote the restricted strict-winning region by $W_V$; $W$ remains the unrestricted original strict-winning region. Any strict cooperative strategy remains in $V$: cooperation supplies legal continuations, and strict safety supplies safe responses to every legal input. Its reachable residual states therefore form a post-fixed point of the defining operator. In the total bipartite arena $H_V$, completed states are $Q_0 = V$; $\delta_0$ retains every original-legal input, and $\delta_1$ retains every original-safe response into $V$.

**Normalization preserves the semantic cooperation region** Here $K$ denotes the existence of a strict, legal-prefix cooperative strategy, before identifying it with any algorithmic region. If $V = \varnothing$, this region is empty. For nonempty $V$, the arena above retains all original-legal inputs and all safe responses inside $V$. **Lemma (cooperative normalization).** *Restricting to this arena and replacing $G$ by $G \wedge A$ preserves $K$.*

*Proof.* Fix a strict cooperative strategy. Collect its source states reachable by legal prefixes, keeping each prefix's residual strategy and memory. Cooperation supplies a legal continuation at each prefix; strict safety supplies a safe response to every legal input, back into that reachable set. This set is a post-fixed point of the nonblocking safety restriction, hence lies in $V$. No legal behavior of this strategy is removed. Conversely, a strict cooperative strategy on the restricted arena extends to the original game: all original-legal inputs remain, and illegal inputs release strict duties. Finally, $(A \Rightarrow (G \wedge A)) \equiv (A \Rightarrow G)$; normalization changes neither legal prefixes nor assumption-fair continuations under the same memory. Thus both inclusions hold. □

**Lemma L.1** (Round interpretation)**.** *For nonempty $V$, put all justice labels of finite total $H_V$ on $Q_0$ only. Replace the one-edge predecessors in [6, Eq. (15)] by* $\mathrm{CPre} = \forall\delta_0\,\exists\delta_1$ *and* $\mathrm{EPre} = \exists\delta_0\,\exists\delta_1$*, with* $\mathrm{CondPre}(U, X) = \mathrm{EPre}(U) \cap \mathrm{CPre}(U \cup X)$*. Each final round vector component equals the $Q_0$ restriction of the corresponding alternating component.*

The full proof is in Supplement A. The correspondence identifies final values, not all inner approximants or extracted ranks. Empty justice lists are padded by $V$ in rounds and $Q_0$ in $H_V$; if $V$ is empty, both regions are empty. Within $V$, normalization establishes (L.4), the premise of the non-conflicting soundness statement in [6, Theorem 3]. Lemma L.1 and the union definition in [6, Appendix B.1] connect the normalized algorithm to that theorem; its soundness and completeness [6, Theorem 3 and Appendix B.2] identify its computed region with the semantic cooperation region $K$.

**Correcting the guard does not ensure cooperation** In *the transformed arena*, either complete helper guard lets a strict-winning strategy answer a legal flip. Monotonicity then keeps $b = 1$, releasing system safety but making $\mathsf{GF}\neg b$ impossible. Normalization after helper conversion therefore cannot generally repair the pipeline. This neither refutes cooperation of an original-contract helper-false view nor alone disproves (L.4), which may hold vacuously when the system objective is impossible.

**Justice: preservation without cooperation** Let environment Boolean $e$ and system Boolean $p$ have true initial predicates and

$$\rho_e = (e' = p), \quad \rho_s = \neg p, \quad A = \mathsf{GF}e, \quad G = \mathsf{GF}\neg p. \tag{L.5}$$

Current-state safety is intentional: strict winners keep $p = 0$, forcing $e' = 0$; at $p = 1$, a legal input exists but safety fails. Thus $W = C = W_C = \{\neg p\}$: preservation holds, but no cooperative strategy exists. The all-zero play satisfies $G$ and violates $A$, falsifying (L.4). Substitution yields raw region $\{\neg p\}$, or $\neg p \vee b$ under either helper guard, with positive initialization. The supplement gives the recurrence; this does not refute the conditional theorem of [6].

**Complete fixed-graph admission criterion** The audit explores the fixed graph with reached private memory. For each reachable node and every original-legal next input, it checks the response and updates against the original equations. The graph edges used below are these checked original-legal, safe fixed responses. Fairness then becomes a graph question: can each node reach a cycle region that visits every ENV justice? Formally, such a strongly connected component (SCC) has a directed cycle, including a self-loop for a singleton, and meets every ENV justice set. Cooperation requires reachability of such an SCC from every reachable node. Strict liveness additionally excludes, for each SYS justice, an ENV-fair SCC in the subgraph avoiding that justice. Initialization, safety, and liveness are checked separately.

A graph interpretation derived from a transformed model is accepted only after typed equality of the corresponding original predicates: initialization, safety, justice, and auxiliary updates. For

helper-based views this comparison is on the helper-false slice, with any extension beyond that slice qualified separately. A direct audit can instead use original source predicates on the saved graph. In a helper-based graph, its *original-contract view* keeps the helper false and retains every original-legal input, source monitor, native node identity, and reached private memory. A failed view refutes that output; it does not quantify over other controllers.

### L.3 Implementation identifiers and trace locations

The two gate identifiers are `GR1SeparatedGame` and `isEmptyController`. The latter checks a stored justice-level $Z$ set before generic default-ZYX extraction uses the same model and memory. Section C.4 records the historical environment-only initializer caveat; D the complete saved-relation checks; G–H the published-replacement capture limits, followup histories, and graph evidence. Section H separately reports a nonrelease extraction variant and a paired comparison sharing one solved model and fixed-point memory; the release identifiers below continue to describe the baseline.

The configured route uses Spectra `1.0.0.202402072123` with JTLV, syntactic Piterman reduction, `optChoice=none`, and `TraceInfo.NONE`. A headless loader replaces IDE resource access; the route excludes the Eclipse GUI and default CUDD backend and differs from the historical C-backed coverage driver [9]. The initial test uses the raw region, and the empty-controller gate rejects only an empty stored justice-level region. Loaded-class hashes match the release archive with the inspected sources; this is not an independent source-to-bytecode proof.

The saved `SynthesisAction` bytecode checks `synthesisPitermanReduction`, then invokes `PitermanSyntacticReduction.doReduction` at offset 487 and `TranslationProvider.translate` at 596. Headless `ReleasedKindJobProducer` follows this order before BDD generation (`SOURCE_ROUTE_TRACE.tsv`), separately from Section B's later-snapshot comparison.

### L.4 Mechanism controls

Table 18 compares constructed or reconstructed controls: Guard (Section B), Justice ((L.5)), Mixed, Traffic P, and Choice/Toggle. These are not independent public applications.

For *Choice*, let environment Boolean $e$ and system Boolean $p$ have

$$\theta_e = \neg e, \quad \theta_s = \neg p, \quad \rho_e = (e' \Rightarrow p),$$
$$\rho_s = \top, \quad A = \mathsf{GF} e, \quad G = \mathsf{GF} \neg p.$$

From all four typed states, $p' = \neg p$ guarantees $G$; after every legal prefix, choosing $e' = p$ supplies an assumption-fair suffix under the same strategy. Thus $W = V = K = C = D$ and $I_M(K) = \mathit{true}$ independently of fixed-point computation. Toggle changes only system safety to $p' = \neg p$, retaining these initializers and justice objectives.

Mixed has environment Boolean $e$, system Booleans $d, y$, $\rho_e = \neg d$, $\rho_s = \mathit{true}$, $A = \mathsf{GF} e$, and $G = \mathsf{GF} y$. Its initial predicates are $\theta_e = \neg e$ and $\theta_s = \neg d \wedge \neg y$. Its cooperative strategy can keep $d = 0, y = 1$ while the environment may choose $e = 1$ after any prefix. Traffic P reconstructs a repaired traffic-light contract from [10], using its original initializer rather than a residual entry from repair history.

Guard's source has environment Boolean $x$, system Boolean $p$ (called $y$ in Section B), trivial initial and justice predicates, and $\rho_e = \rho_s = x'$. The constant strategy $p = 0$ is strict-winning; from every history, choosing $x' = 1$ forever supplies a legal continuation. Thus $W = V = K = D$ and $I_M(K) = \mathit{true}$, independently of the mixed-current-invariant convention. The saved job result binds its input hash to this source and reports native refusal with unchanged initializers, one solver call, and no extraction or printing. A new complete saved-arena check enumerates all 16 original and 64

Table 18: Constructed and reconstructed mechanism controls. Sizes are states/edges; original-contract checks use the helper-false view. A dash means no returned graph.

| Input | Result | Native | View | Strict | Coop. |
|---|---|---|---|---|---|
| Guard | False | — | — | — | — |
| Justice | Graph | 4/10 | 2/2 | Pass | Fail |
| Mixed | Graph | 5/16 | 3/6 | Pass | Pass |
| Traffic P | Graph | 52/496 | 16/80 | Pass | Pass |
| Choice | Graph | 3/7 | 1/1 | Pass | Fail |
| Toggle | Graph | 7/18 | 3/4 | Pass | Pass |

helper current/next assignments, including all four helper quadrants. Source equations and decoded saved predicates agree everywhere. The helper relations are the complete literal form: at any $b = 0$ state, ENV can choose $b' = 1, x' = 0$, and both values of $p'$ violate SYS safety. Finite safety iteration has cardinalities $8 \to 4 \to 4$, retaining exactly the $b = 1$ slice. The saved raw four-fixed-point region agrees extensionally on this fixture and has a false quantified initializer. This establishes the first-flip obstruction in the saved configured run. It does not claim general equivalence of the raw core and strict synthesis, and no new native call was made. The historical partial certificates and new full check remain separately identified in the evidence index (Section K).

Justice's helper-false graph keeps $p = 0$ and then $e = 0$: strictness passes, cooperation fails, and the preceding argument separately excludes a cooperative alternative.

Mixed and Traffic P pass both checks with complete input coverage, despite transformed unfair prefixes. Traffic P's single result also appears in the separate eleven-candidate comparison (Sections G–H).

## L.5 Detailed decision evidence

Table 19: Decisions from completed checks or marked written derivations. Passing graph certificates cover legal-prefix cooperation, not all-history kind strategies. Supp. locates supplementary sections.

| Case | Evidence under the original contract | Permitted decision |
|---|---|---|
| Guard (Supp. B, L) | An original kind strategy exists; the configured released route refuses. The complete saved arena has an unavoidable first-flip safety failure, and its saved raw region agrees on this instance. | The refusal cannot justify discarding the original contract. |
| Humanoid variants (Supp. D, J) | Complete literal relations lose strict initial feasibility (also in released arena 044); cooperative initialization already fails. | Literal refusal alone does not establish original strict infeasibility. |
| Vehicle (188) (Supp. E) | A checked policy reaches $s \in K$; completed full native strict queries confirm $s \in W$ but $(s, 0) \notin W_R$, with both initializers true. Cooperation and reachability have separate policy proofs. | Transformed strict loss at full state $s$ cannot justify rejecting its original cooperative continuation. |
| Elevator (032) (Supp. C) | $K \neq \varnothing$, but $I_M(K)$ is false; an allowed initial input forces an environmental dead end. | Reject cooperative initialization. Restricting initialization alone would not repair this returned graph. |
| PCar (058) (Supp. B, C) | $I_M(W_V)$ is true, $K = \varnothing$, and two strict graphs fail cooperation. | Reject these graphs. A separate source argument excludes cooperative initial realizability. |
| Arbiter P (Supp. G) | Published learned repair: $K = \varnothing$, yet the released route returns a strict, non-cooperative controller. | Reject cooperative admission. Neither strictness nor vacuous $K \subseteq C$ certifies cooperation. |
| Lift P (Supp. H) | Whole-model correspondence fails; a separate direct graph audit passes strictness and fails cooperation. A written argument excludes cooperative initialization. | Reject cooperative admission using this original-contract evidence; whole-model verdict transfer remains UNKNOWN. |
| Lift T (Supp. H) | $I_M(K)$ is true, with a separate checked positive policy; the returned graph reaches an environmental dead end. | Reject this returned graph, while retaining the contract as cooperatively feasible. |
| Choice / Toggle (Supp. I) | $K = D$ for both; Choice returns a non-cooperative self-loop, while the matched Toggle output passes. | Membership in $K$, even for every reached state, cannot replace a fixed-strategy cooperation check. |
| Passing graphs | Complete graphs pass strictness and cooperation, including initialization and input coverage. | The graph witnesses existence within its verified state and private-memory scope. |
| SmartHome1 (223) | Resource limit in the released run; other paths return passing graphs. | UNKNOWN: no semantic ground for rejecting the contract. |

## L.6 Complete literal-preservation statements and proofs

This section gives the complete proofs, propagation and initialization examples, and constructed Guard counterexample for the statements in the main paper's section on exact continuation preservation. The hypotheses below concern strict games under complete relations; they do not establish the semantics of a raw cooperative-core verdict or certify an extracted controller. Local numbering gives the theorem, corollary, and corrected-guard proposition. Section L.1 covers retained helper-true constraints.

**Game and semantic scope** Let $M$ be a finite Mealy game with nonempty typed input and response carriers $E, Y$ and state space $D = E \times Y$, including all logical monitors. A round receives $e'$ before choosing $y'$, reaching $s' = (e', y')$. Environmental safety $\rho_e(s, e')$ is independent of the next response; $\rho_s(s, e', y')$ includes all system constraints and monitor updates. Initial predicates are $\theta_e(e)$ and $\theta_s(e, y)$. Under strict semantics, system safety is required on every edge whose environmental safety prefix, including that edge, remains valid. An environmental safety violation releases system safety on that edge and thereafter. On infinite safety-compliant plays, liveness requires $A \Rightarrow G$; failure of $A$ does not excuse earlier required safety. Neither relation need be total. Invalid type encodings are outside $D$; $A, G$ are generalized Büchi objectives. Strategies retain private memory and give typed responses even to illegal inputs. Define

$$\begin{aligned}\mathrm{CPre}(T) = \{s \in D \mid \forall e' \in E : \rho_e(s, e') \Rightarrow \\ \exists y' \in Y : \rho_s(s, e', y') \wedge (e', y') \in T\}.\end{aligned}$$

An *active* continuation has met environmental initialization, all prior environmental safety, and required system duties. $W \subseteq D$ contains states admitting a strict-winning continuation with some private memory. It includes unreachable typed states and vacuously winning environmental dead ends; initialization applies separately below.

For the helper variants, $b$ denotes a fresh, environment-controlled Boolean observed together with the next original input. Table 1 distinguishes four variants. For retained constraints, $U$ is the helper-true strict safety-winning region, and $F$ contains states from which every legal first flip admits a safe response into $U$. Seven of the eight completed helper-arena audits have literal first-flip timing but retain system constraints when $b = 1$. Thus a matching first flip does not establish complete literal relations. Supplement B gives the full comparison; D gives relation checks.

**The literal reduction** Kind writes each converted safety assumption as $\rho_n^e \vee \mathrm{next}(\mathit{envHelp})$ and each safety guarantee as $\rho_n^s \vee \mathit{envHelp}$ [7, Sec. 4.1, p. 4]. Its Theorem 1 states: "Then $M$ is realizable if and only if $M'$ is realizable" [7, p. 5]. Writing the helper as $b$, conjoining the respective clauses, and including helper persistence gives the following complete relations:

$$\begin{aligned} \theta_e^R &= \theta_e \wedge \neg b, & \theta_s^R &= \theta_s, \\ \rho_e^R &= (\rho_e \vee b') \wedge (\neg b \vee b'), & \rho_s^R &= \rho_s \vee b, \\ A^R &= A \wedge \mathbf{GF}\neg b, & G^R &= G. \end{aligned} \tag{L.6}$$

The guards apply to the *entire* safety relations, including logical auxiliary constraints. This semantic hypothesis must be checked for a syntactic route after all translations.

With $b = b' = 0$, both original safety relations apply; with $b = b' = 1$, both are released. The mismatch occurs on the first flip, $b = 0, b' = 1$: any typed $e'$ is allowed, including one disallowed by $\rho_e$, while $\rho_s$ still applies. Define

$$C(s) \;\equiv\; \forall e' \in E \; \exists y' \in Y : \rho_s(s, e', y'). \tag{L.7}$$

There is no environmental-safety antecedent in $C$. Passing this one-state test is insufficient: a winning strategy must keep it true at later states, as the propagation example below shows. Let $M_C$ have the original environmental relations and justice objectives, but system safety

$$\rho_s^C(s, e', y') \;\equiv\; C(s) \wedge \rho_s(s, e', y') \wedge C(s'). \tag{L.8}$$

Define $W_C = C \cap \mathrm{Win}_{\mathrm{str}}(M_C)$ independently of initialization. If $M_C$ is initialized, use system initializer $\theta_s \wedge C$. The outer intersection with $C$ excludes outside-$C$ states that win vacuously at environmental dead ends; changing system safety alone does not.

**Theorem L.2** (Exact strict-winning continuation region)**.** *Under the stated Mealy, strict, and complete-relation hypotheses, the reduction* (L.6) *has, for every $s \in D$,*

$$W_R(s, 0) = W_C(s), \qquad W_R(s, 1) = \mathit{true}, \tag{L.9}$$

*identifying a region with its characteristic predicate.*

*Proof.* At $b = 1$, legal edges retain the helper, system safety is released, and every infinite legal play violates $\mathsf{GF}\neg b$; hence all states win.

For necessity, a winning strategy at $(s, 0)$ must answer every legal first flip $(e', 1)$ with a $\rho_s$-safe response, so $s \in C$. Restrict it to original-legal histories with $b = 0$: these have exactly the original obligations. After each such finite prefix its residual strategy still wins, because required safety holds and Büchi conditions are prefix-independent. Repeating the first-flip argument gives $C$ at every reached successor. The restricted strategy therefore wins $M_C$, so $W_R(\cdot, 0) \subseteq W_C$.

Conversely, start from $s \in W_C$ and follow a winning $M_C$ strategy while $b = 0$ and inputs are original-legal. It maintains $C$, which supplies a safe response to any first flip. The successor has $b = 1$ and is winning by the first paragraph. Without a flip, original safety and liveness coincide; transformed-illegal inputs release strict duties. Thus $W_C \subseteq W_R(\cdot, 0)$. □

**Corollary L.3** (Full-region preservation)**.** *The literal reduction preserves the entire original strict-winning region at $b = 0$ if and only if $W \subseteq C$.*

*Proof.* $W_C \subseteq W \cap C$ and Theorem L.2 give necessity. If $W \subseteq C$, every original-legal prefix of an original winning strategy has a winning residual strategy. Every reached state therefore belongs to $W \subseteq C$, so that strategy satisfies (L.8) and wins $M_C$. Hence $W \subseteq W_C$, proving sufficiency with the reverse inclusion. □

**Loss can propagate** The exact lost region is $W \setminus W_C$, which can strictly contain $W \setminus C$. For example, take Boolean input $x$, response $y$, trivial justices, $\rho_e = x'$, and $\rho_s = (\neg y \wedge y') \vee (y \wedge x')$. Here $W = D$ and $C = \{y = 0\}$, but every original-legal response from $C$ leaves $C$, so $W_C = \varnothing$. An initial test of $C$ alone misses this delayed loss.

**Preservation from the original initializer** For a continuation region $T$, write

$$I_M(T) \;\equiv\; \forall e \in E : \theta_e(e) \Rightarrow \exists y \in Y : \theta_s(e, y) \wedge T(e, y). \tag{L.10}$$

Original initial realizability is $I_M(W)$; Theorem L.2 and initial $b = 0$ give transformed realizability $I_M(W_C)$. For an originally realizable contract, preservation requires each allowed initial input to admit a $\theta_s$-compatible response in $W_C$. Since $W_C \subseteq W$, original unrealizability is preserved. Full-region preservation is sufficient but unnecessary: replacing the first term of the preceding example by $\neg y \wedge \neg y'$ gives $W = D$ and $W_C = C = \{y = 0\}$. With $\theta_s = \neg y$, both initial decisions are true although the regions differ.

**An original feasible contract can be rejected** The Guard control has Boolean input $x$, response $y$, trivial initial and justice predicates, and $\rho_e = \rho_s = x'$. The original game has $W = V = K = D$: a constant response is strict-winning, and from every history the continuation with $x' = 1$ remains available. However, $C = \varnothing$. At the initial helper value $b = 0$, the first input $(x', b') = (0, 1)$ is legal in the transformed game, while $\rho_s \vee b$ is false for every response. Thus the complete Section 4.1 clauses make a realizable specification unrealizable.

**The existing first-transition correction** When it adds a helper, the released semantic implementation uses the additional first-flip guard $b \vee (\neg b \wedge b')$, equivalent to $b \vee b'$ [13]. Consider the corresponding complete semantic variant $R^+$, replacing only the system safety of (L.6) by

$$\rho_s^{R^+} = \rho_s \vee b \vee b'. \tag{L.11}$$

This existing variant is not a new synthesis algorithm.

**Proposition L.4** (Corrected-guard preservation)**.** *Under the same hypotheses, $W_{R^+}(s, 0) = W(s)$ for every $s \in D$, and $R^+$ preserves initial realizability.*

*Proof.* Follow an original winning strategy while $b = 0$ and original safety assumptions hold. The first flip releases system safety *on that edge* through $b' = 1$; every later legal edge is released through $b = 1$, and the added environmental justice is false. Without a flip the original obligations coincide. Conversely, keeping $b$ always zero restricts any transformed winning strategy to an original winning strategy. The unchanged initial predicates together with initial $b = 0$ then give equality in (L.10). □

**Counterexample scope and released evidence** Guard refutes preservation for the complete printed Section 4.1 relations; Appendix A and Kind's separate well-separation theorem are outside its scope. The complete semantic correction differs from configured syntactic arenas whose constraints can remain after the helper becomes true. Sections D–E and J report the applicable strict-preservation checks.

The configured syntactic route also refuses Guard natively. A later complete saved-arena audit [3] checks every original and helper transition assignment and identifies the literal relations exactly. On the first flip, $x' = 0$ is legal and neither SYS response is safe. Independent finite safety iteration gives an empty helper-false region and the full helper-true region. The saved raw four-fixed-point predicate is exactly this region on Guard; its quantified initializer is false, matching the native refusal before extraction. This identifies the first-flip obstruction in this saved run, without equating the raw cooperative core and strict solving on arbitrary games. The distinct complete semantic correction is not this configured route.

**When syntactic constraints remain** Section L.1 gives the full generalized statement and proof. With $U$ and $F$ as defined above, under the stated original-relation, one-way-helper and justice hypotheses, the active strict region is $W_F$: original strict-winning continuations that maintain $F$. Full preservation holds exactly when $W \subseteq F$; unchanged initialization gives $I_M(W_F)$. Thus safe entry and continued safety must both be checked. The complete literal case has $U = D, F = C$, and the corrected case has $U = F = D$. A retained partial constraint can invalidate either equality.

**Limits of the preservation claim** A syntactic route needs complete-relation and helper-true safety checks: translation can leave partial constraints unguarded or change invariant timing. AUX presence alone is inconclusive; a total deterministic update need not obstruct a response. The state retains all logical monitors throughout the residual-strategy arguments. Conditions using $W$ require the original winning region; no low-cost runtime precheck is established.

## M. Residual Elevator continuation census

This experiment asks whether the fixed final-choice preference used for Lift changes extraction from exact continuation states of Elevator 032. The source and its earlier cooperation failure were known before selection; the complete eleven-state cohort and preference were frozen before these new outcomes. This is a state census on one previously inspected source, not a held-out contract evaluation.

**Source and original feasibility.** The unchanged `ElevatorLTL_386_Elevator.spectra` from [9] declares environment variable $f = \mathit{floor} \in \{\mathsf{LOWER}, \mathsf{MIDDLE}, \mathsf{UPPER}, \mathsf{NONE}\}$ and system variable $m = \mathit{elevMot} \in \{\mathsf{FWD}, \mathsf{BWD}, \mathsf{STOP}\}$, with no logical auxiliaries. On these declared domains, its complete clauses are

$$\begin{aligned}
\theta_e &= \mathit{true}, \qquad \theta_s = (m = \mathsf{STOP}),\\
\rho_e &= (m = \mathsf{STOP} \Rightarrow f \neq \mathsf{NONE}), \qquad \rho_s = \mathit{true},\\
A &= \mathit{true},\\
G &= \mathsf{GF}(m = \mathsf{FWD})\\
&\quad \wedge \mathsf{GF}(f = \mathsf{LOWER} \Rightarrow m = \mathsf{FWD})\\
&\quad \wedge \mathsf{GF}(f = \mathsf{UPPER} \Rightarrow m = \mathsf{BWD}).
\end{aligned}$$

The assumption reads current values; it does not constrain $f'$ directly. The three system justices retain their displayed order. The previously certified region is $K = V = D \setminus \{(\mathsf{NONE}, \mathsf{STOP})\}$, where $|D| = 12$. Hence the original initializer verdict remains $I_M(K) = \mathit{false}$: the allowed initial input $f = \mathsf{NONE}$ forces $m = \mathsf{STOP}$, with no legal continuation. The cohort contains all eleven states of $K$, including three original-initializer-compatible states and eight incompatible states (Table 20).

**Exact entries and paired construction.** For each frozen state $s = (f_s, m_s)$, only the initial predicates are replaced: $\theta_e^s = (f = f_s) \wedge \neg h$ and $\theta_s^s = (m = m_s)$, where $h$ is the native helper. Every source value is fixed, and new controller private memory starts at rank 0; no old controller memory is inherited. The original initializer is saved separately and is not reapplied. The released syntactic route on Spectra `1.0.0.202402072123`/JTLV is unchanged, without semantic correction or justice normalization. One clean Java compilation is followed by eleven fresh JVM processes, one per entry, each solving once and extracting the released baseline and the fixed preference variant from the same game and fixed-point memory objects. There are 22 returned graphs and no retries or substitutions. Per pair, ten body BDDs and the field/domain/justice structure agree before and after the initializer replacement. Snapshots of 34 model and memory BDDs, including the winning and $X/Y/Z$ sets, agree before either extraction and after each; variable order and memory structure are retained. Objects are not shared across different entries.

**Two graph-checking routes.** Each returned graph receives two certificates, giving 44 checks. The source-specific route transcribes the clauses above and the helper encoding, uses its own native-BDD text reader, and compares the saved predicates on all 24 typed transformed states and 576 transition pairs. Its separate strongly connected component implementation checks graph fairness. This is an oracle for this pinned source, not a general source compiler. The saved-model route instead uses the independently decoded original BDD model, whole-domain correspondence, and the generic finite-graph fairness checker. It shares the original parser/translation exports and does not independently establish source compilation. Actual class loading, process completion, and captured-file identities are separate recorded provenance obligations.

Both routes bind the machine graph to the complete native printer output, retain node identities and private rank, and derive the helper-false graph reachable from the exact singleton. They check every original-legal input, unique responses and memory updates, system safety, and strict liveness. No restriction to $V$ or $K$, edge repair, or state merging is applied. Cooperation requires every reachable node to reach an assumption-fair cycle. With no original environment justice, a cycle is still required: a finite dead end cannot pass.

Table 20: All eleven frozen entries. Raw and original graph sizes are nodes/edges and agree between the two outputs in each pair. "Init." means compatibility with the original source initializer. Checks and refinements count final candidate sets examined by the preference variant.

| Entry | Floor | Motor | Init. | Raw | Original | Checks | Refined |
|---|---|---|---|---|---|---|---|
| s01 | LOWER | BWD | no | 11/68 | 6/24 | 68 | 0 |
| s02 | LOWER | FWD | no | 16/96 | 8/32 | 96 | 0 |
| s03 | LOWER | STOP | yes | 11/68 | 6/24 | 68 | 0 |
| s04 | MIDDLE | BWD | no | 11/68 | 6/24 | 68 | 0 |
| s05 | MIDDLE | FWD | no | 16/96 | 8/32 | 96 | 0 |
| s06 | MIDDLE | STOP | yes | 11/68 | 6/24 | 68 | 0 |
| s07 | NONE | BWD | no | 11/68 | 6/24 | 68 | 0 |
| s08 | NONE | FWD | no | 16/96 | 8/32 | 96 | 0 |
| s09 | UPPER | BWD | no | 11/68 | 6/24 | 68 | 0 |
| s10 | UPPER | FWD | no | 13/72 | 5/20 | 72 | 0 |
| s11 | UPPER | STOP | yes | 11/68 | 6/24 | 68 | 0 |

**Observed scope.** Both checking routes establish strictness and cooperation for all 22 original-contract graphs. Across 836 examined candidate sets, there are zero refinements and zero empty-preference fallbacks; each pair has identical raw graphs and printer text. All eleven comparisons are inactive. Thus this census supplies accepted residual controllers, including eight entries outside the source initializer, but no additional preference rescue, regression, or evidence of general effectiveness. It does not change $I_M(K)$, execute a handoff from an old controller, transfer reached private memory, or establish arbitrary-state migration. Graph construction from a prescribed state and execution of a replacement remain distinct evidence scopes.

## N. Intruder: source impossibility and a controlled repair

The unchanged developmental snapshot and its repair answer different questions. The original contract cannot jointly satisfy strictness and legal-prefix cooperation for all permitted initial inputs. A limited repair admits both properties, but the compared controllers have equal source-legal responses. The earlier resource-limited attempts remain preserved technical UNKNOWN records; they are not the current semantic conclusion or negative game certificates.

**Fixed source, history, and intended scope.** The original is the 2,071-byte `syn23/1673286053.spectra` from [9], also present as a developmental CatchingIntruder snapshot. It is not the later completed-task source. The six coordinates lie in $\{0, \ldots, 7\}$: ENV position $I$ and SYS positions $R_0, R_1$. The history bit $h$ records `ONCE(robot0AtButtomLeftCorner)`: $h \leftrightarrow [R_0 = (0, 7)]$ initially and $h' \leftrightarrow (h \vee [R_0' = (0, 7)])$. Robot 1's move permission uses current $h$, not $h'$. The original typed

carrier has $8^6 \cdot 2 = 524,288$ states; a separate helper doubles it. The sole SYS justice is recurring simultaneous orthogonal adjacency of both robots to the intruder. There is no ENV justice. The source does not encode four consecutive stationary states and bottom-row respawn from the task description. Our repair does not claim to restore that complete task.

Let $Clear(u)$ exclude obstacles $(2,1),(3,1),(4,1),(5,1)$, $Step(I,u)$ mean a stay or orthogonal unit move within the domain, and $Adj$ mean orthogonal adjacency. With

$$E_i(s,u) = Adj(R_i, I) \Rightarrow (u_x \neq R_{i,x} \wedge u_y \neq R_{i,y}),$$

the original ENV relation is $\rho_{e0}(s,u) = Clear(u) \wedge Step(I,u) \wedge E_0(s,u) \wedge E_1(s,u)$. The conjunction in each avoidance consequence is retained; replacing it by noncollision's disjunction would change the contract.

**Qualification of original and translated availability.** The syntactic helper insertion precedes invariant translation. At helper $b = b' = 0$, the obstacle test consequently constrains current $I$ instead of next $u$:

$$\rho_{e0}(s,u) = Clear(u) \wedge Step(I,u) \wedge E_0 \wedge E_1,$$
$$\rho_e^{impl,00}(s,u) = Clear(I) \wedge Step(I,u) \wedge E_0 \wedge E_1.$$

For example, at $I = (3,0), R_0 = (2,0), R_1 = (7,0), h = b = 0$, the only stay/unit input satisfying avoidance is $(3,1)$, an obstacle. Thus translated availability is true but original availability is false. This typed example does not itself establish reachability. The later source-Q qualification evaluates the original relation, including next obstacles, and checks 1,048,576 truth assignments across two BDD orders. The selection predicate lifts this actual source Q by helper-true; helper-false alone is not a test of original legality.

**What fails in the original returned controller.** The saved original-legal graph has 1,079 states and 3,353 edges, of which 49 states lack cooperative continuation. Eighteen have no legal next move because perpendicular adjacent robots impose incompatible avoidance; ten have false original Q although translated Q is true; the remaining 21 have current Q true but lead inevitably into dead ends. This acyclic bad region reaches a dead end within five edges. Replacing only final selection by exact source Q leaves the rooted legal responses unchanged. A canonical rank-safe early preference changes the graph to 1,089 states and 3,369 edges with 52 non-cooperative states. All three outputs pass strictness. These graph failures motivate but do not prove contract impossibility.

**Independent contract countercertificate.** A source-based checker validates an ENV counterstrategy containing 276,829 states and all 2,415,931 legal SYS responses to its chosen moves. The certificate contains no SYS-goal state. Every response stays in the goal-free certified set or reaches a state with no legal ENV next move. Staying forever violates SYS justice; leaving in the latter way violates cooperation. The permitted ENV initializer $I = (0,7)$ has a unique legal SYS/history response, which is covered. This excludes every strict cooperative SYS strategy for that initializer, including private memory, and therefore refutes the universal-initialization contract. It proves neither strict-only impossibility nor loss from every one of the eight initial inputs. The independent checker uses original finite source semantics, without relying on the native winning set or generator's transition table. Seventeen malformed/invalid-certificate controls are rejected as specified; ten include complete semantically mutated files.

**A narrowly defined repair.** Write $Q_0(s) = \exists u : \rho_{e0}(s,u)$ and $\Delta(s,u) = \neg Q_0(s) \wedge (u = I)$. Repair G replaces each avoidance clause by $E_i \vee \Delta$. All types, obstacles, initializers, SYS rules, history and justice stay unchanged. Whole-domain qualification establishes

$$\rho_{eG} = \rho_{e0} \vee (\neg Q_0 \wedge Clear(I) \wedge u = I),$$
$$Q_G = Q_0 \vee Clear(I).$$

Old enabled moves are preserved; at an old dead end with clear current cell, exactly a stay is added. The actual source predicate is checked on all 262,144 coordinate valuations, including obstacle and collision states. There are 1,352 added coordinate stays, 1,240 on physically legal configurations. Minimality is only within this fixed repair class: preserving old moves and totalizing clear-cell states by stays. No global minimum repair or reconstructed task correctness is claimed. The narrower capture-only alternatives remain independently refuted; they are not retried until one succeeds.

A separately generated source strategy is independently checked over 407,172 certified states, including 966 goals, with maximum rank 35. Every one of 1,625,884 legal ENV moves has a legal SYS response into the certified set. Goal states have rank 1 and every certified non-goal state has rank at least 2; the chosen response to every legal ENV move strictly decreases rank from a non-goal state. Every state has Q true, and the ENV justice list is empty. Closure therefore supplies an infinite legal continuation from every legal prefix, while descent proves recurring SYS goals. All eight initial ENV positions $(x, 7)$ are covered, each with the required unique SYS and history initialization. The unlabelled 117,116 typed states receive no winning or losing classification. Twenty-six rejection controls cover semantic and representation errors.

**Actual repaired outputs and the no-effect explanation.** One native solve supplies both saved extractors. Independent integer semantics match the original and helper models' complete initial, safety, and ordered justice predicates. The raw baseline and preference graphs have respectively 7,893/508,660 and 8,076/520,372 states/edges. Both source-legal views have 1,099 states and 3,383 edges, cover all eight initial inputs and every legal next input, and pass safety, strict liveness, and legal-prefix cooperation. A second reachability calculation agrees with the SCC-based checks. A complete synchronous product of 1,099 pairs and 3,383 inputs finds no source-response difference.

This equality has an additional local explanation. Any fixed G-legal next input is clear, so every SYS completion satisfies $Q_G$; therefore $C \cap Q_G = C$ for every rank/heuristic subset $C$ at that input. Helper-true is likewise permitted by the lifted selection predicate. The identity excludes direct Q filtering on legal-input candidates. It does not by itself imply identical raw graphs or identical controllers under arbitrary traversal: the compared extractors also differ in rank handling, and helper/illegal-input traversal can affect existing-node preferences. Actual legal-response equality is established separately by the product check. G is a specification-repair success, not evidence of Q superiority.

**Memory and historical execution limits.** First-choice extraction avoids an unused prefetched BDD iterator suffix; owned iterator temporaries are freed and quiet runtime is retained. The selected Boolean assignment preserves increasing numeric-variable-ID, false-first order before projection. Primitive, tiny-model, and Lift regressions support the change. Original Intruder then completes with a 2-GiB heap in 60.09 seconds, with wait4 peak RSS 2,493,972,480 bytes; its controllers still fail cooperation. The G pair completes in 64.37 seconds with peak RSS 2,476,359,680 bytes, and its independent check in 87.24 seconds with 1,335,902,208 bytes. These are individual measurements, not speedups, total-memory bounds, or online guarantees. Post-GC occupancy in the original trial reaches 98.05% of heap.

The earlier two attempts retain their original stopping records: the shared pair hits 44.07 seconds before extraction; the normalized arm fails at a 16-MiB relation file after 24.44 seconds, before native solve. Their three controller judgments remain UNKNOWN, and the partial file and recorded viability iterations are not adopted as certificates. The previously public bounded snapshot preserves all 135 outputs. Later evidence is a separate local extension, not a silent update of that archive or of the 11/14/59 cohorts. Across the completed later runs, input pins, owned-child reaping, and final process-group absence are checked. Memory improvements are verification infrastructure and do not turn technical completion into controller correctness.

**Complete certificate delivery and relocated replay.** The public archive described in Section K, `intruder-certificate-evidence-20261023-v1.zip` supplies the complete binary payloads, exact source/import closure, standard-library qualifiers and verifiers, and both rejection suites. Its root is `seams2027-intruder-certificate-evidence-v1/`; the inner kit is `seams2027-intruder-certificate-replay-v1/`. The outer README provides the six commands and a later replay summary. The inner construction manifest is preserved with its original pre-replay status; it is not the subsequent execution receipt.

A byte-checked copy at a separate temporary location passes original qualification and verification, G qualification and verification, and the 17- and 26-case rejection suites. After driver setup, Python audit-event monitoring in the first four executions confines observed file reads to packaged scientific inputs, result output, and the Python standard library; writes are confined to result output, and selected process-launch and network-connect audit events are denied. This is an application-level audit hook, not an operating-system sandbox or an audit of launcher startup. The rejection suites spawn checker processes and therefore do not have that file-access audit coverage. All six runs complete under process supervision and leave no process group. This checks certificate replay, not fresh synthesis or extraction. The original certificate covers its single selected allowed initializer; the G certificate covers all eight. Neither classifies unlabelled states. Source and scalar semantic references are shared trust inputs, and the implementations were mutually visible. The new archive is distinct from the earlier public bounded snapshot and does not replace its UNKNOWN records.

## O. Matched Port extraction and one-step-Q limits

PortHandshake and Polling are constructed mechanism examples outside the published-replacement and public-input cohorts. Their sources and initializers were fixed before the matched Port experiment; no ordering or source search was used to obtain an effect.

**Exact Port source.** ENV owns Boolean $ack$; SYS owns Boolean $request, gate$. Initial values are $ack = request = 0, gate = 1$. ENV safety is $request \Rightarrow ack'$ and $\neg gate \Rightarrow \neg ack'$; SYS has no additional safety and requires $\mathsf{GF} request$. The unchanged 204-byte source gives $Q_{ref} = \neg request \vee gate$ exactly. After initialization, always choosing $request = gate = 1$ is a strict cooperative source policy. The historical released/current-rank-safe outputs have source-legal sizes 3 states/2 edges and 3 states/4 edges, with strict/non-cooperative and strict/cooperative judgments. Their different rank handling prevents attributing that contrast to Q alone.

**Matched intervention and diagnostic separation.** R and RQ use one solve and shared fixed-point memory, the same canonical rank calculation, admissible progress rules, Z/Y/X ordering, goal/existing-state preferences, initializers, BDD order, and first-choice runtime. All Q-preference sites switch together: TRUE for R and $Q_{ref} \vee helper_true$ for RQ. This qualified Port profile

contains a helper; the source predicate remains unlifted. The unlifted source $Q_{ref}$ is separately retained in both arms; Q-off never changes its diagnostic truth to TRUE. The historical released B is a reference, not a third fresh arm or a same-memory control for the new pair.

A local witness requires a common original-legal prefix and the same full source/history state, input, canonical rank/progress obligation, and existing-state context before Q acts. Both Q-good and Q-bad successors must belong to that pre-intervention progress universe. For the Y-rank-decrease rule, changes in the selected layer within $k < r$ are distinguished from within-layer selection; changes in the selected Z-successor Y layer are classified separately. The final post-Q pools need not coincide. A second `satOne` call on a raw unpreferred pool alone does not reproduce the full Q-off heuristic sequence.

**Bounds and judgment.** The Port native envelope remains 512 MiB heap, 128 MiB direct memory, 2 GiB RSS, and 120 seconds. Scalar diagnostics use already computed outcomes. Additional witness inspection is bounded to 16 contexts, including attempted contexts, 16 records, and 256 KiB aggregate output across both arms. Added BDD-operation and live-temporary budgets, source/helper qualification, and recording-on/off parity passed before the primary pair; BDD references are not retained in logs. The final judgment separates reachable mixed opportunities, matched-context selection changes, and complete output strictness/cooperation. Partial diagnostic coverage leaves unrecorded scope UNKNOWN; a fully validated local witness can still establish its own selection effect.

**Qualified implementation and execution.** A frozen source revision compiled once in two build jobs. Two prerequisite pairs ran with recording off and on; a separate audit confirmed complete controller topology, native identity/rank/order, model, and fixed-point snapshot parity in each arm. All 71 saved BDD DAGs per arm matched under node-ID renaming. Source and selection Q were checked on all 16 typed helper-inclusive states. Eight in-memory rejection controls passed: wrong source, missing typed field, invalid private rank, duplicate edge, missing legal response, source Q replaced by lifted Q, incomplete memory roster, and invalid causal rank/context. They did not alter saved inputs or launch extra native jobs. These qualification pairs are separate from the single primary pair and are not additional outcome samples.

The primary pair used one solve and two extractions. Before/between/after snapshots certify the same model and full fixed-point memory; BDD order remained unchanged. A saved-data checker made no native solver or BDD library call. The raw graphs have 5 states/12 edges for R and 5/14 for RQ. Their complete original-legal helper-false views have 3/2 and 3/4, respectively, with original node identities and private ranks retained. Both cover the sole original initializer and every reachable legal input, satisfy source safety, and pass strict liveness. R fails cooperation at all three reachable nodes; RQ is cooperative everywhere. SCC/reverse-reachability and independent integer transitive-closure checks agree. Helper-true continuations are outside these source-graph judgments.

**Observed local selection effect.** The common original-legal prefix comparison reaches the shared initial pair and compares both legal inputs, $ack' = 0, 1$. Their responses differ for both inputs; histories are not merged after divergence. Only $ack' = 0$ also has a matched pre-Q existing-state context and qualifies as a controlled local witness. At $(ack, request, gate, helper) = (0, 0, 1, 0)$, both private justice indices are 0 and canonical Y ranks are 1. The permitted pre-Q Y-layer 0 contains exactly $(request', gate') = (1, 0)$ and $(1, 1)$ with $ack' = helper' = 0$. Their scalar codes are 2 and 6 and their source-Q values are false and true. R selects code 2; RQ selects code 6. Both selected progress layers are 0. A separate finite replay validates the whole actual state-creation/DFS order

and the recorded candidate sets against saved model and rank DAGs. This is a Q-induced response change within one permitted layer. Because all preference sites switch together, it does not identify a sole causal Y-filter site. The $ack' = 1$ output difference is not promoted to a second same-context witness.

**Diagnostic coverage and measured resources.** The first eight raw decisions per arm produce 16 complete retained records, out of 12 R and 14 RQ decisions. Five records meet the original source scope (two R, three RQ); one context matches across arms and contains both Q-good and Q-bad candidates. There are no failed or omitted records. The normal attempt cap is reached; unrecorded causal scope remains UNKNOWN. This does not limit the separate complete graph judgments. Instrumentation charges 4,240 added BDD calls, peaks at three owned diagnostic temporaries, and writes 13,778 bytes, below the 65,536-call, four-temporary and 256-KiB pair caps; the per-context call cap is 4,096. No extra `satOne`, counterfactual controller, retained BDD reference, operation-cap stop, or byte-cap stop occurs.

The primary native process completes in 0.938 seconds with wait4 peak RSS 212,107,264 bytes (202.28 MiB), within the 512-MiB heap, 128-MiB direct memory, 2-GiB RSS and 120-second envelope. Process reaping and teardown pass. This single recorded macOS measurement is not a minimum-memory, speedup, or online-latency result.

**Saved identities and interpretation.** The unchanged 204-byte Port source has SHA-256 prefix `311394bc0dda`; the frozen manifest is `ad5789a0b7a1`. Extractor, diagnostics, and producer prefixes are respectively `345b86ef5855`, `ec83cf318cf8`, and `bcf755934dc4`; the independent causal replay is `705e2b272332`. The qualification gate is `9ac9de4d0ccf`, primary audit result `cc57621d429d`, and diagnostic record `7fc329796f9f`. The saved manifests retain full SHA-256 values and bind sources, reviewed plan, dependencies, class origins, process receipts, and prerequisites. This predeclared constructed instance establishes a controlled selection effect and a checked output improvement. It adds no necessity claim, general superiority, application generality, benchmark success rate, or executed handoff.

**Polling is a fairness countercontrol.** ENV owns $ack$, SYS owns $poll, heartbeat$; all start false. Safety $\neg poll \Rightarrow \neg ack'$ is always satisfiable, so $Q = Alive = D$ on all eight typed states. ENV justice is $\mathsf{GF}ack$; SYS justice is $\mathsf{GF}heartbeat$. The source permits the cooperative policy $poll = heartbeat = 1$ after initialization, and the checked source regions are $V = K = D$. Nevertheless the three saved released, preference, and normalized outputs are strict and non-cooperative. The normalized arm starts from the original initializer and restricts the original arena to $V$, adds no helper, and copies ENV justice $ack$ after SYS justice $heartbeat$, giving the ordered SYS list $[heartbeat, ack]$. It uses `GR1SeparatedGame` and the unchanged generic ZYX constructor. Its saved final region equals $K = D$, but the complete graph is $0 \to 1 \to 2 \to 2$, with $poll = ack = 0$ throughout; $heartbeat$ is respectively $0, 1, 0$. The terminal SCC cannot satisfy ENV justice. Nodes 0 and 2 retain distinct private ranks despite equal source values. This refutes the selected output, not the existence region. The historical helper-based preference pair inspects ten candidate sets with zero refinements and identical source-legal responses; it is not the matched Port experiment above. The preference has no discriminating one-step predicate; strictness does not exclude preventing ENV fairness. The normalized 8-of-8 success on the eleven replacement contracts is therefore a cohort result, not a universal constructor guarantee. No execution handoff or new external-application generality is established by these constructed sources.

# P. Constructed request-enable update

## P.1 Frozen contracts and routes

This author-controlled semantic study has Boolean ENV inputs $r, a$ (request and ready) and SYS phase $p \in \{I, P, X\}$ (idle, prepare, execute). Its 12-state typed carrier retains every combination. Both models initialize $(r, a, p) = (0, 1, I)$ and have empty justice lists. Their environment relations are

$$\rho_e^{\text{new}} = (p = X \Rightarrow a') \wedge (p \neq I \Rightarrow \neg r'), \qquad \rho_e^{\text{old}} = \rho_e^{\text{new}} \wedge \neg r'.$$

The only update removes the final request-disabling conjunct. The shared system relation is

$$\rho_s = (p = I \Rightarrow ((r' \wedge p' = P) \vee (\neg r' \wedge p' = I))) \wedge (p = P \Rightarrow p' = X) \wedge (p = X \Rightarrow (a' \wedge p' = I)).$$

Ready is required on the edge *out of* execute, not the preceding prepare edge. Inputs are chosen before the response. The source ASTs, full two-model/three-route roster, fixed boundary, routing order, environment scripts, and qualification controls were frozen before execution. There was no parameter search or held-out application.

The three routes are the original strict game, the complete literal helper, and the complete first-flip-corrected helper. The helper is ENV-owned, initially false, persistent once true, and adds $\mathsf{GF}\neg b$ to ENV justice. Literal and corrected relations are exactly those proved in Section L.6; no translated residual constraints are added. Finite safety iteration computes the original 12-state and helper 24-state regions. Separately implemented predecessor and certificate checks compare every state. A losing-state certificate supplies one legal spoiling input per losing state and checks that *every* safe SYS response descends to an earlier losing layer. It does not require every legal input to spoil.

## P.2 Complete region and graph outcomes

A supplied source policy covers every legal input safely from each of the twelve typed roots, with a legal infinite continuation from every reached prefix. These twelve complete root checks certify $W = K = D$ in both original models. For literal conversion, a flip at $X$ can choose $a' = 0$, leaving no safe response; hence $C = F = \{p \neq X\}$ and $U = D$. Old idle stays in $F$, whereas a new request forces $I \to P \to X$, so $W_F^{\text{old}} = \{p = I\}$ and $W_F^{\text{new}} = \varnothing$. Corrected conversion has $F = U = D$ and preserves all twelve winners. Table 21 reports every frozen cell.

Table 21: All six request cells. Entry is $s_1 = (0, 0, I)$; Init uses the declared initializer. Graph counts are reachable states/legal-input responses of the separately checked original-source view from $s_1$; both means strict and legal-prefix cooperative. No graph is returned at a losing entry.

| Model | Route | $\lvert F\rvert$ | Surviving $\lvert W\rvert$ | Entry | Init | Source graph |
|---|---|---|---|---|---|---|
| Old | Original | — | 12 | Yes | Yes | 2/4, both |
| Old | Literal | 8 | 4 | Yes | Yes | 2/4, both |
| Old | Corrected | 12 | 12 | Yes | Yes | 2/4, both |
| New | Original | — | 12 | Yes | Yes | 6/14, both |
| New | Literal | 8 | 0 | No | No | No entry policy |
| New | Corrected | 12 | 12 | Yes | Yes | 6/14, both |

Every returned helper graph is strict but fails helper cooperation: a legal flip makes the added ENV justice impossible. This is kept separate from original-source admission. The source view

restricts to $b = 0$ and original-legal inputs, then checks input coverage, safety, strict liveness, and legal-prefix cooperation anew. All five returned source views pass. The new literal cell has no winning entry policy; an empty table there is a refusal, not a bad returned controller. The old source graph checked against the new model lacks responses on four newly legal request inputs at its two reachable idle states. Its behavior is undefined there. The study does not assert that a separately totalized old implementation would choose an unsafe response.

### P.3 Seven fixed episodes and limits

Every episode begins with the same old legal step $(0, 1, I) \to (0, 0, I)$ and pauses at the noninitial full state $s_1$. The model is updated while retaining the same source object and all three source fields. Policies are precomputed; the execution does not invoke a native solver at the boundary. The three old-route episodes make one post-update idle-loop commit. The new literal episode remains paused with zero commits. The new original, new corrected, and new routing-policy episodes each execute the fixed next-input script

$$(r', a') = (1, 0), (0, 0), (0, 1), (0, 0), (0, 0),$$

giving phases $I \to P \to X \to I \to I \to I$ and five commits. Thus the newly enabled request is exercised. The routing policy checks the literal route first, rejects its empty exact entry region, and then installs the corrected route only after its source-view audit. An event-level audit checks all 66 events across the seven episodes. All 54 finite-game qualification controls and six source-mutation controls have their expected outcomes.

The complete literal $F$ functions are equal between the two models, but the original ENV relations differ, so their $W_F$ regions differ. This is consistent with the decomposition: $F$ alone does not define $W_F$ without the original game. Within a fixed game, equal complete $F$ functions must yield equal $W_F$. The simpler global condition $W \subseteq F$ fails for both literal models and already supports avoiding both routes. Their initializer verdicts also differ. Accordingly the study establishes an executable state-specific distinction beyond a local response test, with no advantage over initializer checking, direct original solving, global avoidance, or a complete graph audit. Trivial source justices limit the fairness claim. No compiler, BDD engine, or native synthesizer is involved. The runtime equations and checker are separately written but author-visible and share the source AST; this is not a blind independent implementation. Section K locates all sources, policies, state tables, certificates, and event records.

## Q. Complete Humanoid first-flip guard census

### Q.1 Question, fixed entry, and intervention

At a fixed noninitial Humanoid 044 state, the local response test accepts the unchanged syntactic helper, but exact continuation does not. Relaxing either the obstacle-response group or the history-response group on the first flip is sufficient at that state. This study asks which of eight declared policies preserve an entry or the whole strict region. It does not change the original source contract or seek a cooperative controller: the original $V = K = \varnothing$.

The source is the archived HumanoidLTL_460_Humanoid contract and its saved 044 original/helper models. We retain the full typed carrier of 18,816 states: two ENV fields (Obstacle and InputMoveMode), four motor/mode outputs, and two distinct Boolean PREV auxiliaries. All original guarantees are current-state predicates in this saved translation. Their trace IDs are partitioned into three frozen groups: obstacle $O = \{6, 10\}$, history $H = \{8, 9\}$, and mode change

$M = \{7, 11, 12, 13, 14, 15, 16\}$. An arm relaxes exactly the chosen source conjuncts in the helper quadrant $b = 0, b' = 1$. The other three quadrants, typing, history updates, initial predicates, objectives, and source relation at $b = b' = 0$ remain fixed. All eight subsets were specified before their outcomes were computed; source and boundary were not selected by this census's results.

At the fixed initialized state $u$, Obstacle is CLEAR and every mode and motor is FWD; both auxiliaries are false. One legal original transition changes only the auxiliaries to $(0, 1)$, yielding the fixed noninitial state $r$. Both $u, r$ belong to original $W$. This is an existential original-legal prefix, not a claim about an installed controller's reachable private graph.

## Q.2 Complete source predicates and history order

Let $z$ mean Obstacle=BLOCKED; CLEAR is $\neg z$. Write $i, o$ for InputMoveMode and OutputMoveMode, and $\ell, r_m, h$ for LeftMotor, RightMotor, and HeadMotor. Abbreviate FWD, BWD, TURN_LEFT_1, TURN_LEFT_2 by $\mathsf{F}, \mathsf{B}, \mathsf{T}_1, \mathsf{T}_2$; CALIB_FWD, CALIB_TURN, TURN_FWD, TURN_BWD, STOP by $\mathsf{C}_F, \mathsf{C}_T, \mathsf{R}_F, \mathsf{R}_B, \mathsf{S}$. The mode domain is $\{\mathsf{F}, \mathsf{B}, \mathsf{T}_1, \mathsf{T}_2\}$; each leg motor has domain $\{\mathsf{F}, \mathsf{B}, \mathsf{C}_F, \mathsf{C}_T, \mathsf{R}_F, \mathsf{R}_B, \mathsf{S}\}$; the head has domain $\{\mathsf{F}, \mathsf{B}, \mathsf{S}\}$. Together with Boolean $z, p_0, p_1$, these give $2 \cdot 4 \cdot 7^2 \cdot 3 \cdot 4 \cdot 2^2 = 18,816$ states.

The ordered auxiliaries are $p_0 = \mathrm{PREV}(i = \mathsf{T}_2)$ and $p_1 = \mathrm{PREV}(i \neq \mathsf{T}_2)$. Both initialize false; their updates are

$$p_0' = [i = \mathsf{T}_2], \qquad p_1' = [i \neq \mathsf{T}_2].$$

They are distinct variables: although complementary after a transition, the initialized pair is $(0, 0)$. The original initializer imposes only this pair; all other fields are free within their types. The sole ENV safety predicate is $(i = \mathsf{F} \wedge p_0) \Rightarrow \neg z$. There are no original ENV or SYS justices.

Below $g_k$ denotes the current-state predicate with saved trace ID $k$. Conjunctions such as $\ell = r_m = \mathsf{C}_T$ require both equalities.

$$\begin{aligned}
O: \quad & g_6: \ (i = \mathsf{F} \wedge z) \Rightarrow (o = \mathsf{T}_1 \wedge \ell = r_m = \mathsf{C}_T), \\
& g_{10}: \ z \Rightarrow o \neq \mathsf{F}; \\
H: \quad & g_8: \ (i = \mathsf{F} \wedge p_0) \Rightarrow \ell = r_m = \mathsf{C}_F, \\
& g_9: \ (i = \mathsf{F} \wedge \neg z \wedge p_1) \Rightarrow \ell = r_m = \mathsf{F}; \\
M: \quad & g_7: \ (i = \mathsf{B} \wedge \neg z) \Rightarrow (o = \mathsf{F} \wedge \ell = \mathsf{F} \wedge r_m = \mathsf{B}), \\
& g_{11}: \ (i = \mathsf{T}_1) \Leftrightarrow (o = \mathsf{T}_2), \\
& g_{12}: \ (i = \mathsf{T}_1) \Rightarrow (\ell = \mathsf{F} \wedge r_m = \mathsf{R}_F), \\
& g_{13}: \ (i = \mathsf{T}_2) \Rightarrow (\ell = \mathsf{B} \wedge r_m = \mathsf{R}_B), \\
& g_{14}: \ (i \notin \{\mathsf{T}_1, \mathsf{T}_2\}) \Rightarrow \\
& \qquad \neg(\ell = \mathsf{R}_B \vee r_m = \mathsf{R}_B \vee (\ell = \mathsf{R}_F \wedge r_m = \mathsf{R}_F)), \\
& g_{15}: \ (i \in \{\mathsf{T}_1, \mathsf{T}_2\}) \Rightarrow h = \mathsf{S}, \\
& g_{16}: \ (i = \mathsf{T}_2 \wedge \neg z) \Rightarrow o = \mathsf{F}.
\end{aligned}$$

All eleven are evaluated on the *current* state. Original SYS safety conjoins them with the two AUX updates. The 00 quadrant keeps this relation and the original ENV predicate; 01 allows every typed next input and conjoins the unrelaxed $g_k$ with the same AUX updates; 11 retains the AUX updates, with every typed next input allowed; 10 is forbidden. The helper starts false and adds ENV justice $\mathsf{GF}\neg b$. Thus the printed predicates determine every arm's first-flip and post-flip relation.

At $r = (\neg z, i, \ell, r_m, h, o, p_0, p_1) = (1, \mathsf{F}, \mathsf{F}, \mathsf{F}, \mathsf{F}, \mathsf{F}, 0, 1)$, all current clauses hold, explaining the common local pass. Without relaxing $O$ or $H$, future input choices can force incompatible obstacle

and history duties; the invariant computation detects this loss beyond the current state. The full eight-arm table reports which invariants survive, rather than inferring them from this local pass.

### Q.3 Whole-function checks and all outcomes

Let $S_{01}^{J}$ be the first-flip SYS relation after relaxing groups $J$. It conjoins the remaining source clauses with the unchanged auxiliary updates. We compute $U$, $F$, $W_F$, and the direct helper winning region for every arm. Exact BDD function comparisons agree with separately written finite-set safety iteration on all typed states; cardinality agreement alone is not used. Every arm has $U = D$. Complete source checks also compare all eleven clause predicates, initialization, and all helper quadrants. Five mutations test clause grouping, current/next timing, history update, initialized auxiliaries, and accidental changes to the 00 relation. A separate two-state propagation control has $|W| = 2, |F| = 1, W_F = \varnothing$.

Table 22: All eight Humanoid policies. Clauses counts relaxed source conjuncts, not runtime or repair quality. Local $F(r)$ is true in every row. Exact is $r \in W_F$; Init is the transformed quantified initializer. Original $|W| = 2,387$ and $V = K = \varnothing$.

| Relaxed | Clauses | $\lvert F\rvert$ | $\lvert W_F\rvert$ | Exact | Init | Whole $W$ |
|---|---|---|---|---|---|---|
| None | 0 | 1,211 | 0 | No | No | No |
| $O$ | 2 | 2,277 | 1,229 | Yes | Yes | No |
| $H$ | 2 | 2,144 | 1,217 | Yes | Yes | No |
| $O, H$ | 4 | 3,816 | 1,841 | Yes | Yes | No |
| $M$ | 7 | 12,966 | 0 | No | No | No |
| $O, M$ | 9 | 15,924 | 1,235 | Yes | Yes | No |
| $H, M$ | 9 | 14,712 | 1,217 | Yes | Yes | No |
| $O, H, M$ | 11 | 18,816 | 2,387 | Yes | Yes | Yes |

The local minimum at $r$ is no relaxation; the exact minimum among the eight candidates has two clauses, with both $O$ and $H$ tied. The predefined bit-mask tie break chooses $O$, not a uniquely superior repair. Only the all-group endpoint preserves every original strict winner. Full preservation is already equivalent to $W \subseteq F$; computing exact $W_F$ adds entry-specific scope rather than a stronger global Boolean test. No runtime, implementation cost, or optimality outside this eight-policy set is measured.

All 28 unordered policy pairs retain complete XOR predicates and witnesses. Twelve pairs have at least one state where both local tests pass but exact membership differs. Their common affected cohort has 1,211 states; it is not twelve independent copies of that population. Of these, 993 are existentially reachable from the complete original initializer through safety-compliant original edges, including 457 noninitial states and 536 initial states. Every such pair compares one of the two initializer-failing policies (none or $M$) with an initializer-passing policy. Thus no stronger local-plus-initializer distinction is observed. Only the none and $M$ arms have propagated loss $(W \cap F) \setminus W_F$, of size 1,211 in each; all other arms have $W_F = W \cap F$.

The finite and BDD checks share saved translated predicates and use mutually visible implementations. The hand-written source factorization provides a separate finite comparison, not an independent compiler. This is strict transformation maintenance on one selected source, with no native solver run, cooperative rescue, executed deployment, or application-prevalence estimate. The all-group endpoint still retains typing and AUX updates; region equality does not assert relation identity with every complete semantic correction.

## R. Native handoff with pending work and a newly enabled input

This separate study addresses two limits of the ten-boundary experiment in Section H: its nonzero monitors already had matching responses, and its fixed continuations never exercised the newly enabled button. Here an unanswered third-floor request survives installation, and the first post-change input requests the newly enabled first floor.

All six fixed executions complete. The two admission policies select different controllers but produce identical source-state sequences on the chosen script; no observed service advantage is claimed.

**Construction and selection.** The new contract is the same 2,218-byte reconstructed Lift T source. The constructed old contract appends only $\mathsf{G}\neg b_1$; its operational relation requires both current and next $b_1$ to be false. All original initializers, system clauses, monitor updates and ordered justice lists remain unchanged. An atomic pause removes exactly this restriction. The source, restriction and script were chosen after earlier diagnosis and inspection of the existing graphs. The complete three-boundary roster, prefixes, three candidates, candidate order and two policies were then frozen before the new certificates and executions. This is an outcome-informed mechanism study on one source, not an independent application sample, fresh synthesis experiment or exact-region comparison.

Both checkers first require old-mode strictness, cooperation, full legal input coverage and system/AUX safety from both original initial choices, native nodes 0 and 1 (respectively $c = 0$ and $c = 1$). They agree on the complete 50-state, 238-edge closure, including its edge relation. All 16 typed inputs are examined at each reached state. The complete closure has exactly three private nodes at which a monitor is $S1$, its matching floor response is false, and cancellation is false. Those three nodes form the entire study cohort. Every listed source valuation is noninitial. A false, incomplete, UNKNOWN or disagreeing old certificate would have stopped the study without another restriction search. No such outcome occurred. The saved field called `rank` is the private justice selector $j$ for $\mathsf{GF}\neg m_{j+1}$, not a decreasing proof rank.

Table 23: Complete pending-boundary roster. Values are $(b, c, f, m)$ with triples ordered by floor and $S0 = 0, S1 = 1$. Entries give graph-local node/stored rank. Every native old rank is 1; no rank is overwritten.

| Old node | Source/history state | Old prefix | Normalized | Preference |
|---|---|---|---|---|
| 524 | $(001, 0, 010, 001)$ | $0 \to 4 \to 524$ | 72/1 | 524/1 |
| 526 | $(011, 0, 010, 001)$ | $0 \to 4 \to 526$ | 74/1 | 526/1 |
| 756 | $(011, 0, 010, 011)$ | $0 \to 12 \to 756$ | 88/0 | 590/0 |

**Candidate certificates and installation.** The fixed order is retain the reached native instance, install the saved normalized graph, then install the saved preference graph. The last two choices create a new private instance at the exact certified root and its actual stored rank. They preserve the same modeled plant, source object and all ten source/AUX values; they do not copy arbitrary old private memory, reset monitors or reapply the source initializer. The additional nonzero monitor at node 756 already has its second-floor response, but its third-floor obligation remains genuinely unanswered.

Each route checks all nine rooted complete suffixes under the new contract. All nine are strict and have complete source input coverage and source/AUX safety. Each retained-native suffix has

154 states and 1,170 edges and fails cooperation. Each normalized or preference suffix has 154 states and 1,176 edges and passes cooperation. The two routes agree on these judgments and the complete edge relations for each candidate. A complete cooperative-false result is an admission result; an incomplete or disagreeing result is not silently treated as false.

Strict-only admission retains each reached native instance. Adding cooperation rejects that candidate and installs normalized roots 72, 74 and 88 with stored ranks 1, 1 and 0. The preference candidates qualify but are not selected because normalization precedes them. Thus there are nine candidate checks per route and six actual episodes, not nine executed candidate arms.

**Executed prefixes and five fixed inputs.** Each episode starts at native node 0 and actually steps and commits its two old-mode edges before pausing. The prefix inputs in $(b_1, b_2, b_3, c)$ order are $0010, 0010$ for node 524; $0010, 0110$ for node 526; and $0110, 0110$ for node 756. The old controller is therefore executed to the boundary, not seeded there. The mode change, certificate-based admission and installation occur while paused; no step commits during that interval.

The same fixed post-change inputs are offered to both policies: $1010, 1000, 1000, 0000, 0000$. Each is checked for source legality at the actual reached state before a unique graph response commits. The first input is illegal under the old $b_1$ restriction. It introduces $b_1$ while retaining the unanswered $b_3$ request; cancellation is never used. There is no alternative input script after a refusal.

Table 24: All six executed continuations, beginning at the retained or installed private root. Each executes five transitions, including the last self-loop. Source states agree between the two policies at every step, although private node identifiers and some ranks differ.

| Boundary | Strict-only | Strict and cooperative |
|---|---|---|
| 524 | $524 \to 546 \to 568 \to 492 \to 0 \to 0$ | $72 \to 78 \to 80 \to 68 \to 0 \to 0$ |
| 526 | $526 \to 546 \to 568 \to 492 \to 0 \to 0$ | $74 \to 78 \to 80 \to 68 \to 0 \to 0$ |
| 756 | $756 \to 608 \to 568 \to 492 \to 0 \to 0$ | $88 \to 92 \to 80 \to 68 \to 0 \to 0$ |

All six runs preserve $m_3 = S1, f_3 = c = 0$ at installation. Their first post-change commit sets $f_3 = 1$ while $m_3$ remains $S1$; the second clears $m_3$ and sets $m_1 = S1$. The third sets $f_1 = 1$; the fourth clears $m_1$; the fifth traverses the final self-loop with all three monitors $S0$. This one-edge response/clearing delay follows the current-state update $m_i' = (m_i \vee b_i) \wedge \neg(f_i \vee c)$. There are 12 old-prefix commits, 30 continuation commits and 98 recorded events. All six rows complete with no refusal, abstention, failure, interruption or UNKNOWN. A finite self-loop traversal is observed; infinite fair repetition uses the separately checked graph and justice conditions. Lift has no original ENV justice and three SYS justices.

**Qualification, event audit and limits.** Before candidate checking, 48 interface checks pass, including nine exact root bindings, nine synthetic installation fixtures, current/next $b_1$ and pending-$b_3$ cases, and targeted invalid bindings. The synthetic fixtures are qualification only and are not counted as study episodes. Negatives include wrong roots or ranks, Boolean/float ranks, lost or swapped monitors, helper-true roots, missing responses, the old $b_3$ mode, stepping while paused, and stale or disagreeing certificates.

The source-event auditor does not import the runtime. It reconstructs all 98 events, both prefixes and suffixes, source/AUX transitions, exact stored ranks, object continuity, admission order and lifecycle changes. The saved positive trace passes; all 16 deliberately mutated event copies are rejected for the expected audit failures. They include missing/duplicate commits, missing resume, duplicate admission, changed rank types, reset pending state, wrong mode or input, replaced source

object, false response summaries and stale certificate bindings. All ten execution stages ran once in order; resource-supervised children finished and their process groups were absent before progression.

The source route uses hand-derived source equations and a finite-set closure algorithm. The saved-model route checks typed source equality against compiler-exported BDDs and uses a separate graph-fairness algorithm. They share the pinned source and saved graph/model evidence; they are not independent native compiler or solver pipelines. These results establish bounded pending-work continuity and a different admission choice at the three declared boundaries. The retained graphs' cooperation defects occur elsewhere in their complete suffixes, so the equal successful traces do not contradict those certificates. No speed, general deployment, arbitrary-memory transport or service-success advantage follows.

**Public replay and reader procedure.** The standalone archive `pending-native-handoff-evidence-20261023-v1.zip` is available at `pending-native-handoff-evidence-20261023-v2`. Its 175 files include the scientific dependency closure, complete saved results, qualification controls, and explicit original/delivered identities. The public download was retrieved and matched byte-for-byte to the qualified local archive. Earlier archives and tags retain their bytes and historical scopes.

Begin with the archive's `README.txt`. Lift specification text is omitted: the offline restoration helper requires the exact earlier public Lift archive and publisher HTML identified there by byte counts and hashes. It reconstructs the two declared source contents; binding retains four separately named source inputs, including the saved models' original aliases. No upstream JAR is needed. After inspecting the immutable manifest, bind a fresh workspace and explicitly review its plan, runtime, dependencies and metadata changes. The README gives the complete command sequence and required review fields. A new reader's session binds the corrected comparer directly; the saved amendment is release history, not an additional reader step or author-operated permission service.

In the author-run relocated replay, all ten controlled stages completed once, with each child reaped and its process group absent before progression. The comparison covers 44 complete records, including both graph-checking routes, all 98 events, and the positive and 16 negative event controls. It preserves scientific values, scalar types, keys, list order and deliberately malformed fixtures. Only verified file bindings, recorded clocks and validated process/object identities are normalized. The initial comparison matched 42 records; two audit summaries still contained unnormalized controller-instance identifiers. The correction binds those two summary aliases to the already validated objects in the same episode. All 44 records then agree, without repeating any scientific stage. The original failed comparison, original comparer and assembly manifest, unchanged ten attempts, and explicit amendment are preserved in the qualification history.

The revised input manifest retains its pre-replay status; a separate release manifest covers every final member except itself. These checks were performed after relocation on the recorded macOS/Python 3.10.9 host, using the standard library plus POSIX process facilities and `ps`. They do not establish external replication, another operating system's compatibility, fresh native synthesis, or a service advantage.

### R.7 Extended pending-work execution: later divergence

The preceding five-input study establishes equal service of the request pending at the update. A separate extension uses all three same boundary entries and the same saved native, normalized and preference candidates. No source clauses, graphs or candidate roots are changed. The full root-specific workloads were frozen before these nine new executions. With input order $(b_1, b_2, b_3, c)$,

node 524 receives

$$1010, 1000, 1010, 0110, 0010, 1000, 1010, 1010, 0010, 0010, 0000, 0000.$$

Nodes 526 and 756 receive the same sequence except that its fourth input is 0010. The retained native arm stops immediately when no legal typed ENV input remains; the alternative-only suffix is not forced onto it. All arms therefore share their ordered input prefix while legal, rather than being compared through illegal continuations.

| Saved candidate | Runs | Commits per run | Old service / clear | Final outcome |
|---|---|---|---|---|
| Retained native | 3 | 8 | 1 / 2 | No legal input |
| Normalized | 3 | 12 | 1 / 2 | Fair self-loop |
| Availability preference | 3 | 12 | 1 / 2 | Fair self-loop |

All original third-floor requests are served at step 1 and clear at 2. The native runs later reach node 680 with a new first-floor request still pending. This finite observation alone is not a liveness counterexample; zero legal inputs is the certified cooperation failure. The six alternatives execute their twelve inputs and enter a fair self-loop. The separate event auditor checks 180 events and 114 commits (18 old-prefix and 96 continuation commits). Its positive case passes and eight focused malformed controls are rejected, including missing stop evidence, phantom legal inputs, an old-monitor reset, malformed private rank, wrong fixed input and false response summaries.

The workloads arose from exploratory shared-input product analysis and were then frozen. These executions provide a consequence of the already identified selection defect, not independent samples or a new contract-level refusal mechanism. The normalized candidate obtains the same benefit as the preference candidate. The old request's successful service in every arm must be distinguished from the later pending first-floor request. The earlier five-input evidence and its 98 events remain a separate study.

# S. Pending patrol continuation under temporary closures

## S.1 Source, change, and exact entry

The new source is the unchanged `Patrolling` specification, corpus entry 277 of the Kind artifact [9], with its imported `DwyerPatterns` definitions. Its 3,304 source bytes have SHA-256 `cde1b981a103047ddefbe8a963f1188d57904659a30fd0e7b678c771213be402`. The constructed old source appends one initial layout condition, $A = (0, 4), B = (2, 4), C = (4, 4)$, and three permanent-open assumptions. The update removes only these ENV additions. It changes neither SYS/AUX clauses nor types. Coordinate persistence retains the actual layout after the update. No source initialization is reapplied at the boundary.

All six checkpoint coordinates and two robot coordinates range over $\{0, \ldots, 4\}$. ENV also controls three blocked flags and `isCharged`. The charging counter ranges over $0, \ldots, 3$; the logical request monitor has states $S0, S1, S2$. This gives 75,000,000 typed current source states and 250,000 typed next ENV valuations. The finite old/new controller closures use one reached layout; this does not restrict the typed next coordinates quantified by the helper test.

In the constructed old-source scenario, the policy selects initial robot position $(0, 0)$; its added permanent-open assumptions initialize the blocked flags false. These are scenario/old-model choices, not initial restrictions of the unchanged public new source. The latter permits all 25 robot positions and arbitrary initial blocked/charged flags, with logical initial conditions $c = 0, m = S0$. The actual

old BOOT step leaves position and counter unchanged and changes the monitor to $S2$ on low power. The reached entry has private patrol phase 0 and is noninitial because its logical monitor differs from the initializer. The charging response has not occurred: robot position is not $(4, 0)$. The old policy graph contains both initialized power valuations and the actual BOOT edge.

### S.2 Complete source equations and policy

Let $p = (x, y)$ be robot position, $t_i$ the three checkpoint positions, $d_i$ their blocked flags, $z$ the charged flag, $c$ the counter, and $m \in \{S0, S1, S2\}$ the logical monitor. Write $H = (4, 0)$ and $\mathrm{near}(p, t) \equiv \|p - t\|_\infty \le 1$. The new ENV relation is

$$\rho_e(s, e') = \bigwedge_i (t_i' = t_i) \wedge \bigwedge_i (\mathrm{near}(p, t_i) \Rightarrow d_i' = d_i) \wedge (c = 3 \Rightarrow z').$$

The old relation additionally requires $\bigwedge_i \neg d_i'$; the pure-ENV invariant translation also initializes these flags false. There are three ENV justices $\mathsf{GF}\neg d_i$. The SYS relation requires

$$\|p' - p\|_1 \le 1, \qquad (p = H \wedge c < 3) \Rightarrow p' = p, \qquad \bigwedge_i (d_i \Rightarrow p' \neq t_i').$$

It includes the complete counter and monitor equations. The counter is $c' = \min(3, c + 1)$ when $p = H$, and $c' = 0$ otherwise. For the monitor,

| $m$ | $m'$ |
|---|---|
| $S0$ | $S2$ if $\neg z \wedge c \neq 3 \wedge p \neq H$, else $S0$ |
| $S1$ | $S1$ |
| $S2$ | $S1$ if $c = 3$; $S0$ if $c \neq 3 \wedge p = H$; $S2$ otherwise. |

The five SYS justices require recurring visits to each $t_i$, to $(0, 0)$, and to $m = S0$. The private patrol phase is not a logical source variable.

The manual policy visits the checkpoint row through the standoff row $y = 2$. It waits two cells before a target while its next blocked flag is true. Once that flag is false, entering the one-cell neighborhood freezes it until after the visit and retreat. The policy then traverses the next standoff, returns to $(0, 0)$, and visits $H$. It stays at $H$ until the counter reaches three. A finite phase list fixes every intermediate cardinal movement; the full list and every resulting edge are supplied. For illegal original inputs the strict strategy is completed with a sink only after the first illegal prefix. That convention does not release SYS duties on a legal first helper flip.

### S.3 Source correspondence and graph certificates

The pinned Spectra parser and translator export complete native ASTs for both sources. A separate support enumerator compares every translated clause to the executable source equations: 35 new clauses over 22,899 assignments and 42 old clauses over 38,536 assignments. The seven extra old translated constraints are the layout initializer and the initial/next forms of the three permanent-open assumptions. SYS and AUX ASTs are identical. Enumeration includes each clause's full typed support, rather than only its reachable samples; unused dimensions are factored. This is 77 clause checks and 61,435 support assignments, not full-state enumeration. Fifteen in-memory malformed AST/source controls are rejected for their expected reasons.

| Graph | Roots | States | Edges | States in $C$ | Strict | Cooperative |
|---|---|---|---|---|---|---|
| Old initialized | 2 | 140 | 277 | 140 | Yes | Yes |
| New pending continuation | 1 | 849 | 12,240 | 787 | Yes | Yes |

All nodes are reachable from the declared roots; every legal input occurs exactly once and every response satisfies source safety and the actual private phase update. The independent auditor evaluates all typed SYS outputs where needed, rather than accepting a producer's safe-response list. Its cooperation check uses reachable accepting cyclic SCCs, including non-bottom SCCs; its strictness check excludes ENV-fair cycles avoiding each SYS justice. The graphs establish feasibility at the declared entries, not at all typed layouts. Eight malformed graph controls and three fairness-algorithm controls qualify the auditor.

### S.4 Exact first-flip condition and lost continuation

Let $T(s)$ be the movement/dwell-permitted next robot cells and $N(s) = |T(s)|$. Put $k(s) = \sum_i [d_i]$. The complete literal condition $\forall e' \exists y' : \rho_s(s, e', y')$ is exactly $N(s) > k(s)$. If $N > k$, at most $k$ of the permitted cells can be occupied by next coordinates of currently blocked checkpoints, leaving a safe cell; the counter and monitor have unique typed updates. If $N \leq k$, assign distinct blocked checkpoints to every cell of $T(s)$. This typed next ENV valuation leaves no safe position. The flip's ENV disjunction releases original coordinate persistence, so such relocation is legal for the literal flip even though illegal in the source game.

The full typed check at the pending boundary covers 15,625 coordinate tuples and factors the sixteen irrelevant next-flag valuations, yielding 250,000 next ENV valuations. The 4,800-state fixed-layout slice has 4,596 states satisfying $C$ and 204 outside; these are slice counts, not full carrier cardinalities. Since SYS/AUX relations are identical, $C$ is identical for old and new models. Every reachable old policy node is in $C$, so its winning policy establishes $r \in W_C^{\text{old}}$.

For the new model a complete adversarial certificate establishes $r \notin W_C^{\text{new}}$. While the robot has not reached $H$, keep all blocked flags false and $z = false$. If it avoids $H$ indefinitely, all ENV justices hold and monitor $S2$ never clears, violating a SYS justice. At the first arrival at $H$, the counter is zero. Next legally set $A$ blocked; $A = (0, 4)$ is far from $H$. Dwell forces a stay with counter one, giving $N = 1, k = 1$ and therefore $\neg C$. The one-step flip witness relocates $A'$ to $H$. This is an encoding-only input; the actual source execution never moves $A$.

The countercertificate has 26 states, 25 nonexit rows, and one outside-$C$ exit. The auditor tests all 300 typed SYS outputs per row and checks exact coverage of all 103 safe replies. Every infinite nonexit branch is ENV-fair and violates the pending-monitor justice; the exit has a typed input with zero safe replies. Combined with the main exact-region theorem, this proves loss for the complete literal reduction. It does not assert a released syntactic solver's verdict, whose retained relations would require separate analysis.

### S.5 Actual source handoff and qualifications

The single software execution retains the same plant and controller instances, all fourteen source fields, logical monitor, and private phase. It performs one old BOOT commit, pauses at the pending entry, records the certified diagnosis, changes the active ENV model, and resumes. No source or private state is reset and no controller is replaced. The frozen workload closes $A$ for two offers at its first standoff, then clears it, and closes it once during the first charger dwell. These are three occurrences of one complete ENV valuation, at new-mode steps 3, 4 and 31. They are legal under the new model and illegal under the old permanent-open requirement.

The robot reaches $H$ at step 30, satisfying the old request, and clears its logical monitor at 31. That step is the single observed outside-$C$ state, with monitor already $S0$. It must not be described as an unserved old request. Thereafter the fixed all-clear, charged input repeats a full source-state and private-phase cycle from steps 32 to 67, length 35, visiting all three ENV and five SYS justices.

The complete graph provides the infinite-path guarantee; the observed cycle supplies execution evidence.

The independent event auditor checks all 278 events and all 68 commits (one old, 67 new), each input, response, graph edge, source-monitor update, private update, pause/resume boundary and object identity. The actual positive audit and eight malformed controls pass qualification; the controls corrupt carried source/monitor/private values, event/summary bindings or continuation evidence and must be rejected. These are selected controls rather than exhaustive fault coverage.

### S.6 Decision, alternatives, and reproduction scope

At the same reached pending entry, both models have $C(r)$ and a strict cooperative source continuation. The old complete literal encoding admits it; the new one does not. The result warrants retaining the requirements and qualifying the converted-game refusal. Validated reuse of the old controller already succeeds. Direct original-game solving and the known complete corrected guard are alternatives; no comparative speed or service advantage over those methods is claimed.

The scenario, policy and workload are constructed after exploratory search; they are not a random or prevalence sample. The new published source is unchanged. This is a finite source-model execution, not robot hardware deployment, a measured released-solver false refusal, or an observed human weakening decision. Source, graph and event auditors are separately implemented but author-visible and share recorded semantics. Native AST agreement depends on the pinned compiler. The accompanying saved-evidence package supplies source bytes, ASTs, complete graphs, counter-certificate, event stream, checker code and relative provenance. The replay manifest distinguishes checks of saved ASTs and recorded evidence from any regeneration of finite graphs or execution of source-model scripts. Saved-AST validation does not rerun native parsing or synthesis; none of these checks constitutes physical deployment. Earlier controls and UNKNOWN records remain in their original archives.

## T. Evaluation details and secondary studies

This section consolidates evaluation material moved from the main paper. It reports existing evidence only; the underlying complete studies and qualification records remain in their original supplement sections.

### Release semantics and the Lift preference

The audited route uses Spectra `1.0.0.202402072123` on JTLV. Selecting *Kind* invokes the syntactic Piterman helper conversion, raw four-fixed-point core, gates, generic constructor, and complete printing. Ordinary GR(1) is the default. The separate semantic class with the corrected first-flip guard is not this selected route. Kind names this method/tool route; lowercase *kind* denotes its continuation requirement. Section L records the implementation trace and controls, and Section H the paired Lift intervention.

The pinned default translator leaves current-only assumptions referring to SYS/AUX variables as current safety: it adds neither an initial constraint nor a primed copy. Eligible ENV-only invariants instead add initialization and next-state constraints. This explains the timing of Lift's one-floor assumption. Other conventions define different contracts; the claims use this recorded release semantics.

For a candidate successor $s$ and helper $h$, the final preference is

$$Q(s) = h(s) \vee \exists e' \in E : \rho_e^{\text{native}}(s, e')[h = 0, h' = 0].$$

Here $E$ is the typed next-input carrier. Without a helper, use the supplied native ENV relation. After the unchanged progress and goal/existing-node preferences form a candidate set, intersect it with $Q$ if that intersection is nonempty; otherwise retain the original set. Select a response as before. Node reuse and private-rank update rules remain unchanged. This tests immediate native input availability, not fairness; the native relation also requires source qualification.

At first creation of the failing Lift node, the final floor candidates are $011, 100, 101, 110, 111$, for input $(b', c') = (101, 0)$ and successor justice index $j = 0$. The released choice is 011; 100 still permits four legal next inputs. The safe 001 response has already been removed by the nearest-progress-layer filter. Later existing-node reuse leaves only 011. Section H retains graph-local identifiers, complete candidate sets, and all saved predicates. The preference acts on 726 final candidate sets with no empty-intersection fallback; the two extractions need not share node identities. Its six prespecified successful controls have zero refinements and zero fallbacks. They establish preservation of those successes when the preference is inactive, not generality when it changes the candidate set.

The configured transformed strict game itself permits the bad Lift node. Section H checks its complete residual graph against the saved relations; this is not inferred by identifying the released arena with the complete literal theorem. The matched extraction result establishes that the same upstream data also permit an accepted graph. It does not make extraction the only permissive condition or turn immediate availability into a cooperative-construction theorem.

### Supporting preservation evidence

The constructed request-enable protocol isolates propagation in seven finite episodes (Section P). Humanoid 044's eight first-flip policies agree between direct and decomposed regions on all 18,816 typed states. Every policy passes the local entry test; exact entry preservation needs either the obstacle or history clause pair. Only relaxing all eleven clauses preserves all 2,387 strict winners. Its cooperative region is empty, and its initializer gives the same policy distinctions (Section Q). These studies support exact continuation diagnosis without claiming an executed cooperative update benefit for Humanoid.

Vehicle 188 supplies cooperative continuation loss despite preserved initialization. Completed native strict queries now corroborate the fixed state loss; cooperation and reachability retain their separate source-policy proofs. This is not an executed update (Section E).

The 59 certified literal models include 49 with full preservation and ten without it. Nine loss calculations complete and one is resource UNKNOWN. None of the nine removes a state of $K$, and all ten affected contracts already lack cooperative initialization. This stopped, resource-conditioned cohort does not establish the Patrol phenomenon's prevalence. Sections E, J and K retain all attempted inputs, exclusions, and UNKNOWN outcomes.

### Selection, repair, and contract-negative controls

Constructed PortHandshake corroborates the selection mechanism. Its ENV input $ack$ and SYS outputs $request, gate$ satisfy assumptions $request \Rightarrow ack'$ and $\neg gate \Rightarrow \neg ack'$; source input availability is $\neg request \vee gate$. Matched arms hold the game, saved progress information and rank rules fixed. At a shared state and permitted progress layer, one chooses $(request', gate') = (1, 0)$ and blocks ENV; the other chooses $(1, 1)$. Complete graph admission changes from strict-only to strict and cooperative. All availability-selection sites change together, so this is not an ablation of a single site (Section O).

Intruder separates contract repair from selection. The original-source countercertificate excludes joint strict/cooperative winning from one allowed initializer. Repair G adds staying inputs only

Table 25: Separate route runs on two input sets. Both means strict and cooperative. Nonblocking strict means strict synthesis on the original nonblocking arena. Refusal, an empty-region stop, and UNKNOWN are distinct. These are not the matched Lift extractions.

| Route | Both | Strict only | Other outcomes |
|---|---|---|---|
| *11 replacement listings* | | | |
| Released | 7 | 2 | 1 refusal; 1 UNKNOWN |
| Normalized 4FP | 8 | 0 | 3 empty-region stops |
| *14 unchanged public inputs* | | | |
| Released | 11 | 2 | 1 UNKNOWN |
| Normalized 4FP | 12 | 0 | 2 refusals |
| Cooperative Slugs | 12 | 0 | 2 refusals |
| Nonblocking strict | 12 | 1 | 1 refusal |

at old input dead ends and has a positive certificate for all allowed initializers. The two extracted outputs agree on every source-legal response, so the benefit comes from the limited repair. Section N retains the full 276,829-state countercertificate and 407,172-state positive certificate, exact source clauses, progress argument, and replay kit. Unlabelled states are not classified by either certificate.

Source obstructions also explain the comparison roster's genuine contract negatives. Arbiter P has $V = K = \varnothing$: an original-legal input forces every safe response to an environment dead end. Elevator 032 has an allowed initial input with no legal successor under the recorded current-invariant semantics; PCar 058 forces a conflict between forward-motion duty and ENV justice on direction-change monitors. Sections B–H retain the exact clauses and entry quantifiers. Lift P's direct graph audit can fail cooperation while its model-transfer obligation remains UNKNOWN. These source-level and graph-level judgments are not interchangeable.

### Combined route comparison retained from the main text

The following table combines the reconstructed replacement listings and unchanged public-input runs; it is not the matched Lift extraction. For the public nonblocking-strict run, PCar 058 passes $I_M(W_V)$ with witnesses for all six initial inputs but has $K = \varnothing$. Elevator 032 has $W_V = K$ on eleven of twelve states, but one of four initial inputs lacks a response there, so both initializer judgments fail.

## U. Native continuation and contract-sensitivity checks

### U.1 Native Patrol continuation: exact relations, certificates, and limits

**Question and fixed entry.** This additional experiment asks whether the continuation loss in Section S also occurs in the *actual saved arena* produced by the released syntactic conversion. It uses exactly the existing Patrol sources: the unchanged published new source has 3,304 bytes and SHA-256 `cde1b981a103047ddefbe8a963f1188d57904659a30fd0e7b678c771213be402`; the constructed old source has 3,485 bytes and SHA-256 `312843a6d6e6f759627f5e149897d442360ecf5d88f408ea781637e518e996f8`. The imported pattern file is also byte-pinned. No source variant

is substituted. The query fixes the same reached state: checkpoint coordinates $(0,4),(2,4),(4,4)$, all blocked flags false, `isCharged=false`, robot at $(0,0)$, charging counter zero, and logical request monitor $S2$. All fourteen source fields are fixed. The BOOT edge from $S0$ in the original-old native arena establishes reachability; no initializer or monitor is reapplied. Private phase 0 is retained in the existing fixed-controller witnesses, not introduced into the logical winning-region calculation.

**Native exports and full-domain correspondence.** Four fresh model-only runs export old/new original and released arenas. The released route applies `PitermanSyntacticReduction` before the full `DefaultTranslators`, with the pinned released parser, translator, and BDD generator. The runs take 5.7–7.3 s each and make no solver or extraction call; no initializer is manually substituted. Saved files contain all typed fields, their current/next encodings, initialization, safety, and ordered justices. A separate support-table construction composes the independently transcribed source equations into BDDs and compares the complete original predicates: 35 new clauses over 22,899 assignments and 42 old clauses over 38,536 assignments. All initialization, safety, and ordered-justice comparisons pass over their full typed domains. These are factored clause-support counts, not enumerations of the whole state-pair domain.

For the *new* source, exact typed name/value mappings retain every logical auxiliary and establish

$$E_{00} = \rho_e, \quad E_{01} = E_{11} = \top, \quad E_{10} = \bot, \qquad S_{00} = S_{01} = \rho_s.$$

The actual $S_{11}$ is nontrivial: generated counter and request-monitor updates remain. Thus complete-literal relation identity is false. Nevertheless, the independently computed helper safety region is $U = D$, and its safe-entry condition is $F = C$:

$$|D| = |U| = 75{,}000{,}000, \qquad |F| = |C| = 71{,}812{,}500, \qquad r \in C \cap F.$$

Initializers and ordered original justices, with the additional helper justice, also match. A second checker reads the raw native BDD node tables without importing the producer's BDD or mapping code. It confirms the quadrant identities and totality of $S_{11}$: for each of 600 typed assignments to its relevant current fields, all 25 typed next robot positions admit exactly one counter/monitor update, independently of next ENV values. This independently establishes $U = D$; with $E_{01} = \top$ and $S_{01} = \rho_s$, it yields $F = C$.

**Closed continuation domain and independent checks.** The active continuation calculation fixes only the six checkpoint coordinates. It retains all values of the remaining ENV fields, robot position, counter, and request monitor, giving 4,800 source states. An exact relation query proves that every typed next ENV input permitted by $E_{00}$ preserves those coordinates. SYS cannot write them. Consequently this is a closed continuation domain, not a restriction on adversarial choices reachable from $r$. The helper-true domain remains *all* 75,000,000 original states; first-flip inputs may relocate checkpoints. A direct-helper query therefore uses $D \wedge (b \vee \mathit{layout})$, not a layout restriction after the flip.

On the closed active slice, independent symbolic strict solution gives $|W| = 3{,}280$ and $|W_F| = 1{,}104$, with $r \in W \setminus W_F$. A separate finite-set implementation enumerates all 4,800 states, 51,072 legal input choices and 205,056 safe responses in the new source; it agrees on these regions and on the slice's 4,596 states in $C$. These winning-region counts concern the declared slice only. The actual-arena decomposition from Section L therefore excludes $(r, 0)$ from the saved new arena's strict-winning region.

A third check binds the existing finite certificates directly to the saved native predicates. It confirms complete input coverage, safety, strictness and cooperation of the original-old graph (140

states, 277 edges) and original-new continuation graph (849 states, 12,240 edges), including all native justices and the actual BOOT edge. For the countercertificate it checks all 300 typed SYS outputs at each of 25 nonexit rows, obtaining exactly the declared 103 safe replies among 26 states. Every infinite nonexit branch is ENV-fair and violates the pending-monitor justice; the exit permits the actual native helper flip and has zero safe responses. This is a direct strict-loss certificate for the saved arena, independent of a completed native solver run. Malformed-certificate controls and SCC fixtures accompany the checker; they are selected qualifications, not exhaustive fault coverage.

| Route at the same pending entry | Old input | New input |
|---|---|---|
| Original source: strict and cooperative | Yes | Yes |
| Complete literal: strict winning | Yes | No |
| Actual released syntactic arena: strict winning | No | No |
| Complete first-flip correction: strict winning | Yes* | Yes* |

*The complete correction is built from each corresponding *original* relation. The new relation's computed $U = F = D$ agrees with complete-guard preservation; that theorem and both source winning witnesses give these positive results. They are theorem-derived, not observed native frontend verdicts. On the old source slice, $|W| = 3{,}280$ and $|W_C| = 3{,}132$; the pending entry belongs to both.

**Important old-input negative.** The actual old export does *not* satisfy the original-relation premise: its $E_{00}$ and ENV initialization differ from those of the original old source. The three added pure-ENV permanent-open invariants are affected by the order of syntactic guarding and invariant translation. At the pending entry the original old relation allows two next ENV valuations, whereas the released old $E_{00}$ allows sixteen. Its retained conditions test the current blocked flags rather than imposing the original next-flag conditions. The original-old graph therefore cannot certify admission of the released old arena. A separate direct binding shows that the same all-reply countercertificate is already legal in that old released arena, including its final losing helper flip. Thus the old released pending entry is already strictly losing. There is *no old-to-new actual released refusal change* in this original-spelling experiment. Section U.3 separately tests one equivalent old spelling, without replacing this negative result. The source/complete-literal update comparison in Section S retains its stated scope; the additional new-arena result establishes conversion loss at a cooperatively feasible entry. The old correspondence failure alone is not treated as a losing verdict; the separate countercertificate supplies that conclusion.

**Resource outcomes and verdict scope.** All unsuccessful attempts are retained. Six independent full-helper BDD attempts reached the fixed 2,000,000-node allocation cap; they changed cofactor evaluation, exact compaction, or variable order, without raising the cap. The final decomposition-only calculation completed in 22.8 s. The old-source decomposition dispatch stopped on the failed correspondence premise rather than applying its theorem outside its assumptions.

| Attempt(s) | Time (s) | Outcome |
|---|---|---|
| Independent BDD attempts 1–4 | 14,14,22,37 | Node-cap UNKNOWN |
| Helper-first BDD attempts 1–2 | 11,33 | Node-cap UNKNOWN |
| Old correspondence-gated decomposition | 14 | Premise failure; no transfer |
| Native strict three-fixed-point attempt | 136 | Stopped for repeated diagnostic output |
| Native strict retry with lossless RLE | 76 | Decoded-output cap; UNKNOWN |

The native queries use `GR1Game`, not the raw cooperative four-fixed-point core. They regenerate the field universe and check every saved safety, initialization and justice predicate before solving; initializers are not changed and their initial verdict is not the continuation query. The first supervisor omitted its intended stream limit; the owned JVM was stopped after 2.95 GB of repeated diagnostics. The captured bytes were losslessly compressed to 17.2 MB and verified by decompression. One retry restored the pinned lossless RLE wrapper, an explicit 16 MiB stored-stream cap and a 180 s wall cap. It stopped after 76 s at the wrapper's 8 GiB decoded-output cap, storing only 29.6 KB stdout and 2.7 KB stderr. Neither run produced an adoptable strict verdict. No further retry or raw four-fixed-point run was made in this original experiment; Section U.5 records one separately bounded strict follow-up. Accordingly, the claimed strict loss rests on exact saved-arena analysis and independent certificates, not a measured frontend refusal.

**Reproduction and trust boundary.** The local artifact root is `experiment/v64_native_continuation_20261023/`. `RESULT_SUMMARY.json` indexes the unchanged input hashes, successful results, negative findings and UNKNOWN attempts. The immutable summary is linked to the final independent audit by `FINAL_AUDIT_ADDENDUM.json`; `COMPARISON.json` separates direct evidence from theorem-derived comparisons. `native/` contains the four complete exports and commands; `source_bridge/` contains source-equation correspondence; `decomposition/new/` contains the successful calculation and its compact exact BDD functions. `independent_audit/` supplies the raw BDD certificate checks, explicit finite regions, quadrant checks, old obstruction and qualification records. `analysis/`, `direct/`, `native_strict/` and `control/` preserve incomplete attempts, process timings and resource incidents. `FILE_MANIFEST.json` and `SHA256SUMS` bind the final files. No publication or external service run is part of this experiment. Selected saved records and exported predicates from this experiment are now distributed through the current continuation release's Patrol replay (opening index). This later public subset does not distribute the complete original experiment or repeat its native executions or omitted source/runtime provenance checks.

The implementations and qualifications are author-visible research assistance, not blinded replication. They share the pinned native source front end and saved export bytes, while using separately implemented symbolic, finite-set, and raw-node checks. These checks establish the stated finite model results; they do not establish hardware behavior, prevalence, native frontend refusal, or a service advantage over direct source solving or the complete correction.

## U.2 Lift: contract-sensitivity control

**Question and predeclared change.** The Lift T diagnosis depends on the recorded source contract and translation convention. To test that dependence, we declared a second contract before its execution: replace exactly the four `assumption` owner tokens of `one_floor` and `one_floor_only1/2/3` by `guarantee`. Every formula and every other source byte remains unchanged, including button updates, current cancellation and movement guards, initializer assignments, and the existing reconstructed response pattern. This is a post-diagnosis sensitivity control, not held-out validation.

Let $B$ denote the six button-update implications in Section H, and let $\mathrm{One}(f)$ denote exactly one true floor. With superscripts $o$ and $n$ for the frozen and changed contracts, the exact translated original-contract difference is

$$\begin{aligned} \theta_e^n &= \theta_e^o, & \theta_s^n &= \theta_s^o \wedge \mathrm{One}(f), \\ \rho_e^o &= \mathrm{One}(f) \wedge B, & \rho_e^n &= B, \\ \rho_s^n &= \rho_s^o \wedge \mathrm{One}(f'). \end{aligned}$$

The added initial conjunct is redundant because the unchanged initializer sets $f = 100$. Both legal joint initial states, with $b = 000, m = 000$ and either cancellation value, remain equal. The source ENV justice list stays empty; the three SYS justices remain $\mathsf{GF}\neg m_i$. Thus the owner change also changes temporal placement: current ENV safety becomes initial and next-state SYS safety. This comparison does not isolate ownership from timing and does not compare two extractions of the same game across contracts.

**Matched execution within each contract.** Each contract receives one native solve, followed by released extraction and the same final-candidate continuation preference used in Section H. The two extractions share the actual model and fixed-point memory objects; source, translation, solver and rank rules are fixed within that pair. The pinned harness adds only a read-only memory export to the existing paired harness. It captures 36 predicates for the frozen contract and 46 for the changed contract, checks every captured BDD before and after preference extraction, and checks unchanged variable order. The snapshot boundary is *after released extraction and around preference extraction*; it is not a pre-released-extraction snapshot or an independent proof of the solver's fixed points. Object identities, call arguments, class hashes and process receipts bind the common execution. Variant model/BDD files are copied from that common run, so byte equality alone is not the shared-execution evidence. The frozen released arm also reproduces all 23 historical semantic checks, including the complete graph and printed controller.

Table 26: Lift contract sensitivity: two contracts by two extractions. Each cell gives native states/edges; original-view states/edges, followed by strict/cooperative verdicts. Every original view has two initial states. Each controller is judged under its own contract.

| Contract | Released extractor | Continuation preference |
|---|---|---|
| Frozen Lift T | 794/13 874; 154/1170<br>pass / fail | 778/13 624; 154/1176<br>pass / pass |
| SYS one-floor invariant | 778/13 624; 154/1176<br>pass / pass | 778/13 624; 154/1176<br>pass / pass |

**Availability and inactive preference.** Exhaustive enumeration finds 640 of the 1,024 typed original states without an ENV successor in the frozen contract and none in the changed contract. These states include seven public Boolean fields and three logical response monitors. The frozen count is $5 \cdot 16 \cdot 8$: five invalid floor valuations, sixteen input valuations and eight monitor valuations. In $B$, next $b_i = 0$ is required only by $b_i \wedge f_i$, whereas next $b_i = 1$ is required only by $b_i \wedge \neg f_i \wedge \neg c$. These guards cannot conflict, so $B$ has an ENV successor at every typed state.

The saved helper-aware preference $Q$ is true on 1408/2048 native typed states for the frozen contract and 2048/2048 for the changed contract. Its helper-false part equals original ENV successor availability on these inputs; helper-true states remain available. The frozen preference refines 726 of 13,624 candidate checks. The changed preference refines zero of 13,624; neither run uses an empty-preference fallback. Within the changed contract, complete graphs agree after excluding only provenance labels and runtime object identities, and printer bytes are identical. This graph comparison is not a $G \to G \wedge A$ normalization route. Equal counts across contracts establish no game identity.

**Complete admission and separate verification.** Both existing auditors check all four returned controllers. The production auditor rebuilds BDDs from the profile AST; the other traverses saved

decision DAGs with a separate reader and evaluates separately encoded source equations. Its source correspondence covers all 1,048,576 original current/next state pairs per arm, initial/domain identities and ordered justices. Both check complete native graphs, exact legal-input coverage, unique responses and safety. Original views are reachable helper-false subgraphs retaining native IDs, ranks and logical monitors, with 2,464 input obligations per view and no edge or memory repair. Their complete view and justice-label sets agree, as do the verdicts in Table 26.

A further saved-evidence audit uses a fresh source transcription, integer-word BDD traversal and repeated deadend elimination to recheck both full source relations and all four graphs. It confirms that frozen released nodes 680 and 708 are exactly the original-view deadends, non-one-hot nodes and cooperation failures; the other views have none. These are distinct checking implementations, not independently authored experiments or independent human review. The new profile and source oracle were authored together; compiler exports, typing and the release's translation convention remain shared premises.

**Reproduction, outcomes and limits.** The archive `v64_lift_contract_contrast_20261023/` contains the preregistration and its hash, both source listings and exact diff, profiles, complete graphs, saved BDDs, memory snapshots, both auditor implementations and certificates, exact execution arguments and process receipts. `root_independent_audit/FACT_CHECKS.json` records the additional recomputation and trust bounds. All 237 preregistered input pins match; compilation precedes the recorded preregistration, and all ten child processes follow it and complete within their caps. Local chronology is not a third-party timestamp.

Null controllers, caps, correspondence failures and checker disagreements were predeclared reportable outcomes; none occurred here. The retained first descriptive-summary attempt failed on a field-name lookup; correcting it repeated neither synthesis nor graph auditing. The two verbose native stdout files remain locally available with sizes and hashes in the receipts; the additional audit checked their presence and size without rereading their 879,283,576 bytes. A compact anonymous packaging plan retains those anchors and the complete semantic evidence; it is not a published artifact. The result bounds the diagnosis to this contract: it demonstrates no preference advantage after the declared change, no general synthesis repair, and no fairness guarantee from one-step availability alone.

### U.3 Equivalent old-source spelling: an isolated pipeline control

Section U.1 retains the primary original-spelling result: both the old and new actual released arenas lose the pending continuation. The following one-variant, preregistered control does not replace it. It asks whether the old mismatch arises from source spelling and translation order while the compiled original contract stays identical.

**Intervention and mandatory equality gate.** For each $X \in \{A, B, C\}$, replace only the terminal declaration `asm alw !Xblocked;` by `asm ini !Xblocked;` and `asm alw !next(Xblocked);`. All other bytes, including mixed line endings, layout clause, and imported patterns, are retained. The control source SHA-256 is `bd07344f0de8028f9d6470451c68a2842e230ced89cc08e4641b83f233e0c168`. Preregistration precedes source creation and native execution; it allows one variant and requires equality of the compiled original games before any released control export.

That gate passes: exact symbolic comparison, under complete typed name/owner/role/value mapping, establishes equality of both initial predicates, both safety relations, and every ordered ENV/SYS justice predicate. All fourteen source fields and logical auxiliaries remain. The current carrier has 75,000,000 states and the current/next carrier has 5,625,000,000,000,000 pairs; equality is

symbolic, not enumeration of that many pairs. No initializer restriction or AUX projection is used. Only then is the control passed through the same released conversion.

**Actual continuation certificate.** Its exported relations satisfy $E_{00} = \rho_e$, $S_{00} = S_{01} = \rho_s$, $E_{01} = E_{11} = \top$, and $E_{10} = \bot$. Original initialization is preserved with helper false. Original justice predicates retain order and identity for both helper values, with the additional helper justice. The nontrivial $S_{11}$ has no next-ENV dependence. Enumeration over all 600 assignments to its complete current support gives all 25 typed next robot positions, each with a unique AUX update. Hence $U = D$ globally and $F = C$.

The unchanged old graph has exact input coverage at all 140 nodes and 277 edges against this actual $E_{00}$, satisfies actual safety, and passes strict and cooperative justice checks. Every node belongs to $F$ under the full typed flip quantifiers. The actual BOOT edge reaches the same fourteen-field pending state, request $S2$, and private phase 0; the pending state fails the SYS initializer. Following this graph at helper false, answering a first flip into $U = D$, and then following a safety-winning strategy proves the actual old control continuation. No solver or extractor is required for this constructive certificate. Together with the unchanged new loss, this yields a separate control-old-continuation/new-loss contrast.

**Isolated difference and mechanism.** Full-domain comparison of the two released old arenas finds differences only in ENV initialization at helper false and in $E_{00}$; all other ENV quadrants, all SYS quadrants, SYS initialization, and all ordered justice predicates coincide. At the pending entry, legal next inputs change from sixteen to two. One input allowed only by the original spelling closes checkpoint $C$ while retaining the layout and low power. The parsed/reduced/translated inventories explain why: the unreduced pure-ENV state invariant generates an initial clause and a primed safety clause. When helper guarding runs first, it instead becomes current $\neg X blocked \vee b'$ safety and generates no corresponding blocked initializer. Explicit initial/next spelling avoids this sensitivity. This isolates the recorded pipeline; it does not claim all rewrites or all Spectra releases behave this way.

**Resources and evidence.** Only two model-only exports were executed: 6.13 s and 7.55 s, with maximum sampled RSS 467,238,912 and 450,953,216 bytes. Each had a 768 MiB heap, 64 MiB direct-memory cap, 60 s wall cap, 1 GiB process-group RSS cap, and 64 MiB stream cap. Captured streams are losslessly compressed and hash-verified. There are no native solver or controller-extraction calls and no observed frontend verdict. The directory `experiment/v64_old_spelling_control_2026 1023/` contains preregistration, exact source delta, original equivalence, released-control certificate, isolated pipeline comparison, translation inventories, and a manifest of 73 frozen files. The producer reuses the earlier independently written raw-native parser, typed enumerator, and SCC checker. Thus this is not a third independent implementation of those components. The shared native compiler and encoding metadata remain trusted, and private policy-update correctness is inherited from the earlier complete source-graph audit.

An independent claim assembly rechecks the exact three source changes, all original and released relations, global $U/F$, graph and pending root, and the isolated pipeline difference. Fresh typed cofactor equivalence and forall–exists procedures agree; its raw parser and SCC infrastructure remain shared. The positive result is strict winning in the actual released arena. Cooperative fairness is claimed only for the helper-false source graph, not for helper-true continuations. The audit also confirms that all 62 prior old/new native tables and all frozen negative results remain unchanged.

### U.4 Executing complete-graph admission at a paused software boundary

**Question and scope.** This bounded demonstrator connects the admission judgment to an actual software pause, decision, and continuation. It reuses the frozen Patrol controller, Lift graphs, source predicates, and complete graph checkers; it adds orchestration and timing instrumentation. It does not generate a controller or recompute a winning region. A gate runs after an executed old prefix, binds the reached source values, logical monitors, and private controller root, and admits a candidate only when complete source and saved-native checks establish safety, strict liveness, and legal-prefix cooperation. An incomplete or unknown check cannot establish admission. A withheld candidate remains paused. The two checker routes examine complete rooted graphs, not sampled executions. The subsequent workload shows that the decision is enforced by this software adapter; it is not the proof of the graph properties.

**Frozen inputs, order, and amendments.** All records are under the local directory `experiment/v64_admission_runtime_20261023/`. The predeclaration fixes 197 input/code pins, three fresh Python processes, case/candidate order, expected outcomes, and resource caps. Each process attempts Patrol first and then Lift boundaries 524, 526, and 756, in each case testing retained native, saved normalized, and saved preference. There is no warmup, cache flush, or additional exploratory run. "Fresh" means new process/module state; it does not mean a cold operating-system filesystem cache.

Two amendments are preserved. Before any worker execution, comparison with the frozen Patrol result corrected a transcription of the expected trace to response step 30, monitor-clear step 31, 67 new commits, and cycle interval $[32, 67)$. `PREDECLARATION_INITIAL_UNEXECUTED.json` and `PREEXECUTION_CORRECTION.json` retain that correction; `PREDECLARATION_ATTEMPT01.json` is the declaration used by the first execution. That execution completed Patrol, then raised `KeyError('graph')` before any Lift episode or gate: the inherited runtime loader normalizes candidate descriptors. The wrapper was corrected to read graph/root/rank from the same frozen raw plan, after checking that its boundary order agrees with the loader. `EXECUTION_AMENDMENT_01.json`, `attempt01_code/`, and the first run preserve the failure and executed code. `PREDECLARATION.json` binds the corrected wrapper used by the two remaining, already declared trials. No fourth trial replaced the failure. Thus Patrol has three completed measurements; each Lift candidate at each root has two, with one preceding setup-only failure disclosed.

**Patrol gate and executed continuation.** The instrumented runtime executes the same BOOT input and pauses after one old commit at the fourteen-field pending state of Sections S and U.1, with request monitor $S2$ and private phase 0. The gate rechecks the source/file bindings and saved translated-AST correspondence: 77 clauses and 61,435 typed support assignments. The existing source checker then validates the complete old graph (140 nodes, 277 edges), new graph (849 nodes, 12,240 edges), and exact old-to-new source/private root. For every reachable node it checks all legal ENV inputs exactly once, source SYS/AUX safety, and the frozen controller's private-phase update.

A separate saved-native-table parser/evaluator loads the original old and new native predicates and checks the same complete graphs against them. It enumerates all typed next-ENV fields with exact decision-DAG pruning; the legal-input check is not a sample of the 250,000 typed next-ENV valuations. It also checks the actual old BOOT edge and that the pending state fails the new SYS initializer. The new query never reapplies that initializer. Both routes perform complete SCC fairness checks. Strictness excludes an ENV-fair infinite suffix avoiding a SYS justice; cooperation requires an ENV-fair infinite continuation from every reachable legal prefix. The checks consider cyclic SCCs, including nonbottom SCCs.

Both routes pass, so the gate records `ADMIT` while the runtime is still paused. The environment is then updated and the same plant and controller objects resume, with source values, monitor, and phase preserved at the boundary. The unchanged workload makes 67 post-update commits, reaches the old pending response at step 30, clears its monitor at step 31, and reaches a repeatable fair cycle with start 32 and end 67. Its three newly enabled closure inputs and the one original-safe state outside $C$ occur at exactly the previously frozen steps. The complete source/private trail equals the earlier executed trail. The controller remains the existing handwritten, post-diagnosis policy; this experiment does not turn it into an independently discovered or previously deployed legacy controller.

After successful continuation, separate diagnostic stages reverify the 4,800-state source $C$ certificate (4,596 in $C$, 204 outside) and the 26-state losing certificate (25 nonexit rows, 103 safe replies, one exit). The native check covers every safe reply and confirms the actual released first-flip zero-response witness. These stages do not determine original-controller admission and are excluded from paused-gate latency. No frontend refusal, new native solver result, or full-region recomputation is observed in this demonstrator.

**Lift gate and private-root binding.** The three entries use the unchanged two-commit old prefixes. All have an unanswered third-floor obligation: $m_3 = S1$, $f_3 = \mathsf{false}$, and cancel false. Reached native nodes 524, 526, and 756 each carry rank 1. Setup verifies the frozen source/native model bindings and existing old-gate evidence. The actual old prefix is checked while it executes. At the paused boundary, both current-candidate routes *recompute* the entire original-legal reachable closure, every legal response obligation, safety, and SCC fairness. The source route examines all sixteen typed ENV inputs per reachable node; the saved native route uses the bound native predicates. Graph/source/model byte pins and exact roots are rechecked at each gate. Prior verdict files alone do not authorize the new continuation.

| Reached old root | Retained native root/rank | Normalized root/rank | Preference root/rank |
|---|---|---|---|
| 524 | 524/1 | 72/1 | 524/1 |
| 526 | 526/1 | 74/1 | 526/1 |
| 756 | 756/1 | 88/0 | 590/0 |

Each candidate is tested in its own episode from the same frozen prefix; the wrapper does not search for an alternative within the paused episode after withholding. A retained candidate must be the very same reached controller instance and private memory. A replacement is a new graph controller at its exact certified stored root/rank, matching every live source and monitor value. The rank changes in the last table row are therefore explicit replacement initialization, not arbitrary transport of old private memory.

At every entry, retained native is complete and strict but noncooperative, so it is withheld with zero post-change commits and the same paused runtime snapshot. Normalized and preference are complete, strict, and cooperative; each is admitted and executes the exact existing twelve-input workload. The pending response occurs at step 1 and its monitor clears at step 2. All six admitted episodes reach the previously recorded fair self-loop without cancel. Their offered-input sequences, source trajectories, private node/rank traces, and observations equal the corresponding frozen executions. The plant and mutable source-state object retain their identities across each replacement.

**Observed gate costs.** The host is an eight-CPU Apple M2 with 24 GiB RAM, macOS 15.7.9 arm64, and CPython 3.10.9. Wall durations use `perf_counter`; RSS uses process `ru_maxrss`, with separate process-group sampling by the supervisor. Table 27 reports median and observed

range, not statistical uncertainty or a latency guarantee. Lift's shared source/native model and runtime loading/binding costs 0.357–0.481 s (median 0.364 s across all three setups, including the first setup-only failure) in addition to the per-candidate gate.

Table 27: Complete rooted graph and paused-gate cost. Each Lift row refers to one candidate at one actual entry. Times are seconds.

| Candidate/entry | Nodes/edges | Decision | Median [range] |
|---|---|---|---|
| Patrol retained | 849/12,240 | Admit | 4.584 [4.448–5.207] |
| Lift native / 524 | 154/1,170 | Withhold | .051 [.043–.059] |
| Lift normalized / 524 | 154/1,176 | Admit | .048 [.040–.056] |
| Lift preference / 524 | 154/1,176 | Admit | .049 [.041–.057] |
| Lift native / 526 | 154/1,170 | Withhold | .050 [.042–.057] |
| Lift normalized / 526 | 154/1,176 | Admit | .048 [.040–.056] |
| Lift preference / 526 | 154/1,176 | Admit | .049 [.041–.058] |
| Lift native / 756 | 154/1,170 | Withhold | .049 [.042–.057] |
| Lift normalized / 756 | 154/1,176 | Admit | .048 [.039–.056] |
| Lift preference / 756 | 154/1,176 | Admit | .049 [.041–.058] |

Table 28: Patrol stage costs: seconds, median [observed range] over three executions. Coverage/safety excludes the separately measured SCC block.

| Stage | Median [range] |
|---|---|
| Source file / saved native AST binding | 1.418 [1.409–1.509] |
| Source old graph coverage and safety | .015 [.014–.017] |
| Source new graph coverage and safety | .305 [.301–.355] |
| Source old graph SCC fairness | .001 [.001–.001] |
| Source new graph SCC fairness | .016 [.016–.020] |
| Exact source boundary binding | .021 [.020–.025] |
| Native table parsing and source binding | 1.829 [1.687–1.998] |
| Native old graph coverage and safety | .052 [.052–.062] |
| Native new graph coverage and safety | .920 [.919–1.198] |
| Native old graph SCC fairness | .001 [.001–.001] |
| Native new graph SCC fairness | .016 [.015–.021] |
| Separate source $C$/losing-certificate check, after continuation | .442 [.335–.445] |
| Separate native losing-certificate check, including released-model parsing | 1.081 [1.077–1.380] |

| Lift per-candidate stage | Seconds, median [range] |
|---|---|
| Exact root and byte binding | .001 [.000–.002] |
| Source coverage and safety, excluding SCC | .014 [.011–.017] |
| Source SCC fairness | .004 [.003–.004] |
| Saved-native coverage and safety, excluding SCC | .028 [.023–.033] |
| Saved-native SCC fairness | .002 [.002–.003] |

The Lift stage rows pool eighteen completed gates descriptively; they are not eighteen independent replications of a single case. Rounding a small binding duration to .000 does not mean that no check ran. Function wrappers time the original source and Lift fairness routines. For Patrol's native checker, a preserved mechanical diff adds timing around its existing complete fairness block. Coverage/safety duration is the enclosing graph-call duration minus this nested block. No predicate, edge, or fairness check is bypassed. Overall gate duration additionally includes root checks, recording, and orchestration. Medians of stages need not sum to the median of their total.

**Event audit and resources.** A separate read-only audit imports neither runtime nor semantic checker. It checks pause–gate–decision–resume order, that no continuation commit precedes admission, and that withheld Lift candidates never install or resume. It checks exact source/native closure equality for Lift, source/monitor preservation, object identities as recorded above, certified replacement roots/ranks, and complete equality with the frozen admitted workload traces. It also checks that the derived Patrol `Plant` and `Controller` class ASTs equal their originals. All 21 episode checks pass: three Patrol episodes and eighteen Lift candidate episodes across the two complete trials. A further separately written standard-library audit checks all event records, 246 file anchors, full source/private trace equality, static semantic preservation, and every timing aggregate. All 25 checks pass with the metadata caveat below. Removing the documented flow/path/report changes reproduces the original Patrol runtime byte-for-byte; removing two timer lines reproduces the native checker byte-for-byte. Neither record audit reimplements all semantic algorithms. Its report is in `independent_audit/`.

Each worker has monitored limits of 120 s wall time, 1 GiB process-group RSS, and 16 MiB for captured streams/output; the supervisor samples every 0.1 s and terminates its owned process group on a cap. No cap is reached, no resource UNKNOWN occurs, and cleanup records no owned process remaining. The three supervised process durations are 6.695, 7.042, and 8.416 s; the first ends with the disclosed orchestration failure. Whole-process peak RSS values are 334.4, 337.4, and 329.6 MiB. These are cumulative suite high-water marks, not incremental memory attributed separately to Patrol, Lift, or an individual stage.

A metadata defect is retained explicitly: in each original `PROCESS.json`, the field named `started_utc` was sampled while writing the completed receipt, so it records approximately the receipt-completion instant, not the start instant. The original files are preserved. This field must not be used as evidence of start time or predeclaration order. The reported elapsed durations are independently computed from monotonic clocks around execution and are unaffected. The predeclaration/amendment records, executed-code snapshots, event sequence numbers, and event monotonic timestamps provide the relevant binding and within-run order evidence; no replacement absolute start timestamp is invented.

**Trust, interpretation, and evidence entry points.** The gate trusts the saved native compiler/exporter and encoding metadata, the source-to-native bindings with their stated scope, and the reused finite-graph algorithms. Complete graph checks establish a judgment about the candidate rooted at the matched state; the live snapshot and installation checks establish that binding for this adapter. They do not establish the corresponding bridge for a physical robot or a native executable controller runtime. The three Lift roots have two distinct source valuations and are not a prevalence sample. Patrol's controller was handwritten after diagnosis; no advantage over a separately deployed legacy controller is established. Filesystem caches and host activity are uncontrolled. No real-time bound, scalability result, comparative performance claim, arbitrary memory-transport guarantee, or cost for synthesis/full-region diagnosis follows from these measurements.

`REPORT.md` gives the descriptive tables; `RESULT_SUMMARY.json` and `AUDIT_COMPLETION.txt` contain the 21-check audit and all individual timings. The two mechanical code diffs are `code/PATROL_RUNTIME.diff` and `code/NATIVE_CHECKER.diff`. Each directory under `runs/` retains `PROCESS.json`, stdout/stderr, and its output/event records, including the failed first Lift attempt. `MANIFEST.json` anchors the preserved local experiment. This complete U.4 runtime packet has not been placed in the published evidence archives; they remain the additional local snapshot indexed at the start of this supplement.

### U.5 One bounded native strict query after diagnostic-transport qualification

**Motivation and scope.** The two native strict attempts in Section U.1 ended because of diagnostic output: the first was manually stopped at 136.103 s after capturing 2.95 GB; the second stopped at 76.739 s when its lossless wrapper reached the 8 GiB decoded-output cap. Source inspection and disassembly of the pinned JTLV class confirm that `BddCache_lookupI` prints its cache class unconditionally, without a configuration switch. This motivated one separately authorized, bounded follow-up on the unchanged new actual arena. Neither previous failure demonstrated computational intractability.

**Exactly two logging hookups.** The copied harness replaces `LosslessStdoutRle` with `ExactDiagnosticPrintStream`, including its receipt filename, and replaces the enclosing UTF-8 `PrintStream` installation with `System.setOut(capture)`. Every scientific harness statement and the pinned JTLV/solver classes remain unchanged. The standalone wrapper counts only exact matching diagnostic lines, preserves their positions through ordered dictionary/run records, and retains every other stdout byte. It neither expands nor hashes repeated raw bytes; the stored stream has its own SHA-256. Stderr remains separate.

Before the native attempt, logging-only fixtures checked byte-for-byte roundtrip of mixed lines, Unicode, partial writes, blank lines and final runs; ten million repetitions; stderr separation; and encoded-cap failure. The first cap fixture exposed an inconsistent completion flag. Its records were retained, the receipt flag corrected, and all fixtures passed before any scientific execution. These qualifications test log transport, not solver correctness.

**Same saved query.** Before solving, all thirteen saved initialization, safety and ordered justice predicates matched. A separate raw-node audit also confirms function equality with the previous attempt for the query domain, pending singleton and both solver relations. All fourteen source fields and logical auxiliaries remain, with request $S2$ and helper false at entry; no initializer or monitor is reapplied. The domain remains $D \wedge (\mathit{layout} \vee b)$: 4,800 helper-false states plus all 75,000,000 helper-true states. The strict three-fixed-point `GR1Game` call must compute its complete winning region before the pending-state membership query. This is not a raw four-fixed-point or native frontend query.

**Caps and observed result.** The sole attempt used a 120 s total budget, TERM at 118 s and KILL at 120 s if needed; 768 MiB Java heap, 64 MiB direct memory and 1 GiB sampled process-group RSS; 16 MiB per stored stream, 64 MiB per nonstream artifact and 128 MiB aggregate output. BDD table/cache starting sizes stayed 10,000/1,000. TERM occurred at 118.031 s and exit at 118.115 s, without KILL. Peak sampled RSS was 777,437,184 bytes; stdout/stderr occupied 117,195/2,898 bytes. The complete encoded log represents 1,950,438,989 diagnostic lines and 105,323,709,165 reconstructable bytes. Counts, record order, receipt and encoded hash agree; the raw stream was neither materialized nor assigned a raw hash.

The result is `UNKNOWN_WALL_CAP`: no winning region or adoptable strict verdict was produced. The last in-process `RUNNING` checkpoint is preserved, while the supervisor records termination. Removing the earlier output-cap abort did not yield completion within this budget; it establishes neither hardness nor frontend refusal. Section U.1's separate saved-arena proof and certificates remain the strict-loss evidence. No further attempt was made.

The closed packet is `experiment/v64_native_strict_diagnostic_20261023/`, indexed by `OUTCOME.json`, `REPORT.txt`, `CLOSURE.json`, `FILE_MANIFEST.json` and `SHA256SUMS`.

### U.6 Native strict solve of the theorem-reduced game

**A separately fixed query.** Following Section U.5, one additional experiment queried the original-new Patrol game $M_F$ from the helper-decomposition theorem. This is a native strict solve of a theorem-reduced game; the full-helper attempt remains `UNKNOWN`. Before execution, `PLAN.json` fixed the query, inputs, resource limits and whole-region comparison. A raw-node preparation derived $F = C$ globally from $\rho_S$ by universal quantification over every typed next ENV valuation and existential quantification over every typed next SYS valuation, including logical auxiliaries. This uses the previously checked actual-new premises $U = D$ and $F = C$; it does not infer $F$ from a known winning region. The prior comparison file was hashed for pinning, but its membership was neither parsed nor used to construct the query. The full carrier has 75,000,000 states and $F$ has 71,812,500.

**Exact predicates and reached state.** Let $Q$ fix only the six ENV coordinates to the checkpoint layout. Its 4,800 states retain the other eight fields, and $|F \cap Q| = 4,596$. Every original-legal ENV input preserves this layout. The native ENV relation is unchanged $\rho_E$; the native SYS relation is exactly

$$\rho_S \wedge F(s') \wedge Q(s) \wedge Q(s').$$

The $Q$ conjuncts represent the declared closed domain. There is no current-state $F(s)$ guard. Before solver construction, regeneration matched all fourteen field/value encodings and ownership records, both original initialization and safety predicates, and the ordered three ENV and five SYS justices. Native typed quantification also matched the independently prepared $F$, and every executed query predicate matched its saved expected BDD. No helper, private phase, monitor reset or initializer replacement is introduced. The full pending $S2$ valuation lies in $F$ but fails the source SYS initializer. The recorded initializer return is false and is ignored for reached-state membership.

**Completed result and full-set check.** The sole pinned `GR1Game` call completed with complete strict memory and normal exit in 78.625 s; peak sampled process-group RSS was 801,538,048 bytes. The saved raw region $W_{\text{raw}}$ gives $|Q \cap W_{\text{raw}}| = 1,152$ and $|F \cap Q \cap W_{\text{raw}}| = 1,104$. Thus the required $F$ intersection removes 48 states; raw $W_{\text{raw}}$ must not be assumed already contained in $F$. Its unfiltered Boolean count includes valuations outside $Q$ and outside the typed carrier, and is not reported as a global state count. After execution, comparison by complete typed valuations matched $F \cap Q \cap W_{\text{raw}}$ against the earlier independent explicit $W_F$ for *all 4,800 states*, not merely their cardinalities. The exact pending root is outside both saved $W_{\text{raw}}$ and $F \cap Q \cap W_{\text{raw}}$.

**Bounds, audit and interpretation.** One compilation and one native attempt were made, with no retry. Limits were 120 s total wall time with TERM at 118 s; 768 MiB heap, 64 MiB direct memory and 1 GiB sampled RSS; 16 MiB per stored stream, 64 MiB per nonstream artifact and 128 MiB aggregate output. No limit triggered. The unchanged qualified diagnostic wrapper retained 74,343

encoded stdout bytes; record totals and the encoded hash passed, without claiming a raw-stream hash.

A separate auditor derived the typed domains and $C$ with fresh Boolean quantification, checked all saved query predicates, and independently matched the complete state sets. It shares the raw-table parser and native encoding evidence; compiler/exporter, pinned solver correctness and recorded process provenance remain trust boundaries. This strengthens the theorem-reduced strict-loss evidence, while providing neither a completed full-helper solve nor a released-frontend or raw-four-fixed-point verdict. The closed packet is `experiment/v64_native_reduced_query_20261024/`, indexed by `OUTCOME.json`, `REPORT.txt`, `CLOSURE.json`, `FILE_MANIFEST.json` and `independent_audit/AUDIT_RESULT.json`. Selected saved query data and the all-4,800-state membership binding are available in the continuation release's `reduced` stage. That stage checks saved Boolean query identities and winning-output memberships; it does not recompute the native winning fixed point.

**A public initialized-negative witness from saved membership.** This inference uses the unchanged public new source's full initializer; it does not reuse the pending $S2$ entry as an initial state. ENV requires three distinct checkpoints, each different from $O = (0, 0)$ and $H = (4, 0)$, and leaves its blocked and charged flags initially unrestricted. SYS imposes no initial robot position; the logical auxiliaries require $c = 0, m = S0$. Thus the admissible input

$$e^* : (A, B, C) = ((0, 4), (2, 4), (4, 4)), \quad d = 000, \quad z = 0$$

has exactly 25 SYS initial choices, one per grid position, including $H$. All lie in the complete saved 4,800-state layout $Q$. Every one is outside the explicit $W_F$ and the independently matched native $F \cap Q \cap W_{\text{raw}}$. Although all 25 belong to original $W$, this does not settle original initialization for other admissible ENV values.

Original-legal $E_{00} = \rho_E$ moves preserve the six checkpoint coordinates, so no legal helper-false continuation escapes $Q$. The global checks $U = D$ and $F = C$ retain every typed first-flip input, including next coordinates outside $Q$. The retained-duty preservation proposition therefore gives the full released membership $W_R(s, 0) = W_F(s)$ on these states; the released initializer preserves the original predicates and adds a false helper. One admissible ENV input with no winning SYS initial choice proves $I_M(W_F) = \mathit{false}$, without computing the global region. The bare false initializer return of the restricted native query is not used for this inference. Original public $I_M(W)$ and $I_M(K)$ remain unestablished. Full-helper native queries remain UNKNOWN, and no frontend refusal or false refusal of an initialized-feasible public contract is inferred.

The existing continuation archive [1] contains the sufficient saved inputs; the following paths are relative to its root. Under `replay/data/evidence/native/`, the roster is `independent_audit/new_explicit_regions.json`, and the original/released `model.json`, `original_thetaE.bdd` and `original_thetaS.bdd` are in `native/new-original/` and `native/new-released/`. The native counterpart `native_strict_W_FQ.bdd` and its model are in `replay/data/evidence/native_reduced_query/single_attempt/native_result/`. Under `records/patrol/`, `RETAINED_CONSTRAINT_PREMISES.json` records global $U, F$ and relation correspondence, while `EXPLICIT_4800_REGION_RESULT.json` and `REDUCED_NATIVE_INDEPENDENT_AUDIT.json` bind the closed domain and all memberships. Evaluate the saved initial predicates using each model's field encoding at $e^*$ over all 300 typed SYS/AUX choices, then match the 25 accepted complete valuations to the roster. Their indices are $12k$ for $0 \le k \le 24$, and none belongs to `W_F_ids`. The existing `patrol reduced` replay stages check saved premises and region bindings, not native fixed points. Raw Patrol source/import files are omitted from this archive; the source hash, inventory and saved semantics retain the stated compiler/exporter and historical-provenance trust boundary. This is not a fresh source compilation or global solve.

## U.7. Patrol decisions at the retained entry

Table 29: Patrol comparisons at the same reached state $r$, with an unanswered charging request. The graph check and reuse retain the same private controller memory. Region results and certificate-backed arena losses are distinct from native solver observations.

| Check or alternative | Result and evidence | Decision supported |
|---|---|---|
| Local flip test $C(r)$ | True before and after (source/certificate check). | A flip can be answered now; future preservation is unsettled. |
| Exact literal region $W_C$ | Old pass, new fail (complete countercertificate and Theorem L.2). | Diagnose conversion loss; do not infer source-contract infeasibility. |
| Actual released arenas | Original old spelling: fail/fail; equivalent old spelling: pass/new fail (native relations and certificates). | Isolate pipeline sensitivity; old contract unchanged. No frontend verdict is inferred. |
| Validated source graph and reuse | Strict and cooperative under both models (complete checks); old request served at 30, cleared at 31 (execution). | Retain the contract and reuse this controller at its retained root. |
| Available complete correction | Preserves this strict continuation (theorem consequence; not a native solver result). | A preserving conversion is available; any returned graph still needs cooperative admission. |

### U.8 One longer, unrestricted full-helper query

**Declared domain and preserved model.** A further, separately preregistered attempt queried the exact actual-new Patrol arena on all 150,000,000 typed states: fourteen source fields, including logical auxiliaries, plus the ENV helper. This differs from Section U.5, whose SYS relation included the closed-domain guard $D \wedge (\mathit{layout} \vee b)$ and therefore retained 75,004,800 states. The new query adds *no* current/next domain, layout, $Q$ or $F$ guard to either native transition relation. The ordered actual fairness roster is four ENV justices—the three original conditions plus the helper condition—and five SYS justices. Both actual initialization predicates remain unchanged.

Before solver construction, all fifteen field/value encodings and ownership records, both safety and initialization predicates, and all nine ordered justices matched the frozen native export. The reached singleton contains the complete pending $S2$ valuation with helper false; it is not reset or reconstructed from initialization. The plan required separate reporting of this singleton's membership and the actual initialized verdict from the same completed query. Typed, helper-false and checkpoint-layout masks were reserved for post-solve inspection. If a region had completed, the plan also required checking every one of the 4,800 checkpoint-state memberships against the earlier independent $W_F$ set. Neither that set nor the reduced query was used to construct this arena.

**Execution and retained outcome.** One compilation and one unchanged pinned `GR1Game` strict call used the already qualified `ExactDiagnosticPrintStream`. No solver, compiler, JTLV class, cache setting or logging semantics changed. The budget was 600 s hard wall time, with TERM at 598 s; other limits remained 768 MiB heap, 64 MiB direct memory, 1 GiB sampled process-group RSS, 16 MiB per stored stream, 64 MiB per nonstream artifact and 128 MiB aggregate output.

Scheduled TERM occurred at 598.038 s and process exit at 598.112 s, without KILL. Peak sampled RSS was 795,492,352 bytes. The result is `UNKNOWN_WALL_CAP`: no complete winning region, pending-state verdict or initialized verdict was produced. Consequently the planned 4,800-state comparison is unavailable for this attempt. The retained `RUNNING` native receipt is the last checkpoint, not an active process. The complete encoded stdout occupies 518,359 bytes and represents 9,377,319,076 repeated diagnostic lines and 506,375,233,862 decoded bytes; counts, ordering and encoded hash passed independently. No digest of an expanded raw stream is claimed.

**Interpretation.** The separate audit passed 72 binding, integrity and resource checks while preserving the UNKNOWN outcome. This is bounded noncompletion, not a proof of intractability, a solver limitation or an observed frontend refusal. This native query leaves its initialized verdict `UNKNOWN`; Section U.6 independently derives $I_M(W_F) = \mathit{false}$ from saved membership. That analytic result is not a completed full-helper solve. All earlier attempts remain intact; no retry followed. Compiler/exporter, field encodings, native solver correctness and recorded process provenance remain trust boundaries. The closed packet is `experiment/v64_native_full_helper_extended_20261024/`, indexed by `OUTCOME.json`, `REPORT.txt`, `CLOSURE.json` and `independent_audit/AUDIT_RESULT.json`. The continuation release preserves the selected outcome as `records/patrol/FULL_HELPER_UNKNOWN.json`; its saved-replay entrypoints do not retry this query.

### U.9 Sequential candidate admission on one paused Lift runtime

**Question and preregistration.** This bounded software experiment tests the manager loop directly: can it withhold one candidate, preserve the actual pending boundary, check the next candidate, and continue after the first passing decision? Each of the three previously frozen Lift boundaries has one runtime object, one mutable source-state object, and one executed old prefix.

The candidate priority is fixed before execution as released native, saved normalized, then saved preference; selection stops at the first complete strict-and-cooperative pass. The separate packet is `experiment/v64_lift_manager_loop_20261024/`. Its `PLAN.json` fixes 70 exact input/code pins, the candidate order, the three existing twelve-input workloads, and one scientific execution, with a 120-second wall bound, 1 GiB sampled RSS bound, 16 MiB per-file bound, and 128 MiB output-tree bound.

An initial supervisor launch failed in a macOS address-space-limit setter before the worker was executed. No checker or runtime episode ran: the failure occurred in `preexec_fn`, and the worker's output directory was never created. The original plan, supervisor, and empty streams are retained. A separately frozen `PREEXEC_AMENDMENT.json` removes the unsupported address-space setter while retaining the RSS, wall, and output limits, and places the actual execution in `run02/`. The worker code is byte-identical across this amendment. There is one scientific execution and no outcome-based retry.

**Fresh checks and one-object selection.** Before execution, both existing checker routes freshly validate the complete old-mode closure: 50 nodes and 238 edges, with input coverage, source safety, strictness, and cooperation all passing. The runtime then executes the frozen two-edge prefix and pauses at the actual pending state. At each candidate attempt, the manager rechecks source and model bindings, saved-native/source predicate correspondence, the exact ten source/monitor values, and the candidate's stored private node and rank. Both routes traverse the entire candidate-rooted graph and recompute coverage, safety, strictness, and cooperation. Admission requires agreement and all five Boolean fields to be true. No saved candidate verdict or certificate loader supplies the decision; an audit hook rejects reads of the prior execution certificates. Raw extraction-result files referenced by the inherited input manifest are hashed for integrity only, not interpreted as admission verdicts.

All three native attempts are complete and strict but noncooperative (154 nodes, 1,170 edges), so the manager withholds them. Each following normalized attempt is complete, safe, strict, and cooperative (154 nodes, 1,176 edges), so it passes and is installed:

| Reached native node/rank | First decision | Installed node/rank | Commits |
|---|---|---|---|
| 524/1 | Withhold | 72/1 | 12 |
| 526/1 | Withhold | 74/1 | 12 |
| 756/1 | Withhold | 88/0 | 12 |

Across each pair of checks, the runtime identity, source-state-object identity, all source/monitor values, old-controller identity, private memory, and commit count remain unchanged; the runtime stays paused. The inherited installation adapter then checks the passing certificate's exact graph digest, node, rank, and complete source/monitor match. Only the selected controller object changes at installation. The preference candidate is explicitly recorded as unattempted after the first pass; this experiment makes no new claim about its verdict.

**Continuation and scope.** All three selected controllers execute their frozen twelve inputs. The old floor-3 request is answered at post-update step 1 and its monitor clears at step 2. No cancellation occurs, no genuinely pending obligation remains, and the final executed transition is a repeatable self-loop with all three monitors clear. The run has six fresh candidate checks, three installations, 36 post-update commits, and 69 saved events; no candidate check is unknown. The supervisor reports 0.650 s overall wall time and the worker reports 74,399,744 bytes peak RSS. These are

measurements of one instrumented run, not a latency benchmark. `run02/output/RESULT.json` and `run02/output/EVENTS.jsonl` preserve every attempt, before/after identity, certificate, installation, and trace.

This extends the separate candidate episodes of Section U.4 with a sequential rejection–admission loop on the same paused state. It demonstrates enforcement by the software adapter using the existing frozen graphs and predicates. It performs no synthesis, contract weakening, winning-region recomputation, source/monitor reset, or initializer reapplication at handoff. The complete-graph checks establish the candidate properties; the twelve-step workloads demonstrate the recorded continuation after those decisions.

An independent saved-artifact audit checks all 69 events, exact object identities, rooted graph coverage, safety, and strongly connected components; it also compares the source predicates on all 1,048,576 source-state pairs using separate truth-table evaluation and native-node parsing. The audit confirms the sequential decisions and continuations without rerunning the manager or synthesis. Native exporter metadata and process execution provenance remain trust boundaries; the audit record is `independent_audit/AUDIT_RESULT.json` in this packet. The continuation release supplies selected unchanged records under `records/manager/` and the qualified `manager` saved-replay stage. It rechecks graphs, source predicates and the 69-event sequence; it does not execute the manager worker or revalidate the omitted original source/runtime pins and worker call-site/read-path provenance.

# V. Primary premises and the selected released route

This section distinguishes an executed controller failure from the premises of the abstract synthesis theorem. The primary-text reading and static release-code audit add no solver call or controller extraction. The final subsection evaluates one preregistered fixed lasso on saved native predicates; it is not a new synthesis or search experiment.

## V.1 Supported input and the property that fails

Kind's GR(1) definition permits environment safety to depend on current input and output variables and next input variables [7, Sec. 3.1]. Its analysis explicitly considers system-only assumptions and their potential to deadlock the environment [7, Sec. 5.1 and Appendix A]. Thus Lift's current One($f$) assumption lies within that stated syntax. The released validator also permits current SYS/AUX references in ENV safety, while rejecting next SYS/AUX dependencies; the invariant translator keeps the recorded current-only assumption. Neither fact establishes what a physical lift should mean.

The saved Lift graph has an original-legal prefix ending where no legal next ENV input exists. It therefore violates Kind's Definition 3 for that fixed output and memory, under either the legal-prefix or the stronger all-history reading. A passing legal-prefix certificate alone does not establish the stronger all-history property.

Kind's output-kindness theorem relies on the correctness of the non-conflicting synthesis construction [7, Theorem 3 and Appendix A]. The extended TACAS statement binds the rank/mode strategy of equations (16) and (18), and its non-conflicting conclusion uses

$$L(H, G) \subseteq L(H, A),$$

as specified in equation (13c) [6, Theorem 3]. Replacing $G$ by $G \wedge A$ is the stated way to establish that sufficient premise when needed. Region membership or a raw four-fixed-point call alone does not bind an arbitrary extracted controller to that theorem. The premise is sufficient, not necessary; its failure does not prove that cooperative synthesis is impossible.

### V.2 Selected configuration and shipped default

The pinned released Eclipse plugin exposes *Spectra: Synthesize Concrete Controller* and the preference *Kind realizability: do not force any assumption violations (reduction plus 4 fixed-points)*. The tested settings select JTLV, no optimizations, no reordering and JTLV text output. The shipped defaults are instead standard GR(1), CUDD, all optimizations, reordering and CMP output. This is a mapping of public configuration and released calls, not a new GUI execution or evidence of default-route behavior.

The validating headless parser and GUI parser both use the released source-to-game translator; their resource-access wrappers differ. The selected path invokes syntactic helper reduction before the default translations, generates the game, invokes `GR1SeparatedGame` on the JTLV branch, and passes its memory to `GR1ConceteControllerConstructionSkip` with default ZYX order. Its goal index, saved $X, Y, Z$ progress filters and final `CoreUtil.satOne` choices are not an established implementation binding to the TACAS rank/mode strategy. The selected raw chain does not replace $G$ by $G \wedge A$; the saved model, preconstructor and solver justice lists agree on one ENV helper justice and three SYS monitor justices. The paired modified extractor remains an explicit intervention.

The preserving `PitermanSemanticReduction` class is present in the release [13]. A direct-reference scan of twenty plugin JARs finds its class name only in its own class; no supported frontend choice or default invocation was established. Reflection and external callers were not exhaustively excluded. Its complete-relation rule, the printed literal reduction, the appendix's unprimed-ENV guard, and the actual syntactic exports must not be identified.

### V.3 One fixed lasso refutes the premise on this arena

The preregistered trace uses the frozen transformed Lift arena. Its initial state has buttons 000, cancellation 0, floors 100, all three monitors $S0$ and helper $h = 0$. The next state changes only $h$ to 1, and repeats forever. Both initial predicates, all type constraints, and both ENV/SYS safety predicates hold on the first edge and self-loop. All three SYS justices hold on the loop, whereas the sole ENV justice $\mathsf{GF}\neg h$ fails. The exact arena therefore has

$$L(H, G) \not\subseteq L(H, A).$$

Evaluation checked 25 frozen input pins and 19 saved decision DAGs, including matching model, preconstructor and solver justice values. A separate source-equation evaluation agrees on the initial values, safety, monitor updates and helper persistence. There was one fixed candidate lasso, no arena enumeration, no solver call and no extraction. The source schema names `original_bdds` and `original_justices` refer here to the transformed native model.

This witness is an arena lasso, not a returned-controller execution. It explains why the sufficient non-conflictingness premise cannot be invoked for this raw arena. The original-contract output failure still rests on the separately audited reachable Lift graph. These facts do not refute an abstract theorem under its own hypotheses.

**Records and limits.** The primary PDFs, relevant pages, hashes and origin checks are indexed by `experiment/v65_submission_execution_20261024/theorem_scope/PRIMARY_SCOPE_REPORT.txt` and its adjacent provenance records. Static release bindings are in `route_scope/ROUTE_SCOPE_AUDIT.json`; the preregistration, evaluation script and exact truth values are adjacent to `route_scope/LIFT_LASSO_RESULT.json`, under the same execution directory. The primary-source and static-route records above remain author-side records. The continuation release [1] separately publishes the fixed-lasso saved replay described in the opening index: 23 semantic payloads, nine

metadata projection values, and 19 DAGs. It does not repeat the original 25-pin check: original Spectra and `final_region` are hash-only omitted, and original paths/runtime provenance are not revalidated. The author-host qualification adds no GUI run, default-failure result, theorem counterexample, returned-controller execution, or independent external reproduction.

# W. Additional source and extraction details

This section gives the exact old Patrol source binding and the stages of the selected Lift extraction. Sections B, H, T and U retain the underlying evidence and its trust boundaries.

## W.1. Equivalent-old Patrol source binding

**An equivalent-source control isolates the old mismatch** The original old spelling already loses: its actual helper-false arena allows sixteen inputs at $r$ instead of two, admitting the same countercertificate. Table 30 isolates the translation order: guarding a state invariant first introduces $b'$ and prevents its expansion into initial/next clauses.

Table 30: Old Patrol source-to-export binding for each blocked flag $d$. Before helper conversion, both forms compile to initial $\neg d$ and safety $\neg d'$. Shown are only these clauses after syntactic helper conversion, at $b = 0$.

| Old source clause | Converted ENV init. | Converted $E_{00}$ |
|---|---|---|
| `alw !d;` | $\top$ | $\neg d$ |
| `ini !d;` and `alw !next(d);` | $\neg d$ | $\neg d'$ |

The preregistered explicit-initial/next control preserves every compiled original predicate over all fourteen fields. Only converted ENV initialization and $E_{00}$ change, restoring two inputs at $r$. Native checks certify the unchanged 140-node graph inside $F = C$ and $U = D$, proving strict continuation. Only this equivalent-old control gives actual win/loss; the original remains loss/loss.

## W.2. Selected Lift route and final-choice filters

**A matched selection diagnosis** The selected Spectra [13] `1.0.0.202402072123`/JTLV route has three stages: syntactic helper conversion and translation; raw four-fixed-point solution by `GR1SeparatedGame`; and generic finite-memory extraction by `GR1ConceteControllerConstructionSkip`. A static release audit maps this headless chain to the public Kind option; shipped defaults use ordinary GR(1), and no GUI execution is claimed. The separate semantic correction has no established frontend option [13]. Identity with the C-backed driver used in Kind's published evaluation is unestablished (Supplements B and V).

**First creation versus later reuse** At construction node 668 (Table 31), input $(b', c') = (101, 0)$ yields candidates with $m' = 101$, private selector $j' = 0$ and false helper. The safe, winning-region-filtered floor set is $\{001, 011, 100, 101, 110, 111\}$; no existing node applies. The nearest progress-layer filter leaves

$$B = \{011, 100, 101, 110, 111\}, \qquad B \cap Q = \{100\}.$$

The released choice 011 first creates node 680; 100 survives the same filters and permits four next inputs. The source-safe 001 was removed earlier. At later reuse $704 \to 680$, existing-node preference

Table 31: Saved initialized Lift prefix to the two-floor response: native node IDs and private justice selector $j$. Rows are completed states; the next row's $(b, c)$ supplies the next input. The helper stays false. The last row is the bad node's construction predecessor, outside this prefix.

| Node | Buttons $b$ | $c$ | Floors $f$ | Monitors $m$ | $j$ |
|---|---|---|---|---|---|
| 0 | 000 | 0 | 100 | 000 | 0 |
| 10 | 010 | 1 | 100 | 000 | 0 |
| 339 | 001 | 1 | 010 | 000 | 0 |
| 438 | 100 | 0 | 001 | 000 | 1 |
| 704 | 101 | 0 | 010 | 100 | 2 |
| 680 | 101 | 0 | 011 | 101 | 0 |
| First-creation predecessor of node 680: | | | | | |
| 668 | 111 | 0 | 010 | 110 | 2 |

leaves $B = \{011\}$. The intervention acts throughout extraction, including first creation; it does not replace only the displayed reused edge. Supplement T gives every preceding filter and candidate.

**A contract-sensitivity control** Moving only the four floor-invariant declarations from ENV assumptions to SYS guarantees removes current One($f$) from ENV safety and adds next One($f'$) to SYS safety. Ownership and timing therefore change together; the two initial states remain unchanged, and each contract has its own solve. All 1,024 source states now admit input (originally 384), so $Q$ is always true. Both graphs are identical and admitted: 154 nodes, 1,176 edges. A separate source checker confirms all four outcomes. This control demonstrates contract sensitivity; it is neither an ownership-only intervention nor a claim that the original contract needs changing (Supplement U.2).

## X. Vehicle: complete saved regions and initialized cooperative reachability

This analysis [14] extends Section E within the same unchanged, previously outcome-selected public source. It evaluates the two saved native strict-winning regions [2]; it invokes no new native solver or game fixed point. The 33,792 typed source valuations retain all twelve source fields. Both independent implementations align fields by name, ownership, type and value encoding; the transformed model adds only the Boolean ENV helper. In particular, speed encodings are read from their value maps, not inferred from the lexicographic value order.

**Complete memberships and correspondence** Each implementation evaluates every source state in the original saved $W$ and in the transformed saved $W_R$ with $b = 0$ and $b = 1$ separately. The direct differences do not assume containment. Independent raw-DAG readers and Boolean operations give the same membership for every state:

| Set of source valuations | Cardinality |
|---|---|
| Typed source carrier $D$ | 33,792 |
| Original strict region $W$ | 8,064 |
| $\{s \in D : W_R(s, 0)\}$ | 6,656 |
| $\{s \in D : W_R(s, 1)\}$ | 33,792 |
| $W \setminus \{s \in D : W_R(s, 0)\}$ | 1,408 |
| $\{s \in D : W_R(s, 0)\} \setminus W$ | 0 |

The two helper slices sum to the earlier native total of 40,448. The helper-true slice is not a set of legal transformed initial states: the transformed ENV initializer requires $b = 0$. Original initial predicates are true on the typed carrier, but winning initialization still requires a winning response to every allowed initial ENV value. All 1,024 initial ENV fibers have such a response, with between three and 33 winning SYS choices per fiber.

Both implementations independently transcribe the source predicates and compare their canonical Boolean forms to the original saved domain, initializers, safety relations, three ENV justices and one SYS justice. The equalities include the exported typed-domain factors and invalid lane/speed encodings. Every state row agrees across implementations when aligned by all twelve named values. This checks saved memberships and source correspondence; it does not independently recompute the native winning fixed points. The native solver, exporter and typed metadata remain shared dependencies.

**Source-specific equality $W = K$** Let $e_{\text{clear}} = (o, p, g, x) = (000, 000, 111, 0)$: all lanes have green lights, no pedestrians or obstacles, and no crosswalk. The complete source ENV safety relation is

$$(o_1 \Rightarrow \neg o_2') \wedge (o_2 \Rightarrow (\neg o_1' \vee \neg o_3')) \wedge (o_3 \Rightarrow \neg o_2').$$

It contains no current or next SYS variable. Thus $e_{\text{clear}}$ is a legal next ENV value from every current value and can repeat forever. All three ENV justice predicates $g_i \wedge \neg p_i \wedge \neg o_i$ hold there and are likewise SYS-independent. Source/BDD equalities and support checks establish these facts; direct evaluation also checks the all-clear transition from all 1,024 current ENV values.

Fix any strict-winning policy from any $s \in W$ and any finite legal prefix compatible with that policy, including its reached private memory. Let ENV choose $e_{\text{clear}}$ forever from the next step. These inputs remain legal and fair regardless of the policy's SYS choices. Under the total-policy semantics used here, a winning policy supplies a safe response on each legal input. Applying that same policy therefore gives a compatible fair continuation, and strict winning supplies SYS justice as well. Since the prefix was arbitrary, the policy is legal-prefix cooperative. Hence $W \subseteq K$; $K \subseteq W$ holds by definition, so $W = K$ for this source. This argument depends on its SYS-independent always-legal fair suffix and is not a general equivalence.

**A positive-length prefix for every winning target** For each $s \in W$, fix the preregistered predecessor

$$p_s = (e_{\text{clear}}; \ell(s), 1).$$

There are only three possible predecessors. Both implementations verify for all 8,064 targets, including all 1,408 losses, that

$$p_s \models \theta_e \wedge \theta_s, \qquad p_s \in W, \qquad \rho_e(p_s, e(s)), \qquad \rho_s(p_s, e(s), y(s)).$$

No alternative predecessor is searched. Separately, the complete initial-fiber check above establishes $\forall e_0 : \theta_e(e_0) \Rightarrow \exists y_0 : \theta_s(e_0, y_0) \wedge W(e_0, y_0)$.

For a fixed target $s$, construct an initialized policy $\sigma_s$. On initial ENV value $e_{\text{clear}}$, choose $p_s$'s SYS value; on the next input $e(s)$, choose $y(s)$ and then a winning continuation from $s$. For other next inputs, use a winning continuation from $p_s$. For every other initial ENV value, choose a certified winning response and its winning continuation. The designated edge is safe, and every branch has a winning tail. A finite prefix does not change the justice conditions, so $\sigma_s$ is initialized strict winning. The preceding all-clear argument makes it cooperative on every legal prefix. The legal ENV choices $e_0 = e_{\text{clear}}$, $e_1 = e(s)$ then yield the one-transition compatible prefix $p_s, s$. Thus the quantifiers are

$$\forall s \in W \ \exists \text{ initialized cooperative winning } \sigma_s \ \exists \text{ compatible legal play reaching } s \text{ at time 1.}$$

The policy and play may depend on the target. One transition is positive length even when it is a self-loop. Neither one common policy visiting all targets nor forced reachability is claimed.

**Evidence and limits** Complete tables retain all 33,792 membership rows, all 8,064 fixed predecessor checks, and every initial ENV fiber. The two implementations agree on every aligned row; no membership, predecessor or initial-fiber check fails. The computations evaluate saved native results, and the cooperation and reachability claims are source-level existence proofs. They do not synthesize or execute the stitched policies. The 1,408 lost states are therefore cooperatively reachable in the stated existential sense within this one selected case. The cardinalities are not an operational probability or a population-frequency estimate.

# Y. Released-route binding and fixed seven-input native comparison

## Y.1. Source-to-query binding

The pinned Spectra release is `1.0.0.202402072123` [13]. The public concrete-synthesis action is `SynthesisAction` under `tau.smlab.syntech.syntCmpAction`. Its public Kind selection is `synthesisPitermanReduction`. The GUI input provider and the headless `SpectraInputProviderNoIDE` both feed the Spectra-to-GameInput translator. The selected chain calls `PitermanSyntacticReduction.doReduction` before `TranslationProvider.translate` with the default translators and `BDDGenerator.generateGameModel`. These are released source/bytecode bindings; they are not evidence that a GUI action completed.

The strict diagnostic branch calls `GR1Game` for the original and transformed games. Raw selected-route runs instead call `GR1SeparatedGame`, test the resulting memory, and pass that memory and the model to `GR1ConceteControllerConstructionSkip` before concrete construction and printing. The class name's spelling is literal. The experiments use headless JTLV settings, not the shipped ordinary-GR(1) defaults. The separate `PitermanSemanticReduction` implements the complete first-flip correction; no public frontend option invoking that class is established. Identity with the C-backed driver used in the published Kind evaluation remains unestablished. Full exported relations, not the printed historical guards alone, bind the main paper's claims.

## Y.2. Fixed selection and protocol

The seven unchanged public inputs below [15] were selected from the previously certified inventory using the already known condition $I_M(K) = \mathit{true}$ and a 65,536-state carrier cap. Their prior symbolic results already had $W = K$ and complete-literal preservation. The roster and commands were frozen before these native comparisons; this is a selected positive/control study, not a blind search. Exactly fourteen scientific JVM/strict-solver invocations ran: one original and one released-route query per input. All completed; no case was UNKNOWN or unattempted. No scientific query was retried. The independent audit used the saved BDDs and source exports, with no native solver rerun.

| Corpus ID | $\lvert D\rvert$ | $\lvert W\rvert = \lvert K\rvert$ | Route source view | Initial ENV fibers | Conversion |
|---|---|---|---|---|---|
| 039 | 11,520 | 2,304 | 2,304 | 1 | helper |
| 040 | 11,520 | 2,528 | 2,528 | 1 | helper |
| 041 | 11,520 | 2,528 | 2,528 | 1 | helper |
| 135 | 65,520 | 63,312 | 63,312 | 60 | skip |
| 148 | 21,600 | 4,032 | 4,032 | 36 | skip |
| 150 | 43,200 | 9,440 | 9,440 | 36 | skip |
| 187 | 33,792 | 7,680 | 7,680 | 1,024 | skip |

### Y.3. Complete membership and initialization checks

All 198,672 typed source-state rows, including AUX variables, were enumerated and saved. For each input, the original native $W$ equals the previous independently computed symbolic $W = K$, and equals the released route's source view. Both directions of every set difference are empty: loss, gain, $K \cap$ loss and $K$ outside the route view are zero. For the three GyroAspect variants 039–041, the converter actually introduces the helper. The route view fixes its current value to false; the full native winning-region sizes are respectively 13,824, 14,048 and 14,048, with helper-true slices of 11,520 each. These true slices count states only; the route initializer requires the helper to be false. For 135, 148, 150 and 187 the released converter skips. All metadata, domains, initial and safety predicates, and ordered justices match the original model. Their view is the whole route region; a helper slice is inapplicable. These four are identity controls.

For every legal initial ENV assignment in the table, a SYS response (including AUX) was checked against the original native region, route region, old symbolic $W$ and old $K$. Every comparison has zero missing fibers, so all four initialization decisions are true on each input. The audit matches typed value encodings and ownership by field name, handles nonmonotone bit renaming in an independent raw-DAG implementation, and checks full predicates, ordered justices, constructor snapshots and the exact helper-false restriction. Raw winning BDDs intersected with their typed domains agree with saved typed regions. The audit rejects residual helper or next-variable dependence in a source view.

### Y.4. Interpretation and dependencies

The new evidence connects completed native strict queries to the old independent symbolic regions and the released conversion on this fixed set. It does not compute native cooperative strategies. Three active inputs share one program family; the other four do not exercise an active helper transformation. The prior cooperative-positive selection and known literal preservation preclude a frequency or blind-discovery interpretation. No target reachability, single-policy coverage, runtime update or deployment result follows. Zero loss here neither refutes the separate Vehicle188 loss nor proves preservation for other inputs. Shared source exports, typed metadata and native solver correctness remain trusted dependencies despite the independent membership implementation.

## Z. Historical report for the unchanged Patrolling input

This section records public evidence confirmed during the current revision. It does not change the earlier outcome-informed selection, constructed old model, post-diagnosis reuse policy or workload of the Patrol study. The evidence is a historical reported result from the original Kind artifact [9], not a new native execution.

**Exact source and result binding** The downloaded artifact ZIP has SHA-256 `65abe41fcca3694a31967d409e642cb2b5785156aed5f464db380c7e3bb98712`. Its source member `artifact/experiments/specs/syn23/1670358790.spectra` has 3,304 bytes and SHA-256 `cde1b981a103047ddefbe8a963f1188d57904659a30fd0e7b678c771213be402`. It is byte-identical to the unchanged new Patrolling source indexed as `corpus277__original` in our roster. The member `artifact/raw_results/4fp_cover/_results_4fp_cover23.csv` has 5,710 bytes and SHA-256 `259adc380dad00301e79c8b18eae4faf9447d8120a666570cb76b7434903805e`. Its header and line 40 are

```
Filename,4FP,R+4FP
1670358790.spectra,TRUE,FALSE
```

The artifact README, lines 112 and 115, identifies the columns as algorithm realizability outputs. Its SHA-256 is `6476e8ad59dadeae771e1344f9e0afcf373c56f80fcc1d4f34b11642b8c81289`. Here `R+4FP` means *reduction plus four fixed points*; it does not denote the preserving $R^+$ defined in the main paper.

**What the packaged driver establishes** A static inspection of the packaged Java driver finds separate `checkRealizability` Boolean answers on the same reparsed source. The paired true/false branch is separate from the caught-exception record; the Java path has no watchdog or timeout-to-false branch. The C-backed game implementation delegates to the factory Boolean. This supports reading the CSV row as the artifact's reported pair of algorithm answers, rather than a timeout token in that driver. It does not replace a historical runtime receipt.

Both arms request CUDD 3.0, SIFT and the same optimization settings. The driver rewrites ENV safety declarations to state invariants, invokes `DefaultTranslators`, and only then applies syntactic reduction before the C-backed four-fixed-point game. That order, backend and platform differ from our fixed-option headless JTLV route. The archive has no loaded-factory/runtime logs establishing identity of the historical execution. Native DLL behavior was not audited; the base factory method can return false if unimplemented. The first-arm TRUE is inconsistent with a simple identical-factory fallback in both arms, but does not prove what was loaded historically. The published CSV uses uppercase Booleans whereas the packaged driver emits lowercase text, so the row is not a byte-for-byte raw execution receipt.

**Supported interpretation** The original public artifact reports different answers with and without reduction for the same source used here. This supplies concrete historical context for the input. It does not establish a default or frontend run, identity with the current strict-query harness, or the strict-semantic cause of the historical difference. The source/export/countercertificate arguments in Sections S and U independently support our current Patrol claims. No historical timing is used to predict present execution time.

## AA. Compiled four-arm comparison on a conditioned eleven-source subset

**Selection and questions.** The fixed roster is drawn from the 59 previously certified completed source cases, not from a new independent population. Eligibility requires at most 262,144 typed source states, including AUX, and a parsed ENV `SAFETY`/`STATE_INV` trigger for helper conversion. The prior public-fourteen and fixed-seven rosters and the separate Vehicle188 witness are excluded. There is no additional filter on source winning or cooperative outcomes, but those prior outcomes and completion/certification status were already known. This is therefore a conditioned extension, not a random or outcome-blind sample, and it supports no incidence or prevalence estimate. Seven of the eleven sources retain AUX fields. Nine prior source cooperative initializer checks are false and two are true.

Each source has four registered arms: A is the original game; B is the selected syntactic helper route; C releases SYS safety on the first helper change only, leaving the other helper quadrants unchanged; D is the existing complete semantic correction. C is a mechanism-isolation intervention and D is an existing correction; neither is a new synthesis method. Source initializers and monitors are retained without reset or projection.

**All registrations and acceptance boundary.** All 44 registrations remain visible. There are 43 saved native results, but only 40 independently audited PASS arms are adopted. The three transformed arms of source 003 exceeded the independent BDD node budget; their saved native values remain unaccepted candidates. Source 059 arm C has no native result: the supervisor's `ps` call timed out before a recorded solver call, and no retry was made. These four terminal UNKNOWNs are not negative synthesis answers. Nine sources have all four arms independently certified. Tables below separate source metadata, native saved values and accepted audit results.

Table 32: Fixed source roster. $|D|$ includes AUX. $|K|$ and $I_M(K)$ are prior source certificates, not newly recomputed cooperative regions.

| ID | Unchanged source filename | $\|D\|$ | AUX | Prior $\|K\|$ | Prior $I_M(K)$ |
|---|---|---|---|---|---|
| 003 | `amba_ahb_realizable_amba_ahb_1.spectra` | 65536 | 0 | 0 | F |
| 044 | `HumanoidLTL_460_Humanoid.spectra` | 18816 | 2 | 0 | F |
| 045 | `HumanoidLTL_462_Humanoid.spectra` | 102400 | 2 | 0 | F |
| 054 | `PCarLTL_Unrealizable_V_2.0_882_PCar_fixed.spectra` | 5184 | 2 | 16 | F |
| 059 | `PcarLTL_552_PCar.spectra` | 20736 | 3 | 1636 | F |
| 060 | `PcarLTL_553_PCar.spectra` | 20736 | 3 | 1636 | F |
| 061 | `PcarLTL_758_PCar_fixed.spectra` | 7776 | 2 | 42 | F |
| 062 | `PcarLTL_763_PCar_fixed.spectra` | 5184 | 2 | 16 | F |
| 164 | `roboticarm_Robot_232.spectra` | 31250 | 0 | 0 | F |
| 225 | `SmartHome1_sprint02_ver2.spectra` | 82944 | 0 | 3510 | T |
| 226 | `SmartHome1_sprint02_ver3.spectra` | 82944 | 0 | 3510 | T |

Table 33: All 44 registered arms. Native $W_0$ denotes source $W$ for A and helper-false winning states for B/C/D; native $W_1$ is inapplicable to A. $I$ is the native strict initializer judgment. Audit columns are populated only for PASS. P=PASS; BUD=independent BDD-budget UNKNOWN; SUP=supervisor UNKNOWN. T/F=true/false; dashes mean inapplicable or unavailable. Daggered native values are not adopted. Loss/gain compare the accepted helper-false region with source $W$.

| ID | Arm | Native $W_0$ | Native $W_1$ | $I$ | Audit | $\lvert U\rvert$ | $\lvert F\rvert$ | $\lvert W_F\rvert$ | Loss | Gain |
|---|---|---|---|---|---|---|---|---|---|---|
| 003 | A | 34575 | — | T | P | — | — | — | — | — |
| 003 | B | 28431$^\dagger$ | 65536$^\dagger$ | $T^\dagger$ | BUD | — | — | — | — | — |
| 003 | C | 34575$^\dagger$ | 65536$^\dagger$ | $T^\dagger$ | BUD | — | — | — | — | — |
| 003 | D | 34575$^\dagger$ | 65536$^\dagger$ | $T^\dagger$ | BUD | — | — | — | — | — |
| 044 | A | 2387 | — | T | P | — | — | — | — | — |
| 044 | B | 0 | 18816 | F | P | 18816 | 1211 | 0 | 2387 | 0 |
| 044 | C | 2387 | 18816 | T | P | 18816 | 18816 | 2387 | 0 | 0 |
| 044 | D | 2387 | 18816 | T | P | 18816 | 18816 | 2387 | 0 | 0 |
| 045 | A | 11232 | — | T | P | — | — | — | — | — |
| 045 | B | 8 | 102400 | F | P | 102400 | 6120 | 8 | 11224 | 0 |
| 045 | C | 11232 | 102400 | T | P | 102400 | 102400 | 11232 | 0 | 0 |
| 045 | D | 11232 | 102400 | T | P | 102400 | 102400 | 11232 | 0 | 0 |
| 054 | A | 470 | — | T | P | — | — | — | — | — |
| 054 | B | 470 | 5184 | T | P | 5184 | 470 | 470 | 0 | 0 |
| 054 | C | 470 | 5184 | T | P | 5184 | 5184 | 470 | 0 | 0 |
| 054 | D | 470 | 5184 | T | P | 5184 | 5184 | 470 | 0 | 0 |
| 059 | A | 5196 | — | T | P | — | — | — | — | — |
| 059 | B | 3596 | 20736 | T | P | 20736 | 4544 | 3596 | 1600 | 0 |
| 059 | C | — | — | — | SUP | — | — | — | — | — |
| 059 | D | 5196 | 20736 | T | P | 20736 | 20736 | 5196 | 0 | 0 |
| 060 | A | 5196 | — | T | P | — | — | — | — | — |
| 060 | B | 3596 | 20736 | T | P | 20736 | 4544 | 3596 | 1600 | 0 |
| 060 | C | 5196 | 20736 | T | P | 20736 | 20736 | 5196 | 0 | 0 |
| 060 | D | 5196 | 20736 | T | P | 20736 | 20736 | 5196 | 0 | 0 |
| 061 | A | 496 | — | T | P | — | — | — | — | — |
| 061 | B | 496 | 7776 | T | P | 7776 | 500 | 496 | 0 | 0 |
| 061 | C | 496 | 7776 | T | P | 7776 | 7776 | 496 | 0 | 0 |
| 061 | D | 496 | 7776 | T | P | 7776 | 7776 | 496 | 0 | 0 |
| 062 | A | 470 | — | T | P | — | — | — | — | — |
| 062 | B | 470 | 5184 | T | P | 5184 | 470 | 470 | 0 | 0 |
| 062 | C | 470 | 5184 | T | P | 5184 | 5184 | 470 | 0 | 0 |
| 062 | D | 470 | 5184 | T | P | 5184 | 5184 | 470 | 0 | 0 |
| 164 | A | 31250 | — | T | P | — | — | — | — | — |
| 164 | B | 31250 | 31250 | T | P | 31250 | 31250 | 31250 | 0 | 0 |
| 164 | C | 31250 | 31250 | T | P | 31250 | 31250 | 31250 | 0 | 0 |
| 164 | D | 31250 | 31250 | T | P | 31250 | 31250 | 31250 | 0 | 0 |
| 225 | A | 3510 | — | T | P | — | — | — | — | — |
| 225 | B | 3510 | 82944 | T | P | 82944 | 3510 | 3510 | 0 | 0 |
| 225 | C | 3510 | 82944 | T | P | 82944 | 82944 | 3510 | 0 | 0 |
| 225 | D | 3510 | 82944 | T | P | 82944 | 82944 | 3510 | 0 | 0 |
| 226 | A | 3510 | — | T | P | — | — | — | — | — |
| 226 | B | 3510 | 82944 | T | P | 82944 | 3510 | 3510 | 0 | 0 |

| ID | Arm | Native $W_0$ | Native $W_1$ | $I$ | Audit | $\lvert U\rvert$ | $\lvert F\rvert$ | $\lvert W_F\rvert$ | Loss | Gain |
|---|---|---|---|---|---|---|---|---|---|---|
| 226 | C | 3510 | 82944 | T | P | 82944 | 82944 | 3510 | 0 | 0 |
| 226 | D | 3510 | 82944 | T | P | 82944 | 82944 | 3510 | 0 | 0 |

**What the completed comparisons establish.** Among the ten independently certified B arms, sources 044, 045, 059 and 060 lose strict source states, with no gained state. Sources 044 and 045 also change the strict initializer judgment from true to false; 059 and 060 retain positive strict initialization despite their region loss. The other six completed B arms preserve the source region. Every completed B arm has $U = D$, including those retaining AUX. All nine completed C arms and ten completed D arms preserve the exact source region: independent canonical equality checks establish set equality, not merely matching cardinalities. Thus these inputs do not demonstrate a proper $U$ caused by retained AUX, or insufficiency of first-change release on this set.

All four B-loss sources have prior $I_M(K) = \mathit{false}$, but that does not mean their cooperative regions are empty: sources 059 and 060 each have 1,636 prior $K$ states. This experiment does not recompute $K$ or establish a new cooperative-initialization loss. It does not replace the prior 59-source cohort's null cooperative-loss result and UNKNOWN boundary. The 11-source, 10-B, 9-C and 10-D denominators answer different questions; none is a population-frequency estimate.

**Execution, independence and availability.** The comparisons use the pinned headless selected configuration and shared compiler outputs; they establish neither GUI/default execution nor identity with the public default route. The independent audit imports saved primitive relations and uses separate Boolean, quantification and fixed-point code. This is not independent compilation or external reproduction. The native constructor normalizes only metadata-empty justice lists to a TRUE justice; this normalization is recorded explicitly, and the remaining predicate-preservation checks remain in force. The bounded failures above remain UNKNOWN, without retries or replacement by favorable native candidate values. There is no natural-adaptation, performance, new-method or deployment-frequency claim. The accompanying release packages these saved records; the opening evidence index identifies the archive and its checking scope. Packaging adds no scientific execution, and the relocated replay adapter remains unexecuted.

## AB. Supporting diagnoses and comparison records

The main paper focuses on exact continuation preservation and two insufficient tests. The following output-level diagnosis and comparisons support a separate graph-installation boundary. They are not additional instances of the same continuation-loss theorem.

### AB.1 Lift: strict existence does not certify a returned graph

Continuation preservation concerns whether a winning strategy exists; installing one returned controller requires its own evidence. Lift reconstructs the published trivial-replacement listing, using the same paper's definition of its omitted response pattern [10].

The Boolean floor invariant, "exactly one current floor," belongs to ENV safety. At floor 2 with both endpoint buttons pressed, a SYS movement disjunct permits next output 011: floors 2 and 3 together. No SYS clause forbids that extra floor. The reached state has no legal next input (Fig. 1). The complete 154-node graph is strict-only, although a source policy retaining one floor and alternating endpoint priority passes strictness and cooperation (Supplement H).

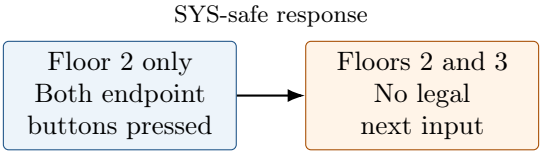



Figure 1: A source-safe Lift response makes the ENV-owned floor invariant false. The returned graph fails cooperation; another policy for the same source contract passes. Full source values and the initialized prefix are in Supplement H.

Table 34: Supporting comparisons. An admitted graph passes complete strictness and legal-prefix cooperation checks. Selection rules and denominators are separate.

| Fixed group | Result | Meaning and full records |
|---|---|---|
| 11 reconstructed listings, five published examples | Selected route: 7 admitted, 2 strict-only, 1 raw negative, 1 correspondence UNKNOWN. Normalized: 8 admitted, 3 empty-region stops. | An admitted alternative for each feasible listing; Supplement H. |
| 14 certified public inputs, at most 4,096 states; no region/output filter | 56 input/route outcomes. Selected: 11 admitted, 2 strict-only, 1 UNKNOWN. Each normalized/cooperative-Slugs route: 12 admitted, 2 negative. Nonblocking strict: 12 admitted, 1 strict-only, 1 negative. | Separate runs; no population or performance comparison; Supplement T. |
| 59 certified models from the stopped 425-unit inventory | 49 satisfy $W \subseteq C$; 10 have strict losses. Nine exact-loss calculations finish with no lost $K$ state; one UNKNOWN. All ten lack cooperative initialization. | No confirmed cooperative-initialization loss; Supplements E and J. |
| 7 fixed cooperative-positive native controls [15] | 14 native queries, 198,672 typed-state comparisons: source $W$, prior $W = K$ and route source view agree; both initial judgments true. | 3 helper-active, 4 identity/skip controls; no loss or gain; Supplement Y. |

A matched diagnosis supplies identical model, solve and stored fixed-point memory to two extractions. A final-choice preference keeps candidates with a legal next input when available, while retaining helper-true candidates; progress, goal, existing-node and private-memory rules remain fixed. Its 154-node graph passes both complete checks. The preference also acts at first creation of a bad node (Supplements H, T and W).

Selection is not the sole cause: the encoded arena already permits the bad state. Moving the invariant to SYS/next makes both outputs pass but changes ownership and timing together. Nor does immediate input availability guarantee ENV fairness, as the constructed Polling control shows (Supplement O). The generic extractor has no established binding to the prescribed cooperative rank/mode construction [6], and Lift's raw arena fails its sufficient language premise (Supplement V). This fixed-output diagnosis does not refute that construction.

### AB.2 Separate comparison groups

The two fixed rosters supply admitted alternatives for all cooperatively feasible contracts: 8 of 11 reconstructed listings and 12 of 14 public inputs. Those are separate route runs, not matched extraction or comparative superiority. Infeasible or correspondence-UNKNOWN examples remain in the records (Supplements H and T).

## AB.3 Historical Patrol reports and executed reuse

For historical context, the original artifact reports `4FP=TRUE` and `reduction+4FP=FALSE` for the byte-identical new input, where 4FP means four fixed points [9]. Those paired reports use a distinct historical driver; they establish neither a current-tool reproduction nor a strict-semantic cause (Supplement Z).

The checked source controller executes 67 new-mode commits without resetting plant/controller objects, source fields, monitors or private phase. It serves the initial request at step 30 and clears its monitor at 31; a repeatable 35-step full-state/private-memory cycle meets all eight source justices. The workload was fixed after diagnosis, before execution. This is validated reuse in the constructed scenario. The known complete first-change correction is another strict-preserving alternative by theorem; a direct full-arena native source solve was not run here (Supplements S and U).

## AB.4. Computation provenance

This section retains the detailed audit and cohort account behind the main paper's evidence summary. It introduces no new execution or result.

**Named computations and separate checks.** The preceding proofs explain the mechanisms. This section and the current evidence index distinguish the computations that corroborate them from other checks that remain incomplete. A result is always attached to a named game or validation procedure. A solve computes a winning region. A saved-region audit checks an existing result. A theorem certificate here derives the claimed region from checked relations and an independently precomputed source region; it does not solve the transformed game anew. An independent fixed-point audit instead attempts to recompute the region.

**Implementation and answer.** The strict winning-region calculations use JTLV and Spectra release `1.0.0.202402072123` [13]. The headless syntactic conversion runs before default translation and BDD construction. Static source and bytecode inspection bind this conversion to the *public opt-in Kind option*; shipped defaults use ordinary GR(1). The winning-region queries here use released `GR1Game` with strict semantics. They establish answers for the checked games. Corresponding Kind frontend answers for Vehicle and Patrol remain unobserved. Section AC reports two separate public Kind realizability checks for Obstacle. AC.7 adds completed ordinary-GR(1) controls for the same exact A/B queries, using JTLV with optimizations and reordering disabled. These use the installed default synthesis mode; they do not test every default setting or backend. Vehicle and Patrol have no corresponding public-mode observations. The separate complete-correction class exists in the release; its supported frontend invocation remains unestablished.

*UNKNOWN* means that the named query or validation attempt did not establish a definitive answer under its recorded stopping rule. It is never a negative synthesis verdict. A native result can be saved while a separate audit remains UNKNOWN. Unattempted work and processing exceptions retain separate statuses and causes.

**Dependencies and test objects.** The checks bind typed domains, source predicates, initializers, ordered justices and helper transitions to the queried relations. In Vehicle, full relation checks establish the complete literal form; two evaluators then inspect the same saved native sets. In Patrol, the hand proof uses the displayed source equations, and separate full-domain export checks establish the identities used in the main paper's Patrol section. Obstacle also has source-equation winning regions computed before the native candidates. A saved-set audit or theorem certificate does not

recompute a winning fixed point. Shared compiler exports and typed-domain metadata remain dependencies of the corresponding checks (Supplements B, E, U, V, W, X and AC).

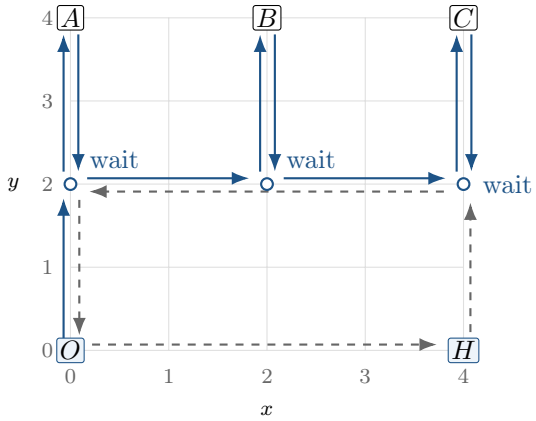
For Patrol, source-legal steps preserve the reached checkpoint layout. The 4,800-state restriction covers source-legal prefixes; the first-change test $F$ still uses every typed input, including relocated checkpoints, through the full-domain relations. Writing $D_L$ for that closed source domain, exact calculations give $|W \cap D_L| = 3,280$ and $|W_F \cap D_L| = 1,104$. A completed native $M_F$ query agrees with a separate explicit calculation and membership audit on all states (Supplements U.1 and U.6). It corroborates the maintained-response calculation. The loss in the actual helper game follows from the source all-response proof, the checked export identities and the main paper's retained-duty preservation proposition. Direct full-helper attempts remain UNKNOWN and supply no observed refusal. Their distinct stopping records are retained in Supplement U.5.

## AB.5. Wider comparisons and selection

A stopped 425-unit inventory yielded 70 completed models and, after eleven certification exclusions, 59 analyzed models without outcome filtering. Ten have literal strict losses; all ten lack cooperative initialization. Of nine completed exact-loss calculations, seven have empty $K$ and two preserve all $K$ states; one remains UNKNOWN. This cohort therefore confirms no cooperative-initialization loss. Vehicle, Patrol 277 and the published Obstacle update pair all lie outside this frozen cohort. The primary Patrol NEW source is byte-identical to corpus 277 (Section Z); its later case-specific proof does not change the 59-model denominator. At selection, Patrol had no eligible authoritative independent region certificate: the recorded status was `UNKNOWN_NO_AUTHORITATIVE_INDEPENDENT_W`, with prior status `UNKNOWN_NATIVE_NOT_ARRIVED` and no BDD worker started for this condition check. A separately recorded native attempt stopped at its raw-output cap; that later resource UNKNOWN is not the cohort-selection reason. The cohort is conditioned on completion and certification under the stopped resource protocol, not a sample establishing corpus-wide incidence.

The remaining 49 cohort models satisfy $W \subseteq C$, so the literal strict encoding preserves all their source winners. In the nine completed exact-loss rows, $K \cap (W \setminus W_C) = \varnothing$: seven have empty $K$, and the other two preserve every $K$ state. Let $K_{\text{all}}$ denote source states admitting Kind's all-history cooperation requirement, as distinct from the legal-prefix set $K$ used in this supplement. Since $K_{\text{all}} \subseteq K$, these existing facts exclude $K_{\text{all}} \setminus W_C$ in 58 cohort models. The exact intersection for the single remaining corpus model 159 (RoboticArm) is UNKNOWN. This is a consequence for the literal strict target, not a zero-of-59 incidence result, a Kind-to-Kind preservation theorem, or an observation about public Kind actions. The existing reachability relation uses legal joint edges; it is not the reached set of a winning or deployed controller and carries no private memory.

A conditioned eleven-source subset has 40 certified results and four UNKNOWNs across 44 encoding variants. Four of ten certified syntactic variants lose strict states, and two also lose strict initialization; all four loss sources already lack cooperative initialization. All ten have $U = D$. The checked first-change and complete corrections preserve their source regions. Supplement AA gives every variant and unresolved operation. These conditioned comparisons supply no incidence estimate; the small proper-$U$ example establishes a logical need, not its frequency in those exports. Supporting comparisons and output diagnoses are in Supplement AB, and fixed-seven native controls in Supplement Y [15]. Claim-specific saved records and scoped replays remain separately identified (see the current evidence index).



Solid: checkpoint visits. Dashed: return, charge, restart.

Figure 2: The source patrol route on the fixed $5 \times 5$ layout. The robot waits two cells below each checkpoint, visits it when entry is safe, then retreats. After $A, B, C$ it returns to $O$, visits charger $H$, dwells, and returns along row 2 to the first waiting point. The main paper's Patrol section gives the complete waiting, safety and fairness argument. This source route witnesses feasibility; the loss proof covers every safe strategy.

### AB.6. Patrol route retained from the main paper

The route shown is the source-policy witness used by the main paper's Patrol proof. Its standoffs, waiting rule, counter/monitor behavior and all-response loss argument retain their original scope. The main paper specifies the coordinates and complete source rules directly.

## AC. Published Obstacle Evasion requirements-update pair

### AC.1 Source provenance and the ordinary-game projection

We use the exact old/update example distributed with Amram et al.'s *Dynamic Update for Synthesized GR(1) Controllers* [16]. The author site and source archive are `https://smlab.cs.tau.ac.il/syntech/dupdate/` and `https://smlab.cs.tau.ac.il/syntech/dupdate/Specifications.zip`. The archive is 6,517 bytes; its SHA-256 is `16483e1d023bdaea9201209639b67eb9b654ab367fa7eac51d62966f2b5916c4`. The original files are `Specifications/ObstacleEvasion/MovingObstacleEvasion.spectra` and `Specifications/ObstacleEvasion/MovingObstacleEvasionUpdate.spectra`. The old source hash is `19c71d970b05fec41f0fa60fce155bf136b33836ffd785a65c2c436a712c3ac1`; the unprojected update source hash is `bf2836d68e9a7b019b8f2c9499ccf1ad8dccdc56f5c31ff78ea7d80e45b1a228`.

The paper motivates changing forbidden and repeatedly visited robot locations while avoiding a moving obstacle. Its update semantics replaces old SYS goals. The old forbidden positions $(3,5),(4,5)$ become $(6,4),(8,7)$; recurring targets $(5,2),(2,3)$ become $(3,7),(4,8)$. ENV transitions and SYS movement/collision clauses are unchanged. This is a published full-SYS requirements replacement, not an ENV-only update, a same-pending-obligation continuation, or an observed industrial workload. No link from this standalone pair to one particular pair in the paper's larger evaluation is established.

The five scalar fields are obstacle coordinates in 1..7, Boolean `obsWait`, and robot coordinates in 1..8, giving $7^2 \cdot 2 \cdot 8^2 = 6,272$ typed states. No source monitor, pattern or AUX field is declared.

The old initializer fixes obstacle $(7, 7)$, `obsWait=false`, robot $(1, 1)$. The update intentionally omits initializers and declares the switch robot=$(5, 5)$. For the ordinary Spectra game parser, exactly the 45-byte switch declaration is removed from the update; the remaining 1,902-byte source has hash `be0eaddf7b165df254d417dcb8cfcd962307f859f1ac3d26514c4a26c4b3954b`. The declaration remains boundary metadata, never an inserted initializer. Old-source projection is identity. Independent source-to-export checks cover domain, initializers, safety and ordered justices, including the current/next component interpretation of array predicates.

## AC.2 Region and initializer evidence

The four registered arms are A: source, B: released syntactic helper, C: release only the $0 \to 1$ SYS obligations, and D: existing complete semantic correction. C leaves the other quadrants and initializer/justice conditions of B unchanged. Both variants use the headless released Spectra/JTLV route described in the main paper; these are strict games, not the separate public Kind checks reported in AC.6 or a default GUI experiment. The original source initializers, including their absence in the new source, are preserved. Helper introduction binds initialization to $b = 0$.

| Source | Arm | $\lvert W_R(\cdot, 0)\rvert$ | $\lvert W_R(\cdot, 1)\rvert$ | Loss | Gain | Initial |
|---|---|---|---|---|---|---|
| old | A | 5,016 | – | 0 | 0 | true |
| old | B | 0 | 6,272 | 5,016 | 0 | false |
| old | C | 5,016 | 6,272 | 0 | 0 | true |
| old | D | 5,016 | 6,272 | 0 | 0 | true |
| new | A | 5,015 | – | 0 | 0 | true |
| new | B | 0 | 6,272 | 5,015 | 0 | false |
| new | C | 5,015 | 6,272 | 0 | 0 | true |
| new | D | 5,015 | 6,272 | 0 | 0 | true |

For A, the first count means source $|W|$ and no helper is present. Every helper-introducing arm has $U = D$. For B, the one-step entry condition is $F = \varnothing$, so $W_F = \varnothing$. For C and D, $F = D$ and $W_F = W$; every state membership, not just the cardinality, agrees with the independently computed source region. The difference between source counts 5,016 and 5,015 does not imply that the regions differ in only one state.

The immediate obstruction is already in both sources. For current robot $(x, y)$, choose next obstacle $(\min(x, 7), \min(y, 7))$. Its occupied $2 \times 2$ square includes the current robot. The source `robotAvoidsObstacle` clause compares the current robot against next obstacle coordinates; it contains no next SYS variable. Thus some avoidance conjunct is false for every response. Under B's checked first change, that input is legal. C removes exactly the conflicting response duty at that transition; its success here is not a universal sufficiency claim when retained helper-true constraints may give proper $U$.

Both source ENV relations are total on typed states and have zero ENV justice assumptions. After any legal prefix of a strict winner, its residual remains winning and ENV has a next legal input. Repeatedly choosing legal inputs supplies an infinite legal continuation; the empty ENV justice conjunction holds automatically. Hence $W \subseteq K$, and $K \subseteq W$ by definition: $K = W$ for both sources. Independently enumerating the authored initializer quantifiers gives $I_M(K) = I_M(W) = \mathit{true}$ in each, but false initialization for B. This argument establishes the paper's legal-prefix cooperation. Section AC.5 separately proves the stronger all-history property for these source contracts. Neither argument identifies the result of a raw four-fixed-point implementation.

### AC.3 A sealed old-only prefix and the complete switch roster

Before any source/compiler/solver calculation, the published switch robot=$(5,5)$ was completed with all 98 obstacle/`obsWait` valuations in fixed lexicographic order. No winning, reachability, safety or initializer filter changes that denominator. Source initialization is not reapplied at the replacement boundary.

Using only the old source and old $W$, a FIFO search with lexicographic ENV/SYS edges selected the following seven-edge legal prefix and sealed it before new-source evaluation. Every state is old-winning. The terminal state is also new-winning and belongs to new A/C/D, but not B. This establishes a possible old winning prefix; it does not establish an extracted controller, its private memory or an installed handoff.

| Step | Obstacle | `obsWait` | Robot | Source ID |
|---|---|---|---|---|
| 0 | $(7,7)$ | false | $(1,1)$ | 6144 |
| 1 | $(6,6)$ | true | $(1,2)$ | 5185 |
| 2 | $(6,6)$ | false | $(1,3)$ | 5122 |
| 3 | $(5,5)$ | true | $(1,4)$ | 4163 |
| 4 | $(5,5)$ | false | $(2,5)$ | 4108 |
| 5 | $(4,6)$ | true | $(3,4)$ | 3411 |
| 6 | $(4,6)$ | false | $(4,4)$ | 3355 |
| 7 | $(3,5)$ | true | $(5,5)$ | 2404 |

The complete boundary partitions into 48 old-W-prefix-reachable states winning in both sources, 30 states winning in both but not old-W-prefix reachable, and 20 states outside both winning sets. The 30 are not an old/new winning-region discrepancy. In each variant A/C/D accept the same 78 boundary states and B accepts zero. The obstruction is present before and after the update; these results do not show update-caused loss.

The following table lists every frozen completion. Robot is $(5,5)$ throughout. "W/ACD" means membership in both source regions and all six helper-false A/C/D source views; these are identical for these 98 rows. "Reach" means old-W-prefix reachability. All 98 helper-false B entries are false in both variants; all helper-true B/C/D entries are true. Thus the table and these constants specify all 784 source/helper-false arm memberships.

| No. | Obstacle | `obsWait` | Source ID | W/ACD | Reach |
|---|---|---|---|---|---|
| 1 | $(1,1)$ | 0 | 36 | 1 | 0 |
| 2 | $(1,1)$ | 1 | 100 | 1 | 0 |
| 3 | $(1,2)$ | 0 | 164 | 1 | 0 |
| 4 | $(1,2)$ | 1 | 228 | 1 | 0 |
| 5 | $(1,3)$ | 0 | 292 | 1 | 1 |
| 6 | $(1,3)$ | 1 | 356 | 1 | 0 |
| 7 | $(1,4)$ | 0 | 420 | 1 | 1 |
| 8 | $(1,4)$ | 1 | 484 | 1 | 0 |
| 9 | $(1,5)$ | 0 | 548 | 1 | 1 |
| 10 | $(1,5)$ | 1 | 612 | 1 | 0 |
| 11 | $(1,6)$ | 0 | 676 | 1 | 0 |
| 12 | $(1,6)$ | 1 | 740 | 1 | 0 |
| 13 | $(1,7)$ | 0 | 804 | 1 | 0 |
| 14 | $(1,7)$ | 1 | 868 | 1 | 0 |
| 15 | $(2,1)$ | 0 | 932 | 1 | 0 |
| 16 | $(2,1)$ | 1 | 996 | 1 | 0 |

| No. | Obstacle | `obsWait` | Source ID | W/ACD | Reach |
|---|---|---|---|---|---|
| 17 | (2, 2) | 0 | 1060 | 1 | 1 |
| 18 | (2, 2) | 1 | 1124 | 1 | 1 |
| 19 | (2, 3) | 0 | 1188 | 1 | 1 |
| 20 | (2, 3) | 1 | 1252 | 1 | 1 |
| 21 | (2, 4) | 0 | 1316 | 1 | 1 |
| 22 | (2, 4) | 1 | 1380 | 1 | 1 |
| 23 | (2, 5) | 0 | 1444 | 1 | 1 |
| 24 | (2, 5) | 1 | 1508 | 1 | 1 |
| 25 | (2, 6) | 0 | 1572 | 1 | 1 |
| 26 | (2, 6) | 1 | 1636 | 1 | 1 |
| 27 | (2, 7) | 0 | 1700 | 1 | 0 |
| 28 | (2, 7) | 1 | 1764 | 1 | 0 |
| 29 | (3, 1) | 0 | 1828 | 1 | 1 |
| 30 | (3, 1) | 1 | 1892 | 1 | 0 |
| 31 | (3, 2) | 0 | 1956 | 1 | 1 |
| 32 | (3, 2) | 1 | 2020 | 1 | 1 |
| 33 | (3, 3) | 0 | 2084 | 0 | 0 |
| 34 | (3, 3) | 1 | 2148 | 1 | 1 |
| 35 | (3, 4) | 0 | 2212 | 0 | 0 |
| 36 | (3, 4) | 1 | 2276 | 1 | 1 |
| 37 | (3, 5) | 0 | 2340 | 0 | 0 |
| 38 | (3, 5) | 1 | 2404 | 1 | 1 |
| 39 | (3, 6) | 0 | 2468 | 0 | 0 |
| 40 | (3, 6) | 1 | 2532 | 1 | 1 |
| 41 | (3, 7) | 0 | 2596 | 1 | 1 |
| 42 | (3, 7) | 1 | 2660 | 1 | 1 |
| 43 | (4, 1) | 0 | 2724 | 1 | 1 |
| 44 | (4, 1) | 1 | 2788 | 1 | 0 |
| 45 | (4, 2) | 0 | 2852 | 1 | 1 |
| 46 | (4, 2) | 1 | 2916 | 1 | 1 |
| 47 | (4, 3) | 0 | 2980 | 0 | 0 |
| 48 | (4, 3) | 1 | 3044 | 1 | 1 |
| 49 | (4, 4) | 0 | 3108 | 0 | 0 |
| 50 | (4, 4) | 1 | 3172 | 0 | 0 |
| 51 | (4, 5) | 0 | 3236 | 0 | 0 |
| 52 | (4, 5) | 1 | 3300 | 0 | 0 |
| 53 | (4, 6) | 0 | 3364 | 0 | 0 |
| 54 | (4, 6) | 1 | 3428 | 1 | 1 |
| 55 | (4, 7) | 0 | 3492 | 1 | 1 |
| 56 | (4, 7) | 1 | 3556 | 1 | 1 |
| 57 | (5, 1) | 0 | 3620 | 1 | 1 |
| 58 | (5, 1) | 1 | 3684 | 1 | 0 |
| 59 | (5, 2) | 0 | 3748 | 1 | 1 |
| 60 | (5, 2) | 1 | 3812 | 1 | 1 |
| 61 | (5, 3) | 0 | 3876 | 0 | 0 |
| 62 | (5, 3) | 1 | 3940 | 1 | 1 |
| 63 | (5, 4) | 0 | 4004 | 0 | 0 |
| 64 | (5, 4) | 1 | 4068 | 0 | 0 |
| 65 | (5, 5) | 0 | 4132 | 0 | 0 |
| 66 | (5, 5) | 1 | 4196 | 0 | 0 |
| 67 | (5, 6) | 0 | 4260 | 0 | 0 |
| 68 | (5, 6) | 1 | 4324 | 1 | 1 |

| No. | Obstacle | `obsWait` | Source ID | W/ACD | Reach |
|---|---|---|---|---|---|
| 69 | $(5,7)$ | 0 | 4388 | 1 | 1 |
| 70 | $(5,7)$ | 1 | 4452 | 1 | 1 |
| 71 | $(6,1)$ | 0 | 4516 | 1 | 0 |
| 72 | $(6,1)$ | 1 | 4580 | 1 | 0 |
| 73 | $(6,2)$ | 0 | 4644 | 1 | 1 |
| 74 | $(6,2)$ | 1 | 4708 | 1 | 1 |
| 75 | $(6,3)$ | 0 | 4772 | 0 | 0 |
| 76 | $(6,3)$ | 1 | 4836 | 1 | 1 |
| 77 | $(6,4)$ | 0 | 4900 | 0 | 0 |
| 78 | $(6,4)$ | 1 | 4964 | 1 | 1 |
| 79 | $(6,5)$ | 0 | 5028 | 0 | 0 |
| 80 | $(6,5)$ | 1 | 5092 | 1 | 1 |
| 81 | $(6,6)$ | 0 | 5156 | 0 | 0 |
| 82 | $(6,6)$ | 1 | 5220 | 1 | 1 |
| 83 | $(6,7)$ | 0 | 5284 | 1 | 0 |
| 84 | $(6,7)$ | 1 | 5348 | 1 | 0 |
| 85 | $(7,1)$ | 0 | 5412 | 1 | 0 |
| 86 | $(7,1)$ | 1 | 5476 | 1 | 0 |
| 87 | $(7,2)$ | 0 | 5540 | 1 | 0 |
| 88 | $(7,2)$ | 1 | 5604 | 1 | 0 |
| 89 | $(7,3)$ | 0 | 5668 | 1 | 1 |
| 90 | $(7,3)$ | 1 | 5732 | 1 | 1 |
| 91 | $(7,4)$ | 0 | 5796 | 1 | 1 |
| 92 | $(7,4)$ | 1 | 5860 | 1 | 1 |
| 93 | $(7,5)$ | 0 | 5924 | 1 | 1 |
| 94 | $(7,5)$ | 1 | 5988 | 1 | 1 |
| 95 | $(7,6)$ | 0 | 6052 | 1 | 0 |
| 96 | $(7,6)$ | 1 | 6116 | 1 | 0 |
| 97 | $(7,7)$ | 0 | 6180 | 1 | 0 |
| 98 | $(7,7)$ | 1 | 6244 | 1 | 0 |

### AC.4 Independent certificate, retained failures and limits

The evidence in this subsection is a distinct post-hoc theorem certificate for saved native regions. Section AC.7 adds later positive rank/response witnesses and ordinary-mode controls without changing these records. Independent source-equation winning regions were computed before the native candidates. Exact canonical-Boolean correspondence checks bind source predicates, typed fields, initializers, ordered justices, helper quadrants and immutable snapshot identities. The certificate proves $U = D$ from typed $S_{11}$ totality, computes one $\mathrm{CPre}_{01}(D)$ per helper arm, and uses only the sufficient endpoints $F = \varnothing$ or $F = D$. It never substitutes $W \cap F$ for $W_F$ at a proper $F$. All typed source/helper states and all Boolean current-state encodings are checked, including raw/typed/domain/helper-false export masks. Initializer enumeration uses one SYS response satisfying both its initializer and winning membership for each admissible initial ENV value.

All eight certificates pass. The method performs six one-step predecessor calls but no new native/source solve or fixed-point iteration. It was introduced after observing native candidates and resource caps, so it is not a preregistered replacement or an independently recomputed transformed fixed point. The eight original independent fixed-point attempts retain UNKNOWN at the fixed 500,000-node limit. A first distinct certificate attempt retains an old/A serialization exception

and seven not-attempted slots. A separately authorized amendment removes the duplicate output keyword and repeats the fixed eight certificates without changing the earlier evidence or its status. Independent pre-execution and result reviews check that separation and the interpretation of the saved records.

Saved records include `FINAL_CERTIFICATE_SUMMARY.json`, `FINAL_CERTIFICATE_TABLE.csv`, `ALL98_FIXED8_MEMBERSHIP.json`, `ALL98_FIXED8_MEMBERSHIP.csv`, and per-arm certificates, membership tables and process receipts. The frozen-input manifest, source/native outputs, original failures and all subsequent checks remain available alongside the fixed protocol. This is one published example pair, not prevalence, external reproduction, comparative performance, default frontend behavior or demonstrated deployment. No controller extraction or private-memory transfer is inferred from finite source-state prefixes.

### AC.5 Source strict winning and all-history kindness

The legal-prefix requirement used for $K$ is generally weaker than Kind's Definition 3 [7]. That definition additionally asks for an ENV-safe, ENV-fair continuation from every position of a play induced by the winning system strategy and an arbitrary ENV strategy, including histories after an ENV violation. The continuation uses the same residual system strategy. It does not reapply the source initializer or repair the already elapsed prefix. For the two Obstacle sources, the following argument establishes this stronger property.

**Lemma (total ENV safety without ENV justice).** Consider a finite Mealy source with nonempty typed domain $D = E \times Y$. Assume that $\rho_e(s, e')$ does not depend on the next SYS response and that every $s \in D$ admits at least one $e'$ satisfying $\rho_e(s, e')$. The latter condition is required on all typed states, not merely on $W$. Assume that the ENV justice list is empty, or contains only identically true padding. Let $S$ be a strict-winning source strategy. If initially partial, extend it to a strategy $\bar{S}$ returning a typed response on every typed history. The extension must preserve its initial response for every input satisfying $\theta_e$, and its responses and ordinary memory updates on histories where the original strict duties apply. For an already total $S$, take $\bar{S} = S$. Then $\bar{S}$ satisfies the additional all-history requirement of Kind's Definition 3. Consequently, strict-winning and kind strategy existence coincide for this source.

*Proof.* Choose a legal ENV selector $f : D \to E$ with $\rho_e(s, f(s))$ for every $s \in D$. Such a selector exists by finiteness and totality. Fix any history $h = (s_0, \ldots, s_i)$ induced by $\bar{S}$ and any ENV strategy. The history may contain an ENV initializer or safety violation. From position $i$ onward, choose $e_{t+1} = f(s_t)$ and let SYS use

$$y_{t+1} = \bar{S}_T(s_0, \ldots, s_i, \ldots, s_t, e_{t+1}).$$

This retains the full prefix and the strategy memory reached along it; it neither invokes the initial selector again nor resets memory. Next-response independence means that SYS cannot falsify the chosen ENV edge. Typed totality puts the successor back in $D$, so the same argument can repeat indefinitely. The resulting suffix is ENV-safe, and the empty or identically true ENV justice list makes it ENV-fair. The construction works from every induced history and position.

Strict realizability is preserved because the extension leaves the initial output and all responses carrying strict duties unchanged. An initially invalid ENV input disables those duties; after a first ENV safety violation, arbitrary typed completion creates no new strict SYS safety or justice obligation. This does not erase initial or pre-violation duties already incurred. Thus $\bar{S}$ remains strict-winning and satisfies the additional kindness condition. The converse follows because Definition 3 already requires strict winning. □

The source strategy functions in the primary definition are already total. The completion condition states what is needed when starting from a partial implementation. It does not require a safe SYS response at every typed state: after an ENV violation the strict source duties have been released. An implementation that instead stops there is not thereby certified as a typed-total strategy.

**A selector for both Obstacle sources.** Let $o = (o_0, o_1) \in \{1, \ldots, 7\}^2$ be the obstacle position, $p = (p_0, p_1) \in \{1, \ldots, 8\}^2$ the current robot position, and $w = \texttt{obsWait}$. If $w$ is true, keep $o' = o$ and set $w' = \mathit{false}$. Otherwise set $w' = \mathit{true}$ and, independently for each coordinate, choose

$$o'_j = \begin{cases} o_j - 1, & o_j > p_j, \\ o_j + 1, & o_j < p_j - 1, \\ o_j, & \text{otherwise.} \end{cases}$$

The decrement case has $o_j \geq 2$; the increment case has $o_j \leq 6$. Both stay within 1..7 and move by at most one. In the remaining case, $p_j - 1 \leq o_j \leq p_j$, so neither directed-motion antecedent applies. Together with the wait flag choice, these cases satisfy every ENV wait, motion and directed-motion clause. They use only current robot coordinates, not the next SYS response. This proves totality over all 6,272 typed states without assuming source winning or reachability. The old and new sources have the same ENV transition relation and empty ENV justice lists. Their initializers differ: the old ENV initializer fixes $o = (7, 7), w = \mathit{false}$; the new source has no ENV initial restriction.

**Application and boundary.** The existing strict source results therefore establish kind strategy existence for both authored initializers. The retained state $r = (o = (3, 5), w = \mathit{true}; p = (5, 5))$, source ID 2404, belongs to both source winning regions. The lemma also applies to a separately defined continuation query fixing that current state. This does not reset to the old initializer, construct an installed controller, or establish a private-memory transfer. The proof preserves the residual memory of the chosen strategy; it says nothing about an arbitrary existing controller memory.

This is a source-game argument. Helper conversion adds $\mathsf{GF}\neg b$: at $b = 1$, the one-way safety rule prevents any ENV-safe suffix satisfying that justice. The lemma therefore does not apply to the transformed game. It neither equates a raw four-fixed-point result $Q$ with source $W$ or $K$, nor establishes a frontend answer or an extracted controller's correctness. The strengthening above is a mathematical consequence of the existing source relations and strict results, not a new solver execution.

### AC.6 Two fixed public Kind realizability checks

The source-kind lemma permits a comparison beyond ordinary strict winning: both queries below admit a strategy satisfying the all-history source requirement of Kind's Definition 3. We separately tested the published tool's *Check Realizability* action on those fixed inputs. Both returned *Specification is unrealizable*. This is an observed answer from a selected public checking path, not a new strict-region certificate or a controller-handoff experiment.

**Queries and exact initial meaning.** The first query uses the unchanged 2,097-byte published old source from AC.1. Its ENV initializer fixes obstacle $(7, 7)$ and $obsWait = false$; its SYS initializer fixes robot $(1, 1)$. The joint state is source ID 6144 and belongs to the accepted old winning region. The ENV initial domain is nonempty, and the required initial response exists for its sole input. Thus the initialized existence claim retains the universal-ENV/existential-SYS initial quantifiers.

The second query uses the 1,902-byte ordinary projection of the update, followed by this fixed 124-byte appendix:

```
asm retainedBoundaryEnv:
obstacle[0] = 3 & obstacle[1] = 5 & obsWait;
gar retainedBoundarySys:
robot[0] = 5 & robot[1] = 5;
```

It fixes exactly the previously selected state $r = (o = (3, 5), w = true; p = (5, 5))$, source ID 2404, without conjoining the old initializer. The input is 2,026 bytes, with SHA-256 `212309df489c3ddc 7bf7ec6a750aed2d0da4fb7e26995e5ee3a0405c9897e1d5`. Its first 1,902 bytes are identical to the accepted update projection; all noninitial source clauses are unchanged. It is not the byte-unchanged public update, nor an execution of that file's switch declaration. The seven-edge old-only prefix in AC.3 fixed this endpoint before new-source evaluation. No input was changed in response to a GUI result.

The accepted new strict region contains $r$. Applying AC.5 to this separately initialized source query establishes a kind continuation. The lemma retains the full history and residual memory of the chosen strategy; it does not show that an arbitrary installed old controller's memory is suitable for the new strategy. No such old-controller memory was extracted or transferred.

**Binding the point query to the accepted predicates.** Before the second public action, a separate NoIDE export generated saved predicates for the fixed input without constructing a solver or computing a winning fixed point. A saved-data comparison checked the source clause inventory, full ENV/SYS safety relations, typed domains, all four helper-transition quadrants, ordered justices, and the helper-false source relation against the accepted records. The ENV and SYS initial predicates were each nonempty; their conjunction fixed exactly $r$ with helper false. The point initializer therefore does not obtain a result through an empty initial domain or an old-source restart.

The export supervisor nevertheless ended with `UNKNOWN_SUPERVISOR_EXCEPTION`: its final process snapshot raced with process exit. The OS return code was not recorded and is not inferred to be zero. That original status remains unchanged. Before the GUI action, a separate review adopted the saved mathematical objects after checking their structure, predicate equality and complete capture: 257 checks and 55 pinned files passed in that review. This was a saved-data acceptance, not a conversion of the supervisor record into PASS. It ran no new Java process, model generation, solver or fixed point.

The NoIDE exporter and Eclipse use the same pinned release translator but different resource providers and Java versions (17.0.19 and 17.0.9). No live GUI BDD model was exported. The

saved predicate correspondence is therefore not an observation that every internal GUI predicate is identical to the NoIDE export.

**Selected public path and completed observations.** Both actions used the installed Spectra release `1.0.0.202402072123`, Kind selected by `synthesisPitermanReduction`, JTLV, no optimizations and no reordering. The public command was `tau.smlab.syntech.checkRealAction`. The installed source and saved bytecode identify the static path: the shipped syntactic reduction precedes the default translators, and the recorded JTLV branch invokes `GR1SeparatedGame`. Both actual consoles displayed that class and `JTLVJavaFactory 1.3`. The separate complete-correction class represented by $R^+$ is not selected on this path. Section AE.1 distinguishes this static fact from live-model equality and runtime class origins.

| Fixed query | Source existence evidence | Public observation |
|---|---|---|
| Old source, authored initialization | Strict source certificate plus AC.5 | Unrealizable; 1,011 ms |
| Updated projection, point initialization at $r = 2404$ | Strict source certificate at $r$ plus AC.5 | Unrealizable; 745 ms |

Each input received one public action, with no retry and no controller extraction. The times are plugin diagnostic computation times, not a performance benchmark. Exact input snapshots, final settings, full console screenshots, action logs, release-file checks and completion records bind the observations. Separate read-only completion audits found no discrepancy.

The old-input logger has one start/done pair and two identical source snapshots. The retained-query logger has one start/done pair in one XML file and one archived source snapshot; its same-second storage does not provide two independently retained snapshots. That snapshot and the prepared/current input all match the fixed hash above. Both GUI runs completed without a recorded exception or resource-stop signal. These GUI completion records are separate from the earlier export supervisor's UNKNOWN and do not alter it. Console text is a visual transcription of the unedited screenshots, not text obtained from the accessibility API.

**Decision consequence and limits.** For the old original initialization and for the fixed updated point query, source-kind strategy existence is established independently of the observed Boolean answers. If the selected public path is used as a source-level continuation-feasibility gate at $r$, its false answer would reject a continuation whose existence is separately proved. This comparison does not rely only on the weaker legal-prefix set $K$. The finite old prefix identifies a possible reached source valuation; it does not establish a running old controller, its private memory, an installed update, or an actual deployment rejecting the candidate.

The old query was already negative, so the update is not shown to introduce the obstruction. The two observations neither export an all-state raw four-fixed-point region $Q$ nor identify $Q$ with $W$ or $K$. The headless correction certificates remain separate evidence for the strict-game mechanism; the GUI observations do not isolate one internal defect as the exclusive cause. They establish neither default or all-backend behavior, incidence, deployment benefit, nor tool-author intent. The eight earlier independent fixed-point UNKNOWNs and the certificate-processing exception with seven then-unattempted slots in AC.4 remain unchanged.

## AC.7 Direct-source strategy certificates and matched public controls

**Scope and exact queries.** This additional study supplies explicit positive strategy certificates for the same two initialized queries used in AC.6, and ordinary-GR(1) public controls on the same input bytes. These are separate from the earlier saved-region/theorem certificates in AC.4. Query A is the unchanged 2,097-byte published old source, initialized at $(o = (7, 7), w = \mathit{false}; p = (1, 1))$. Query B is the 2,026-byte retained-point update query, initialized at $(o = (3, 5), w = \mathit{true}; p = (5, 5))$. Both start the certificate strategy in phase zero.

The checker binds the exact source length, SHA-256, case name, explicit query tuple and initial tuple. Its separate source-binding check also confirms the byte identities: A equals published old; the 1,947-byte published update equals the 1,902-byte ordinary projection plus its 45-byte switch suffix; B equals that projection plus the fixed 124-byte point initializer. Thus B preserves all noninitial source clauses and does not conjoin the old initializer. These identities do not execute the switch declaration or prove a controller handoff.

**An independently implemented source checker.** The checker imports only Python standard-library modules. It transcribes the original source equations directly and imports no generator module, old source evaluator, predecessor or fixed-point routine, saved graph, or candidate-winning-set truth. Certificate provenance is informational, not a proof premise. Its independence is from those implementations; the mathematical argument still depends on the correctness of the source transcription, the adopted declaration semantics, the checker and its runtime. This is not an independent general-purpose Spectra compiler.

The explicit state tuple is $(o_0, o_1, w, p_0, p_1)$, with obstacle coordinates in 1..7, a Boolean wait flag, and robot coordinates in 1..8. There are 98 typed ENV valuations, 64 typed SYS valuations and 6,272 typed states. Integer and Boolean types are checked separately. The ENV predicate is the complete conjunction of wait alternation, waiting-position persistence, obstacle movement by at most one per coordinate, and the four directed-motion implications listed in AC.5.

For SYS, the checker verifies current-state collision avoidance and the two authored forbidden-location safety clauses: $(3, 5), (4, 5)$ for A, and $(6, 4), (8, 7)$ for B. The declarations named `AvoidInf initely` are globally enforced safety clauses, despite their names. Transition checks limit each robot coordinate's change to at most one and preserve `robotAvoidsObstacle`, which tests the *current* robot against the *next* obstacle. Closure also checks successor current-state safety, hence the next robot against the next obstacle. These are distinct obligations. The two recurring SYS goals, in source order, are $(5, 2), (2, 3)$ for A and $(3, 7), (4, 8)$ for B. Both ENV justice lists are empty.

**Certificate obligations and the winning argument.** A certificate supplies a nonempty set $R$ of explicit typed states, one bit of goal phase $q \in \{0, 1\}$, two integer rank functions, and one SYS response for each legal ENV input at each state and phase. Ranks lie in 0..6271; only finite nonnegative values are needed for the well-foundedness argument. Write $G_q$ for the current phase's source goal. The checker requires $\mathrm{rank}_q(s) = 0$ exactly when $s \in G_q$.

For each $s \in R$ and each phase, the checker independently enumerates all 98 typed next ENV tuples. The supplied response keys must equal the entire set satisfying the source ENV predicate: missing, duplicate or extra inputs are rejected. This legal-input set must be nonempty. Every response is typed, satisfies the complete source SYS transition relation, and reaches a state $s' \in R$. Outside $G_q$, it must retain phase $q$ and satisfy

$$\mathrm{rank}_q(s') < \mathrm{rank}_q(s).$$

At $G_q$, the next phase must be $1 - q$; no descent is required across that phase change.

Consequently, every source-legal play stays safe in $R$. A phase cannot avoid its goal forever, because that would produce an infinite strictly decreasing sequence of nonnegative integers. The next step changes phase, and the same argument repeats, giving infinitely many visits to both source goals. This covers every legal ENV branch, not a selected execution or merely one cycle.

Initialization is checked separately over all 98 ENV assignments and, for each allowed initial ENV value, all 64 SYS assignments. Every allowed ENV value must have a source-initial SYS witness in $R$. Both exact queries have one admissible initial pair, present with phase zero. The accepted results are:

| Query | $\lvert R\rvert$ | Policy edges over two phases | Maximum rank | Initial pairs | Checker result |
|---|---|---|---|---|---|
| A | 5,016 | 12,656 | 17 | 1 | accept |
| B | 5,015 | 12,672 | 17 | 1 | accept |

Both receipts report `ACCEPT_POSITIVE_EXISTENCE`. They certify positive strategy existence for the exact queries and the checked strategy on $R$. They do not prove that $R$ is a maximal winning region or exclude any state outside it. The coincidence of these cardinalities with the earlier region counts does not change that scope; complete-region claims retain their earlier computational premises.

**Typed completion and the separate all-history bridge.** The strategy has an ENV-violation latch in addition to the goal phase. Both are private strategy memory, not additional source variables. On an invalid ENV initial value or after the first ENV safety violation, it latches the violation and forever returns typed SYS value $(1, 1)$ with phase zero; it does not resume the certificate when ENV later becomes legal. Otherwise unreachable histories use the same specified fallback. The checked policy applies throughout every induced legal prefix, preserving valid initial outputs and all responses and ordinary memory updates carrying strict duties. Completion creates no new SYS safety obligation after an ENV violation and erases none already owed.

The positive winning check does *not* itself prove global ENV totality. A separate check, bound to both exact sources, enumerates all 6,272 typed states and all 98 next ENV tuples at each state, including states outside $R$ and states reachable by fallback. It reports `ACCEPT_SOURCE_ENV_TOTALITY_ONLY`: 9,536 legal ENV edges, with minimum outdegree one and maximum nine. The source ENV relation is independent of the next SYS response and has no justice assumptions.

Combining this totality result with the typed-total certificate strategy and AC.5 gives the source-specific all-history Kind property: after any induced history, repeatedly select a legal ENV input while retaining the same strategy, full prefix and reached phase/latch memory. Typed successors remain in the domain, so this yields an ENV-safe infinite suffix; empty ENV justice makes it fair. Initialization is not restarted. The strict positive certificate and the ENV-totality argument are logically separate premises. Neither supplies a general Kind-preservation theorem or transfers arbitrary memory from an installed old controller.

**Fixed malformed-certificate qualification.** Two unmodified positive controls were accepted. For each source, the frozen qualification roster supplied 25 malformed certificates; all 50 were rejected with their expected reason codes. The roster covers:

| Group | Mutations | Per source |
|---|---|---|
| ENV coverage | Missing input, extra illegal input, duplicate input | 3 |
| Safety and closure | Unsafe robot step, unsafe current state, missing successor | 3 |
| Ranks | Negative, oversized or Boolean rank; zero off goal; no decrease | 5 |
| Phase | Wrong phase, phase stuck at a goal, missing phase policy | 3 |
| Query binding | Wrong query, case, source hash or source path; initial state missing | 5 |
| Value types | Integer wait value, Boolean SYS coordinate | 2 |
| State and schema | Duplicate state, extra metadata key, duplicate JSON key | 3 |
| Completion | Wrong completion declaration | 1 |

These checks measure observed rejection sensitivity of this fixed roster, not complete checker correctness and not negative source realizability.

Forty-eight rejection receipts directly contain the checked input hash. The two duplicate-JSON-key inputs fail during parsing before the input hash is returned to the receipt, leaving `certificate_sha256` null. Their retained compressed inputs were checked against the exact frozen mutation construction and the qualification driver's archive hashes; the saved receipts and output report `DUPLICATE_JSON_KEY`. Thus those two rejections have an explicit driver/input provenance chain, not a populated receipt input-hash field.

Six frozen supervised stages completed once each: the two generations, the two positive checks, the separate totality/binding check, and qualification. Each positive-check stage and the totality/binding stage had 180-second wall and per-process CPU budgets. The qualification stage had 300-second wall and per-process CPU budgets. The owned-process supervisor monitored the process tree against a 1-GiB sampled RSS budget. RSS samples, particularly for short stages with only two samples, do not establish instantaneous peak memory or a continuous hard RSS bound. These records support no performance claim.

**Matched public ordinary-GR(1) controls.** Using the same A/B bytes and installed release as AC.6, the public *Check Realizability* action was run once per query in *Original GR(1)* mode, with JTLV, optimizations disabled and reordering disabled. Both answers were *Specification is realizable*. The previous selected Kind-mode answers remain false:

| Query | Ordinary GR(1) | Previous Kind | Ordinary console time |
|---|---|---|---|
| A | true, one action | false, one action | 10,391 ms |
| B | true, one action | false, one action | 14,221 ms |

The times are diagnostic provenance, not a benchmark. The ordinary consoles identify `GR1GameMemoryless`, `JTLVJavaFactory 1.3`, and the diagnostics `env 7 / sys 6 / aux 0`. Exact source archives, one start/done XML pair per query, screenshots and owned-process monitoring bind the observations. The preference UI, installed default-initializer bytecode and runtime console establish ordinary-mode selection; the serialized `synthesisMethod` preference is absent because ordinary GR(1) is the installed default.

This is a matched-input mode contrast. Mode selection changes both reduction and solver, so the contrast does not isolate one exclusive internal cause. No live GUI intermediate BDD or predicate export was obtained. The installed-code mapping and saved source correspondence are distinct from observing those live intermediate predicates. A was already negative in Kind mode; the update is not shown to introduce that obstruction.

**Evidence identity and retained limits.** The checker file `check_certificate.py` has SHA-256 `9618f0cbbe7dafb92f963e200f2304f802629aef248f563de0e039eda23ce28d`; the qualification driver `qualify_rejections.py` has SHA-256 `4ff6c3beb4e04050e87b0c401ae6bf6fc90a6bfcd0bb7b4a10ffad8c7f5be4cf`. The original execution-input A and B certificate hashes are respectively `25a6412edbf3ef5a3024eb2b0e3854a688572417c867b5e7af13342b2e07dc60` and `d9ccee3264c87f09cd3b71ee981bb6fc925eb3dc18c4259ffd9207fbd9dc7641`. The receipts bind the original execution inputs and exact sources. The public certificates change only local provenance metadata; their different byte hashes are listed separately in the public manifest and transformation ledger. The hashes above identify the original inputs.

The fifth package in the current-evidence index contains the positive certificates, checker, qualification records, ordinary controls and prior Kind results at the fixed document release. It omits upstream source bodies, raw screenshots and AX captures, supplying restoration pins and recipes, authored console transcriptions and logger XML. Its guide separates validation status from the original scientific executions. A later existing-host replay of the exact public v1 inputs rechecked A/B, all-typed ENV totality and the fixed controls. Section AE.2 gives its adopted scope and distinguishes the v1 inputs from the later v2 records.

The eight earlier independent exported-game fixed-point attempts remain UNKNOWN. The new certificates prove existence, not maximal regions or new negative-game results. Public controls establish neither raw four-fixed-point regions nor corpus incidence, installed replacement, deployed rejection or arbitrary private-memory handoff. The finite old prefix, old native evidence and failure statuses are unchanged.

# AD. All-history source-kind completions

This section strengthens two source-strategy existence statements without changing any saved winning region or execution record. For Vehicle, a universal fair input extends each target-specific initialized policy in Section X. For Patrol, a specified completion extends the *new-source continuation* at the exact retained entry and reached private phase in Sections S and U. The latter completion is a mathematical construction; it has not been implemented, executed or mechanically qualified.

## AD.1 All-history cooperation keeps the same strategy

Let a complete source state be $s = (e, y)$ in the nonempty typed carrier $D = E \times Y$. ENV offers the next input $e'$ before SYS chooses $y'$. A typed-total strategy supplies a typed response on every typed input history, including histories after an ENV assumption violation.

Kind's Definition 3 [7, Def. 3, p. 4] requires a strict-winning system strategy $S$ to satisfy an additional condition: for every ENV strategy and every position $i$ of the induced play, alternative future ENV choices can produce an infinite suffix from the reached state $s_i$ satisfying

$$\mathsf{G}\rho_e \ \wedge \ \bigwedge_n \mathsf{GF} J_n^e.$$

There is no ENV initializer $\theta_e$ in this suffix formula. The strategy functions in [7, p. 3] retain the history: at a future time $t$ the response is $S_T(s_0, \ldots, s_i, \ldots, s_t, e_{t+1})$. A finite-memory representation therefore uses its actual reached memory and the same strategy's specified updates. Replacing future ENV behavior neither reapplies initialization nor discards source monitors or private memory.

For clarity, the separate strict requirement of [7, Def. 1] is

$$\begin{aligned}(\theta_e \Rightarrow \theta_s) \ \wedge \ \big(\theta_e \Rightarrow \mathsf{G}((\mathsf{H}\rho_e) \Rightarrow \rho_s)\big) \\ \wedge \ \left[\left(\theta_e \wedge \mathsf{G}\rho_e \wedge \bigwedge_n \mathsf{GF} J_n^e\right) \Rightarrow \bigwedge_j \mathsf{GF} J_j^s\right].\end{aligned}$$

The historical operator $\mathsf{H}$ includes the current edge. At the first ENV-safety violation its antecedent becomes false and remains false thereafter; the whole-play liveness antecedent containing $\mathsf{G}\rho_e$ is also false. If $\theta_e$ failed initially, its guarded duties were absent from the start. A later locally legal suffix does not repair that history or restore those antecedents. This permits strict-preserving completion outside assumption compliance. It does not make an arbitrary absorbing completion kind.

The legal-prefix set $K$ used elsewhere in this supplement concerns only source-legal prefixes. The all-history condition also concerns positions after ENV violations. The following implications are source-specific; they do not assert a general equality between these notions or characterize a raw four-fixed-point solver region.

## AD.2 A universal fair-input lemma

**Lemma AD.1** (Universal fair input)**.** *Assume that $\rho_e(s, e')$ is independent of the next SYS response. Suppose some typed ENV value $e_*$ satisfies*

$$\forall s \in D : \rho_e(s, e_*), \qquad \forall y \in Y \ \forall n : J_n^e(e_*, y).$$

*Then every typed-total strict-winning source strategy is Kind. In particular, any typed-total completion of a strict-winning policy that preserves its strict obligations has the all-history property.*

*Proof.* Fix the completed strategy $S$, an arbitrary ENV strategy and any induced position $i$. The current state need not lie in the source-winning region or follow a legal prefix. Keep the full history and $S$'s reached memory, and offer $e_*$ forever from the next input onward. The first edge is ENV-legal from the actual $s_i$, independently of SYS's response. Totality supplies a typed successor in $D$, so the same argument applies inductively. The resulting suffix is infinite and satisfies $\mathsf{G}\rho_e$. Every ENV justice holds at every state from the first successor onward, independently of the SYS output, hence every $\mathsf{GF}J_n^e$ holds even when $J_n^e$ was false at $s_i$. This is exactly Definition 3's additional condition. Strict winning is the separate premise, retained by the stipulated completion. No restoration of SYS fairness after an earlier ENV violation is used. □

**General selector form relocated from the main paper.** The following state-dependent form and its proof were in the previous main paper. They are retained here without changing the special-case lemmas or their historical numbers.

**Lemma** (Fair ENV selector)**.** *Suppose $f : D \to E$ is ENV-legal at every typed source state: $\rho_e(s, f(s))$. From every typed starting state, suppose every infinite path choosing $e' = f(s)$, with arbitrary typed SYS responses, satisfies all ENV justices. Then every typed-total strict-winning strategy for the source query is Kind.*

*Proof.* Fix such a strategy, any induced history and a position in it. Keep the full prefix and reached memory, and choose $f(s)$ thereafter. The first edge is ENV-legal independently of SYS's response. Typed totality leaves a state in $D$, so this construction continues indefinitely. The selector premise makes the suffix ENV-fair. The same SYS strategy therefore satisfies the all-history condition; strict winning remains its separate premise. No initializer is reapplied. □

With empty ENV justice, all-domain ENV totality supplies such a selector. A universally legal constant input satisfying all ENV justices independently of SYS is another instance. These are the existing arguments in Supplements AC.5 and AD.2. A partial winning policy must first be completed without changing initializer responses or response/memory updates that still carry strict duties. Patrol instead used a strategy-specific completion; the lemma is not being applied to arbitrary Patrol strategies.

## AD.3 Vehicle: initialized Kind reachability for each target

The unchanged Vehicle source in Sections E and X has ten Boolean ENV fields, a three-valued lane and speed in 0..10, with no logical auxiliary fields or explicit initializer [2, 14]. Write $o_i, p_i, g_i$ for obstacle, pedestrian and green-light inputs for lane $i$, and $x$ for the crosswalk input. Its complete ENV safety relation is

$$\rho_e = (o_1 \Rightarrow \neg o_2') \wedge (o_2 \Rightarrow (\neg o_1' \vee \neg o_3')) \wedge (o_3 \Rightarrow \neg o_2'),$$

and its three ENV justice predicates are $g_i \wedge \neg p_i \wedge \neg o_i$. The all-clear value

$$e_{\text{clear}} = (o, p, g, x) = (000, 000, 111, 0)$$

makes every consequent in $\rho_e$ true from *every typed current state* and makes all three justices true for every typed SYS output. Lemma AD.1 applies on the full source carrier, including states outside $W$. The complete source-predicate correspondence and support checks in E and X, and the existing all-clear check over all 1,024 current ENV values, bind these equations to the saved study. This argument does not reuse Obstacle's empty-ENV-justice premise.

For each saved source-winning target $s$, Section X supplies the predecessor

$$p_s = (e_{\text{clear}}; \text{lane}(s), 1)$$

that is initially allowed and in $W$, with an original-legal, original-safe edge to $s$. Its initial-fiber check also supplies a winning SYS response for every allowed initial ENV value. Form the target-specific strategy $S_s$ as follows. On initial $e_{\text{clear}}$ choose $p_s$. On the next input $e(s)$ take the certified edge and follow a winning continuation from $s$. On other next inputs use a winning strategy from $p_s$; on other initial ENV inputs choose their certified winning response and continuation. Every branch has a winning tail, and the finite designated safe prefix does not change a Büchi objective. Thus the initialized strategy is strict winning.

Complete $S_s$ on ENV-illegal histories without changing its valid initializer responses or any response and ordinary memory update still carrying a strict duty. Such a typed-total completion exists: after the first ENV violation choose typed lane and speed values, preserving the entire legal behavior; there is no source AUX update to override. Lemma AD.1 makes the completed strategy Kind. In particular, its designated legal edge is unchanged. The exact conclusion is

$$\forall s \in W \quad \exists \text{ initialized Kind strategy } S_s \quad \exists \text{ compatible legal play } \sigma : \quad \sigma_1 = s.$$

Here membership in the saved $W$ remains the computational premise described in Section X. Its 8,064 states include the 1,408 saved helper-false losses. The strategy and play may depend on $s$; this is neither one shared strategy visiting all targets nor forced reachability against every ENV choice. The proof adds no native computation, maximality certificate or controller execution. The saved solver/exporter/type-map dependencies and outcome-based case selection are unchanged.

## AD.4 Patrol source equations and the retained seed

Use the *new* public Patrol source [9] with checkpoint positions $t_1, t_2, t_3$, blocked flags $d_1, d_2, d_3$, charged flag $z$, robot position $p$, charging counter $c$ and request monitor $m$. Position coordinates range over $\{0, \dots, 4\}$, $c \in \{0, 1, 2, 3\}$ and $m \in \{S0, S1, S2\}$. Let $H = (4, 0)$ and $\text{near}(p, t) \equiv \|p - t\|_\infty \le 1$. The complete new-source ENV relation is

$$\rho_e(s, e') = \bigwedge_{i=1}^{3} (t_i' = t_i) \wedge \bigwedge_{i=1}^{3} \big(\text{near}(p, t_i) \Rightarrow d_i' = d_i\big) \wedge (c = 3 \Rightarrow z'),$$

with the three ENV justices $\mathsf{GF}\neg d_i$. The deterministic source updates are

$$F_c(s) = \begin{cases} \min(3, c+1), & p = H, \\ 0, & p \neq H, \end{cases}$$

and, writing $T \equiv \neg z \wedge c \neq 3 \wedge p \neq H$,

$$F_m(s) = \begin{cases} S2, & m = S0 \wedge T, \\ S0, & m = S0 \wedge \neg T, \\ S1, & m = S1, \\ S1, & m = S2 \wedge c = 3, \\ S0, & m = S2 \wedge c \neq 3 \wedge p = H, \\ S2, & m = S2 \wedge c \neq 3 \wedge p \neq H. \end{cases}$$

These are unique typed values for every typed *current* state. They do not depend on the newly offered ENV value or on the selected next robot position. In particular, the $c = 3$ test has priority in state $S2$, and $S1$ remains absorbing.

The retained seed is

$$\begin{aligned} r: \quad & (t_1, t_2, t_3) = ((0,4),(2,4),(4,4)), \\ & (d_1, d_2, d_3) = (0,0,0),\ z = 0, \quad p = (0,0),\ c = 0,\ m = S2, \qquad \mu_r = 0. \end{aligned}$$

Here $\mu_r$ is the private route phase actually reached at the boundary. Sections S and U certify a strict-winning, legal-prefix cooperative continuation at this seed, with the same phase updates. Its complete new-source legal graph has 849 states and 12,240 edges: legal-input coverage, safety, strict fairness, private-phase identity and an ENV-fair continuation from every graph state are checked. Section U.1 binds the source rules and AUX updates to the native predicates. We use these existing facts as premises; no graph is recomputed here. The source monitor remains $S2$ at $r$ and is not reset to its original initializer $S0$.

### AD.5 A completion for the new Patrol continuation

**Lemma AD.2** (Patrol completion at the retained seed)**.** *The certified new-source policy from $(r, \mu_r)$ has a typed-total completion that is strict winning and satisfies Definition 3's all-history ENV-suffix condition at every position induced from that seed. It preserves every legal-history output and private-phase update, and the exact counter and monitor update equations on every transition.*

*Proof.* Let $Q = \{(0,0),(0,4),(4,0),(4,4)\}$. A checkpoint can be near at most one member of $Q$: distinct corners have Chebyshev distance four, whereas a point near both would bound that distance by two. Three checkpoints therefore leave at least one corner outside all three neighborhoods. Let $q(t_1, t_2, t_3)$ select the first such corner in a fixed order. This is a total deterministic function for every typed layout, including coincident checkpoints and layouts violating the original initializer.

Keep the original private route memory $\mu$ and add a private Boolean $\beta$, initially false at the retained seed. Equivalently, $\beta$ records an actual ENV-safety violation in the complete history. It is strategy memory, not a logical source field or the reduction's helper. Given current $s$ and the offered typed input $e'$, set

$$\beta' = \beta \vee \neg\rho_e(s, e').$$

If $\beta' = 0$, use exactly the certified policy's response and private-phase update. If $\beta' = 1$, choose

$$p' = q(t_1', t_2', t_3'), \qquad c' = F_c(s), \qquad m' = F_m(s), \qquad \mu' = \mu.$$

Once true, $\beta$ remains true; the original phase is frozen, never reset or resumed at an invented phase. The rule is defined for every typed input, including repeated checkpoint relocations. Every resulting state with $\beta = 1$ has $p = q(t_1, t_2, t_3)$, outside all current checkpoint neighborhoods. The current-state AUX equations remain exact, including an absorbing $S1$.

Before any actual ENV violation, the completion is precisely the certified strict-winning policy. At the first violating edge and thereafter, the historical strict-safety antecedent and the whole-play liveness antecedent containing $\mathsf{G}\rho_e$ are false. The possibly nonlocal response to $q$ is used only on those edges. It does not release movement, dwell or avoidance duties on a source-legal prefix. All earlier required responses are unchanged, and the AUX equations are preserved even after strict duties are released. Thus the completed strategy is strict winning from the seed.

Now fix an arbitrary induced history and position $i$ and keep this same completed strategy and its reached memory $(\mu_i, \beta_i)$. If $\beta_i = 0$, the position belongs to a legal prefix of the certified policy.

Its legal-prefix cooperation supplies an ENV-safe, ENV-fair suffix from precisely that reached phase. Along this suffix the completion follows the same policy and $\beta$ remains false.

If $\beta_i = 1$, retain all checkpoint coordinates and choose all next blocked flags false and the next charged flag true, forever. On the first edge the neighborhood-freeze antecedents are false, coordinate persistence holds, and $c = 3 \Rightarrow z'$ holds. Because the layout is retained, the next robot position is the same selected corner. Induction gives an infinite ENV-safe suffix. Every state from its first successor onward satisfies all three $\neg d_i$, so all ENV justices hold. This works for every reached $c, m$ and private phase: $z' = 1$ handles $c = 3$, and no ENV condition constrains $m$. The same strategy keeps $\beta = 1$ and the reached phase. No source initializer, monitor initializer or private-memory initializer is reapplied when ENV changes its future choices. The suffix need not restore SYS fairness after the past violation and does not make that past legal. Both cases establish Definition 3's condition. □

## AD.6 Initialization, execution and proof-review limits

Lemma AD.2 concerns a continuation from the actual retained state and reached phase. An initialized mathematical form uses the explicit point-continuation predicates

$$\theta_e^r(e) \equiv e = e_r, \qquad \theta_s^r(e, y) \equiv y = y_r$$

and the original *new-source* transition and justice relations. For the matching ENV initial value choose the seeded output and $\mu_r$ with $\beta = 0$. For any other typed ENV initial value $\theta_e^r$ is false: choose $p = q(t_1, t_2, t_3)$, retain $c_r, m_r, \mu_r$, and set $\beta = 1$. Its point-initializer-guarded strict duties are absent and the bad-initialization branch has the same fair suffix proved above. This is the initialized form of the same continuation-existence statement, not reapplication of the public source initializer or a runtime reset.

The public new initializer admits other checkpoint layouts; the certificate at $r$ does not provide winning responses for all of them. We assert neither full original-initializer Kind realizability of that source, nor old-source initialized all-history Kind, nor kindness of arbitrary sinks. For example, a constant-position sink can remain near a blocked checkpoint forever, forcing its blocked flag to stay true on every legal ENV suffix; then its $\mathsf{GF}\neg d_i$ assumption cannot hold. Exact AUX updates alone do not remove that obstruction. This example does not diagnose the completion of an uninspected executable controller.

The new $\beta/q$ completion has not been implemented, executed or mechanically qualified. The recorded graph and software handoff certify the original policy's legal behavior, not execution of this new completion. Its nonlocal move after an ENV violation is formal off-assumption behavior, not a physical-recovery claim. Vehicle likewise retains its target-dependent existential quantifiers and its saved-region trust boundaries. Neither proof supplies a transformed-game Kind identity, a raw-four-fixed-point or public-GUI answer, a new native region, a maximality certificate, an incidence estimate or a resolution of any recorded UNKNOWN.

These source-level arguments were checked in the coordinating Codex review and a separate Codex proof review against the primary Kind definition and source equations. They are proof-only additions, not proof-assistant verification, independent human review or new scientific execution.

# AE. Public-path correspondence and exact-input replay

## AE.1 What the selected public Kind path establishes

Kind's printed algorithm applies four fixed points to a reduced specification [7, Sec. 4.1, p. 4]. Its authored safety guards are $\rho_e \vee b'$ and $\rho_s \vee b$. On the first change $b = 0, b' = 1$, these formulas release ENV safety but retain SYS safety, although the accompanying prose describes release from the change onward. This timing deduction concerns the printed formulas; it does not identify the strict region of the reduced game with the final cooperative region.

The installed source and saved bytecode evidence for release `1.0.0.202402072123` identifies the selected public action more precisely than its preference label alone. The `Check Realizability` action tests `synthesisPitermanReduction`, invokes the shipped `PitermanSyntacticReduction` on the game input, and then applies the default translators before constructing the game model. For the recorded JTLV configuration, the job invokes `GR1SeparatedGame`; the exact A/B consoles corroborate that solver branch. The default translation list includes state-invariant, pattern, monitor, counter and other normalizing translations.

The separate shipped `PitermanSemanticReduction` class implements the complete-relation correction whose SYS relation is $\rho_s \vee b \vee b'$ [13]. It is not selected on this identified static action path. The syntactic class acts on authored constraints before the default translators; therefore a particular complete exported relation still needs its own gate for types, logical auxiliaries, initialization and ordered justices. The case-specific gates in AC.4 and AC.6 provide that separate saved-model evidence. The presence of a correction class in a release does not establish that a public preference calls it.

The adopted static clarification rechecked the retained release/source bindings: 20 installed JARs, 16 pinned members, six bundled source copies and nine saved disassemblies matched their pins. A bounded constant-pool inspection of 690 classes found the public action's direct reference to the syntactic class, with no direct reference to the separate semantic class in that set. This last absence does not exclude reflection, another bundle or runtime class substitution. Static control flow is not a historical trace of actual class-definition origins.

No intermediate *live GUI* game model was captured for the historical A/B actions. The separately saved NoIDE predicates therefore are not retroactive observations of that GUI object. Both paths use the pinned Spectra input translator but have distinct resource-provider initialization and recorded Java versions. The ordinary control in AC.7 changes both reduction and solver: its observed `GR1GameMemoryless` differs from the selected Kind path. Consequently the mode contrast does not isolate an exclusive cause of the public negative answers.

These distinctions also constrain the negative-answer contract. The complete literal characterization and $W \subseteq C$ certify *strict-region* preservation for the stated encoding. They do not alone certify the soundness of a negative public `GR1SeparatedGame`/Kind answer for either strict or source-kind existence. A contract for a particular oracle must state its returned region, the initialization quantifier and the required semantic correspondence. The job's final quantified initial test does not by itself identify its raw four-fixed-point region $Q$ with $W$, legal-prefix $K$ or all-history Kind. The source-kind positive premises for exact A/B remain the independently checked certificates and source-specific bridge in AC.5–AC.7. The headless same-*strict*-solver comparisons in AC.2 remain a distinct experiment. Section AF preserves the historical exact-B false/UNKNOWN diagnostic. Section AG reports a distinct completed pair with the same observer and fixed raw four-fixed-point solver: unchanged false, sole typed-$S_{01}$ release true. This local intervention effect leaves the historical public-action cause unresolved.

## AE.2 Recorded replay of exact public v1 inputs

After the original AC.7 stages, the coordinating task performed one recorded restoration and the four planned checker stages on the existing host, using the exact public v1 checker, qualifier, certificates and fixed control bytes. This is a later scoped scientific recheck of those inputs. The adopted outcomes were:

| Stage | Accepted scope |
|---|---|
| Positive A | Positive strategy existence; 5,016 certificate states and 12,656 policy edges. |
| Positive B | Positive strategy existence; 5,015 certificate states and 12,672 policy edges. |
| ENV totality / binding | All 6,272 typed states and 98 candidate ENV moves per state; 9,536 legal edges, outdegree 1–9; exact source binding. |
| Fixed qualification | Two unmodified positive controls accepted and all 50 predetermined malformed inputs rejected with the expected codes. |

A separate record audit passed 1,455 of 1,455 checks. The 50 malformed-input rejections measure the fixed checker's observed sensitivity, not negative source realizability. The earlier scientific records, original certificates, proof text, scientific programs and controls remain unchanged.

**Which bytes were replayed.** The replayed v1 ZIP has 2,347,279 bytes and SHA-256

`e1359c08454430ff19c8af2a8322a426d089b8a1c0cd180a2c631f267782ac8b`.

Its 312 payload files are identified in `replay/input_identity/V1_PAYLOAD_MAP.json` of the later record package. Five source inputs were restored once from the already-local official ZIP using the exact member selection, 45-byte projection and 124-byte point-initializer recipe. All five lengths and hashes matched the frozen inputs. No network retrieval was used for that restoration. Actual explicit source arguments identify the restored files read by the checker. The unchanged `source.path` fields in scientific receipts are historical logical labels, not the actual read paths when an explicit source override is supplied.

The later v2 record package retains these prior scientific payload bytes and adds the replay records; the v2 archive itself was not scientifically replayed. It is identified as `v71-r6-positive-certificates-and-controls-20261026.zip`, 2,729,454 bytes, SHA-256

`7a71e382afb1bca414b107545c8b410ff5b8b9437a6b76aaf67f119937f218a3`.

Its top directory is `r6_positive_certificates_and_controls/`. Within it, `replay/VERIFICATION_ADDENDUM.txt` has SHA-256

`252cfe0914383b376dcfe8cb79da03cf37b8f87326d3abd913153ed7ab0c5beb`.

These version-specific identities distinguish the later records from the immutable v1 release referenced in the opening evidence index. The scientific replay was adopted; publication binding of the v2 package is a separate decision. We do not infer an unrecorded public tag or claim that the old v1 tag identifies the v2 bytes.

**Retained replay records and source correspondence.** The package's `replay/records/` includes the root adoption, input freezes, restoration receipt, execution audit and exact argument records. `replay/runs/` retains the four stage records and the 52 new qualification receipts. `replay/TRANSFORMATION_LEDGER.json` maps original record lengths and hashes to their location-sanitized public derivatives. Only private location prefixes were replaced; embedded hashes continue to identify original records and are resolved through that ledger. The original science records in `c1/` and `c2/` were not rewritten.

All 50 actual replay malformed gzip inputs match the existing public controls in both compressed and decompressed bytes. `replay/QUALIFICATION_REPLAY_INPUT_MAP.json` binds those reused controls to the new receipts. Forty-eight receipts directly contain their input hash. The two duplicate-key cases fail during parsing before that receipt field is assigned, so their hash field remains null. Their raw inputs, frozen driver construction and archive/summary bindings retain their provenance; the null fields are not replaced with invented values. The document source ZIP contains none of these experiment bundles, private logs or upstream source bodies.

**Resource and execution limits.** The four replay stages had wall budgets of 180, 180, 180 and 300 seconds, respectively, with CPU limits enforced *per process* by `RLIMIT_CPU`; they were not aggregate hard CPU limits across the process tree. Each qualifier child checker had at most 180 seconds of wall and per-process CPU time within the driver's remaining wall budget. The 64-MiB per-file limit was distinct from the sampled 1-GiB process-tree RSS, 128-MiB stage-output and 512-MiB total-output caps.

All 70 supervisor samples were below their configured caps. RSS and directory sizes were sampled, not continuously measured peaks or hard memory isolation. The configured interval was 0.25 seconds and the largest observed gap was about 0.2941 seconds. Stage and total directory scans were sequential, so their values were neither simultaneous nor necessarily monotonic during compression and deletion. These diagnostics establish no performance advantage.

The four outer scientific invocations used `-I -S -B`. The adapter started under ordinary Python, and the frozen qualifier's 52 child checker invocations did not inherit those flags. The cleaned environment therefore does not establish fully hermetic child startup. The single-attempt statement concerns the recorded reservations, ordered invocations and owned run roots, not a machine-wide execution census.

**Checking the distribution and reading the result.** From the v2 package root, the two commands

```
python3 -I -S -B verify_integrity.py
python3 -I -S -B verify_replay_integrity.py
```

perform static integrity and replay-record-binding checks. They do not execute a scientific checker, qualifier, source restorer or GUI action. Their own zero-replay counters do not erase the separately recorded historical replay. The package's `FUTURE_REPLAY_COMMANDS.txt` describes optional future scientific interfaces; this manuscript reports the completed historical run only.

The added evidence is existing-host rechecking of the exact public v1 scientific inputs and their direct-source correspondence. It establishes no fresh-host or external reproduction, fully hermetic execution, maximal winning region, negative realizability proof, closure of the eight previous exported-game UNKNOWNs, exclusive GUI cause, deployed handoff or performance advantage. Packaging the v2 records performed no further source restoration, scientific rerun or public action.

# AF. Historical same-solver diagnostic at exact B

**Question and scope.** This historical v5 diagnostic asked whether releasing the first-change SYS duty alters the answer of one fixed released raw four-fixed-point solver at the exact retained query B of AC.6. The unchanged arm returned *false*. The release arm entered the solver but stopped at the predeclared raw-output capture cap before returning a Boolean; its answer is *UNKNOWN*. That interrupted comparison left the effect unresolved; Section AG reports a distinct completed pair. The result is separate from AC.2's completed same-*strict*-solver comparisons and from the historical public GUI observations.

## AF.1 Fixed input, intervention and pre-call checks

Both arms use the 2,026-byte query from AC.6, SHA-256

`212309df489c3ddc7bf7ec6a750aed2d0da4fb7e26995e5ee3a0405c9897e1d5`.

It consists of the 1,902-byte ordinary projection of the published update and the fixed 124-byte point-initializer appendix. The joint initial point is obstacle $(3, 5)$, `obsWait=true`, robot $(5, 5)$, helper $b = 0$; its source ID is 2404. The ENV and SYS initial predicates are separately nonempty, and initialization retains universal ENV and existential SYS quantification. No old initializer or controller memory is transferred. AC.1 and AC.6 specify the original archive, projection and endpoint selection.

Each arm constructs its model through the same pinned released syntactic reduction and default translators in a private NoIDE provider. The sole intervention replaces the complete SYS relation in the typed quadrant $b = 0, b' = 1$ by unrestricted typed responses. It adds edges, removes none, and changes no edge outside that quadrant. The unchanged arm has zero delta. In both arms $S_{00}, S_{10}, S_{11}$ are preserved, with $S_{11} = \top$. The full ENV relation, typed domains, point initializers and ordered justices are fixed. The ENV justice is $\neg b$; the two SYS justices remain visits to $(3, 7)$ and $(4, 8)$ in that order. This intervention does not invoke the separate semantic-correction class or replace the complete translation pipeline.

Before either call, the recorded comparisons bind the full relations, all helper quadrants, source correspondence, point initialization, ordered justices and exact-B baseline to the adopted saved predicates. There are 61 successful native comparison records per arm, including checks against its adopted arm before and after solver construction. The recorded dispatch uses JTLV's Java backend, `SINGLE_FUNC` ENV/SYS relations and both simultaneous-conjunction flags false. Both calls use Java 17.0.19 and the same released `GR1SeparatedGame` class, whose SHA-256 is

`6ce9b49dc638e4b03de10662c5dcd28ef470019f25cb7e2151afac77d0d7b502`.

The later independent audit checked saved guard records, source ordering, runtime identities, hashes and manifests. It did not recompute these predicates or run a solver. A saved-predicate cross gate passed before the two serial solve attempts.

## AF.2 Completed answer and interrupted output

| Arm | Answer | Recorded completion |
|---|---|---|
| Unchanged | false | Solver returned; complete native raw-output capture. |
| Typed $S_{01}$ release | UNKNOWN | Call entered; no Boolean and no return event; raw-output cap interrupted capture. |

The unchanged status is `PASS_ONE_RAW_FOUR_FP_RESULT`. Its Boolean is `realizable=false`, and a call-returned event is present. The complete raw capture has 62,483,773 bytes. The release record has status `UNKNOWN_DIAGNOSTIC_CAPTURE`; the `realizable` field is absent. Its exception is `RAW_STDOUT_CAP` during `ONE_RAW_FOUR_FP_CALL_BEGIN`, inside the raw solver call.

The raw-output limit was 64 MiB (67,108,864 bytes). The saved release prefix contains 67,108,819 raw bytes; the next whole 54-byte diagnostic record exceeded the 45 remaining bytes. Independent decoding verified the complete unchanged stream and every recorded release-prefix byte, respectively. Their raw SHA-256 values are

| | |
|---|---|
| Unchanged | `c2c67f96ced727d211f90d69dec3b74a3ef5b60f3fb6b06ee44d4c6b1e193f17` |
| Release prefix | `2b27620bcbfc9c9d3a4dd6830ad74cb0c21defe34a4d8104b7008cbc24b455c4` |

The lossless encodings have 4,219 and 4,220 bytes, respectively, with 78 records each. A complete final encoding frame proves integrity of the saved prefix, not completion of the interrupted solver output. Likewise, `PROCESS_FINISHED`, completed teardown and complete supervisor pipe draining describe cleanup and transport; none supplies the missing release-arm Boolean. A counter set on entry to the winning-region routine does not prove that routine returned.

The fixed solve bounds were 300 seconds wall time, 240 seconds CPU *per process*, 512 MiB Java heap, sampled 1 GiB process-tree RSS, and 64 MiB stream caps. The raw-output cap was the observed interruption; no time- or memory-infeasibility conclusion follows. Sampled RSS and per-process CPU are not hard aggregate process-tree peak guarantees. No controller extraction or native strict-solver call occurred.

### AF.3 Retained history and interpretation limits

The earlier cross-gate attempt in preparation v4 remains `UNKNOWN_STAGE`, with exit code 2. Its native output, manifest and launch receipt are preserved. It was not relabeled successful when a later saved-predicate cross revalidation in v5 passed. The legacy v4 solve slots were never entered. Only the subsequent two v5 raw calls were entered, in the fixed order unchanged then release, once each. The history contains five model constructions: one preliminary model, two original gate models and the two solve models. Of the two entered raw calls, only one returned. No result was selected away. This protocol ended without a retry, alternative arm or cap increase.

This historical observation is false/UNKNOWN, not a completed two-Boolean comparison. It establishes neither the unreturned answer nor an intervention magnitude, eventual termination, global equality of the raw region with $W_F$, $K$ or all-history Kind, or correctness of an extracted strategy. The private headless provider is not a recovered historical GUI game or dispatch trace. This result does not isolate a solver-only cause of the public A/B negatives, demonstrate the complete corrected pipeline, refute source-kind existence, or show an installed controller handoff. Earlier Obstacle fixed-point, Patrol full-helper and cohort-159 UNKNOWNs remain unchanged.

**Evidence locator and publication boundary.** The stable evidence identifier is `AF-R7-SAME-SOLVER-DIAGNOSTIC`, with this section as its logical appendix locator. The accompanying release [18] identifies its separate experimental archive, `results/archives/same-solver-diagnostic-20261026-v1.zip`, and diagnostic guide. The guide gives the exact package identity and checking scope; earlier public tags do not identify these records. The document source ZIP includes this account, not private experimental logs, runtime inputs or an executable diagnostic bundle. No public replay, fresh-host reproduction or additional scientific execution is claimed here.

# AG. Completed same-observer pair at exact B

The sole typed-$S_{01}$ release changed the returned answer of the fixed raw four-fixed-point solver at exact query B from *false* to *true*. Both fresh runs completed under one predeclared observer and finite budget. This is a new paired experiment, separate from AF's historical false/UNKNOWN diagnostic, AC.2's strict-region comparisons, and the public-mode observations in AC.5–AC.7.

## AG.1 Exact query and sole intervention

The input is `B_NEW_POINT_R2404`: the same 2,026 bytes pinned in AF.1, comprising the 1,902-byte ordinary projection of the updated source and the 124-byte point-initializer appendix. Its SHA-256 is

`212309df489c3ddc7bf7ec6a750aed2d0da4fb7e26995e5ee3a0405c9897e1d5`.

The joint point is obstacle $(3, 5)$, `obsWait=true`, robot $(5, 5)$, and helper $b = 0$, with source ID 2404. Both separate initial predicates are nonempty; initialization retains universal ENV and existential SYS quantification. No old initializer or controller memory is transferred. This is the updated point query, not query A or a whole-update-region calculation. AC.1 and AC.6 retain the source and endpoint binding.

Each arm constructs a fresh model with the same pinned released syntactic reduction and default translators through the private NoIDE provider. The intervention changes only the complete SYS relation on the typed quadrant $b = 0, b' = 1$: it permits every typed response there. It adds edges, removes none, and changes no edge outside $S_{01}$. The unchanged arm has zero delta. The other quadrants $S_{00}, S_{10}, S_{11}$ are preserved; typed $S_{11}$ is already $\top$. The full ENV relation, typed domains, six-field metadata and bit order, point initializers and ordered justices are fixed. The single ENV justice is $\neg b$; the two SYS justices are visits to $(3, 7)$ and $(4, 8)$ in that order. Neither source/monitor reset nor $G \wedge A$ normalization is applied. The intervention does not select the separate semantic-correction class or substitute a corrected public translation pipeline.

Both arms use Java 17.0.19, the pinned `net.sf.javabdd.JTLVJavaFactory` backend, and the same released `GR1SeparatedGame` class with SHA-256

`6ce9b49dc638e4b03de10662c5dcd28ef470019f25cb7e2151afac77d0d7b502`.

The observed ENV/SYS dispatch is `SINGLE_FUNC`; both simultaneous- conjunction/abstraction flags are false. The recorded path uses `yieldStatesOrig`/`predSingleTrans`, without decomposed transition partitions. Each arm makes one raw `checkRealizability` call. No native strict-solver call or controller extraction occurs.

## AG.2 Semantic checks and serialized-byte scope

All 39 recorded raw/typed in-model comparisons pass per arm: ten for the exact original B baseline, nine for preserved components, and ten each for the adopted arm before and after solver construction. These checks bind the complete relations, helper quadrants, initialization, domains and ordered justices to the original adopted gate. They support the specified intervention and preserved components within each newly generated model. The later saved-record audit checked the executed comparisons and their provenance; it did not independently reconstruct saved predicates or generate another model.

All four new exported `model.json` files are byte-identical to their corresponding old-v5 solve exports. Both baseline exports and the unchanged arm also match their original native-gate exports

byte for byte. The release arm differs from its original gate only in the SHA values of the serialized `domain` and `domain_ENV_next` descriptors. Descriptor sizes, node counts, flags, support and other metadata agree. Thus the successful complete in-model raw/typed comparisons support original-gate semantic identity; universal serialized-byte identity is not claimed. Static file hashing is a separate provenance check, not a replacement for those executed comparisons.

### AG.3 Completed runs and predeclared bounds

The two fresh processes ran in the fixed order unchanged, then release. The second launch required completion of the first, regardless of its Boolean value. Each process generated one model, entered one raw call and returned from that call. There was no retry or selection of a preferred outcome.

| Arm | Boolean | Completion |
|---|---|---|
| Unchanged | false | Normal call return, OS exit 0, normal teardown, full capture. |
| Typed $S_{01}$ release | true | Normal call return, OS exit 0, normal teardown, full capture. |

No limit or stop signal occurred. Finalized native counters and the ordered launch, return and cleanup records agree with these completions.

Both arms used the same lossless observer and the same limits declared before either new run: 64 GiB (68,719,476,736 bytes) raw output, 64 MiB (67,108,864 bytes) each for encoded stdout and stderr, 300 seconds wall time, 240 seconds CPU per process, 512 MiB Java heap, and sampled 1 GiB process-tree RSS. Sampled RSS and per-process CPU do not establish hard aggregate process-tree peaks. The new raw ceiling belongs to this distinct pair; AF's 64-MiB protocol and interrupted release remain unchanged. The new release result is paired with the new unchanged run, not with AF's earlier unchanged run.

### AG.4 Complete capture integrity

| Retained stream measure | Unchanged | $S_{01}$ release |
|---|---:|---:|
| Raw bytes | 62,483,773 | 3,578,789,826 |
| Encoded bytes | 4,219 | 7,899 |
| Encoding records | 78 | 146 |
| Repeated diagnostic lines | 1,157,084 | 66,273,862 |
| Literal bytes | 1,237 | 1,278 |

Each encoding contains exactly one terminal end, with no unparsed or unflushed bytes. Independent totals agree with both the native receipts and the terminal records. The raw SHA-256 values are

| | |
|---|---|
| Unchanged | `c2c67f96ced727d211f90d69dec3b74a3ef5b60f3fb6b06ee44d4c6b1e193f17` |
| Release | `45c2dca9246b60e089163977156674f931ff6e8da7971128d8d36b0246ec8a67` |

The corresponding encoded SHA-256 values are

| | |
|---|---|
| Unchanged | `71f5e374f4c5ad2fa5a6be8da8015be16fffff63abe9151ef3b010ef3f749c7e` |
| Release | `7b66ac436a64519eca7fd618047dbd2dddc26dff6edf4b56980e3e0db8415ebc` |

The coordinating task executed the reviewed finite streaming QA once per arm under separate pinned allowances. Each guardian/worker completed with exit 0, confirmed cleanup and no limit, error or signal; all 24 listed completion conditions passed. The streaming check hashed every admitted raw byte in order and matched both raw and encoded digests. The independently reviewed code and

actual execution evidence establish complete coverage of 3,641,273,599 raw bytes. Neither expanded raw stream was written to disk, and the independent reviewer did not repeat the raw expansion. Tiny encoded storage alone is not the integrity argument. These are saved-output checks, not additional solver calls or a performance comparison.

Native stderr is nonempty and retained: 288 bytes/four lines unchanged, 7,913 bytes/108 lines for the release. Every byte was classified against the pinned loaded JTLV default garbage-collection format, with sequential counters and consistent cumulative times. The relevant source and loaded class identities were checked; no non-GC stderr was ignored.

### AG.5 History, inference and evidence boundary

| Diagnostic lineage | Models | Calls entered | Calls returned |
|---|---|---|---|
| Historical v5 (AF) | 5 | 2 | 1 |
| Distinct completed pair (AG) | 2 | 2 | 2 |
| Cumulative | 7 | 4 | 3 |

AF's earlier v4 cross-gate failure and v5 false/UNKNOWN observation remain in their original history. All new scientific slots and the two separate QA allowances were consumed once; the remaining model/call slots are zero and no further execution or retry is authorized.

The supported conclusion is a positive local intervention effect: at this exact B and fixed configuration, releasing only typed $S_{01}$ changes the raw-solver Boolean from false to true. It does not identify an exclusive cause or live-model identity for the historical GUI negatives, establish universal raw/strict/Kind equivalence, validate the full corrected public pipeline or extracted/transferred controller, or demonstrate operational gate benefit, installation, incidence or performance gain. The eight other Obstacle exported-game audits, Patrol full-helper attempts and cohort-159 UNKNOWNs retain their distinct statuses and reasons.

The protocol identifier is `v71-r9-distinct-same-observer-exact-B-pair-v1`; it is an evidence identifier, not an appendix label. The root scientific adoption and independent result review bind the exact inputs, native run records, streaming-QA receipts and byte manifests. The accompanying release [18] identifies the separate experimental archive `results/archives/r9-completed-same-observer-pair-20261026-v1.zip` and its completed-pair guide, which records the exact package identity and checking scope. The document source ZIP contains this account and its build inputs, not private logs, runtime inputs or an executable scientific bundle. No publication, fresh-host reproduction or new scientific execution is performed by this document revision.

## Acknowledgments

OpenAI Codex [17] contributed generated and revised prose, formal arguments and internal proof checks, case/design exploration, experiment and checker code, protocol preparation, execution orchestration, audits, result analysis, and artifact preparation. This involvement extends through the main paper and this supplement; it was not limited to language editing. Separate agent reviews were internal checks within this workflow. The all-history source-kind additions in Section AD received a coordinating Codex review and a separate Codex proof review against the primary definition and source equations; they added no scientific execution. Section AG incorporates a separately authorized two-process experiment and finite saved-output QA, with separate internal agent review and coordinating scientific adoption. This document revision performed no additional scientific execution. Claude Fable 5.1 and ChatGPT Pro (whose underlying model was not exposed) supplied main-paper-only AI review comments without the supplement or earlier reviews. Both reviewed successive Codex manuscripts and an alternate candidate independently authored by Claude Code Opus 5.5. Fable separately reviewed the supplement and the primary Kind paper without rescoring the earlier main-paper-only review. They do not establish independent human review, formal proof-assistant verification, or external reproduction. The mathematical arguments, source/certificate checks, controlled executions, and recorded failures have the specific scope stated in each section; automated agreement alone does not validate a claim beyond that scope.

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